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Preface

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1 Introduction

1.1 Overview

This document is a normative supplement to the UAF DMM document (c4i/19-06-16).

This document specifies a UAF profile to enable practitioners to express architectural model elements and organize them in a set of domains, model kinds, and view specification (specified in the UAF DMM) that support the specific needs of end users in defense and commercial industry.

UAFP 1.1 defines a set of stereotypes and model elements and relationships to satisfy the requirements of the UPDM 3.0 RFP and the UAF DMM. The profile specification documents the language architecture in terms of UML profiling mechanism.

A number of UAFP stereotypes inherit from SysML stereotypes where reuse of SysML semantics is necessary. The reusable portions of the SysML specification are not included directly in the specification but are made explicit through the stereotype inheritance.

2 Additional Information

2.1 Language Architecture

The UAFP specification reuses a subset of UML 2.5.1 and SysML 1.5 and provides additional extensions needed to address requirements in the UPDM 3.0 RFP Mandatory Requirements. Those requirements form the basis for this document. This document describes the language architecture in terms of the UML 2.5.1 and SysML 1.5 parts that are reused and the defined UML 2.5.1 extensions; and specifies how to implement UAFP. This clause explains design principles and how they are applied to define the UAFP language architecture.

2.2 Core Principles

The fundamental design principles for UAFP are:

- **Requirements-driven:** UAFP is intended to satisfy the requirements of the UPDM 3.0 RFP Mandatory Requirements.
- **UAF Domain Metamodel (DMM) driven:** The DMM served as a foundation for profile development.
- **Reuse of existing specifications:** UAFP reuses UML/SysML wherever practical to satisfy the requirements of the UAFP 3.0 RFP and leverage features from both UML and SysML to provide a robust modeling capability. Consequently, UAFP is intended to be relatively easy to implement for vendors who support UML 2.x and SysML 1.x.
- **Compliance levels:** UAFP has a single compliance level based upon a combination of the reuse of UML and SysML elements. It is expected that the views that are created as result of this profile have frames that reflect the underlying SysML diagram type that is used as the basis for the view. It also expected that the graphical notation used to display elements within those views correspond to the standard SysML graphical notation of the SysML/UML metaclass that the stereotype extends.
- **Interoperability:** UAFP inherits the XMI interchange capability from UML. The UAFP specification reuses a subset of UML 2.5.1 and provides additional extensions needed to address requirements in the UPDM 3.0 RFP Mandatory Requirements.

2.3 Representing Stereotype Constraints

The UAF Profile uses an enhanced standard notation to represent metaconstraints graphically in the UAF profile diagrams to improve readability of the UAF Profile specification and overcome limitations of being unable to visualize constraints diagrammatically in UML.

The metaconstraints appears in the UAFP specification diagrams for visualization purposes only, however the representation in the XMI is as a UML constraint, specified in structured English. These constraints are implementable in a tool, by OCL for example.

A simple UML profile defines these metaconstraints.

The following subsections detail the metaconstraint profile definition within the UAF profile.

2.3.1 Metaconstraint dependency

«metaconstraint» is a stereotype that extends the Dependency metaclass. It is used to specify constrained elements within the profile.

A sample of the «metaconstraint» dependency is a diagram for stereotype extending the Dependency metaclass.

MapsToCapability is a UAFP stereotype that extends Abstraction (a type of Dependency in UML). The constraint on this stereotype is that its client end must be stereotyped by an Activity (which is abstract) and its supplier end must be stereotyped by a Capability. But as

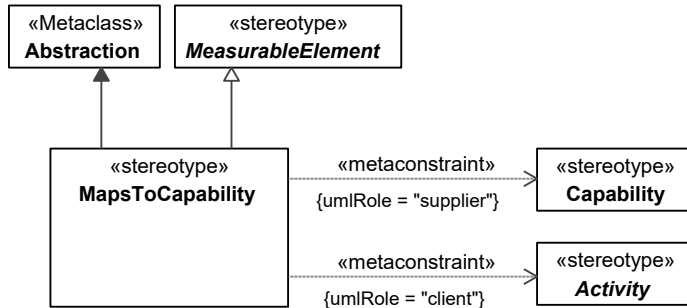


Figure 2:1 – MapsToCapability Stereotype

With the metaconstraint dependency added to the diagram (see Figure) which shows that MapsToCapability is a stereotype extending the Abstraction metaclass, that inherits the properties of a MeasurableElement and is used for modeling a relationship between an Activity (or its specializations) and a Capability (or its specializations). A Dependency stereotyped MapsToCapability must have its values for the client property stereotyped as an Activity, and its values for the supplier property must be stereotyped Capability.

Note – When stereotype extends Connector, the stereotype property umlRole has values "end[0].role" and "end[1].role." For example:

This is done because Connector has no direct "linkage" to the connected element; it links to the Connector Ends, which references the linked element. So, end[n] gives the reference to the ConnectorEnd, and role gives the reference to the linked element.

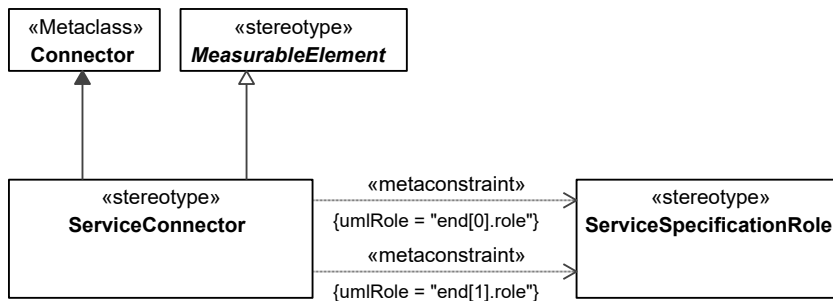


Figure 2:1 – Connector Extension

2.3.2 Metarelationship dependency

«metarelationship» is a stereotype for dependency, showing that certain domain concepts will be implemented using regular UML relationships.

For example: A Capability may depend on other Capabilities or be subtype of a Capability, but this concept cannot be visualized on the diagram:

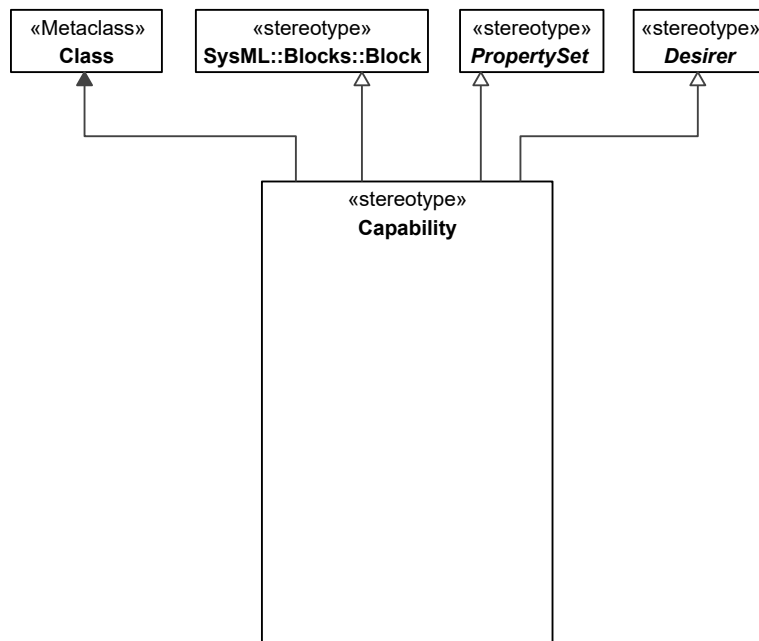


Figure 2:3 – Capabilities Generalization

We are using the «metarelationship» dependency to visualize the dependency and the generalization concept.

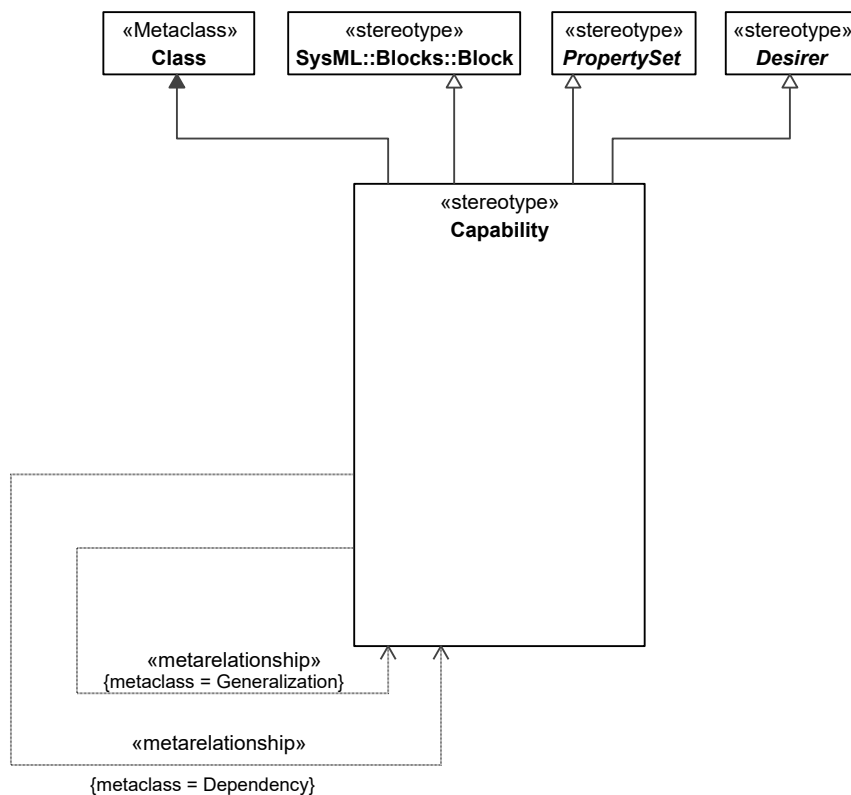


Figure 2:4 – Visualizing «metarelationship»

This diagram should be read as follows:

Capability may have other Capabilities related to it, using the UML Dependency metaclass and it may have sub types of Capabilities related to it, using the the UML Generalization metaclass.

The «metarelationship» dependency will appear only in the specification diagrams, but not the profile XMI.

2.3.3 Stereotyped relationship dependency

Although the «metarelationship» dependency creates a good way to show the constrained ends of the stereotyped relationship, it also creates some overhead when showing the relationship between two stereotypes.

For example, Figure 2:2 below shows that elements of subtype Achiever have a stereotyped relationship called AchievedEffect with elements of type ActualState.

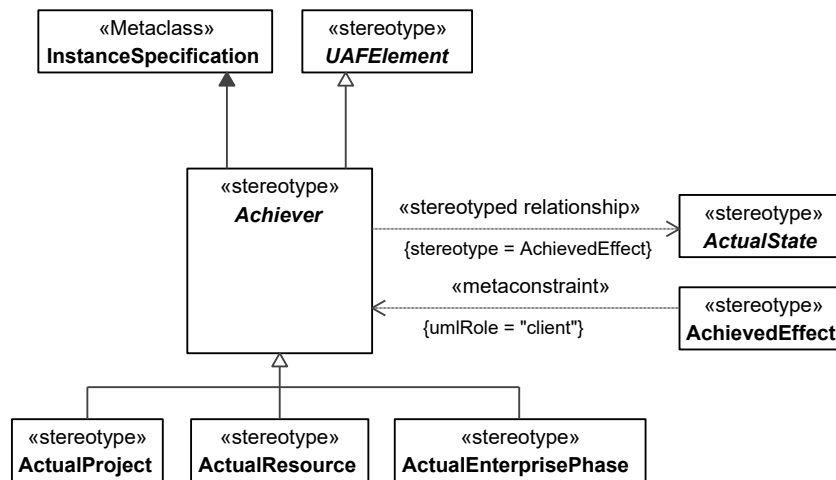


Figure 2:2 – Use of the AchievedEffect «stereotyped relationship» dependency

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3 UAF Stereotypes

UAFP imports the entire SysML profile. This is intended to provide more seamless integration with system modeling using SysML and to be able to fully leverage the capabilities of SysML in UAFP. An example of this is the integration of Requirements into the UAFP and also the use of Parametric Diagrams and integration of elements based upon instance specifications to allow the assessment of measures within an architecture developed using UAFP.

3.1 UAF

UAF is top level profile root.

3.1.1 UAF::Dictionary

Stakeholders: Architects, users of the architecture, Capability Owners, Systems Engineers, Solution Providers.

Concerns: Definitions for all the elements in the architecture, libraries of environments and measurements.

Definition: Presents all the elements used in an architecture. Can be used specifically to capture:

- elements and relationships that are involved in defining the environments applicable to capability, operational concept or set of systems.
- measurable properties that can be used to support analysis such as KPIs, MoEs, TPIs etc.

Alias

Package: Dictionary

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Comment

Description

A metamodel Artifact used to define an alternative name for an element.

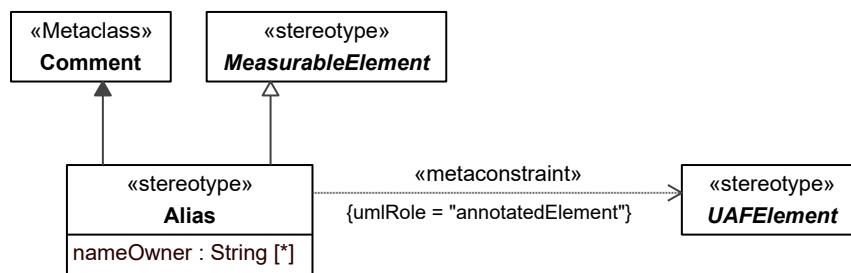


Figure 3:1 - Alias

Attributes

nameOwner : String[*] Someone or something that uses this alternative name.

Constraints

[1] Alias.annotatedElement Value for the annotatedElement metaproperty must be stereotyped by the specialization of «UAFElement».

Definition

Package: Dictionary

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Comment

Description

A comment containing a description of an element in the architecture.

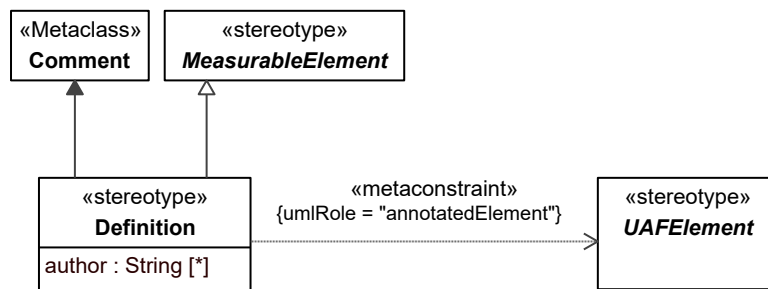


Figure 3:2 - Definition

Attributes

`author : String[*]` The original or current person (architect) responsible for the Definition.

Constraints

[1] Definition.annotatedElement Value for the annotatedElement metaproperty must be stereotyped by the specialization of «UAFElement».

SameAs

Package: Dictionary

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Dependency

Description

A dependency relationship that asserts that two elements refer to the same real-world thing.

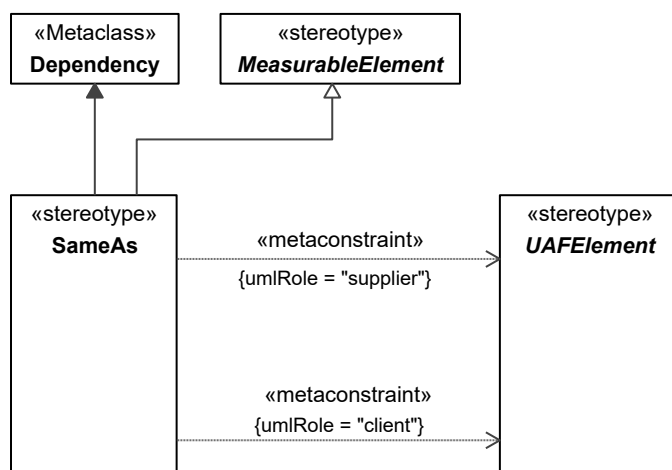


Figure 3:3 - SameAs

Constraints

[1] SameAs.client Values for the client metaproperty must be stereotyped by the specialization of «UAFElement».

- [2] SameAs.supplier Values for the supplier metaproperty must be stereotyped by the specialization of «UAFElement».

3.1.2 UAF::Parameters

ActualCondition

Package: Parameters

isAbstract: No

Generalization: [ActualPropertySet](#)

Extension: InstanceSpecification

Description

An actual situation with respect to circumstances under which an OperationalActivity, Function or ServiceFunction can be performed.

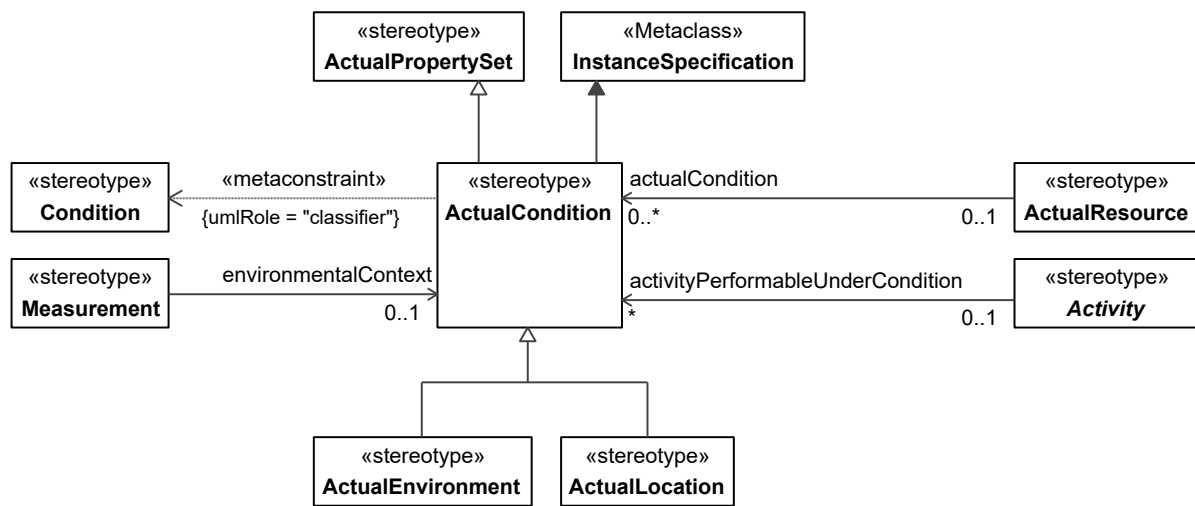


Figure 3:4 - ActualCondition

Constraints

- [1] ActualCondition.classifier Value for the classifier metaproperty has to be stereotyped «Condition» or its specializations.

ActualEnvironment

Package: Parameters

isAbstract: No

Generalization: [ActualCondition](#)

Extension: InstanceSpecification

Description

Actual circumstances of an Environment.

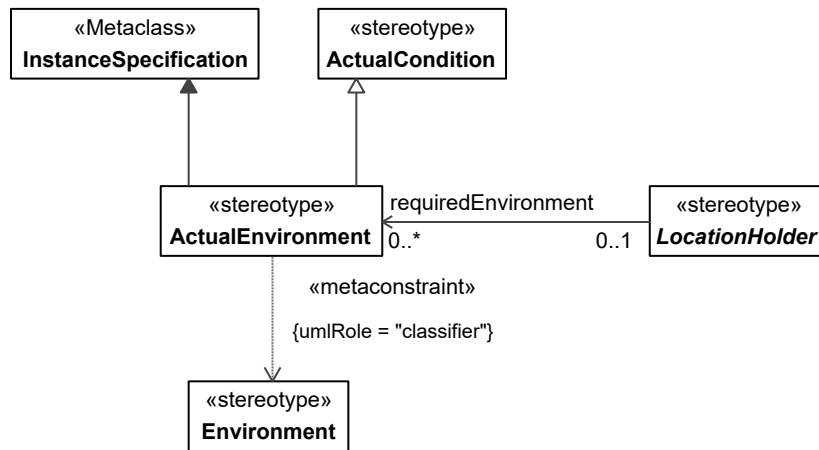


Figure 3:5 - ActualEnvironment
Constraints

[1] ActualEnvironment.classifier Value for the classifier metaproperty has to be stereotyped «Environment» or its specializations.

ActualLocation

Package: Parameters

isAbstract: No

Generalization: [ActualCondition](#)

Extension: InstanceSpecification

Description

A physical location, for example using text to provide an address, Geo-coordinates, etc.

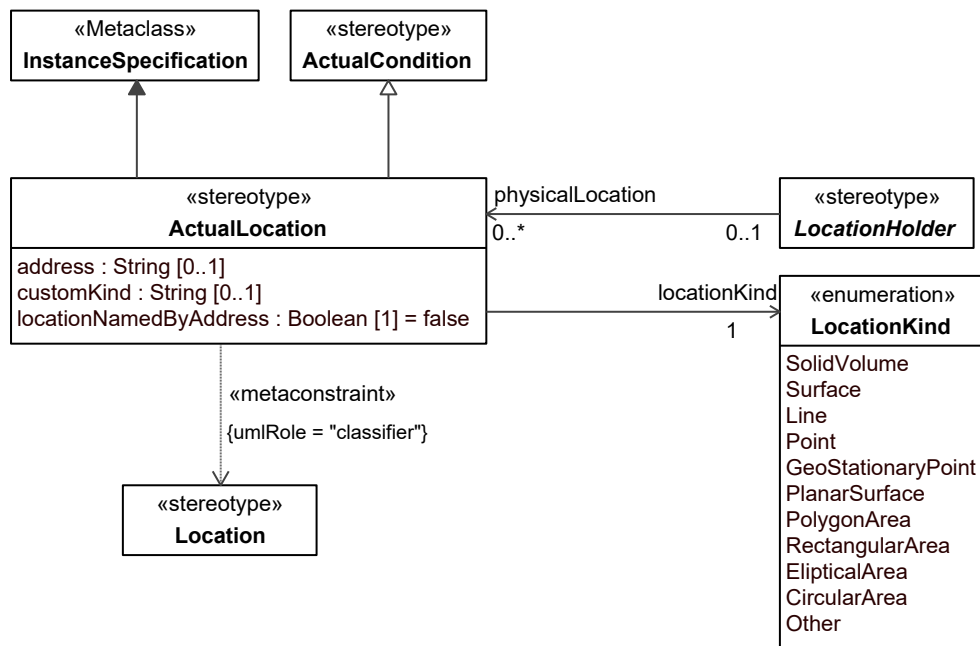


Figure 3:6 – ActualLocation

Attributes

address : String[0..1]	String describing the address of the ActualLocation, i.e. "1600 Pennsylvania avenue", "The White House"
customKind : String[0..1]	String describing a location kind that is not in the LocationKind enumerated list
locationNamedByAddress : Boolean[1]	Boolean that indicates if the ActualLocation address is embedded in the ActualLocation name. By default = false.

Associations

locationKind : LocationKind[1]	Enumerated value describing the kind of ActualLocation.
--------------------------------	---

Constraints

[1] ActualLocation.classifier	Classifier metaproperty value must be stereotyped «Location» or its specializations.
-------------------------------	--

ActualMeasurement

Package: Parameters

isAbstract: No

Generalization: [ActualState](#)

Extension: Slot

Description

An actual value that is applied to a Measurement.

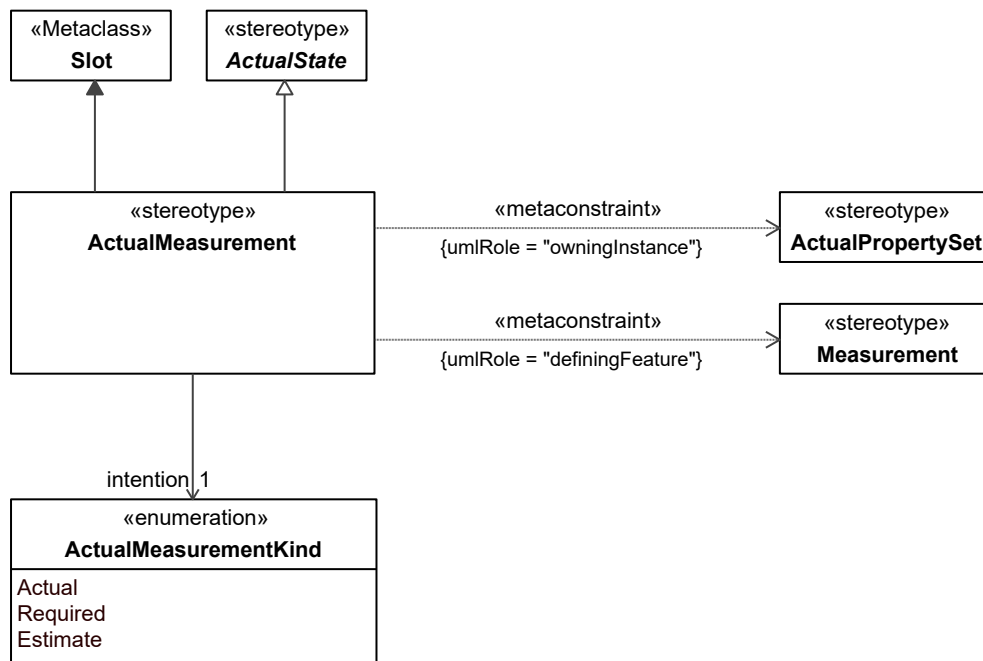


Figure 3:7 – ActualMeasurement

Associations

intention : ActualMeasurementKind[1]	Enumerated value describing the intent of the ActualMeasurement.
--------------------------------------	--

Constraints

- [1] ActualMeasurement.definingFeature Value for the definingFeature metaproperty must be stereotyped «Measurement» or its specializations.
- [2] ActualMeasurement.owningInstance Value for the owningInstance metaproperty must be stereotyped «ActualPropertySet» or its specializations.

ActualMeasurementKind

Package: Parameters

isAbstract: No

Description

Enumeration of the possible kinds of ActualMeasurement. Its enumeration literals are:

- Actual - Indicates that the ActualMeasurement associated with the ActualMeasurementKind is a realworld value.
- Required - Indicates that the ActualMeasurement associated with the ActualMeasurementKind is a value that is expected to be achieved.
- Estimate - Indicates that the ActualMeasurement associated with the ActualMeasurementKind is an estimate of a realworld value.

ActualMeasurementSet

Package: Parameters

isAbstract: No

Generalization: [ActualPropertySet](#)

Extension: InstanceSpecification

Description

A set of ActualMeasurements.

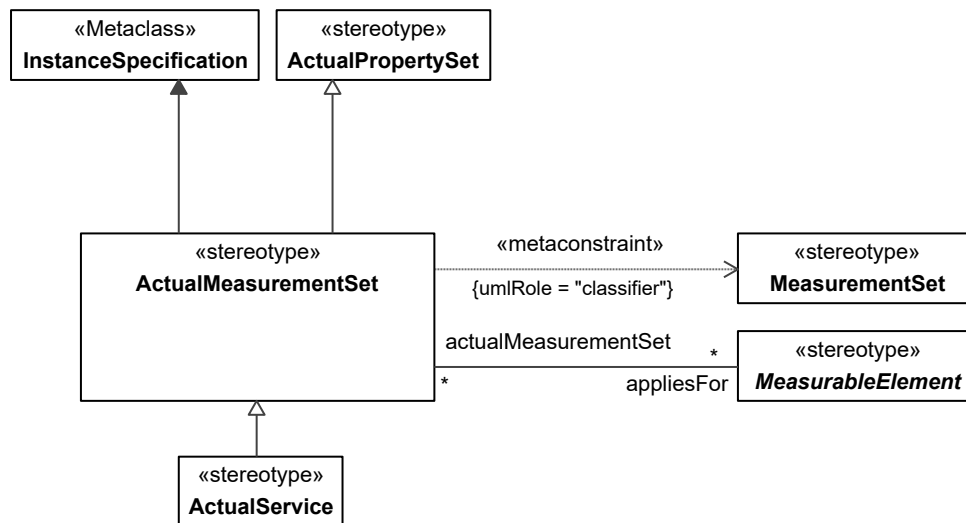


Figure 3:8 - ActualMeasurementSet

Associations

appliesFor : MeasurableElement[*] Relates the ActualMeasurementSet to the elements that are being measured.

Constraints

- [1] ActualMeasurementSet.classifier Classifier metaproperty value must be stereotyped «MeasurementSet» or its specializations.

[2] ActualMeasurementSet.slot

Value for the slot metaproperty must be stereotyped «ActualMeasurement» or its specializations.

ActualPropertySet

Package: Parameters

isAbstract: No

Generalization: [ActualState](#)

Extension: InstanceSpecification

Description

A set or collection of Actual properties.

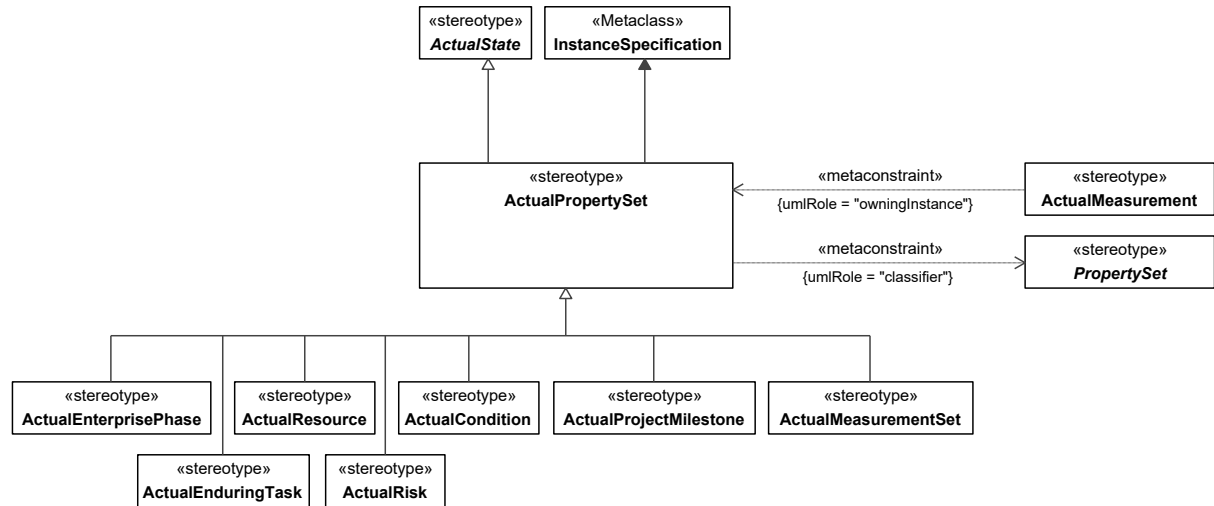


Figure 3:9 – ActualPropertySet

Constraints

[1] ActualPropertySet.classifier Value for the classifier metaproperty must be stereotyped by the specialization of «PropertySet».

Condition

Package: Parameters

isAbstract: No

Generalization: [PropertySet](#), ValueType

Extension: DataType

Description

A type that defines the Location, Environment and/or GeoPoliticalExtent.

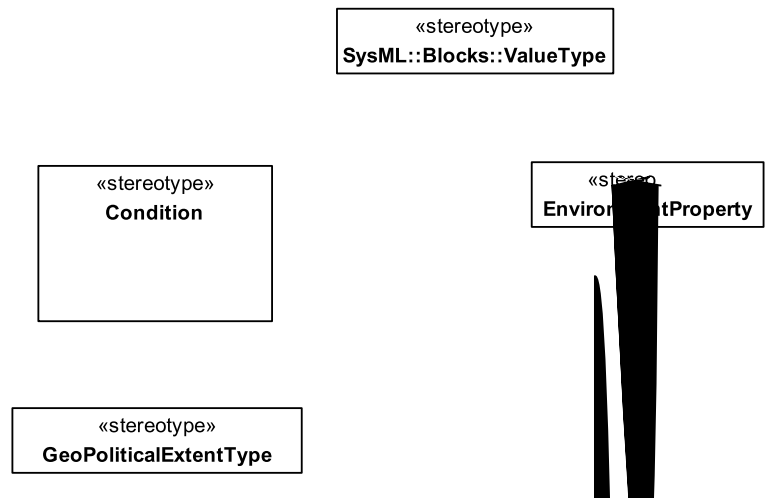


Figure 3:10 - Condition

Environment

Package: Parameters

isAbstract: No

Generalization: [Condition](#)

Extension: DataType

Description

A definition of the environmental factors in which something exists or functions. The definition of an Environment element can be further defined using EnvironmentKind.

Figure 3:11 - Environment

Associations

kind : EnvironmentKind[1] Captures the kind of Environment.

EnvironmentKind

Package: Parameters

isAbstract: No

Description

Enumeration of the possible kinds of Environment. Its enumeration literals are:

- **TerrainType** - Indicates that the Environment associated with EnvironmentKind captures a kind of terrain used to describe the terrain state of an environment at a particular time (e.g., muddy, frozen ground, deep snow, etc.).
- **WeatherConditions** - Indicates that the Environment associated with EnvironmentKind captures a kind of weather condition (e.g., Typhoon, Hurricane, Very Hot, Humid etc.).
- **LightConditions** - Indicates that the Environment associated with EnvironmentKind captures a kind of light condition (e.g., broad daylight, dusk, moonlit, etc.).
- **CBRNEEnvironment** - Indicates that the Environment associated with EnvironmentKind is of a Chemical, Biological, Radiological or Nuclear (CBRN) kind.
- **SituationType** - Indicates that the Environment associated with EnvironmentKind captures a kind of situation used to describe the types and levels of threat (e.g., Corrosive, Fire, Smoke, Peaceful etc.).

EnvironmentProperty

Package: Parameters

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Property

Description

A property of an Environment that is typed by a Condition. The kinds of Condition that can be represented are Location, GeoPoliticalExtentType and Environment.

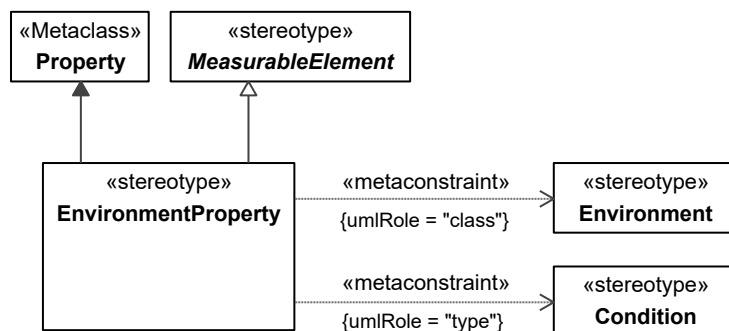


Figure 3:12 - EnvironmentProperty

Constraints

- | | |
|---------------------------------|--|
| [1] EnvironmentalProperty.class | Value for the class metaproperty must be stereotyped «Environment» or its specializations. |
| [2] EnvironmentalProperty.type | Value for the type property must be stereotyped «Condition» or its specializations. |

GeoPoliticalExtentType

Package: Parameters

isAbstract: No

Generalization: [ResourceExchangeItem](#), [OperationalExchangeItem](#), [Condition](#)

Extension: DataType

Description

A type of geospatial extent whose boundaries are defined by declaration or agreement by political parties.

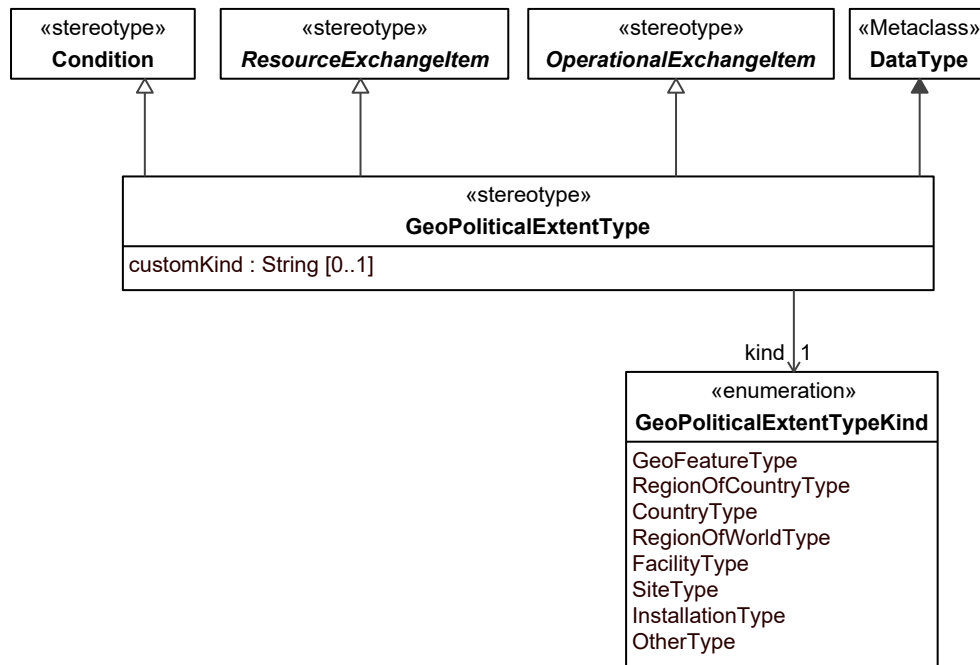


Figure 3:13 – GeoPoliticalExtentType

Attributes

`customKind : String[0..1]` Captures the kind of `GeoPoliticalExtentType` if the `GeoPoliticalExtentTypeKind` has been set to "OtherType".

Associations

`kind : GeoPoliticalExtentTypeKind[1]` Captures the kind of `GeoPoliticalExtentType`.

GeoPoliticalExtentTypeKind

Package: Parameters

isAbstract: No

Description

Enumeration of the possible kinds of `GeoPoliticalExtentType`. Its enumeration literals are:

- `GeoFeatureType` - Indicates that the `GeoPoliticalExtentType` associated with the `GeoPoliticalExtentTypeKind` is a type of object that encompasses meteorological, geographic, and control features mission significance.
- `RegionOfCountryType` - Indicates that the `GeoPoliticalExtentType` associated with the `GeoPoliticalExtentTypeKind` is a type of large, usually continuous segment of a political state, nation or its territory.
- `CountryType` - Indicates that the `GeoPoliticalExtentType` associated with the `GeoPoliticalExtentTypeKind` is a type of political state, nation or its territory.
- `RegionOfWorldType` - Indicates that the `GeoPoliticalExtentType` associated with the `GeoPoliticalExtentTypeKind` is a type of large, usually continuous segment of a surface or space; area.

- **FacilityType** - Indicates that the GeoPoliticalExtentType associated with the GeoPoliticalExtentTypeKind is a type of a real property entity consisting of underlying land and one or more of the following: a building, a structure (including linear structures), a utility system, or pavement.
- **SiteType** - Indicates that the GeoPoliticalExtentType associated with the GeoPoliticalExtentTypeKind is a type of Physical (geographic) location that is or was owned by, leased to, or otherwise possessed. Each site is assigned to a single installation. A site may exist in one of three forms: (1) Land only, where there are no facilities present and where the land consists of either a single land parcel or two or more contiguous land parcels. (2) Facility or facilities only, where the underlying land is neither owned nor controlled by the government. A stand-alone facility can be a site. If a facility is not a stand-alone facility, it must be assigned to a site. (3). Land and all the facilities thereon, where the land consists of either a single land parcel or two or more contiguous land parcels.
- **InstallationType** - Indicates that the GeoPoliticalExtentType associated with the GeoPoliticalExtentTypeKind is a type of base, camp, post, station, yard, center, or other activity, including leased facilities, without regard to the duration of operational control. An installation may include one or more sites.
- **OtherType** - Indicates that the GeoPoliticalExtentType associated with the GeoPoliticalExtentTypeKind is a type not covered by the standard GeoPoliticalExtentTypeKinds.

Location

Package: Parameters

isAbstract: No

Generalization: [ConceptItem](#), [Condition](#)

Extension: DataType

Description

A specification of the generic area in which a LocationHolder is required to be located.

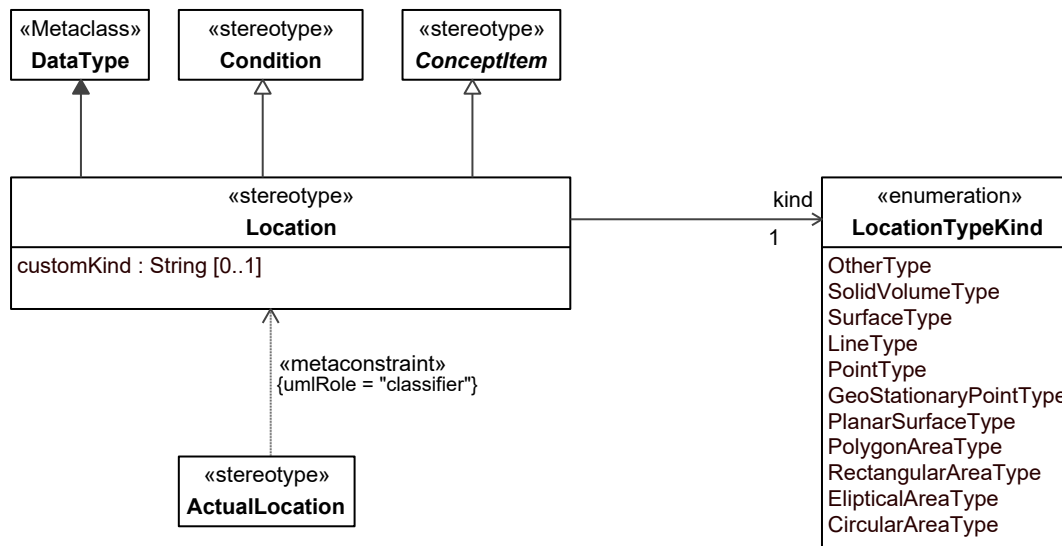


Figure 3:14 - Location

Attributes

`customKind : String[0..1]` Captures the kind of Location if the LocationTypeKind has been set to "OtherType".

Associations

`kind : LocationTypeKind[1]` Captures the kind of Location.

LocationHolder

Package: Parameters

isAbstract: Yes

Generalization: [UAFElement](#)

Extension: Element

Description

Abstract grouping used to define elements that are allowed to be associated with a Location.

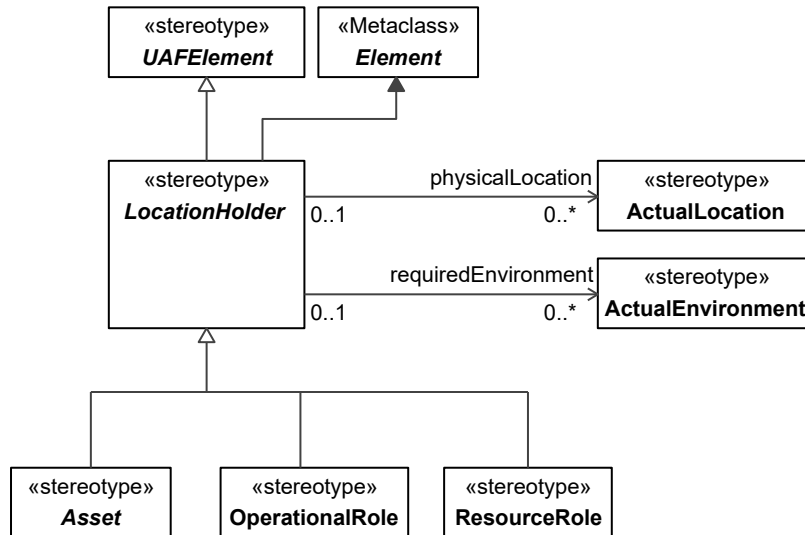


Figure 3:15 – LocationHolder

Associations

physicalLocation : ActualLocation[0..*]

Relates a LocationHolder (i.e. OperationalPerformer, OperationalRole, ResourceRole etc.) to its ActualLocation.

requiredEnvironment : ActualEnvironment[0..*]

Relates a LocationHolder (i.e. OperationalPerformer, OperationalRole, ResourceRole etc.) to the Environment in which it is required to perform/be used.

LocationKind

Package: Parameters

isAbstract: No

Description

Enumeration of the possible kinds of location applicable to an ActualLocation. Its enumeration literals are:

- SolidVolume - Indicates that the ActualLocation associated with the LocationKind is the amount of space occupied by a three-dimensional object of definite shape; not liquid or gaseous.
- Surface - Indicates that the ActualLocation associated with the LocationKind is a portion of space having length and breadth but no thickness or regards to time.
- Line - Indicates that the ActualLocation associated with the LocationKind is a geometric figure formed by a point moving along a fixed direction and the reverse direction.
- Point - Indicates that the ActualLocation associated with the LocationKind is a unidimensional Individual.

- GeoStationaryPoint - Indicates that the ActualLocation associated with the LocationKind is a unidimensional Individual.
- PlanarSurface - Indicates that the ActualLocation associated with the LocationKind is a two-dimensional portion of space.
- PolygonArea - Indicates that the ActualLocation associated with the LocationKind is a space enclosed by a polygon.
- RectangularArea - Indicates that the ActualLocation associated with the LocationKind is a space enclosed by a rectangle.
- EllipticalArea - Indicates that the ActualLocation associated with the LocationKind is a space enclosed by an ellipse.
- CircularArea - Indicates that the ActualLocation associated with the LocationKind is a space enclosed by a circle.
- Other - Indicates that the ActualLocation associated with the LocationKind is a LocationKind that is not on the enumerated list.

LocationTypeKind

Package: Parameters

isAbstract: No

Description

Enumeration of the possible kinds of location type that are applicable to a Location. Its enumeration literals are:

- OtherType - Indicates that the Location associated with the LocationTypeKind describes a type of is a LocationKindType that is not on the enumerated list.
- SolidVolumeType - Indicates that the Location associated with the LocationTypeKind describes a type of amount of space occupied by a three-dimensional object of definite shape; not liquid or gaseous.
- SurfaceType - Indicates that the Location associated with the LocationTypeKind describes a type of portion of space having length and breadth but no thickness or regards to time.
- LineType - Indicates that the Location associated with the LocationTypeKind describes a type of geometric figure formed by a point moving along a fixed direction and the reverse direction.
- PointType - Indicates that the Location associated with the LocationTypeKind describes a type of unidimensional Individual.
- GeoStationaryPointType - Indicates that the Location associated with the LocationTypeKind describes a type of unidimensional Individual.
- PlanarSurfaceType - Indicates that the Location associated with the LocationTypeKind describes a type of is a two-dimensional portion of space.
- PolygonAreaType - Indicates that the Location associated with the LocationTypeKind describes a type of space enclosed by a polygon.
- RectangularAreaType - Indicates that the Location associated with the LocationTypeKind describes a type of space enclosed by a rectangle.
- EllipticalAreaType - Indicates that the Location associated with the LocationTypeKind describes a type of space enclosed by an ellipse.
- CircularAreaType - Indicates that the Location associated with the LocationTypeKind describes a type of space enclosed by a circle.

MeasurableElement

Package: Parameters

isAbstract: Yes

Generalization: [UAFElement](#)

Extension: Element

Description

Abstract grouping for elements that can be measured by applying MeasurementSets to them.

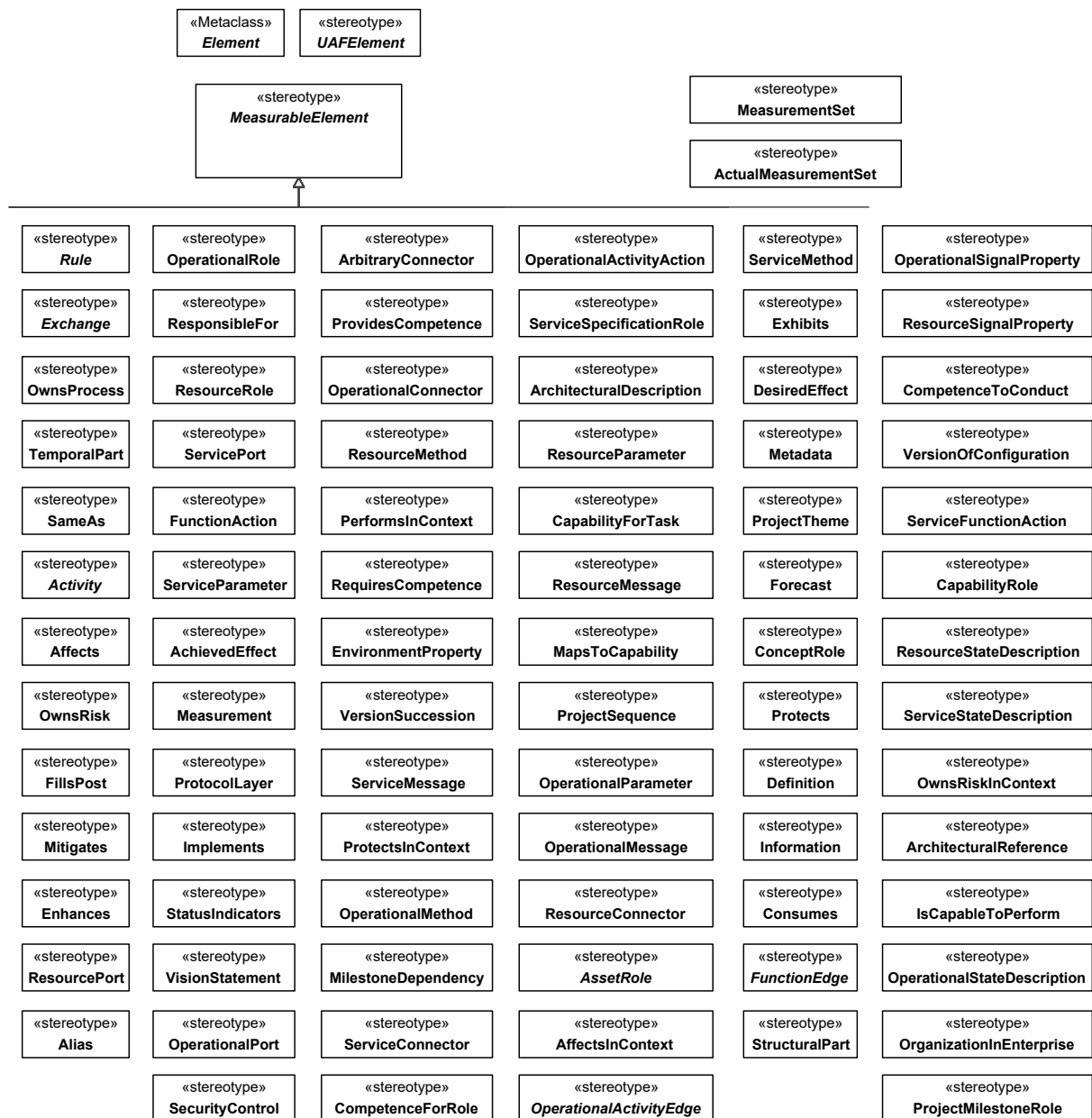


Figure 3:16 – MeasurableElement

Associations

actualMeasurementSet : ActualMeasurementSet[*]

Relates the MeasurableElement to the ActualMeasurementSet that provides its ActualMeasurements.

measurementSet : MeasurementSet[*]

Relates the MeasurableElement to the MeasurementSet that provides its Measurements by which it can be measured.

Measurement

Package: Parameters

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Property

Description

A property of an element representing something in the physical world, expressed in amounts of a unit of measure.

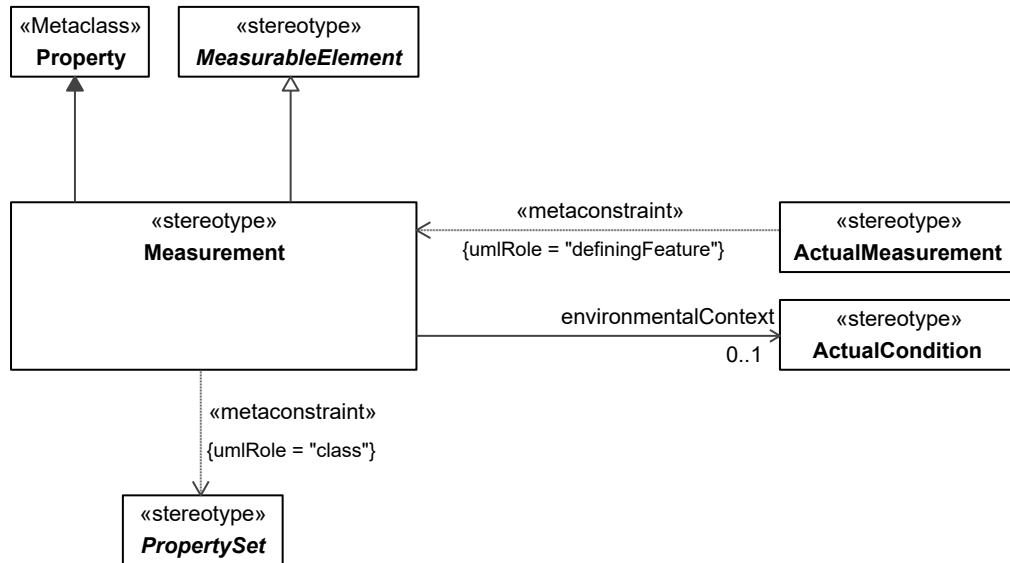


Figure 3:17 – Measurement

Associations

environmentalContext : ActualCondition[0..1] Relates the Measurement to the Condition (which provides the environmentalContext) under which the Measurement is expected to be taken.

Constraints

[1] Measurement.class Value for the class metaproperty must be stereotyped by the specialization of «PropertySet».

MeasurementSet

Package: Parameters

isAbstract: No

Generalization: [PropertySet](#), ValueType

Extension: DataType

Description

A collection of Measurements.

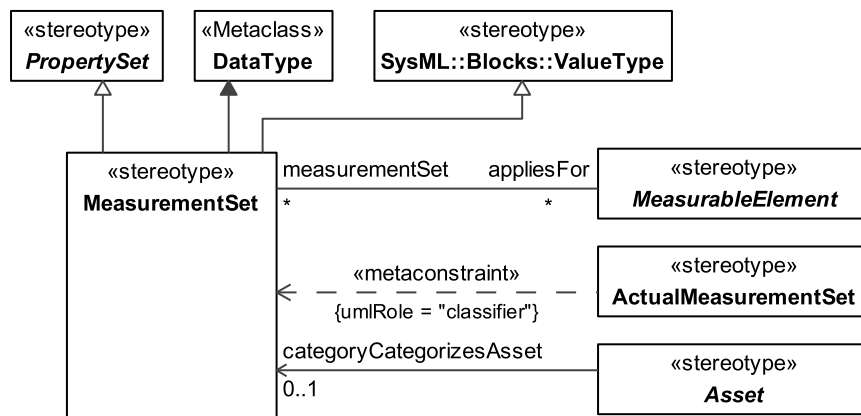


Figure 3:18 – MeasurementSet

Associations

appliesFor : MeasurableElement[*] Relates the MeasurementSet to the MeasurableElement that it is applicable to.

PropertySet

Package: Parameters

isAbstract: Yes

Generalization: [UAFElement](#)

Extension: Element

Description

An abstract grouping of architectural elements that can own Measurements.

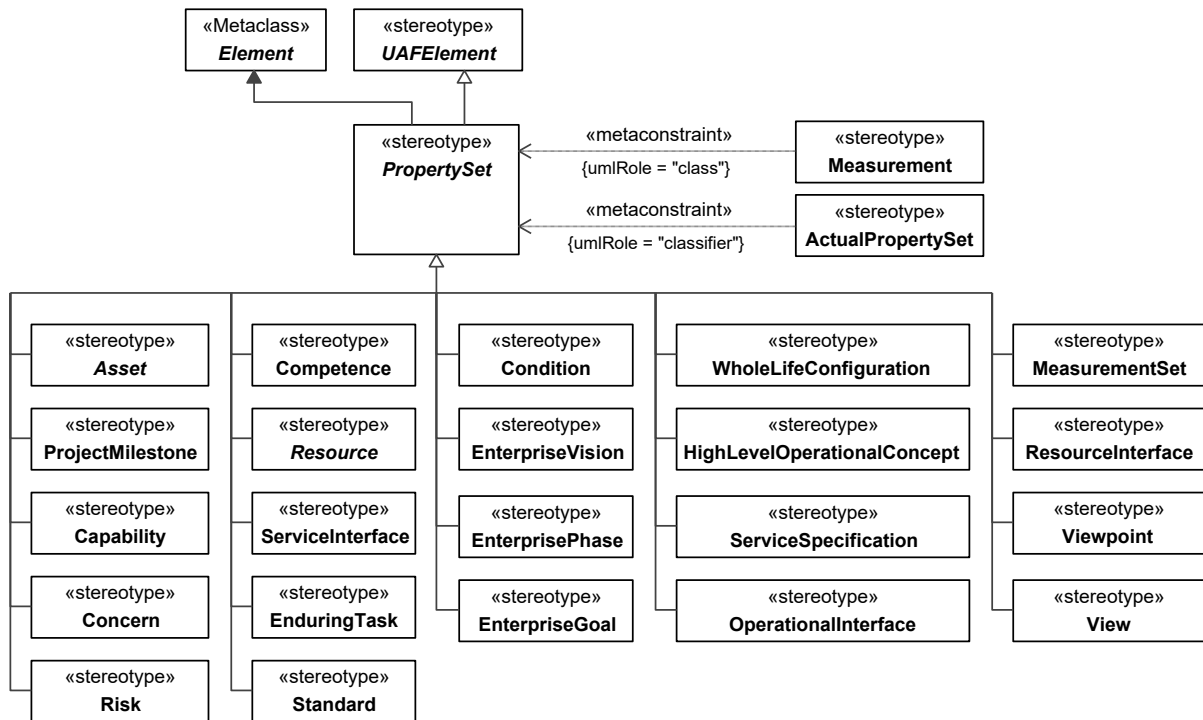


Figure 3:19 - PropertySet

3.1.3 UAF::Metadata

Stakeholders: Enterprise Architects, people who want to discover the architecture, Technical Managers.

Concerns: Captures meta-data relevant to the entire architecture

Definition: Provide information pertinent to the entire architecture. Present supporting information rather than architectural models.

UAF::Metadata::Taxonomy

Contains the elements that contribute to the Metadata Taxonomy Viewpoint.

ActualState

Package: Taxonomy

isAbstract: Yes

Generalization: [UAFElement](#)

Extension: Element

Description

Abstract element that applies temporal extent to a set of elements realized as Instance Specifications.

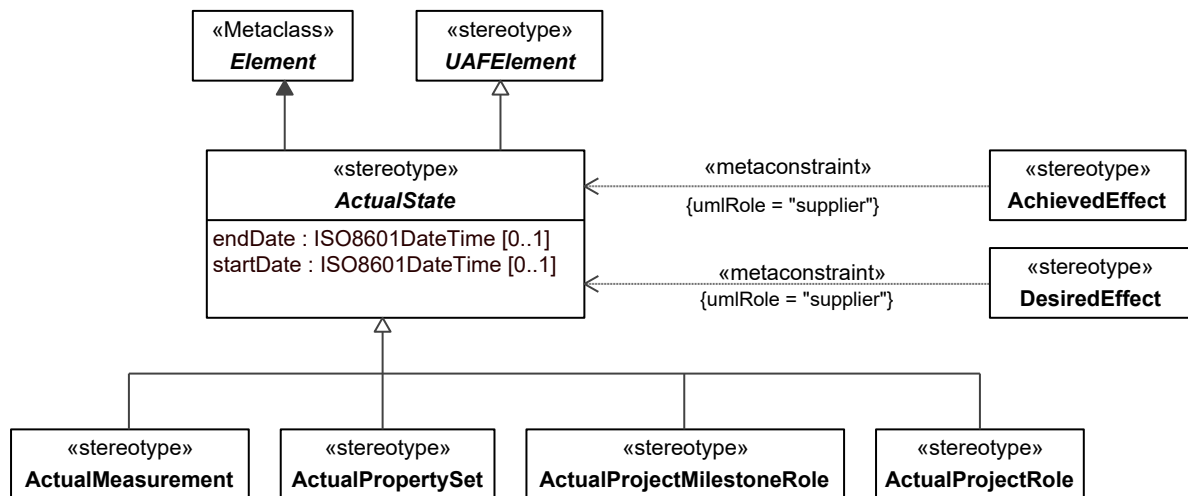


Figure 3:20 – ActualState

Attributes

endDate : ISO8601DateTime[0..1] End time for all "actual" elements.
 startDate : ISO8601DateTime[0..1] Start time for all "actual" elements.

ISO8601DateTime

Package: Taxonomy

isAbstract: No

Generalization: [UAFEElement](#)

Extension: LiteralString

Description

A date and time specified in the ISO8601 date-time format including timezone designator (TZD): YYYY-MM-DDThh:mm:ssTZD.

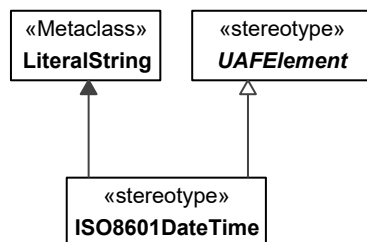


Figure 3:21 - ISO8601DateTime

UAF::Metadata::Connectivity

Contains the elements that contribute to the Metadata Connectivity Viewpoint.

Exchange

Package: Connectivity

isAbstract: Yes

Generalization: [MeasurableElement](#), ItemFlow, [SubjectOfSecurityConstraint](#)

Extension: InformationFlow

Description

Abstract grouping for OperationalExchanges and ResourceExchanges that exchange Resources.

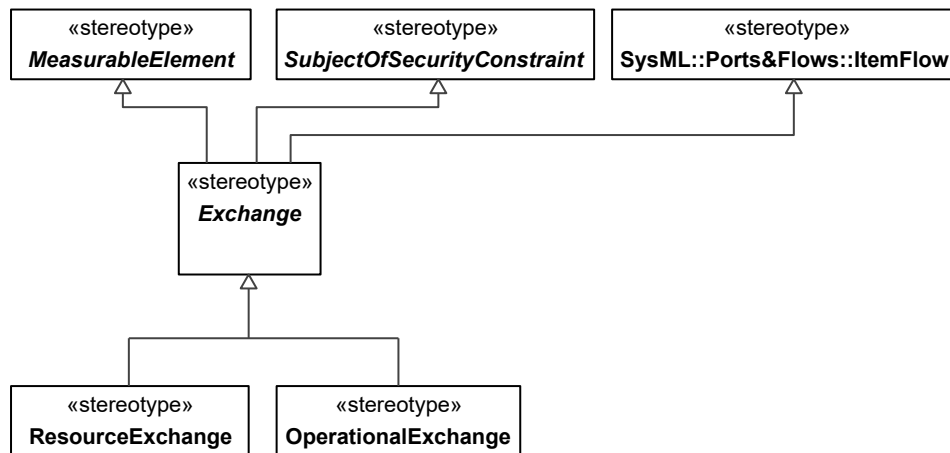


Figure 3:22 - Exchange

Resource

Package: Connectivity

isAbstract: Yes

Generalization: [PropertySet](#)

Extension: Element

Description

Abstract element grouping for all elements that can be conveyed by an Exchange.

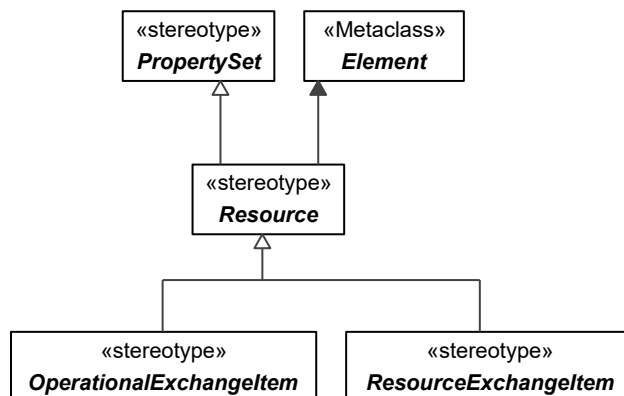


Figure 3:23 - Resource

UAF::Metadata::Processes

Contains the elements that contribute to the Metadata Processes Viewpoint.

Activity

Package: Processes

isAbstract: Yes

Generalization: [MeasurableElement](#)

Extension: Activity

Description

An abstract element that represents a behavior or process (i.e., a Function or OperationalActivity) that can be performed by a Performer.

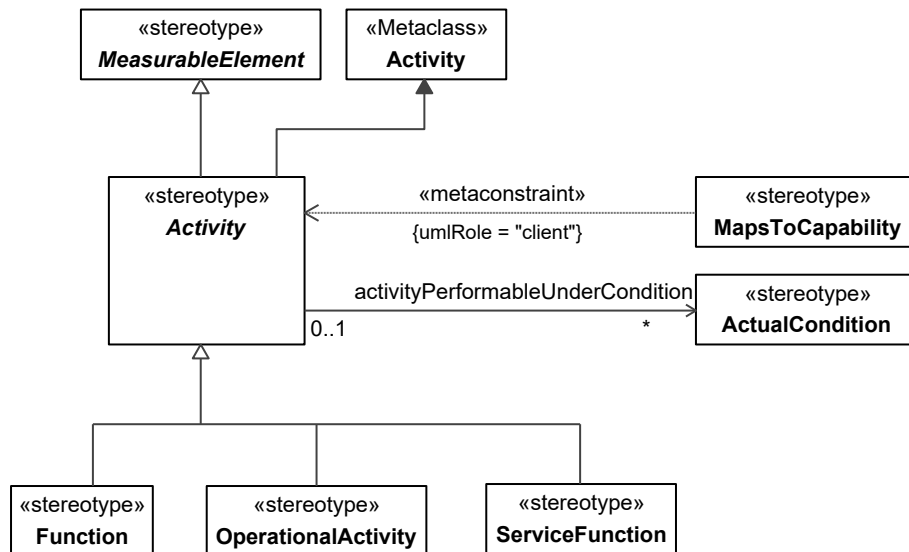


Figure 3:24 – Activity

Associations

activityPerformableUnderCondition : ActualCondition[*] The environment under which an activity is performed.

CapableElement

Package: Processes

isAbstract: Yes

Generalization: [UAFEElement](#)

Extension: Element

Description

An abstract type that represents a structural element that can exhibit capabilities.

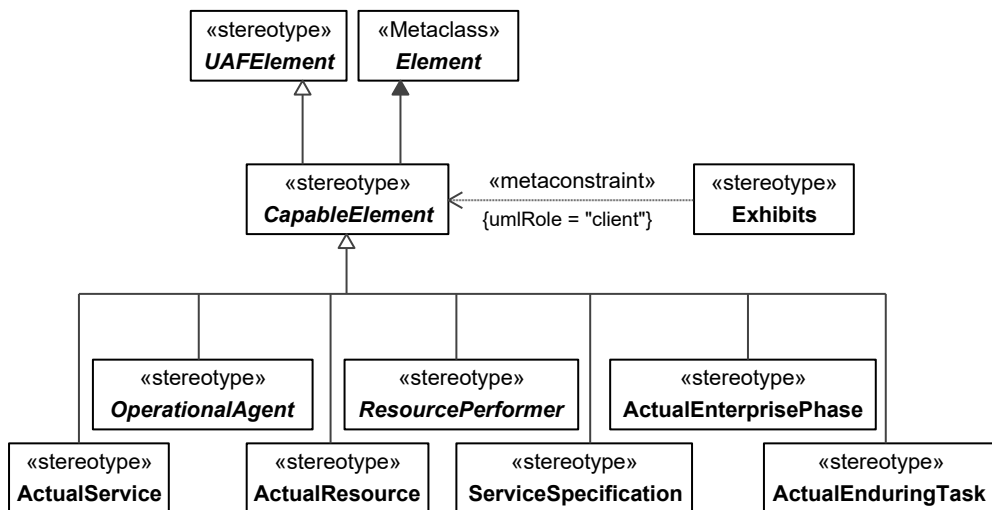


Figure 3:25 - CapableElement

IsCapableToPerform

Package: Processes

isAbstract: No

Generalization: [MeasurableElement](#), Allocate

Extension: Abstraction

Description

An Abstraction relationship defining the traceability between the structural elements to the Activities that they can perform.

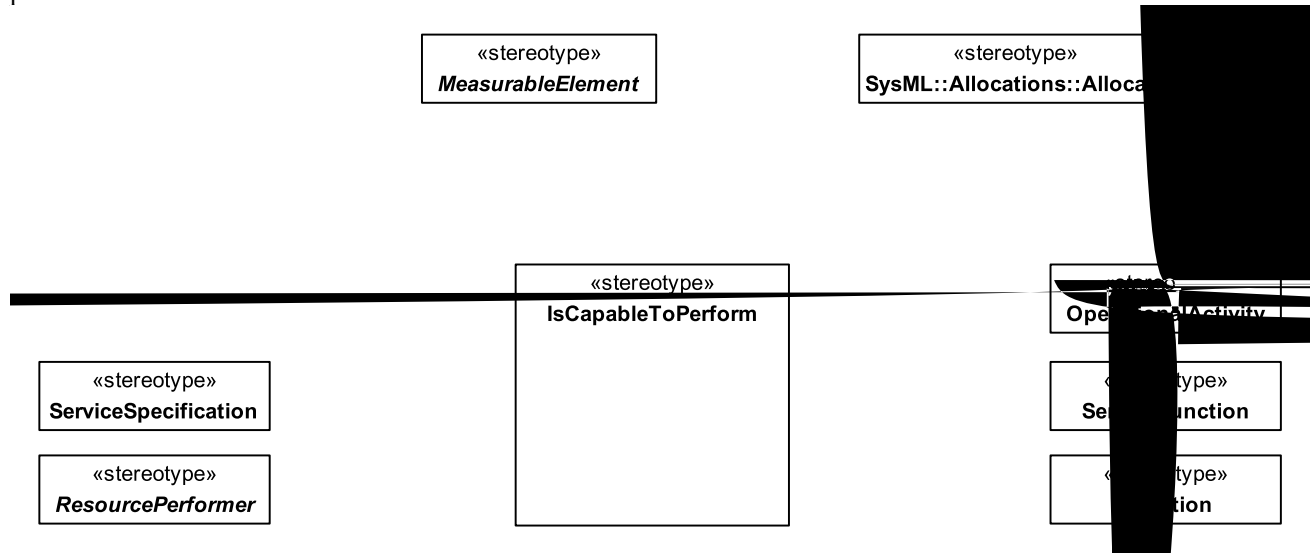


Figure 3:26 – IsCapableToPerform

Constraints

- | | |
|------------------------------------|---|
| [1] IsCapableOfPerforming.client | In case of value for IsCapableToPerform.supplier is stereotyped:
a. «OperationalActivity» or its specializations, values for the client metaproperty must be stereotyped by any of specializations of «OperationalAgent»,
b. «ServiceFunction» or its specializations, values for the client metaproperty must be stereotyped «ServiceSpecification» or its specializations,
c. «Function» or its specializations, except for «ProjectActivity», values for the client metaproperty must be stereotyped by any of specializations of «ResourcePerformer»,
d. «ProjectActivity» or its specializations, values for the client metaproperty must be stereotyped by any of specializations of «Project». |
| [2] IsCapableOfPerforming.supplier | In case of value for IsCapableToPerform.client is stereotyped:
a. by a specialization of «OperationalAgent», values for the supplier metaproperty must be stereotyped «OperationalActivity» or its specializations,
b. «ServiceSpecification» or its specializations, values for the supplier metaproperty must be stereotyped «ServiceFunction» or its specializations,
c. by a specialization of «ResourcePerformer», values for the supplier metaproperty must be stereotyped «Function» or its specializations, except for «ProjectActivity»,
d. by a specialization of «Project», values for the supplier metaproperty must be stereotyped «ProjectActivity» or its specializations. |

PerformsInContext

Package: Processes

isAbstract: No

Generalization: [MeasurableElement](#), Allocate

Extension: Abstraction

Description

An abstraction relationship that relates an OperationalAction to a OperationalRole, or a FunctionAction to a ResourceRole. It indicates that the action can be carried out by the role when used in a specific context or configuration.

Extension: Comment

Description

Information associated with an ArchitecturalDescription, that supplements the standard set of tags used to summarize the Architecture. It states things like what methodology was used, notation, etc.

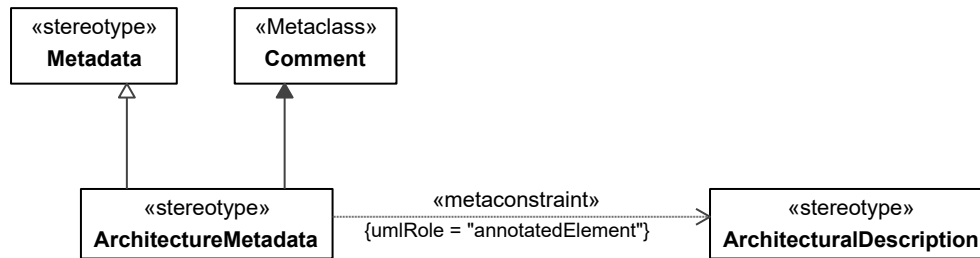


Figure 3:28 – ArchitectureMetadata

Constraints

- [1] ArchitectureMetadata.annotatedElement Value for the annotatedElement metaproperty must be stereotyped «ArchitecturalDescription» or its specializations.

DataModel

Package: Information

isAbstract: No

Generalization: [SubjectOfOperationalConstraint](#), [SubjectOfResourceConstraint](#)

Extension: Package

Description

A structural specification of data types, showing relationships between them. The type of data captured in the DataModel is described using the enumeration DataModelKind (Conceptual, Logical and Physical).

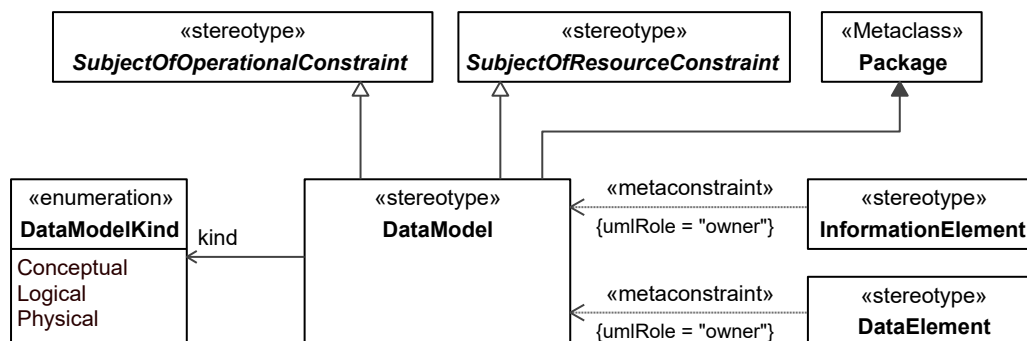


Figure 3:29 – DataModel

Associations

- kind : DataModelKind[] Captures the kind of DataModel being respresented, Conceptual, Logical or Physcial.

DataModelKind

Package: Information

isAbstract: No

Description

Enumeration of the possible kinds of DataModel. Its enumeration literals are:

- Conceptual - Indicates that the DataModel associated with the DataModelKind is a conceptual DataModel that defines the required high-level data concepts and their relationships.
- Logical - Indicates that the DataModel associated with the DataModelKind is a logical data model that allows analysis of an architecture's data definition aspect, without consideration of implementation specific or product specific issues. It details the conceptual data model.
- Physical - Indicates that the DataModel associated with the DataModelKind is a physical data model that is an implementable specification of a data structure. A physical data model realizes a logical data model, taking into account implementation restrictions and performance issues while still enforcing the constraints, relationships and typing of the logical data model.

Information

Package: Information

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Comment

Description

A comment that describes the state of an item of interest in any medium or form -- and is communicated or received.

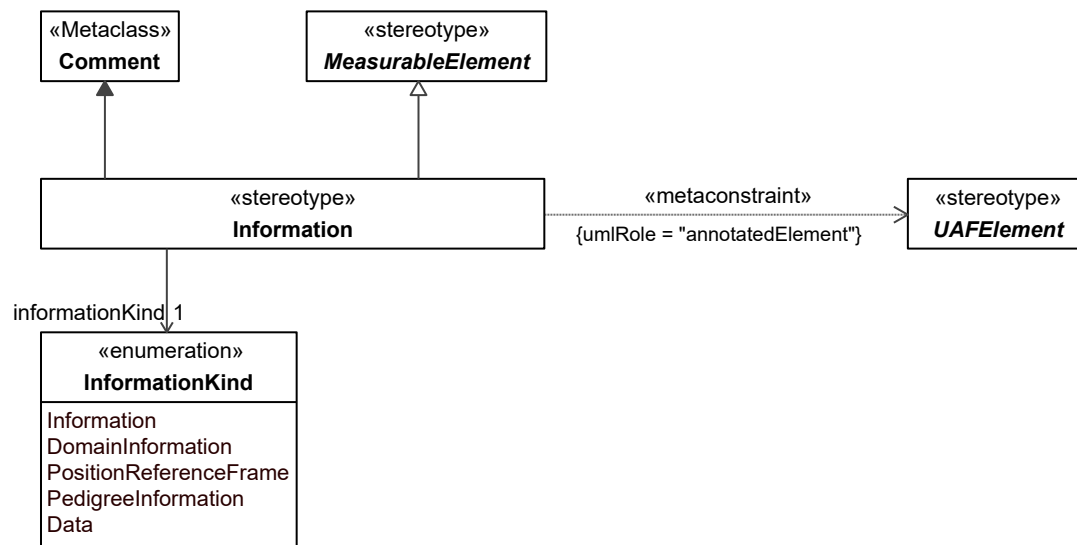


Figure 3:30 – Information

Associations

informationKind : InformationKind[1] Captures the kind of information.

Constraints

[1] Information.annotatedElement Value for the annotatedElement metaproperty must be stereotyped by a specialization of «UAFEElement».

InformationKind

Package: Information

isAbstract: No

Description

Enumeration of the possible kinds of Information. Its enumeration literals are:

- Information - Indicates that the Information associated with the InformationKind describes the state of a something of interest that is materialized -- in any medium or form -- and communicated or received.
- DomainInformation - Indicates that the Information associated with the InformationKind describes information within the scope or domain of the architecture.
- PositionReferenceFrame - Indicates that the Information associated with the InformationKind describes an arbitrary set of axes with reference to which the position or motion of something is described or physical laws are formulated.
- Pedib(d)-6.4TJ±0.002 Tc 0 5 1 Tf0.004 Tc -0.004 Tw 10.4687 -0 0 10.02 108 6t(he002 Tc 073he002 Tc(w 10.156

Attributes

category : String[0..1]	Defines the category of a Metadata element example: http://purl.org/dc/terms/abstract .
dublinCoreElement : String[0..1]	A metadata category that is a DublinCore tag.
metaDataScheme : String[0..1]	A representation scheme that defines a set of Metadata.
name : String[0..1]	The name of the Metadata.

Constraints

[1] Metadata.annotatedElement	Value for the annotatedElement metaproperty must be stereotyped by a specialization of «UAFElement».
-------------------------------	--

UAF::Metadata::Constraints

Contains the elements that contribute to the Metadata Constraints Viewpoint.

Rule

Package: Constraints

isAbstract: Yes

Generalization: [MeasurableElement](#)

Extension: Constraint

Description

An abstract grouping for all types of constraint (i.e. an OperationalConstraint could detail the rules of accountancy best practice).

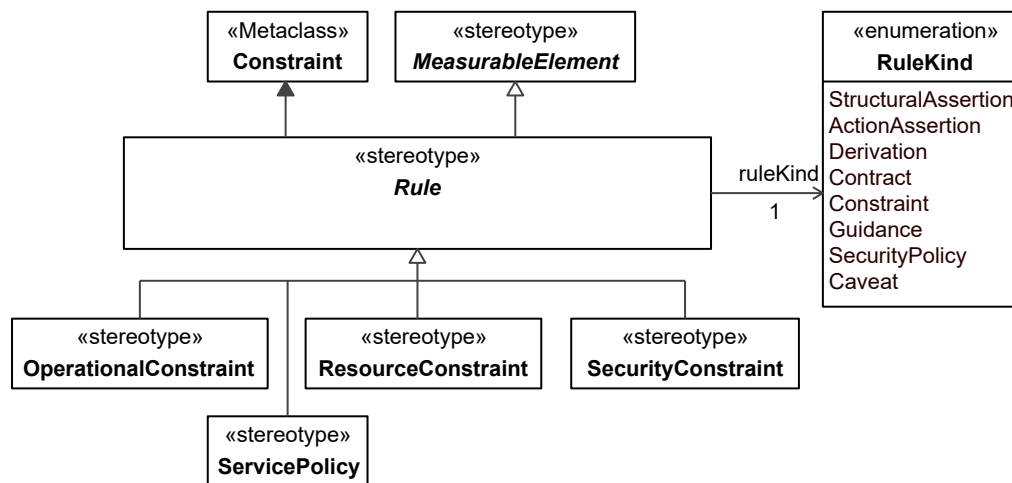


Figure 3:32 – Rule

Associations

ruleKind : RuleKind[1]	Captures the kind of Rule that is being described.
------------------------	--

RuleKind

Package: Constraints

isAbstract: No

Description

Enumeration of the possible kinds of Rules applicable to constraints. Its enumeration literals are:

- **StructuralAssertion** - Indicates that the Rule associated with the RuleKind is a statement that details that something of importance either exists as a concept of interest or exists in relationship to another thing of interest.
- **ActionAssertion** - Indicates that the Rule associated with the RuleKind is a statement that concerns some dynamic aspect.
- **Derivation** - Indicates that the Rule associated with the RuleKind is a statement that details a Rule derived from another Rule.
- **Contract** - Indicates that the Rule associated with the RuleKind is a statement that details a consent among parties regarding the terms and conditions of activities that said parties participate in.
- **Constraint** - Indicates that the Rule associated with the RuleKind is a statement that details a limitation, e.g. business rule, restraint, operational limitation.
- **Guidance** - Indicates that the Rule associated with the RuleKind is a statement that details an authoritative statement intended to lead or steer the execution of actions.
- **SecurityPolicy** - Indicates that the Rule associated with the RuleKind is a statement that details a constraint that specifies policy for information handling, physical security, encryption, etc.
- **Caveat** - Indicates that the Rule associated with the RuleKind is a statement that details alternate conditions under which the rule is not valid.

UAF::Metadata::Traceability

Contains the elements that contribute to the Metadata Traceability Viewpoint.

ArchitecturalReference

Package: Traceability

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Dependency

Description

A dependency relationship that specifies that one architectural description refers to another.

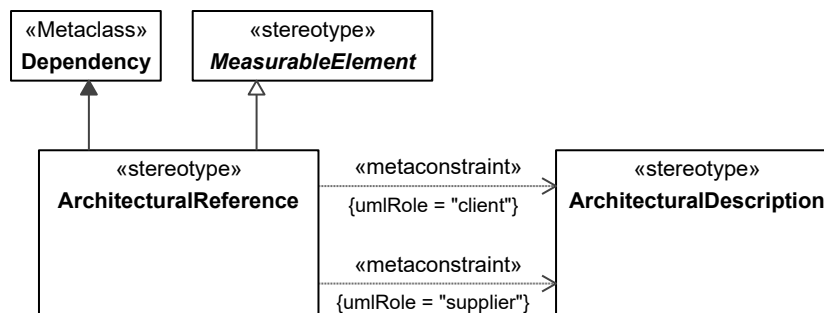


Figure 3:33 – ArchitecturalReference

Constraints

- | | |
|-------------------------------------|--|
| [1] ArchitecturalReference.client | Value for the client metaproperty must be stereotyped «ArchitecturalDescription» or its specializations. |
| [2] ArchitecturalReference.supplier | Value for the supplier metaproperty must be stereotyped «ArchitecturalDescription» or its specializations. |

Constraints

- [1] Implements.client In case of value for Implements.supplier is stereotyped:
- a. by any of specializations of «OperationalAgent», values for the client metaproperty must be stereotyped by any of specializations of «ResourcePerformer»,
 - b. «OperationalActivity» or its specializations, values for the client metaproperty must be stereotyped «Function» or its specializations,
 - c. «ServiceFunction» or its specializations, values for the client metaproperty must be stereotyped «Function» or its specializations,
 - d. «ServiceInterface» or its specializations, values for the client metaproperty must be stereotyped «ResourceInterface» or its specializations,
 - e. «OperationalInterface» or its specializations, values for the client metaproperty must be stereotyped «ResourceInterface» or its specializations,
 - f. «OperationalConnector» or its specializations, values for the client metaproperty must be stereotyped «ResourceConnector» or its specializations,
 - g. «OperationalExchange» or its specializations, values for the client metaproperty must be stereotyped «ResourceExchange» or its specializations,
 - g. «OperationalRole» or its specializations, values for the client metaproperty must be stereotyped «ResourceRole» or its specializations,
 - h. «ResourceConnector» or its specializations, values for the client metaproperty must be stereotyped «ResourceConnector» or its specializations,
 - i. «ActualEnduringTask» or its specializations, values for the client metaproperty must be stereotyped «OperationalActivity» or its specializations,
 - j. «InformationElement» or its specializations, values for the client metaproperty must be stereotyped «DataElement» or its specializations.
- [2] Implements.supplier In case of value for Implements.client is stereotyped:
- a. by any of specializations of «ResourcePerformer», values for the supplier metaproperty must be stereotyped by any of specializations of «OperationalAgent»,
 - b. «Function» or its specializations, values for the supplier metaproperty must be stereotyped «OperationalActivity», «ServiceFunction» or their specializations,
 - c. «ResourceInterface» or its specializations, values for the supplier metaproperty must be stereotyped «ServiceInterface», «OperationalInterface», or their specializations,
 - d. «ResourceConnector» or its specializations, values for the supplier metaproperty must be stereotyped «OperationalConnector», «ResourceConnector» or their specializations,
 - e. «ResourceExchange» or its specializations, values for the supplier metaproperty must be stereotyped «OperationalExchange» or its specializations,
 - f. «ResourceRole» or its specializations, values for the supplier metaproperty must be stereotyped «OperationalRole» or its specializations,
 - g. «OperationalActivity» or its specializations, values for the supplier metaproperty must be stereotyped «ActualEnduringTask» or its specializations,
 - h. «DataElement» or its specializations, values for the supplier metaproperty must be stereotyped «InformationElement» or its specializations.

3.1.4 UAF::Strategic

Stakeholders: Capability Portfolio Managers.

Concerns: capability management process.

Definition: describe capability taxonomy, composition, dependencies and evolution.

UAF::Strategic::Taxonomy

Contains the elements that contribute to the Strategic Taxonomy Viewpoint.

ActualEnterprisePhase

Package: Taxonomy

isAbstract: No

Generalization: [ActualPropertySet](#), [CapableElement](#), [Achiever](#)

Extension: InstanceSpecification

Description

A phase of an actual enterprise endeavor.

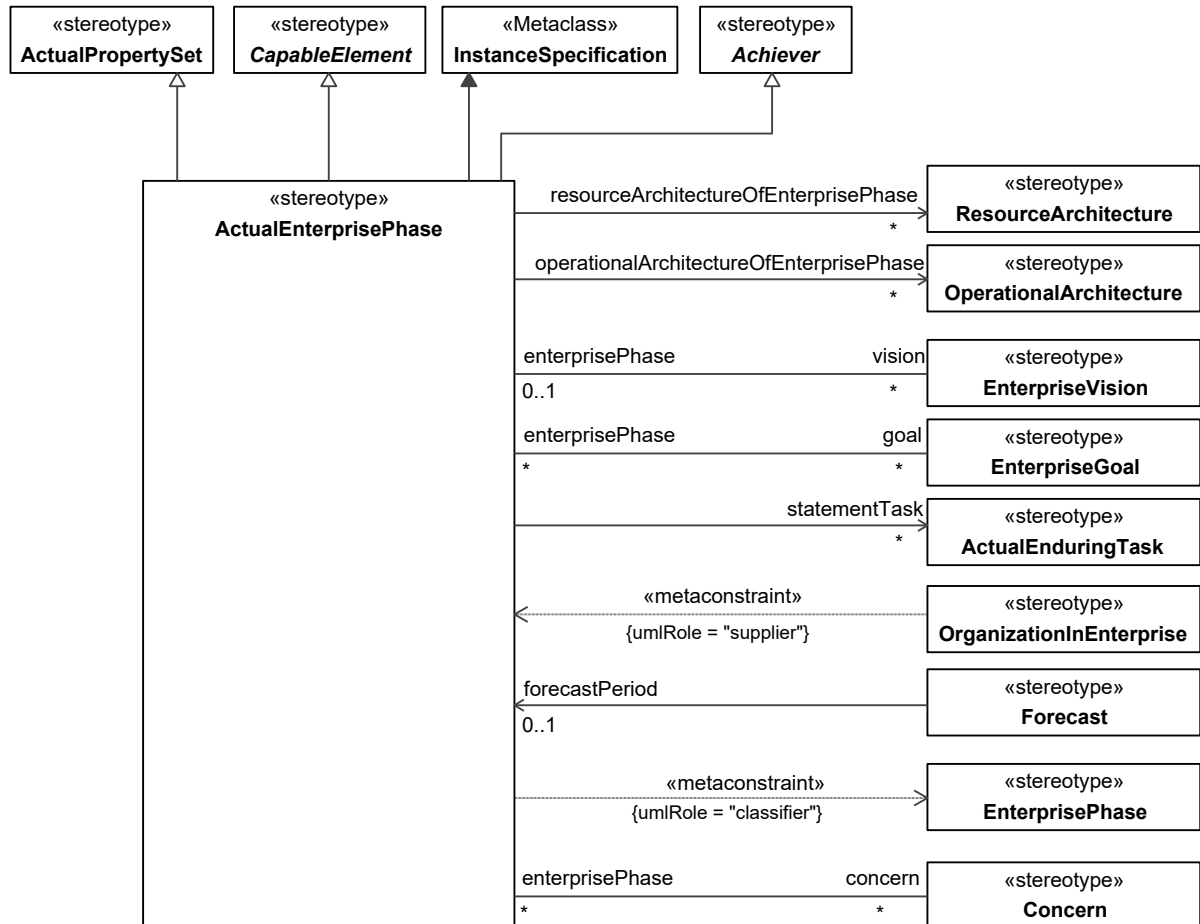


Figure 3:35 – ActualEnterprisePhase

Associations

concern : Concern[*]

goal : EnterpriseGoal[*]

operationalArchitectureOfEnterprisePhase : OperationalArchitecture[*]

resourceArchitectureOfEnterprisePhase : ResourceArchitecture[*]

statementTask : ActualEnduringTask[*]

vision : EnterpriseVision[*]

The Goal towards which this Phase is directed and is in support of.

Relates an ActualEnterprisePhase to its relevant OperationalArchitecture.

Relates an ActualEnterprisePhase to its relevant ResourceArchitecture.

Relates the ActualEnterprisePhase to the ActualEnduringTasks that are intended to be implemented during that phase.

The Vision towards which this Phase is directed and is in support of.

Constraints

- [1] ActualEnterprisePhase.classifier Value for the classifier metaproperty must be stereotyped by «EnterprisePhase» or its specializations.
- [2] ActualEnterprisePhase.start/endDate Must fall within the start and end dates of the enclosing ActualEnterprisePhase having this ActualEnterprisePhase set as a value for a slot.

Capability

Package: Taxonomy

isAbstract: No

Generalization: [PropertySet](#), [Desirer](#), Block

Extension: Class

Description

An enterprise's ability to Achieve a DesiredEffect realized through a combination of ways and means (e.g., CapabilityConfigurations) along with specified measures.

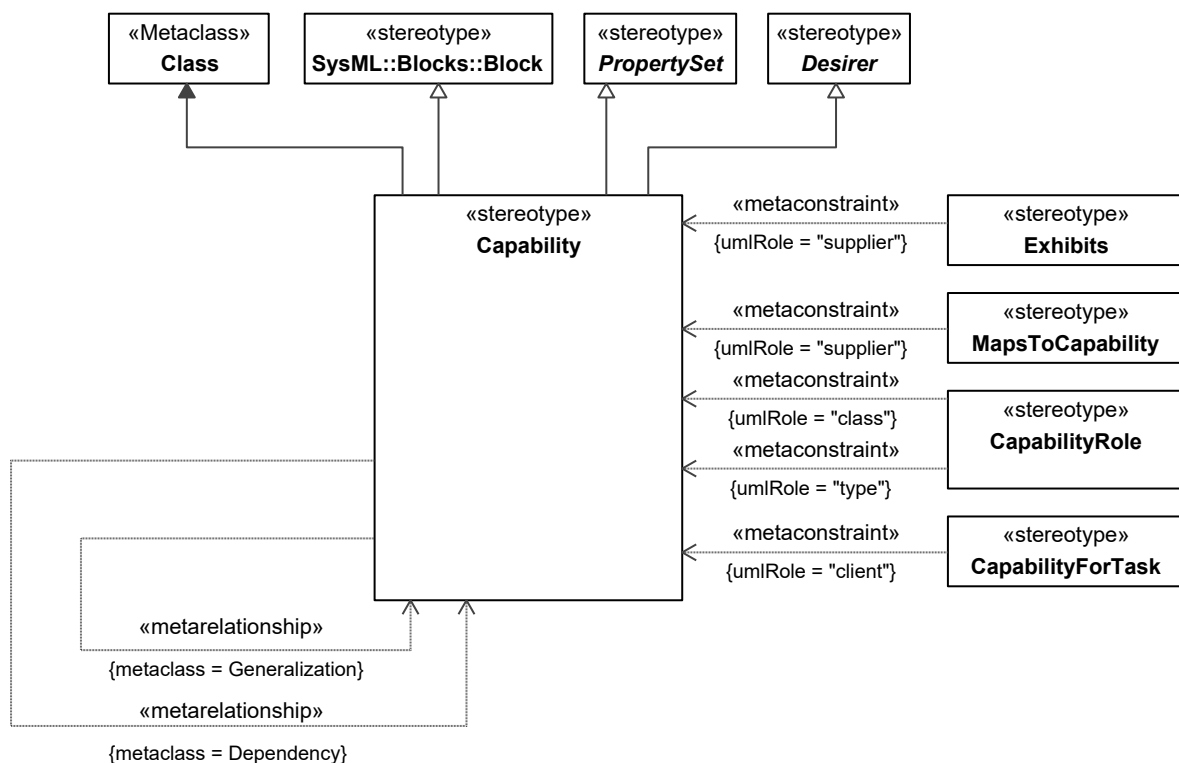


Figure 3:36 - Capability

EnterpriseGoal

Package: Taxonomy

isAbstract: No

Generalization: [PropertySet](#), Requirement

Extension: Class

Description

A statement about a state or condition of the enterprise to be brought about or sustained through appropriate Means. An EnterpriseGoal amplifies an En

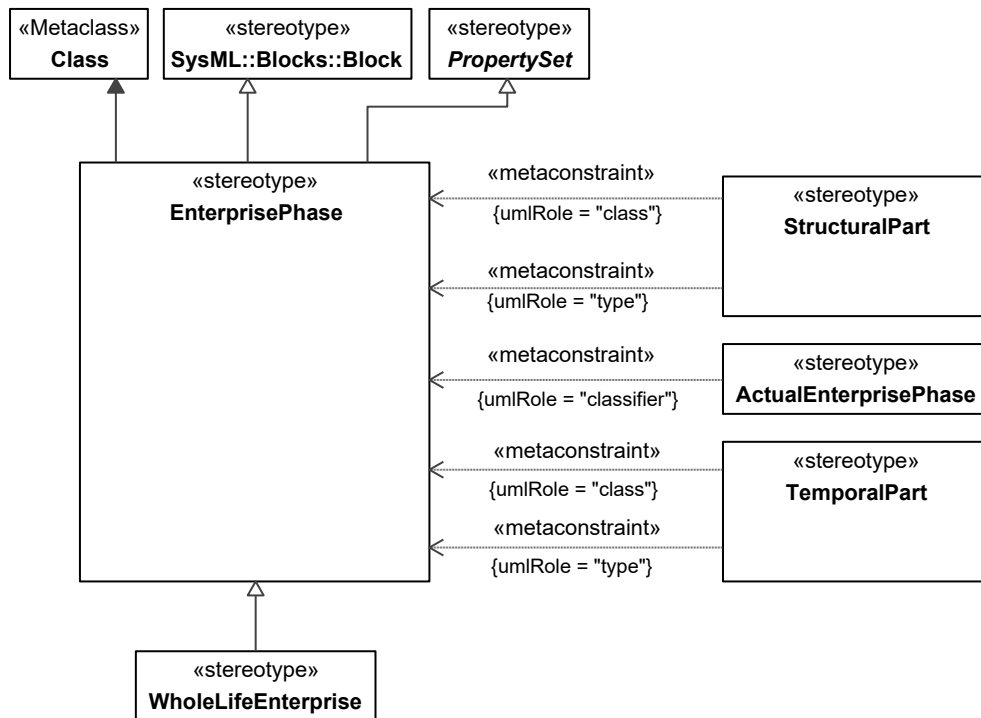


Figure 3:38 - EnterprisePhase

EnterpriseVision

Package: Taxonomy

isAbstract: No

Generalization: [PropertySet](#), Block

Extension: Class

Description

A Vision describes the future state of the enterprise, without regard to how it is to be achieved.

<https://www.omg.org/spec/BMM/1.3/>

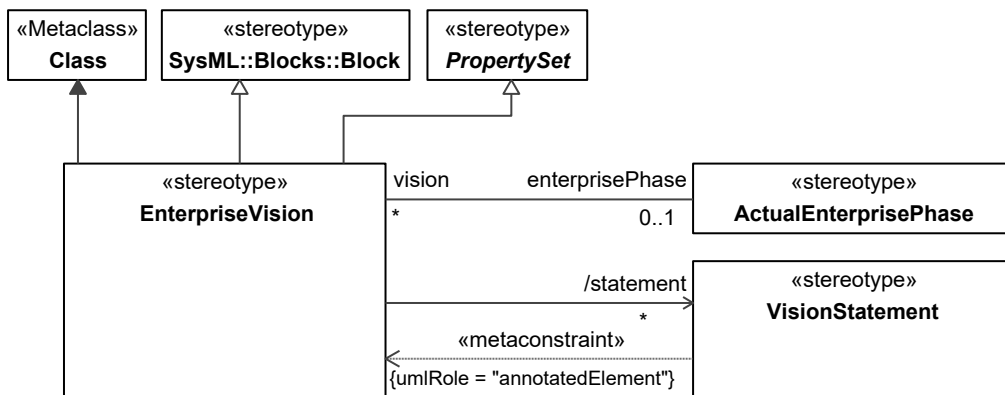


Figure 3:39 – EnterpriseVision

Associations

enterprisePhase : ActualEnterprisePhase[0..1] Relates the EnterpriseVision to the ActualEnterprisePhase in which the EnterpriseVision is expected to be realized.

statement : VisionStatement[*] A description of the Vision.

VisionStatement

Package: Taxonomy

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Comment

Description

A type of comment that describes the future state of the enterprise, without regard to how it is to be achieved.
<https://www.omg.org/spec/BMM/1.3/>

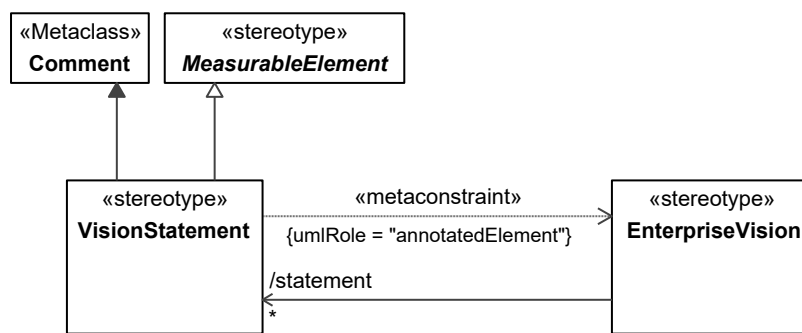


Figure 3:40 – VisionStatement

Constraints

[1] VisionStatement.ownedAttribute Values for annotatedElement metaproperty must be stereotyped «EnterpriseVision» or its specializations.

WholeLifeEnterprise

Package: Taxonomy

isAbstract: No

Generalization: [EnterprisePhase](#)

Extension: Class

Description

A WholeLifeEnterprise is a purposeful endeavor of any size involving people, organizations and supporting systems. It is made up of TemporalParts and StructuralParts.

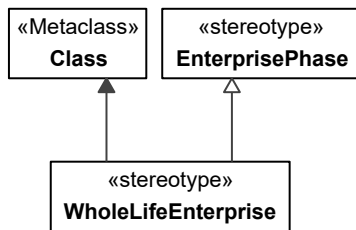


Figure 3:41 - WholeLifeEnterprise

UAF::Strategic::Structure

Contains the elements that contribute to the Strategic Structure Viewpoint.

CapabilityRole

Package: Structure

isAbstract: No

Generalization: [MeasurableElement](#), [Desirer](#)

Extension: Property

Description

Property of a Capability typed by another Capability, enabling whole-part relationships and structures.

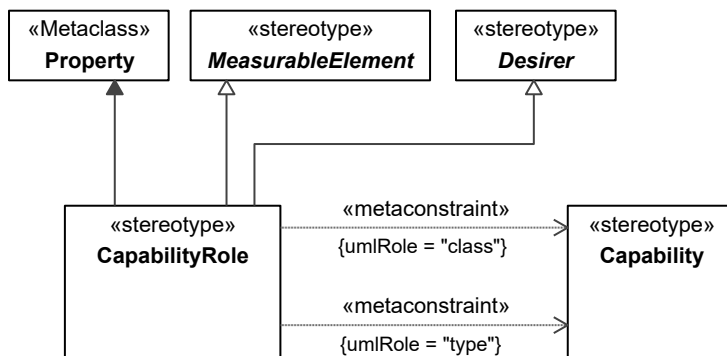


Figure 3:42 – CapabilityRole

Constraints

- [1] CapabilityProperty.class Value for class metaproperty must be stereotyped «Capability» or its specializations.
- [2] CapabilityProperty.type Value for type metaproperty must be stereotyped «Capability» or its specializations.

StructuralPart

Package: Structure

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Property

Description

Usage of an EnterprisePhase in the context of another EnterprisePhase. It asserts that one EnterprisePhase is a spatial part of another. Creates a whole-part relationship that represents the structure of the EnterprisePhase.

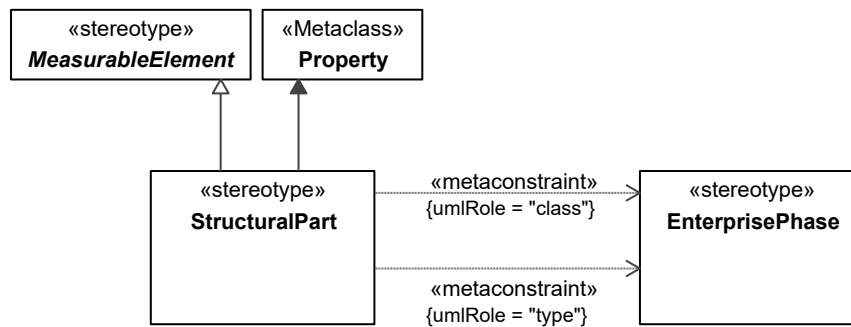


Figure 3:43 – StructuralPart

Constraints

- [1] StructuralPart.class Value for class metaproperty must be stereotyped «EnterprisePhase» or its specializations.
- [2] StructuralPart.type Value for type metaproperty must be stereotyped «EnterprisePhase» or its specializations.

TemporalPart

Package: Structure

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Property

Description

Usage of an EnterprisePhase in the context of another EnterprisePhase. It asserts that one EnterprisePhase is a spatial part of another. Creates a whole-part relationship that represents the temporal structure of the EnterprisePhase.

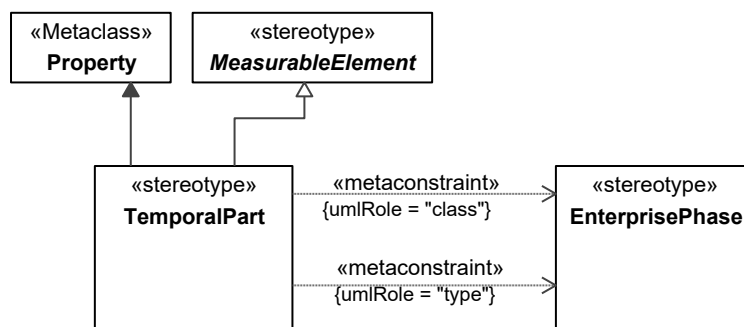


Figure 3:44 - TemporalPart

Constraints

- [1] TemporalPart.class Value for class metaproperty must be stereotyped «EnterprisePhase» or its specializations.
- [2] TemporalPart.type Value for type metaproperty must be stereotyped «EnterprisePhase» or its specializations.

UAF::Strategic::Processes

Contains the elements that contribute to the Strategic Proceses Viewpoint.

ActualEnduringTask

Package: Processes

isAbstract: No

Generalization: [CapableElement](#), [ActualPropertySet](#)

Extension: InstanceSpecification

Description

An actual undertaking recognized by an enterprise as being essential to achieving its goals - i.e. a strategic specification of what the enterprise does.

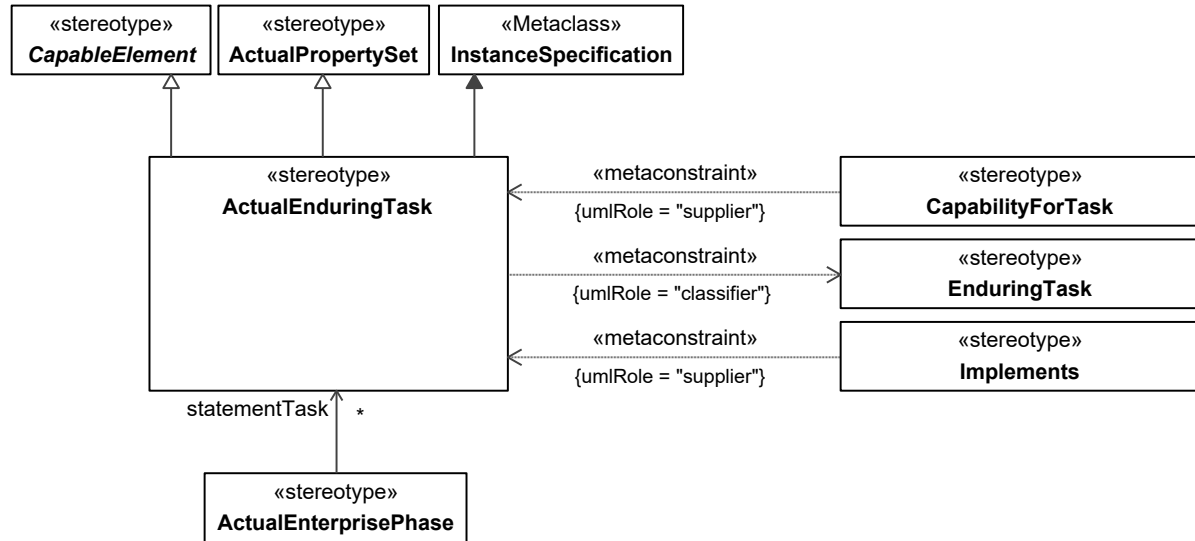


Figure 3:45 – ActualEnduringTask

Constraints

- [1] ActualEnduringTask.classifier Value for the classifier metaproperty must be stereotyped by «EnduringTask» or its specializations.

CapabilityForTask

Package: Processes

isAbstract: No

Generalization: [MeasurableElement](#), Allocate

Extension: Abstraction

Description

An abstraction relationship that asserts that a Capability is required in order for an Enterprise to conduct a phase of an EnduringTask.

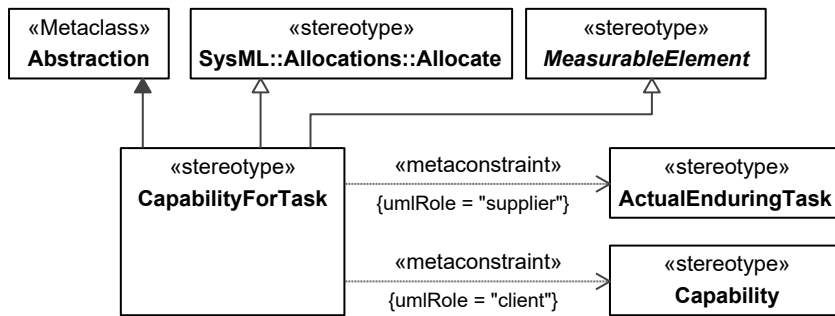


Figure 3:46 – CapabilityForTask

Constraints

- [1] CapabilityForTask.client Value for the client metaproperty must be stereotyped «Capability» or its specializations.
- [2] CapabilityForTask.supplier Value for the supplier metaproperty must be stereotyped «ActualEnduringTask» or its specializations.

EnduringTask

Package: Processes

isAbstract: No

Generalization: [PropertySet](#), Block

Extension: Class

Description

A type of template behavior recognized by an enterprise as being essential to achieving its goals - i.e. a template for a strategic specification of what the enterprise does.

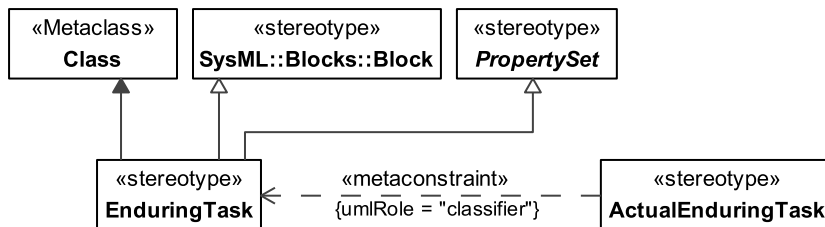


Figure 3:47 - EnduringTask

UAF::Strategic::States

Contains the elements that contribute to the Strategic States Viewpoint.

AchievedEffect

Package: States

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Dependency

Description

A dependency relationship that exists between an ActualState (e.g., observed/measured during testing) of an element that attempts to achieve a DesiredEffect and an Achiever.

Figure 3:48 – AchievedEffect

Associations

desiredEffect : DesiredEffect[*]

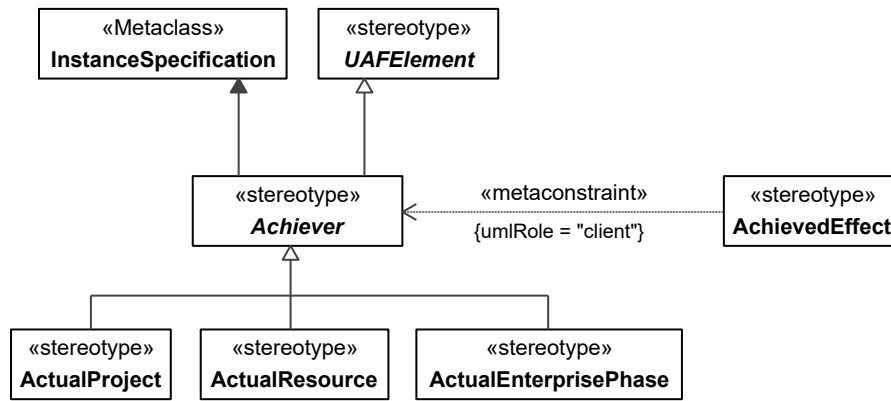


Figure 3:49 - Achiever

DesiredEffect

Package: States

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Dependency

Description

A dependency relationship relating the Desirer (a Capability or OrganizationalResource) to an ActualState.

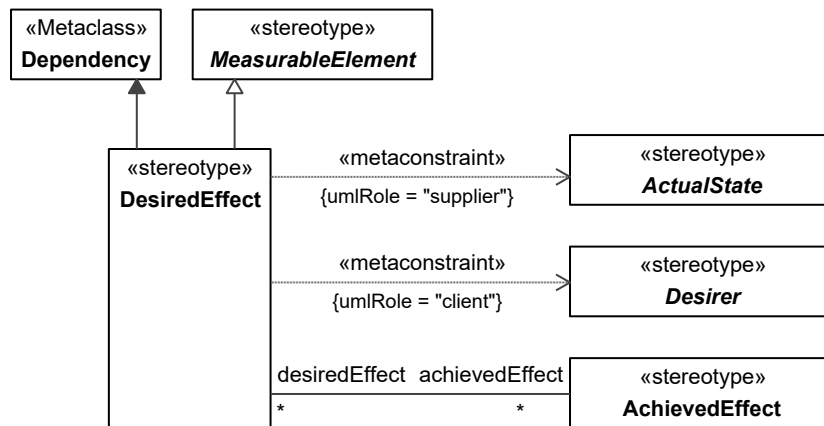


Figure 3:50 – DesiredEffect

Associations

achievedEffect : AchievedEffect[*]

Constraints

- [1] DesiredEffect.client Value for the client metaproperty must be stereotyped a specialization of «Desirer».
- [2] DesiredEffect.supplier Value for the supplier metaproperty must be stereotyped a specialization of «ActualState».

that the

ActualOrganization»

support (exhibit)

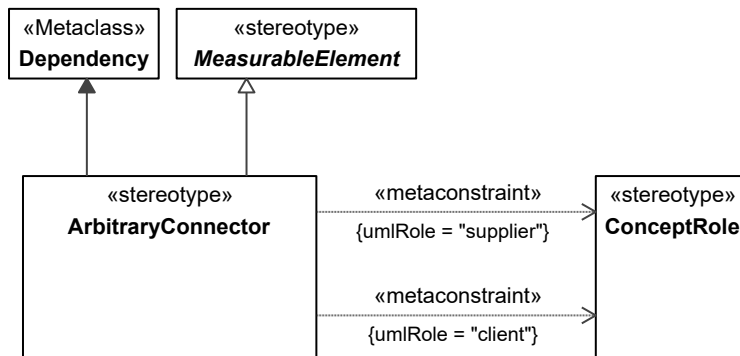


Figure 3:55 – ArbitraryConnector

Constraints

- [1] ArbitraryConnector.client The value for client metaproperty has to be stereotyped «ConceptRole» or its specializations.
- [2] ArbitraryConnector.supplier The value for supplier metaproperty has to be stereotyped «ConceptRole» or its specializations.

ConceptItem

Package: Taxonomy

isAbstract: Yes

Generalization: [UAFElement](#)

Extension: Element

Description

Abstract, an item which may feature in a HighLevelOperationalConcept.

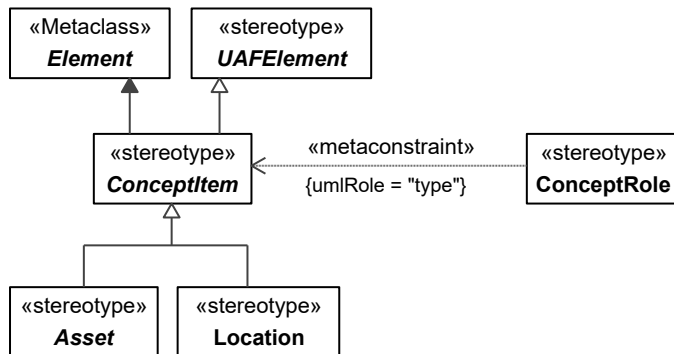


Figure 3:56 - ConceptItem

ConceptRole

Package: Taxonomy

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Property

UAF::Operational::Structure

Contains the elements that contribute to the Operational Structure Viewpoint.

KnownResource

Package: Structure

isAbstract: No

Generalization: [OperationalPerformer](#), [ResourcePerformer](#)

Extension: Class

Description

Asserts that a known ResourcePerformer constrains the implementation of the OperationalPerformer that plays the role in the OperationalArchitecture.

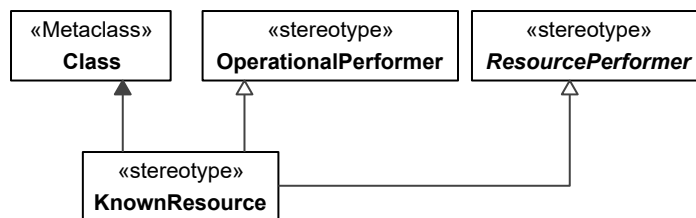


Figure 3:59 - KnownResource

OperationalAgent

Package: Structure

isAbstract: Yes

Generalization: [OperationalAsset](#), [SubjectOfOperationalConstraint](#), [CapableElement](#), [Desirer](#)

Extension: Class

Description

An abstract type grouping OperationalArchitecture and OperationalPerformer.

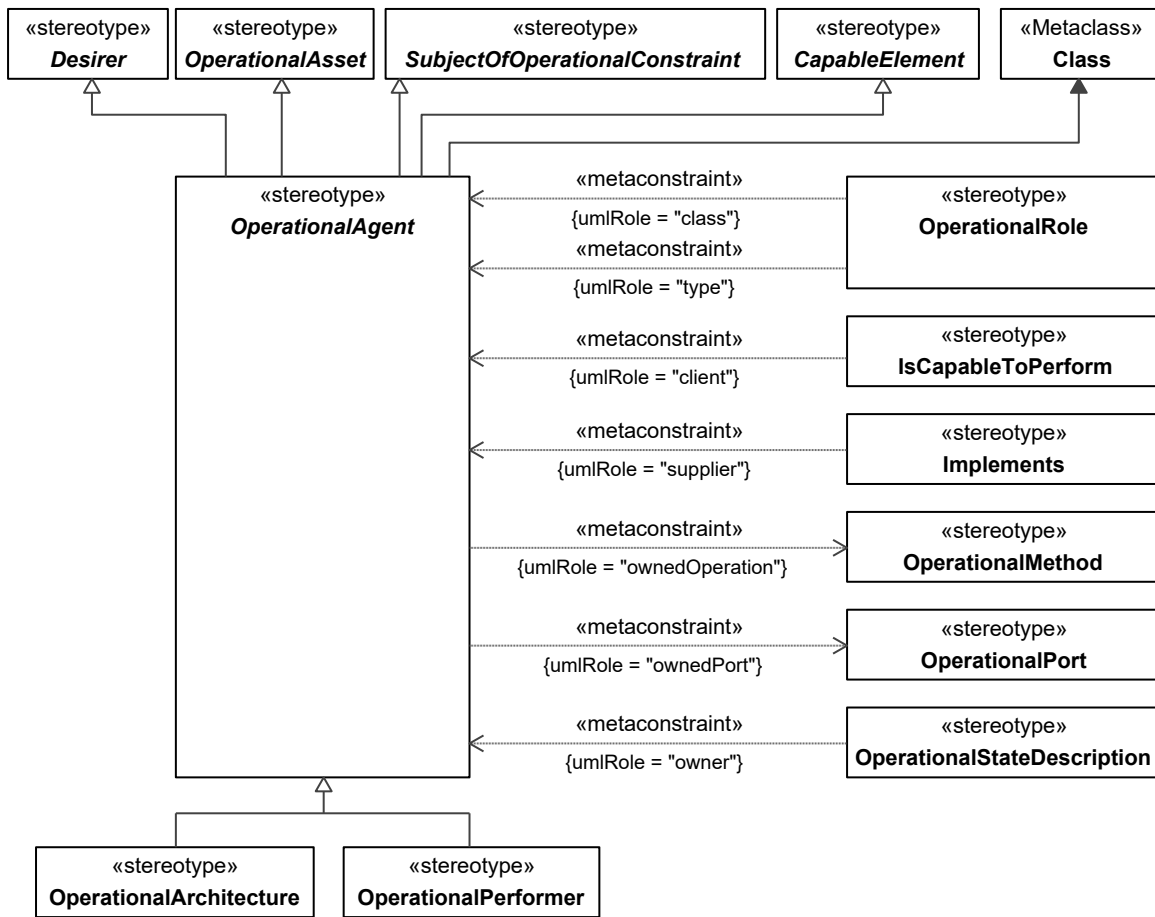


Figure 3:60 – OperationalAgent

Constraints

- | | |
|---|--|
| [1] OperationalAgent.isCapableToPerform | Is capable of performing only «OperationalActivity» elements or its specializations. |
| [2] OperationalAgent.ownedOperation | Values for the ownedOperation metaproperty must be stereotyped «OperationalMethod» or its specializations. |
| [3] OperationalAgent.ownedPort | Values for the ownedPort metaproperty must be stereotyped «OperationalPort» or its specializations. |

OperationalArchitecture

Package: Structure

isAbstract: No

Generalization: [OperationalAgent](#), [Architecture](#)

Extension: Class

Description

An element used to denote a model of the Architecture, described from the Operational perspective.

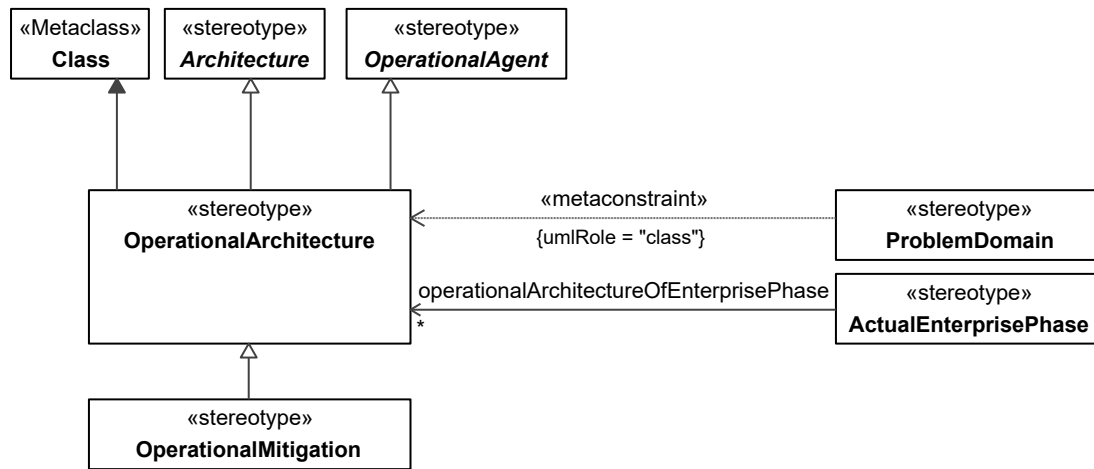


Figure 3:61 – OperationalArchitecture

OperationalMethod

Package: Structure

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Operation

Description

A behavioral feature of an OperationalAgent whose behavior is specified in an OperationalActivity.

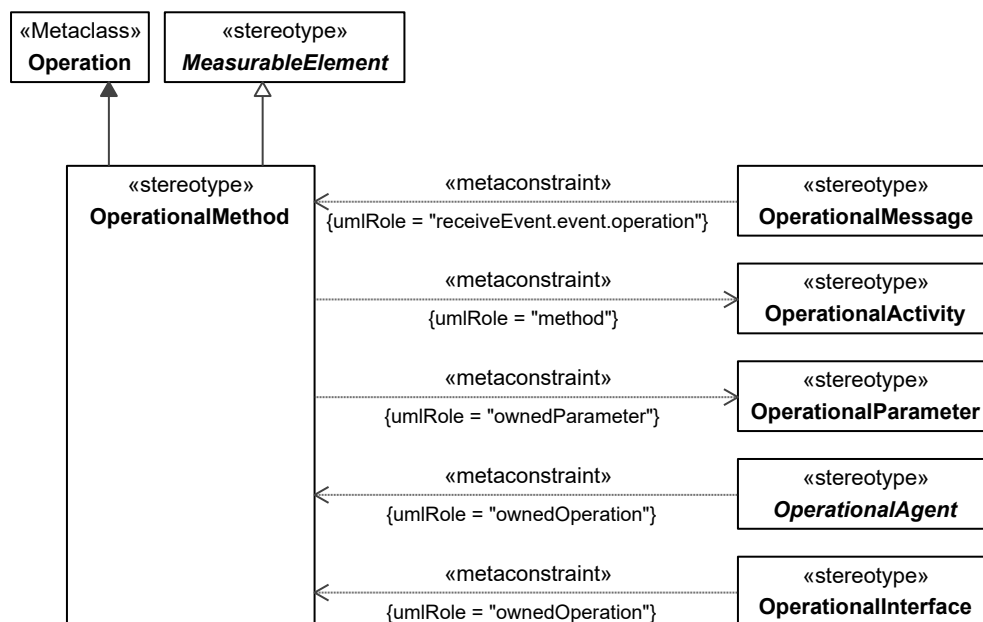


Figure 3:62 – OperationalMethod

Constraints

- | | |
|--------------------------------------|---|
| [1] OperationalMethod.method | Value for the method metaproperty must be stereotyped «OperationalActivity» or its specializations. |
| [2] OperationalMethod.ownedParameter | The values for the ownedParameter metaproperty must be stereotyped «OperationalParameter» or its specializations. |

OperationalParameter

Package: Structure

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Parameter

Description

An element that represents inputs and outputs of an OperationalActivity. It is typed by an OperationalExchangeItem.

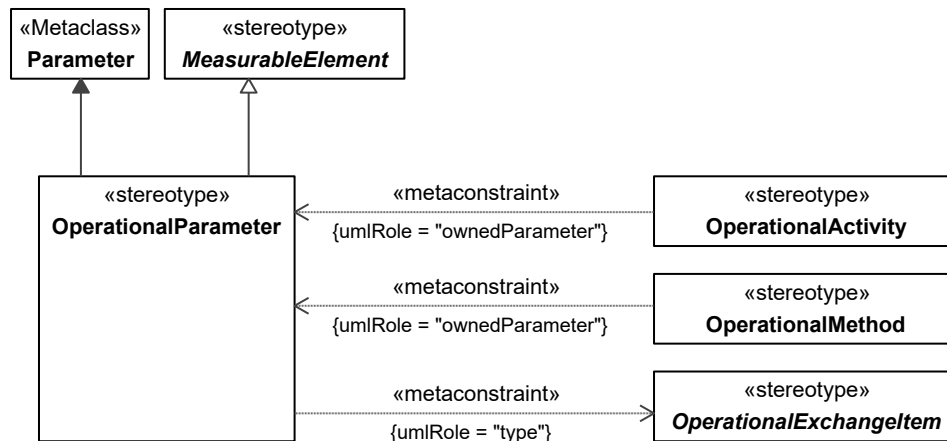


Figure 3:63 – OperationalParameter

Constraints

- | | |
|-------------------------------|---|
| [1] OperationalParameter.type | Value for the type metaproperty must be stereotyped by specialization of «OperationalExchangeItem». |
|-------------------------------|---|

OperationalPerformer

Package: Structure

isAbstract: No

Generalization: [OperationalAgent](#)

Extension: Class

Description

A logical agent that IsCapableToPerform OperationalActivities which produce, consume and process Resources.

OperationalRole

Package: Structure

isAbstract: No

Generalization: [MeasurableElement](#), [LocationHolder](#), [AssetRole](#)

Extension: Property

Description

Usage of an OperationalPerformer or OperationalArchitecture in the context of another OperationalPerformer or OperationalArchitecture. Creates a whole-part relationship.

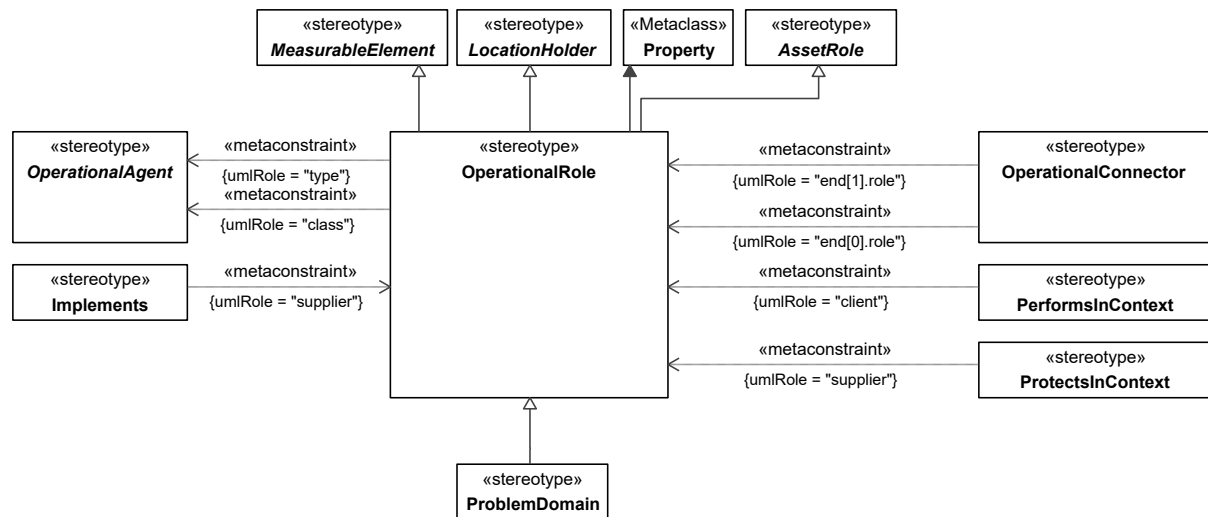


Figure 3:66 – OperationalRole

Constraints

- [1] OperationalRole.class Value for class metaproperty must be stereotyped by a specialization of «OperationalAgent».
- [2] OperationalRole.type Value for type metaproperty must be stereotyped by a specialization of «OperationalAgent».

ProblemDomain

Package: Structure

isAbstract: No

Generalization: [OperationalRole](#)

Extension: Property

Description

A property associated with an OperationalArchitecture, used to specify the scope of the problem.

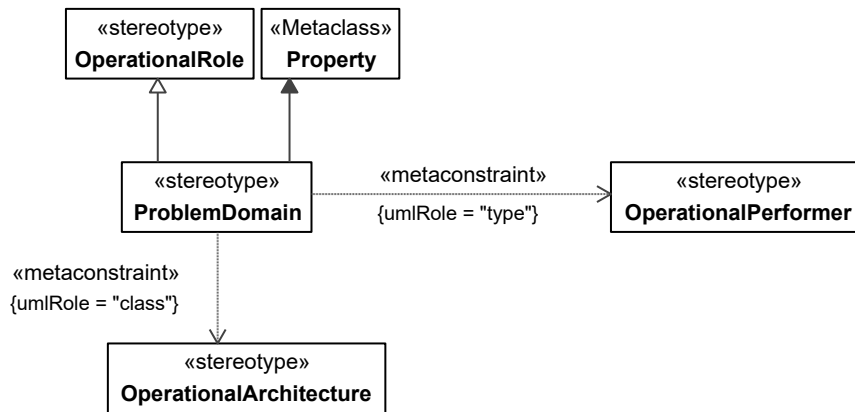


Figure 3:67 – ProblemDomain

Constraints

- [1] ProblemDomain.class Value for the class metaproperty must be stereotyped «OperationalArchitecture» or its specializations.
- [2] ProblemDomain.type Value for the type metaproperty must be stereotyped «OperationalPerformer» or its specializations.

UAF::Operational::Connectivity

Contains the elements that contribute to the Operational Connectivity Viewpoint.

OperationalConnector

Package: Connectivity

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Connector

Description

A Connector that goes between OperationalRoles representing a need to exchange Resources. It can carry a number of OperationalExchanges.

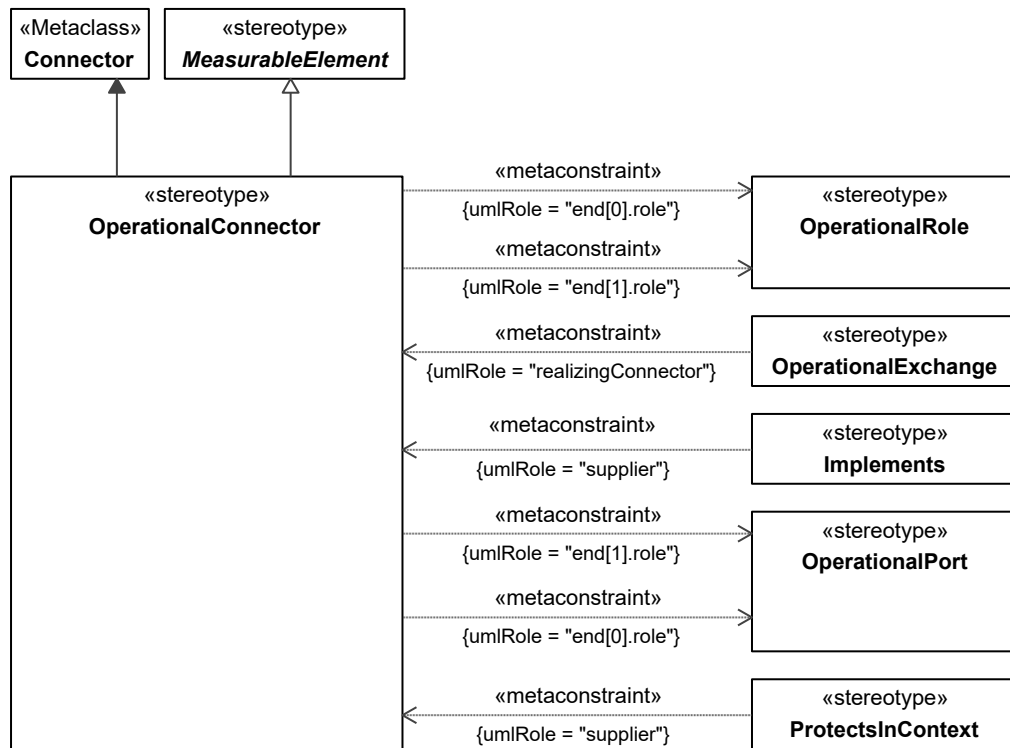


Figure 3:68 – OperationalConnector

Constraints

- [1] **OperationalConnector.end** The value for the role metaproperty for the owned **ConnectorEnd** must be stereotype **«OperationalRole»/«OperationalPort»** or its specializations.

OperationalExchange

Package: Connectivity

isAbstract: No

Generalization: [Exchange](#), [SubjectOfOperationalConstraint](#)

Extension: InformationFlow

Description

Asserts that a flow can exist between **OperationalPerformers** (i.e. flows of information, people, materiel, or energy).

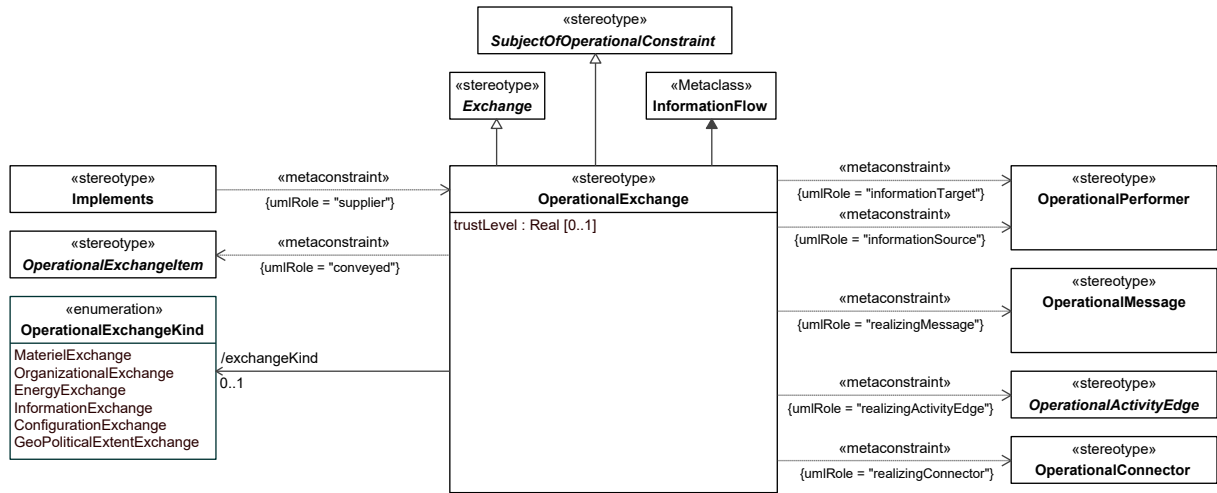


Figure 3:69 – OperationalExchange

Attributes

trustLevel : Real[0..1] Captures the directional arbitrary level of trust related to an OperationalExchange between two OperationalPerformers.

Associations

exchangeKind : OperationalExchangeKind[0..1] Captures the kind of Resource being exchanged.

Constraints

[1] OperationalExchange.conveyed

In case of OperationalExchange.operationalExchangeKind:
 = InformationExchange, the conveyed element must be stereotyped «InformationElement» or its specializations,
 = MaterielExchange, the conveyed element must be stereotyped «ResourceArtifact» or its specializations,
 = EnergyExchange, the conveyed element must be stereotyped «NaturalResource» or its specializations,
 = OrganizationalExchange, the conveyed element must be stereotyped «OrganizationalResource» or its specializations,
 = ConfigurationExchange, the conveyed element must be stereotyped «CapabilityConfiguration» or its specializations, or
 = GeoPoliticalExtentExchange, the conveyed element must be stereotyped «GeoPoliticalExtentType» or its specializations.

[2] OperationalExchange.informationSource

Value for informationSource metaproperty has to be stereotyped «OperationalPerformer» or its specializations.

[3] OperationalExchange.informationTarget

Value for informationTarget metaproperty has to be stereotyped «OperationalPerformer» or its specializations.

[4] OperationalExchange.realizingActivityEdge

Value for realizingActivityEdge metaproperty has to be stereotyped by any specialization of «OperationalActivityEdge».

[5] OperationalExchange.realizingConnector

Value for realizingConnector metaproperty has to be stereotyped «OperationalConnector» or its specializations.

[6] OperationalExchange.realizingMessage

Value for realizingMessage metaproperty has to be stereotyped «OperationalMessage» or its specializations.

OperationalExchangeItem

Package: Connectivity

isAbstract: Yes

Generalization: [Resource](#)

Description

An abstract grouping for elements that defines the types of elements that can be exchanged between OperationalPerformers and conveyed by an OperationalExchange.

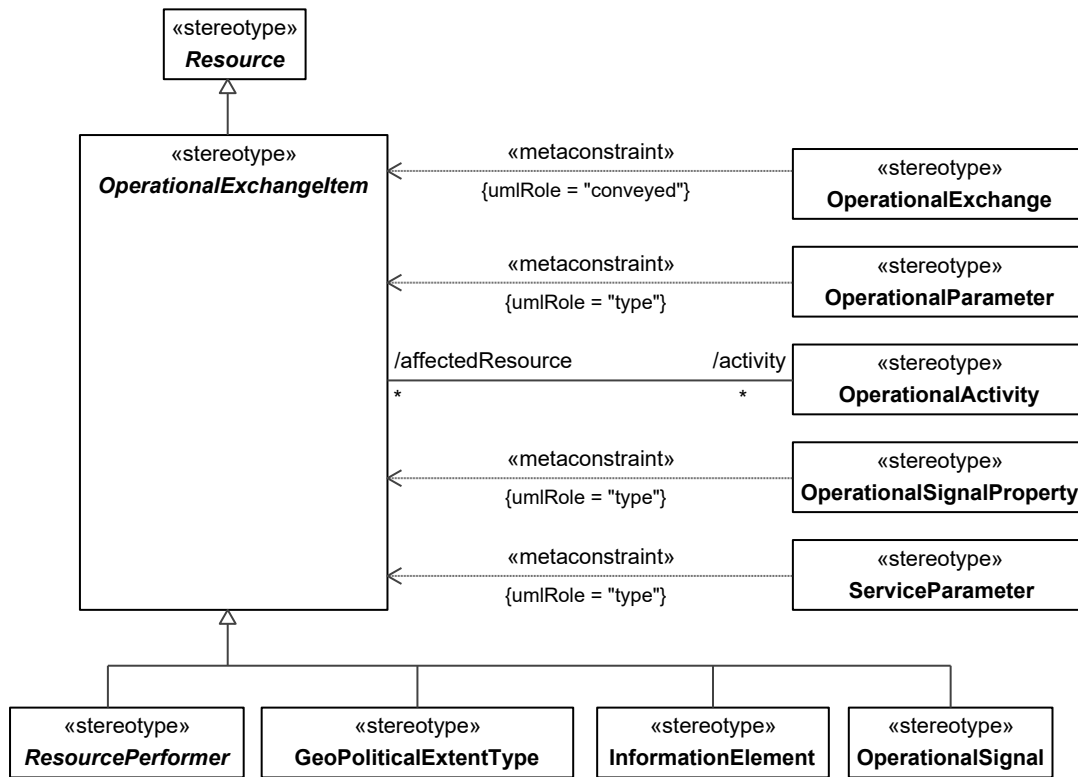


Figure 3:70 – OperationalExchangeItem

Associations

activity : OperationalActivity[*] A collection of OperationalActivities that consume and/or produce the OperationalExchangeItem internally.

OperationalExchangeKind

Package: Connectivity

isAbstract: No

Description

Enumeration of the possible kinds of operational exchange applicable to an OperationalExchange. Its enumeration literals are:

- **MaterielExchange** - Indicates that the OperationalExchange associated with the OperationalExchangeKind is a logical flow of materiel (artifacts) between Functions.

- **OrganizationalExchange** - Indicates that the **OperationalExchange** associated with the **OperationalExchangeKind** is a logical flow where human resources (**PostTypes**, **RoleTypes**) flow between **OperationalPerformers**.
- **EnergyExchange** - Indicates that the **OperationalExchange** associated with the **OperationalExchangeKind** is a logical flow where energy is flowed from one **OperationalPerformer** to another.
- **InformationExchange** - Indicates that the **OperationalExchange** associated with the **OperationalExchangeKind** is a logical flow where information is flowed from one **OperationalPerformer** to another.
- **ConfigurationExchange** - Indicates that the **OperationalExchange** associated with the **OperationalExchangeKind** is a logical flow where **CapabilityConfigurations** flow from one **OperationalPerformer** to another.
- **GeoPoliticalExtentExchange** - Indicates that the **OperationalExchange** associated with the **OperationalExchangeKind** is a logical flow where **GeoPoliticalExtentTypes** (i.e. **Borders**) flow from one place to another.

OperationalInterface

Package: Connectivity

isAbstract: No

Generalization: [PropertySet](#), [InterfaceBlock](#)

Extension: Class

Description

A declaration that specifies a contract between the **OperationalPerformer** it is related to, and any other **OperationalPerformers** it can interact with.

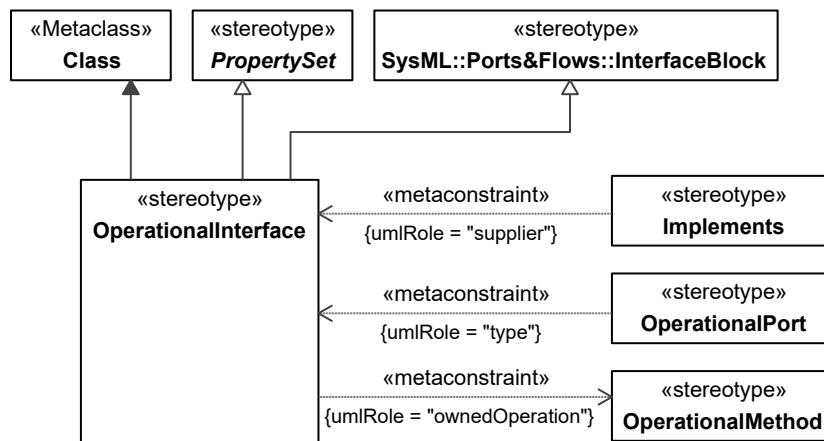


Figure 3:71 – OperationalInterface

Constraints

- [1] **OperationalInterface.ownedOperation** Values for the **ownedOperation** metaproperty must be stereotyped **«OperationalMethod»** or its specializations.

OperationalSignal

Package: Connectivity

isAbstract: No

Generalization: [OperationalExchangeItem](#), [SubjectOfOperationalConstraint](#)

Extension: Signal

Description

An **OperationalSignal** is a specification of a kind of communication between operational performers in which a reaction is asynchronously triggered in the receiver without a reply.

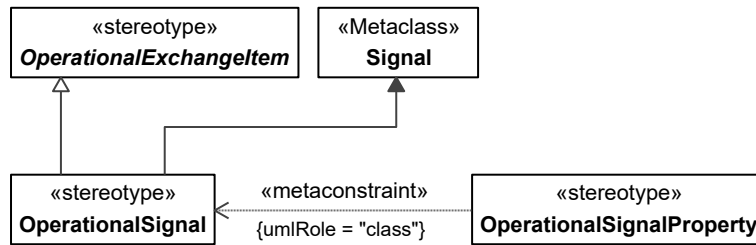


Figure 3:72 - OperationalSignal

OperationalSignalProperty

Package: Connectivity

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Property

Description

A property of an **OperationalSignal** typed by **OperationalExchangeItem**. It enables **OperationalExchangeItem** e.g. **InformationElement** to be passed as arguments of the **OperationalSignal**.

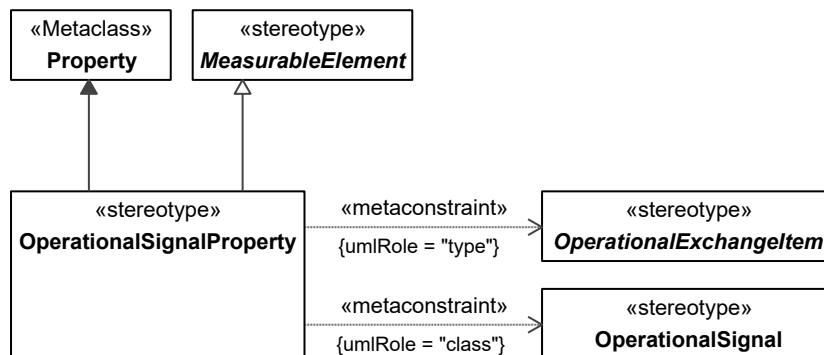


Figure 3:73 – OperationalSignalProperty

Constraints

- [1] **OperationalSignalProperty.class** Value for class metaproperty must be stereotyped «OperationalSignal» or its specializations.
- [2] **OperationalSignalProperty.type** Value for type metaproperty must be stereotyped by a specialization of «OperationalExchangeItem».

UAF::Operational::Processes

Contains the elements that contribute to the Operational Processes Viewpoint.

OperationalActivity

Package: Processes

isAbstract: No

Generalization: [Activity](#), [SubjectOfOperationalConstraint](#)

Extension: Activity

Description

An Activity that captures a logical process, specified independently of how the process is carried out.

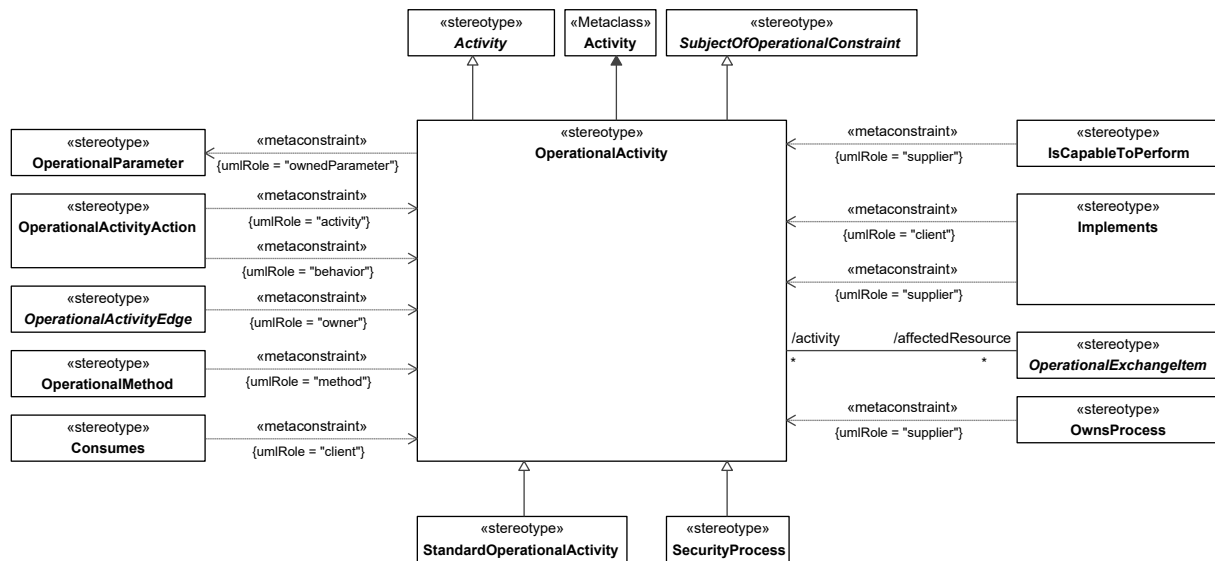


Figure 3:74 – OperationalActivity

Associations

affectedResource : **OperationalExchangeItem**[*] A collection of **OperationalExchangeItems** consumed and produced internally within the **OperationalActivity**.

Constraints

[1] **OperationalActivity.ownedParameter** The values for the **ownedParameter** metaproperty must be stereotyped **«OperationalParameter»** or its specializations.

OperationalActivityAction

Package: Processes

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: CallBehaviorAction

Description

A call of an **OperationalActivity** in the context of another **OperationalActivity**.

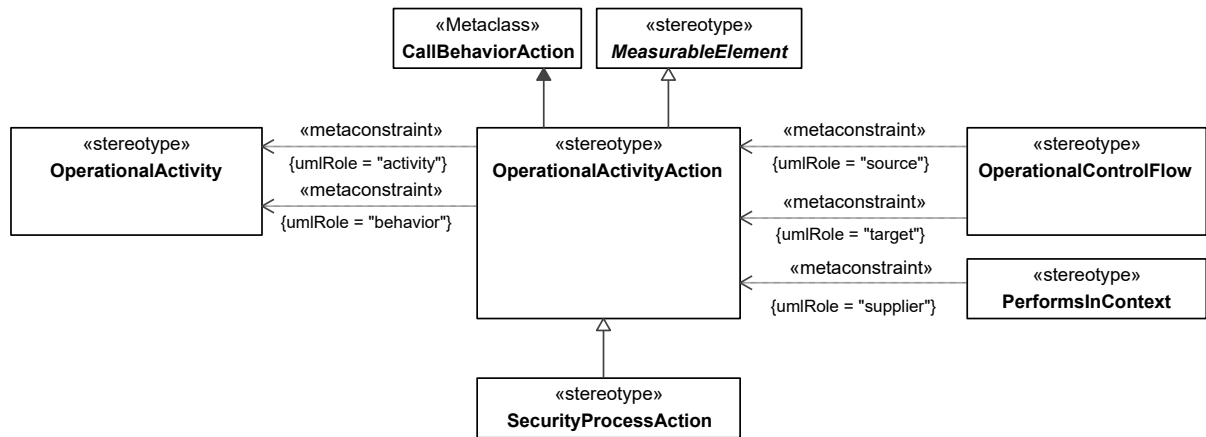


Figure 3:75 – OperationalActivityAction

Constraints

- [1] OperationalActivityAction.activity Value for the activity metaproperty must be stereotyped «OperationalActivity» or its specializations.
- [2] OperationalActivityAction.behavior Value for activity metaproperty must be stereotyped «OperationalActivity» or its specializations.

OperationalActivityEdge

Package: Processes

isAbstract: Yes

Generalization: [MeasurableElement](#)

Extension: ActivityEdge

Description

Abstract grouping for OperationalControlFlow and OperationalObjectFlow.

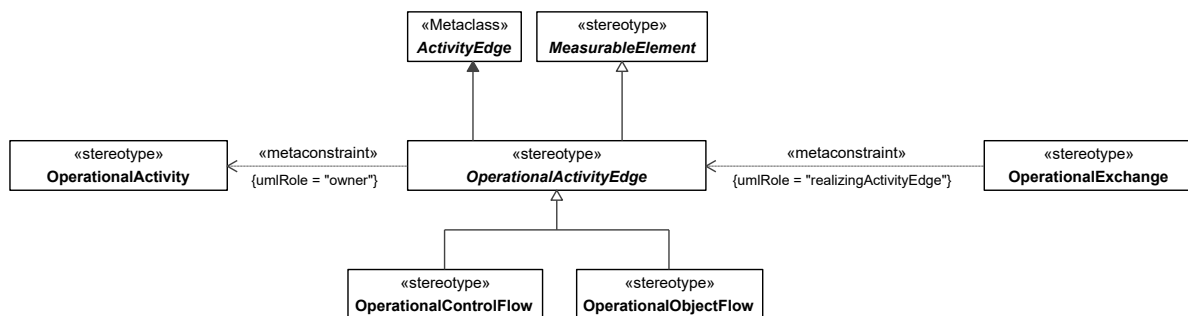


Figure 3:76 – OperationalActivityEdge

Constraints

- [1] OperationalActivityEdge.owner «OperationalActivityEdge» must be owned directly or indirectly by «OperationalActivity» or its specializations.

OperationalControlFlow

Package: Processes

isAbstract: No

Generalization: [OperationalActivityEdge](#)

Extension: ControlFlow

Description

An ActivityEdge that shows the flow of control between OperationalActivityActions.

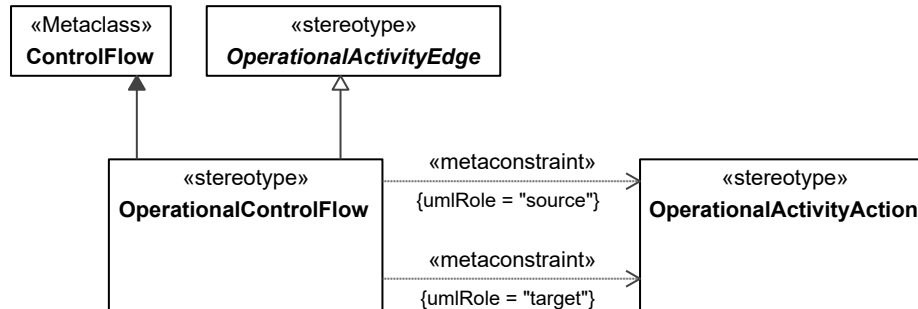


Figure 3:77 – OperationalControlFlow

Constraints

- [1] OperationalControlFlow.source Value for the source metaproperty must be stereotyped «OperationalActivityAction» or its specializations.
- [2] OperationalControlFlow.target Value for the target metaproperty must be stereotyped «OperationalActivityAction» or its specializations.

OperationalObjectFlow

Package: Processes

isAbstract: No

Generalization: [OperationalActivityEdge](#)

Extension: ObjectFlow

Description

An ActivityEdge that shows the flow of Resources (objects/information) between OperationalActivityActions.

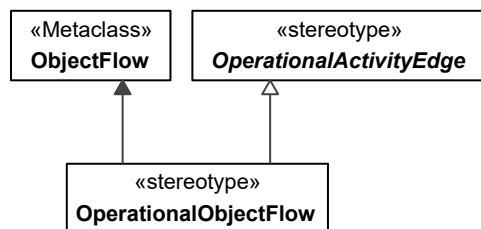


Figure 3:78 - OperationalObjectFlow

StandardOperationalActivity

Package: Processes

isAbstract: No

Generalization: [OperationalActivity](#)

Extension: Activity

Description

A sub-type of OperationalActivity that is a standard operating procedure.

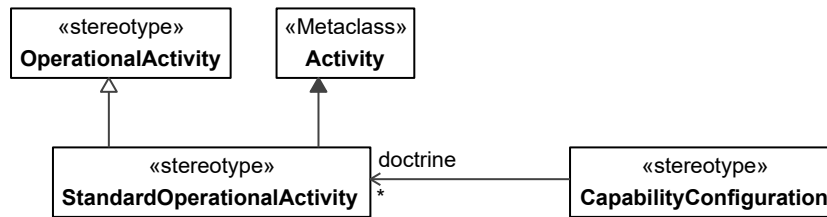


Figure 3:79 - StandardOperationalActivity

UAF::Operational::States

Contains the elements that contribute to the Operational States Viewpoint.

OperationalStateDescription

Package: States

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: StateMachine

Description

A state machine describing the behavior of an OperationalPerformer, depicting how the OperationalPerformer responds to various events and the actions.

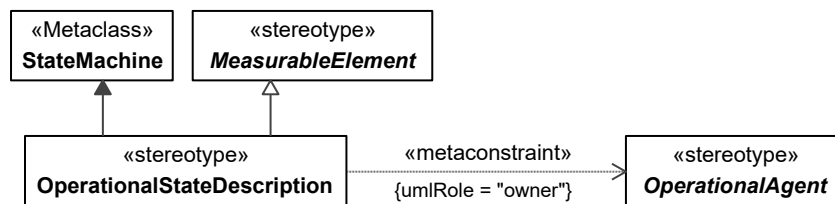


Figure 3:80 – OperationalStateDescription

Constraints

- [1] OperationalStateDescription.owner Values for the owner metaproperty must be stereotyped with specializations of «OperationalAgent».

UAF::Operational::Interaction Scenarios

Contains the elements that contribute to the Operational Interaction Scenarios Viewpoint.

OperationalMessage

Package: Interaction Scenarios

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Message

Description

Message for use in an operational interaction scenario which carries any of the subtypes of OperationalExchange.

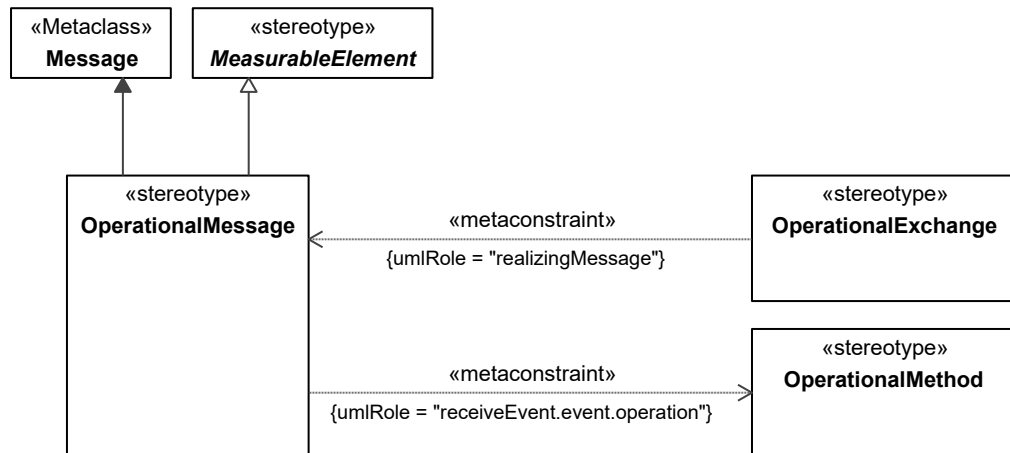


Figure 3:81 – OperationalMessage

Constraints

- [1] `OperationalMessage.receiveEvent.event.operation` Values for the `receiveEvent.event.operation` metaproperty must be stereotyped with **«OperationalMethod»** or its specializations.

UAF::Operational::Information

Contains the elements that contribute to the Operational Information Viewpoint.

InformationElement

Package: Information

isAbstract: No

Generalization: [OperationalAsset](#), [OperationalExchangeItem](#), [SubjectOfOperationalConstraint](#)

Extension: Class

Description

An item of information that flows between **OperationalPerformers** and is produced and consumed by the **OperationalActivities** that the **OperationalPerformers** are capable to perform (see **IsCapableToPerform**).

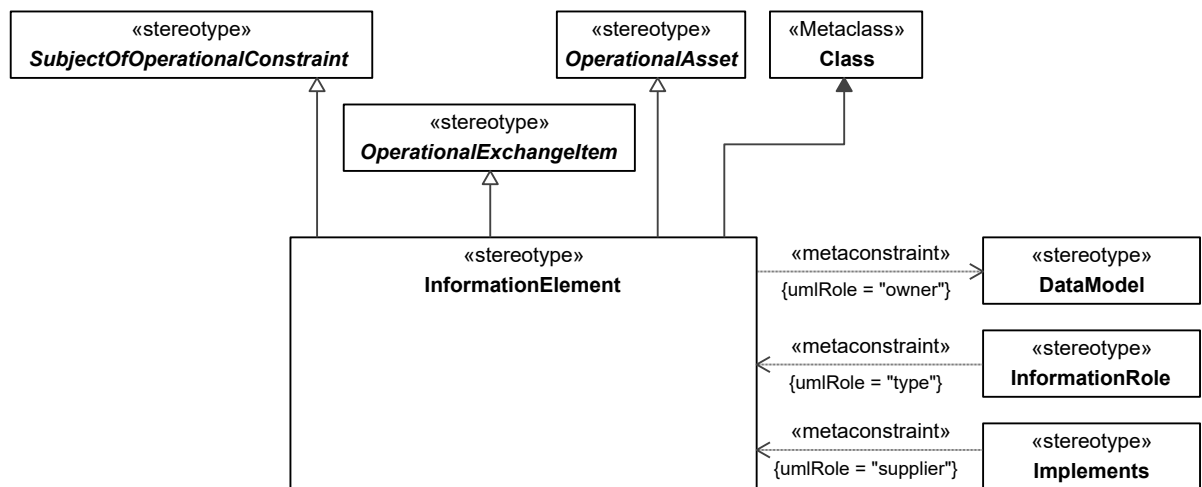


Figure 3:82 – InformationElement

Constraints

- [1] InformationElement.owner Values for the owner metaproperty must be stereotyped «DataModel» or its specializations.

UAF::Operational::Constraints

Contains the elements that contribute to the Operational Constraints Viewpoint.

OperationalConstraint

Package: Constraints

isAbstract: No

Generalization: [Rule](#)

Extension: Constraint

Description

A Rule governing an operational architecture element i.e. OperationalPerformer, OperationalActivity, InformationElement etc.

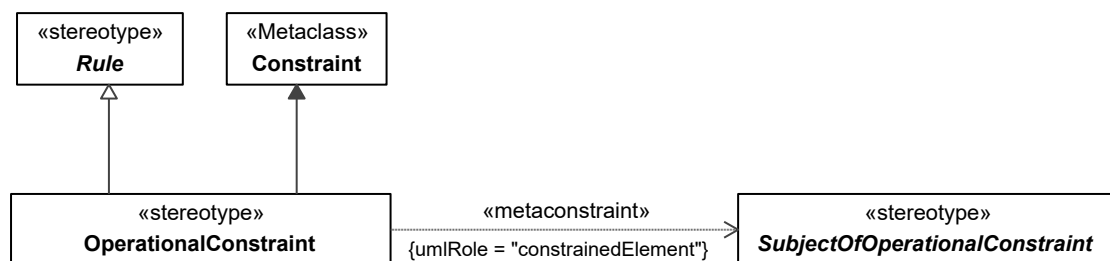


Figure 3:83 – OperationalConstraint

Constraints

- [1] OperationalConstraint.constrainedElement Value for the constrainedElement metaproperty must be stereotyped by any specialization of «SubjectOfOperationalConstraint».

SubjectOfOperationalConstraint

Package: Constraints

isAbstract: Yes

Generalization: [UAFElement](#)

Extension: Element

Description

An abstract grouping of elements that can be the subject of an OperationalConstraint.

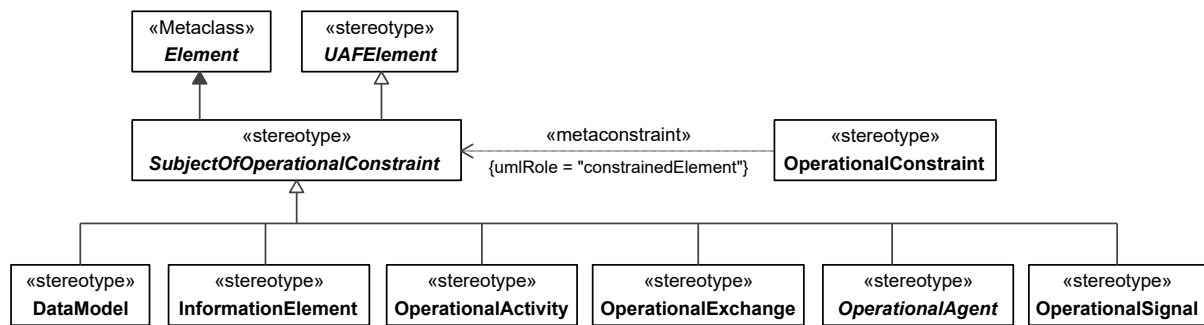


Figure 3:84 - SubjectOfOperationalConstraint

3.1.6 UAF::Services

Stakeholders: Enterprise Architects, Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: specifications of services required to exhibit a Capability.

Definition: shows Service Specifications and required and provided service levels of these specifications required to exhibit a Capability or to support an Operational Activity.

UAF::Services::Taxonomy

Contains the elements that contribute to the Services Taxonomy Viewpoint.

ServiceSpecification

Package: Taxonomy

isAbstract: No

Generalization: [PropertySet](#), [VersionedElement](#), [CapableElement](#), Block

Extension: Class

Description

The specification of a set of functionalities provided by one element for the use of others.

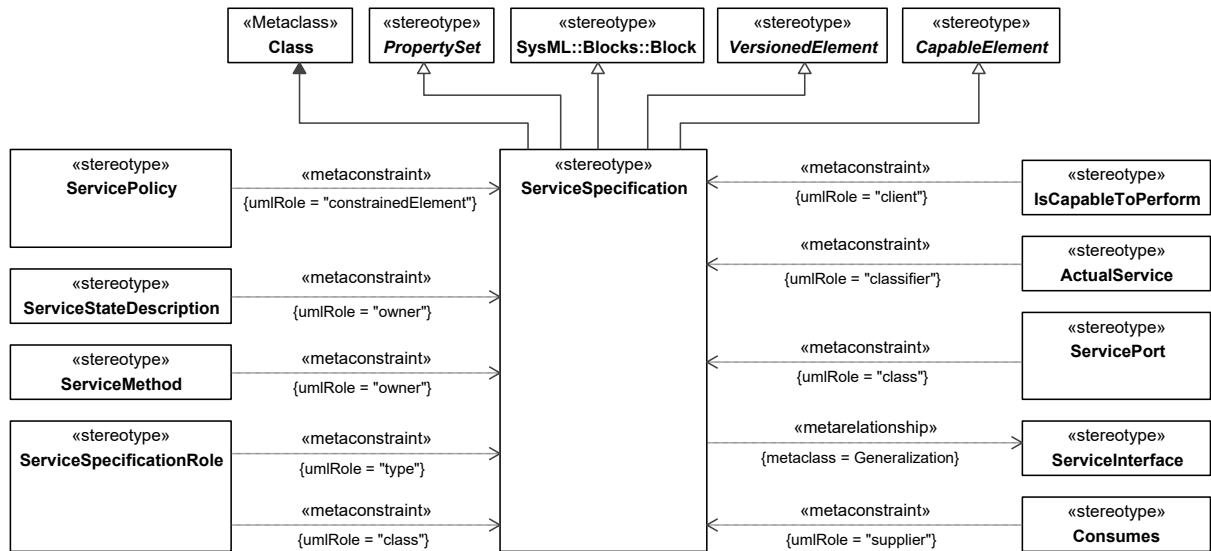


Figure 3:85 - ServiceSpecification

UAF::Services::Structure

Contains the elements that contribute to the Services Structure Viewpoint.

ServiceMethod

Package: Structure

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Operation

Description

A behavioral feature of a ServiceSpecification whose behavior is specified in a ServiceFunction.

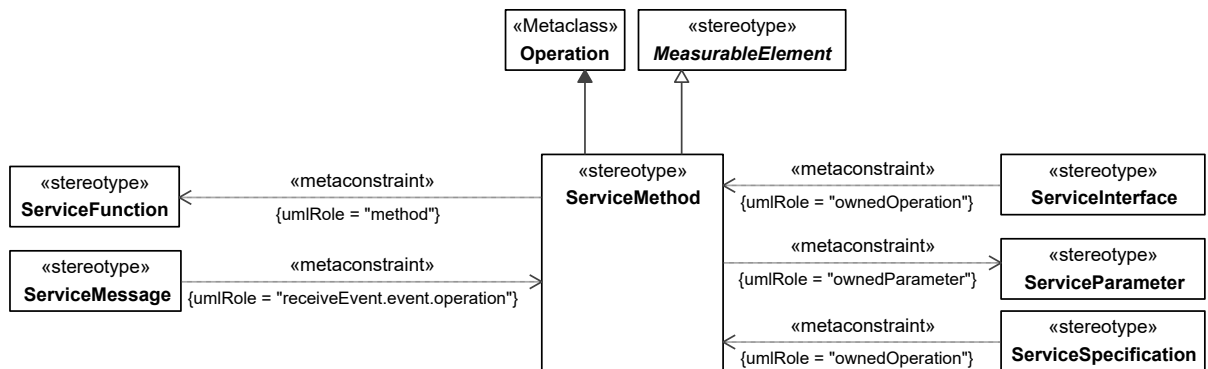


Figure 3:86 – ServiceMethod

Constraints

[1] ServiceMethod.method

Value for the method metaproperty must be stereotyped «ServiceFunction» or its specializations.

- [2] `ServiceMethod.ownedParameter` The values for the `ownedParameter` metaproperty must be stereotyped `«ServiceParameter»` or its specializations.
- [3] `ServiceMethod.owner` The values for the `owner` metaproperty must be stereotyped `«ServiceSpecification»` or its specializations.

ServiceParameter

Package: Structure

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Parameter

Description

An element that represents inputs and outputs of a `ServiceFunction`, represents inputs and outputs of a `ServiceSpecification`.

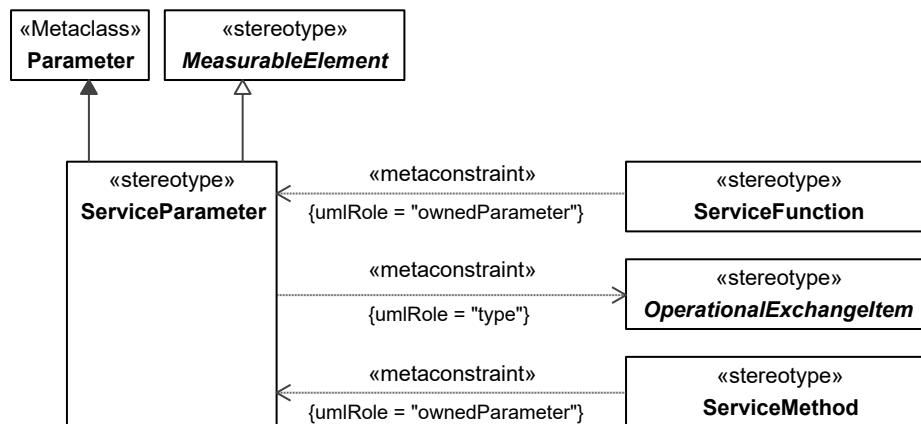


Figure 3:87 – ServiceParameter

Constraints

- [1] `ServiceParameter.type` The values for the `type` metaproperty must be stereotyped a specialization of `«OperationalExchangeItem»`.

ServicePort

Package: Structure

isAbstract: No

Generalization: `ProxyPort`, [MeasurableElement](#)

Extension: Port

Description

An interaction point for a `ServiceSpecification` through which it can interact with the outside environment and which is defined by a `ServiceInterface`.

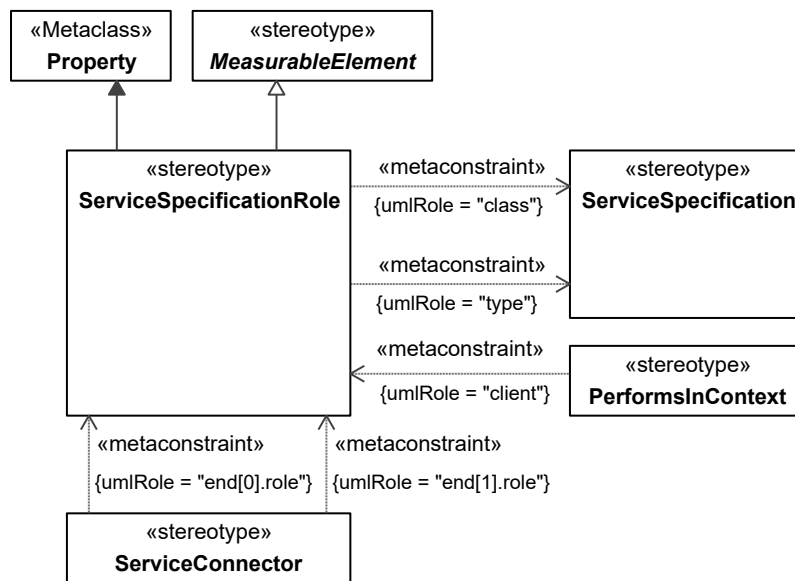


Figure 3:89 – ServiceSpecificationRole

Constraints

- [1] ServiceSpecificationRole.class Value for the class metaproperty must be stereotyped «ServiceSpecification» or its specializations.
- [2] ServiceSpecificationRole.type Value for the type metaproperty must be stereotyped «ServiceSpecification» or its specializations.

UAF::Services::Connectivity

Contains the elements that contribute to the Services Connectivity Viewpoint.

ServiceConnector

Package: Connectivity

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Connector

Description

A channel for exchange between two ServiceSpecifications. Where one acts as the consumer of the other.

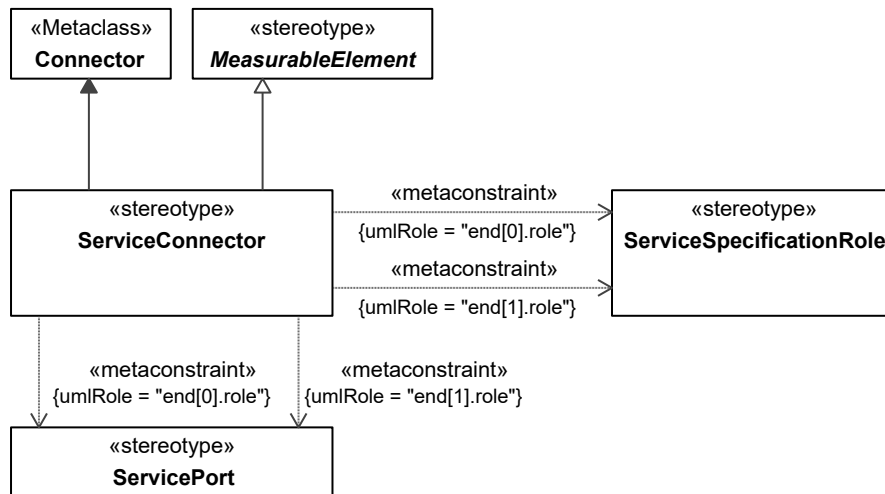


Figure 3:90 – ServiceConnector

Constraints

- [1] ServiceConnector.end The value for the role metaproperty for the owned ConnectorEnd must be stereotyped «ServicePort», «ServiceSpecificationRole» or their specializations.

ServiceInterface

Package: Connectivity

isAbstract: No

Generalization: [PropertySet](#), [InterfaceBlock](#)

Extension: Class

Description

A contract that defines the ServiceMethods and ServiceMessageHandlers that the ServiceSpecification realizes.

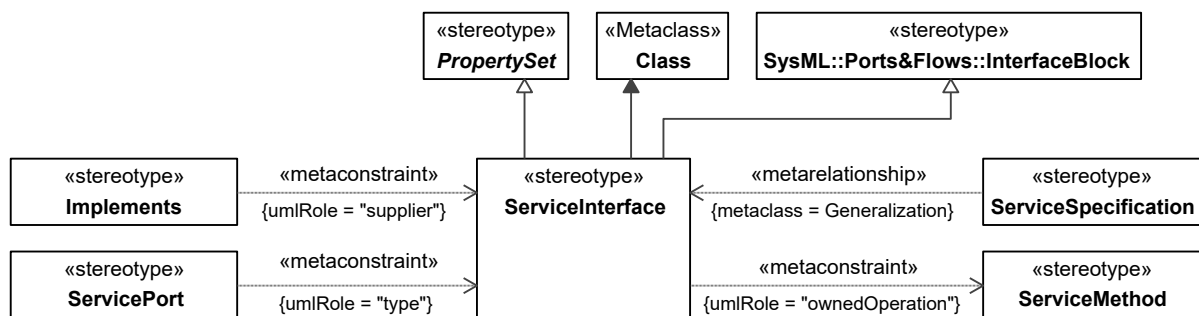


Figure 3:91 – ServiceInterface

Constraints

- [1] ServiceInterface.ownedOperation Values for the ownedOperation metaproperty must be stereotyped «ServiceMethod» or its specializations.

UAF::Services::Processes

Contains the elements that contribute to the Services Processes Viewpoint.

ServiceFunction

Package: Processes

isAbstract: No

Generalization: [Activity](#)

Extension: Activity

Description

An Activity that describes the abstract behavior of ServiceSpecifications, regardless of the actual implementation.

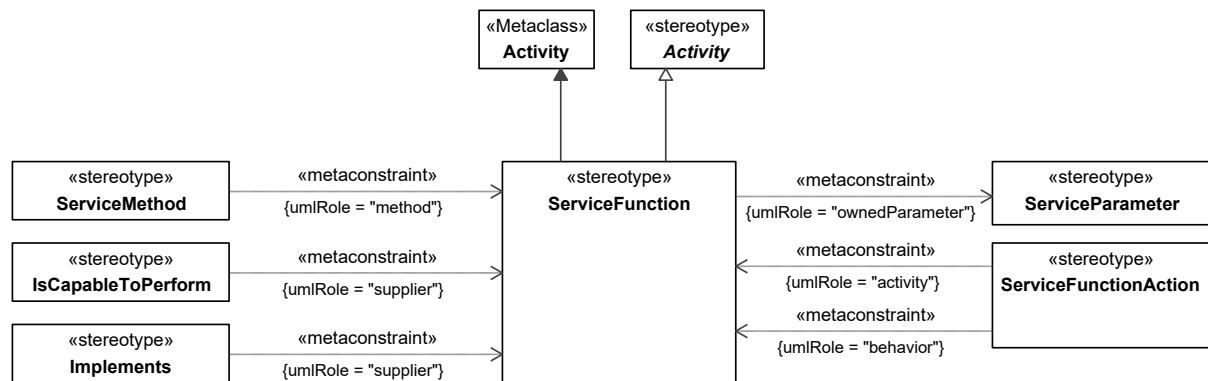


Figure 3:92 – ServiceFunction

Constraints

- [1] ServiceFunction.ownedParameter The values for the ownedParameter metaproperty must be stereotyped «ServiceParameter».

ServiceFunctionAction

Package: Processes

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: CallBehaviorAction

Description

A call of a ServiceFunction in the context of another ServiceFunction.

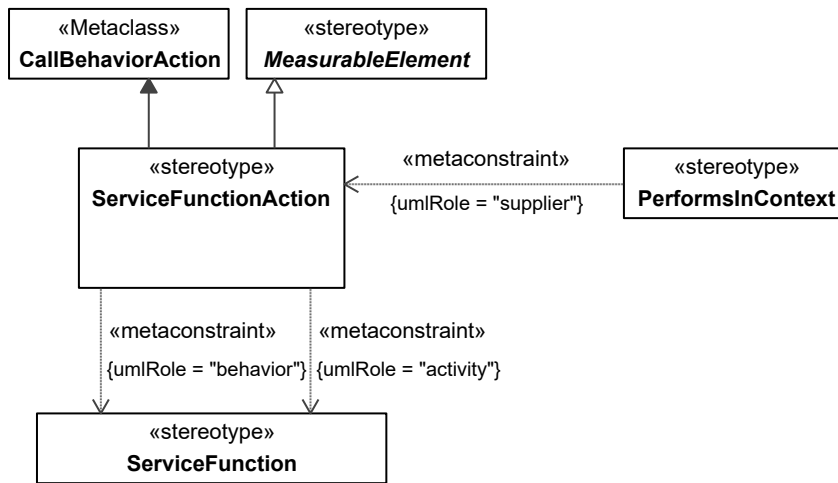


Figure 3:93 – ServiceFunctionAction

Constraints

- [1] ServiceFunctionAction.activity Value for the behavior metaproperty must be stereotyped «ServiceFunction» or its specializations.
- [2] ServiceFunctionAction.behavior Value for the activity metaproperty must be stereotyped «ServiceFunction» or its specializations.

UAF::Services::States

Contains the elements that contribute to the Services States Viewpoint.

ServiceStateDescription

Package: States

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: StateMachine

Description

A state machine describing the behavior of a ServiceSpecification, depicting how the ServiceSpecification responds to various events and the actions.

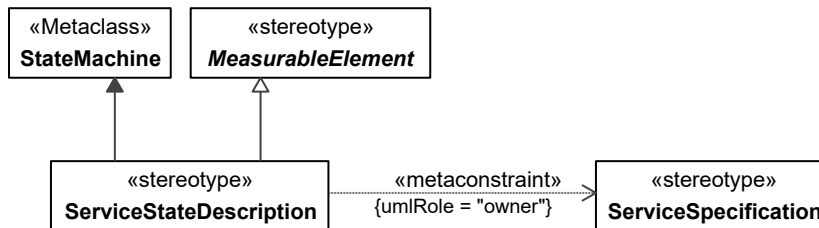


Figure 3:94 – ServiceStateDescription

Constraints

- [1] ServiceStateDescription.owner Values for the owner metaproperty must be stereotyped «ServiceSpecification» or its specializations.

UAF::Services::Interaction Scenarios

Contains the elements that contribute to the Services Interaction Scenarios Viewpoint.

ServiceMessage

Package: Interaction Scenarios

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Message

Description

Message for use in a Service Event-Trace.

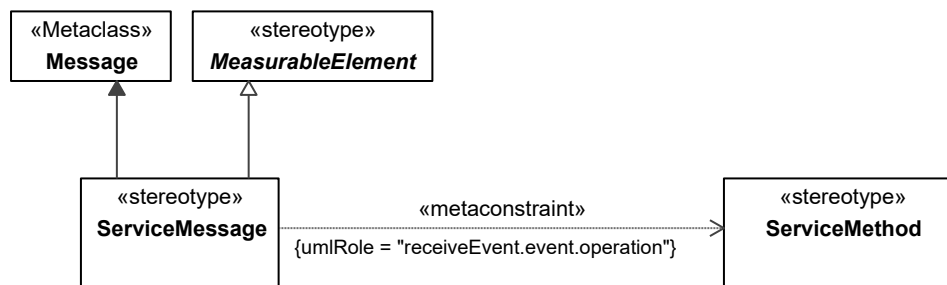


Figure 3:95 – ServiceMessage

Constraints

- [1] ServiceMessage.receiveEvent.event.operation Values for the receiveEvent.event.operation metaproperty must be stereotyped with «ServiceMethod» or its specializations.

UAF::Services::Constraints

Contains the elements that contribute to the Services Constraints Viewpoint.

ServicePolicy

Package: Constraints

isAbstract: No

Generalization: [Rule](#)

Extension: Constraint

Description

A constraint governing the use of one or more ServiceSpecifications.

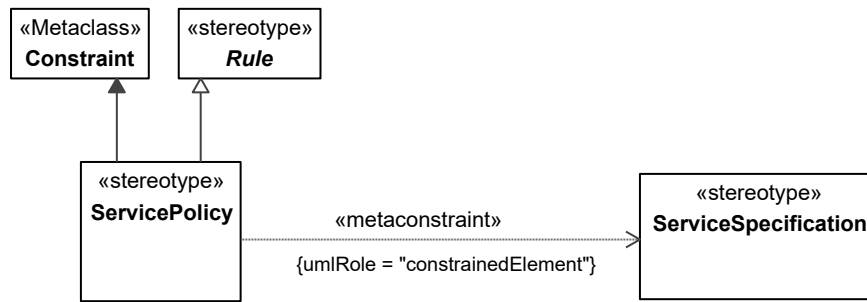


Figure 3:96 – ServicePolicy

Constraints

- [1] ServicePolicy.constrainedElement Values for constrainedElement metaproperty must be stereotyped «ServiceSpecification» or its specializations.

UAF::Services::Traceability

Contains the elements that contribute to the Services Traceability Viewpoint.

Consumes

Package: Traceability

isAbstract: No

Generalization: Allocate, [MeasurableElement](#)

Extension: Abstraction

Description

An abstraction relationship that asserts that a service in someway contributes or assists in the execution of an OperationalActivity.

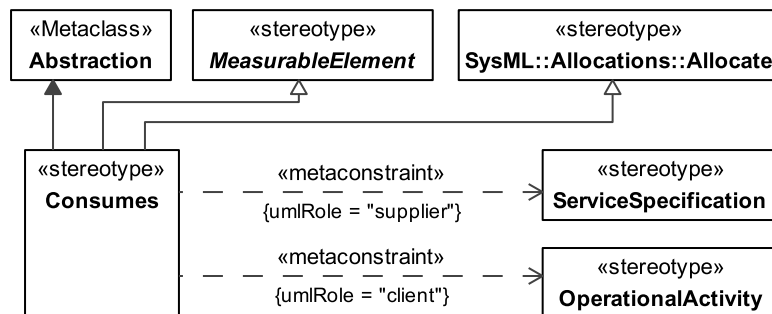


Figure 3:97 – Consumes

Constraints

- [1] Consumes.client Value for the client metaproperty must be stereotyped «OperationalActivity» or its specializations.
- [2] Consumes.supplier Value for the supplier metaproperty must be stereotyped «ServiceSpecification» or its specializations.

3.1.7 UAF::Personnel

Stakeholders: Human resources, Solution Providers, PMs.

Concerns: human factors.

Definition: aims to clarify the role of Human Factors (HF) when creating architectures in order to facilitate both Human Factors Integration (HFI) and systems engineering (SE).

UAF::Personnel::Taxonomy

Contains the elements that contribute to the Personnel Taxonomy Viewpoint.

Organization

Package: Taxonomy

isAbstract: No

Generalization: [OrganizationalResource](#)

Extension: Class

Description

A group of OrganizationalResources (Persons, Posts, Organizations and Responsibilities) associated for a particular purpose.

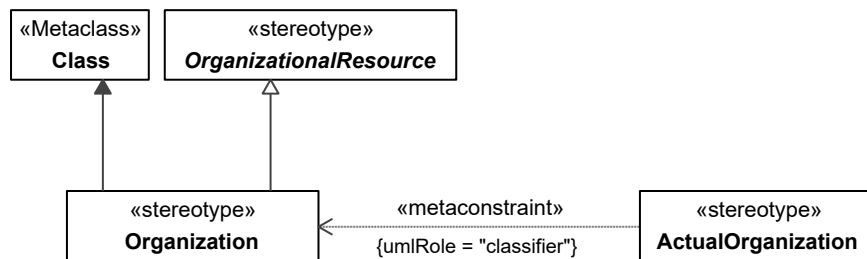


Figure 3:98 - Organization

OrganizationalResource

Package: Taxonomy

isAbstract: Yes

Generalization: [PhysicalResource](#), [Stakeholder](#)

Extension: Class

Description

An abstract element grouping for Organization, Person, Post and Responsibility.

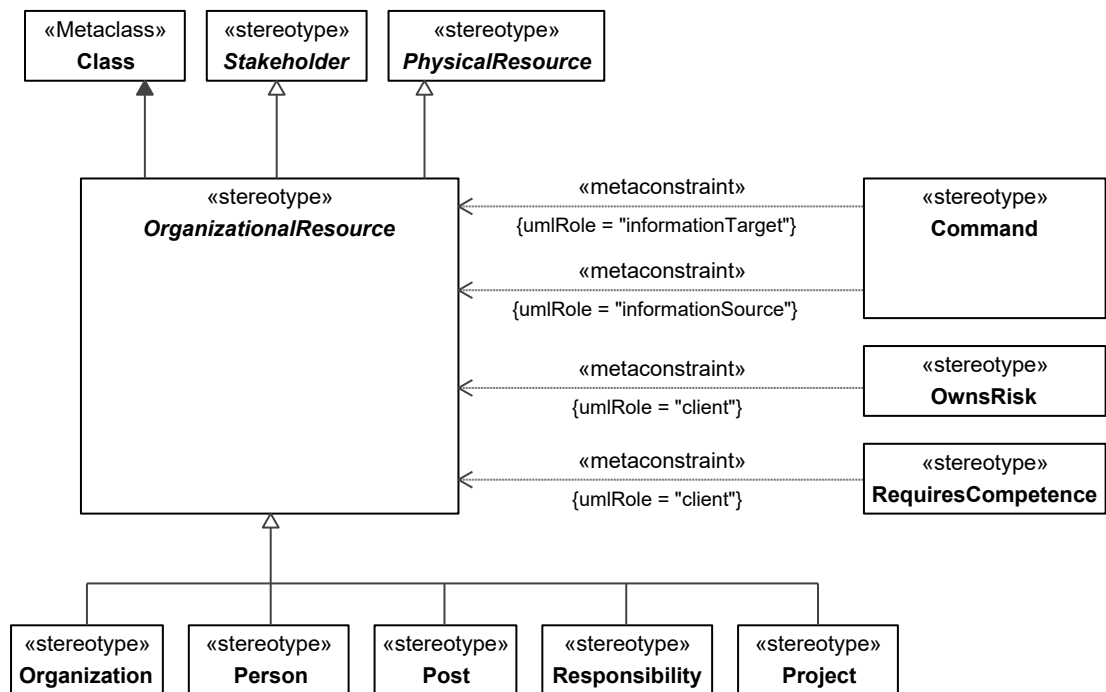


Figure 3:99 – OrganizationalResource

Person

Package: Taxonomy

isAbstract: No

Generalization: [OrganizationalResource](#)

Extension: Class

Description

A type of a human being used to define the characteristics that need to be described for ActualPersons (e.g., properties such as address, telephone number, nationality, etc).

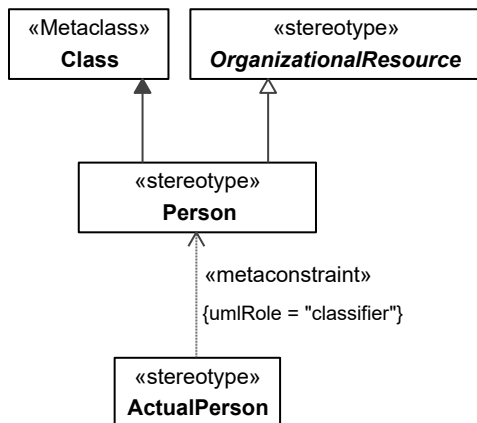


Figure 3:100 - Person

Post

Package: Taxonomy

isAbstract: No

Generalization: [OrganizationalResource](#)

Extension: Class

Description

A type of job title or position that a person can fill (e.g., Lawyer, Solution Architect, Machine Operator or Chief Executive Officer).

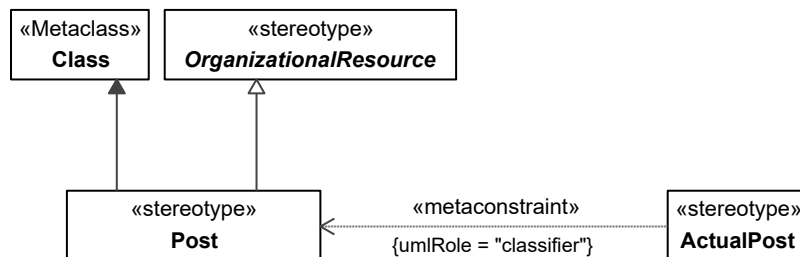


Figure 3:101 - Post

Responsibility

Package: Taxonomy

isAbstract: No

Generalization: [OrganizationalResource](#)

Extension: Class

Description

The type of duty required of a Person or Organization.

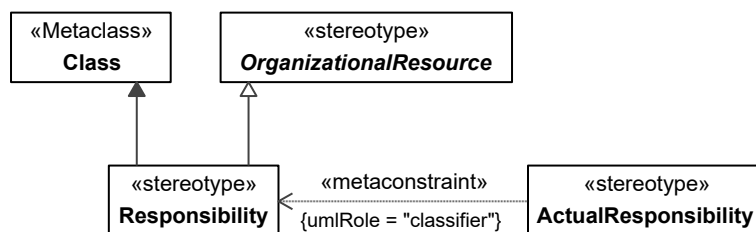


Figure 3:102 - Responsibility

UAF::Personnel::Connectivity

Contains the elements that contribute to the Personnel Connectivity Viewpoint.

Command

Package: Connectivity

isAbstract: No

Generalization: [ResourceExchange](#)

Extension: InformationFlow

Description

A type of ResourceExchange that asserts that one OrganizationalResource commands another.

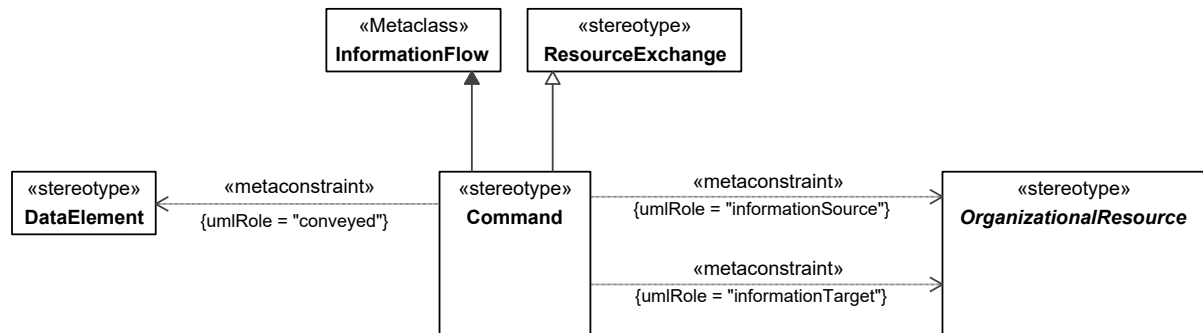


Figure 3:103 – Command

Constraints

- | | |
|-------------------------------|---|
| [1] Command.conveyed | Value for the conveyed metaproperty must be stereotyped «DataElement» or its specializations. |
| [2] Command.informationSource | Value for the informationSource metaproperty must be stereotyped by the specialization of «OrganizationalResource». |
| [3] Command.informationTarget | Value for the informationTarget metaproperty must be stereotyped by the specialization of «OrganizationalResource». |

Control

Package: Connectivity

isAbstract: No

Generalization: [ResourceExchange](#)

Extension: InformationFlow

Description

A type of ResourceExchange that asserts that one PhysicalResource controls another PhysicalResource (i.e., the driver of a vehicle controlling the vehicle speed or direction).

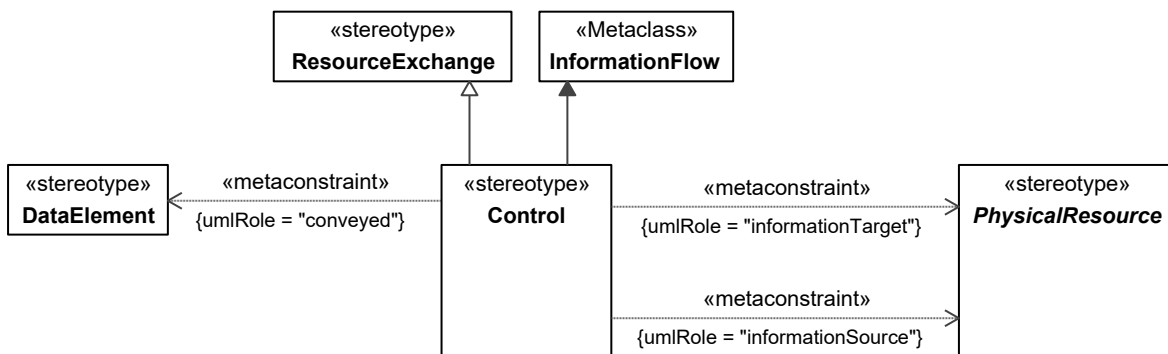
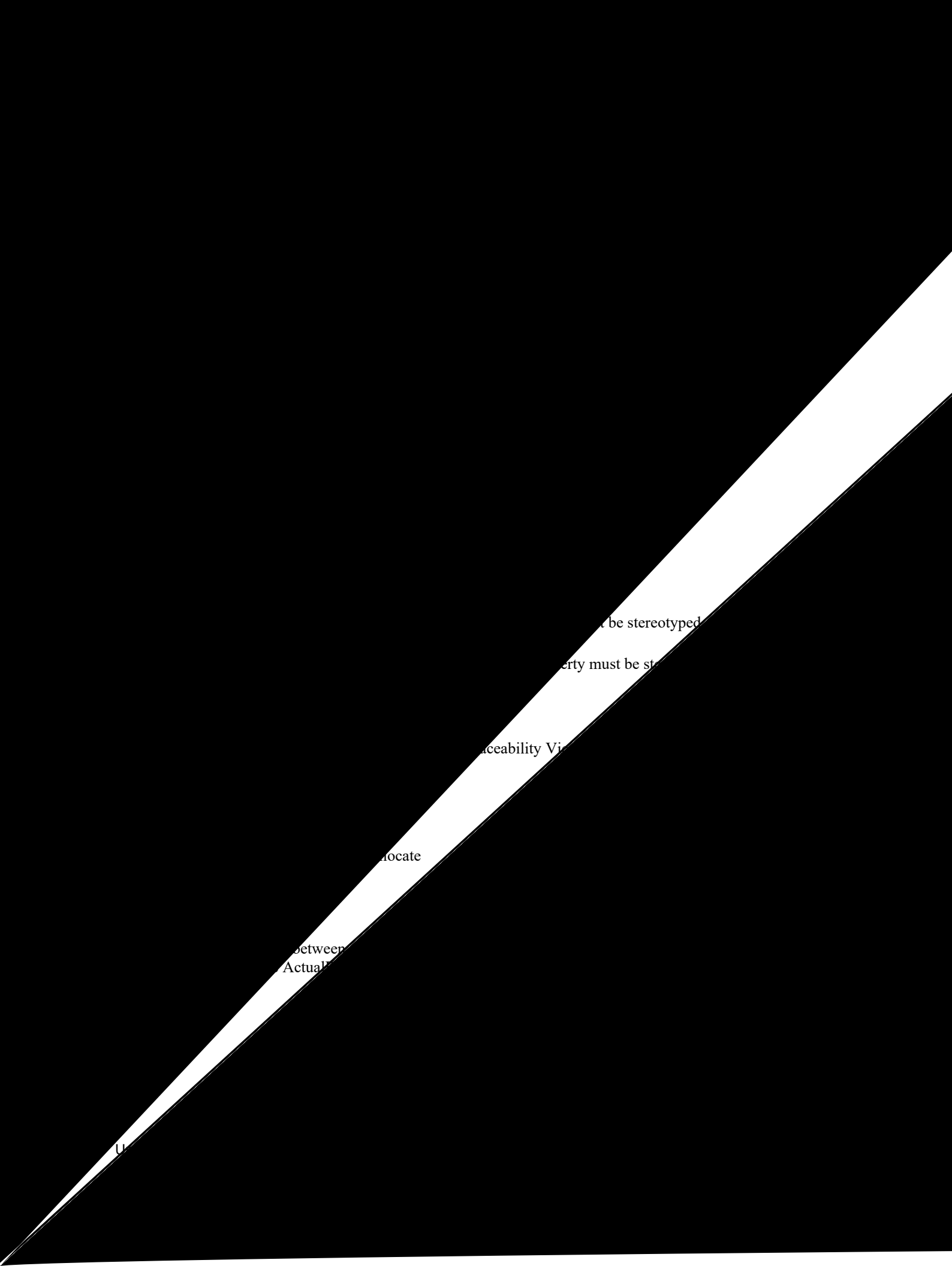


Figure 3:104 – Control

Constraints



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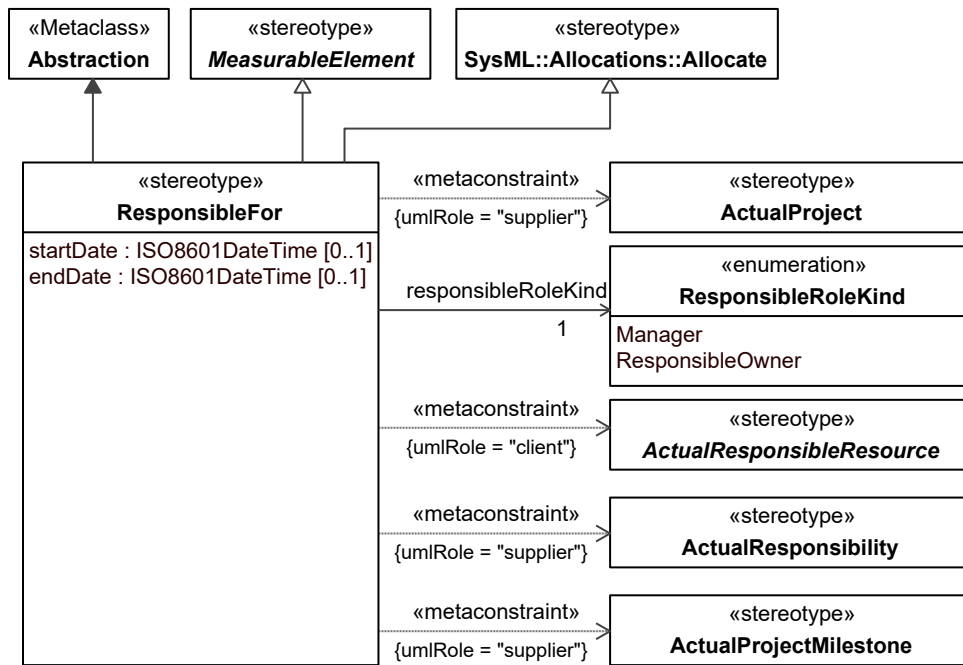


Figure 3:109 – ResponsibleFor

Attributes

- endDate : ISO8601DateTime[0..1] End date of an ActualResponsibleResource being ResponsibleFor and ActualProject or ActualResponsibility.
- startDate : ISO8601DateTime[0..1] Start date of an ActualResponsibleResource being ResponsibleFor and ActualProject or ActualResponsibility.

Associations

- responsibleRoleKind : ResponsibleRoleKind[1] Captures the kind of role (Manager or ResponsibleOwner) responsible for the ActualProject or ActualResponsibility.

Constraints

- [1] ResponsibleFor.client Value for the client metaproperty must be stereotyped by the specialization of «ActualResponsibleResource».
- [2] ResponsibleFor.supplier Value for the supplier metaproperty must be stereotyped «ActualProject», «ActualResponsibility», «ActualProjectMilestone» or their specializations.

ResponsibleRoleKind

Package: Traceability

isAbstract: No

Description

Enumeration of the possible kinds of ResponsibleFor relationship. Its enumeration literals are:

- Manager - Indicates that the ResourceInteraction associated with the ResourceInteractionKind is an implementation of logical flow.
- ResponsibleOwner - Indicates that the ResourceInteraction associated with the ResourceInteractionKind is an implementation of logical flow.

3.1.8 UAF::Resources

Stakeholders: Systems Engineers, Resource Owners, Implementers, Solution Providers, IT Architects.

Concerns: definition of solution architectures to implement operational requirements. Definition: captures a solution architecture consisting of resources, e.g. organizational, software, artifacts, capability configurations, natural resources that implement the operational requirements. Further design of a resource is typically detailed in SysML or UML.

UAF::Resources::Taxonomy

Contains the elements that contribute to the Resources Taxonomy Viewpoint.

CapabilityConfiguration

Package: Taxonomy

isAbstract: No

Generalization: [ResourceArchitecture](#)

Extension: Class

Description

A composite structure representing the physical and human resources (and their interactions) in an enterprise, assembled to meet a capability.

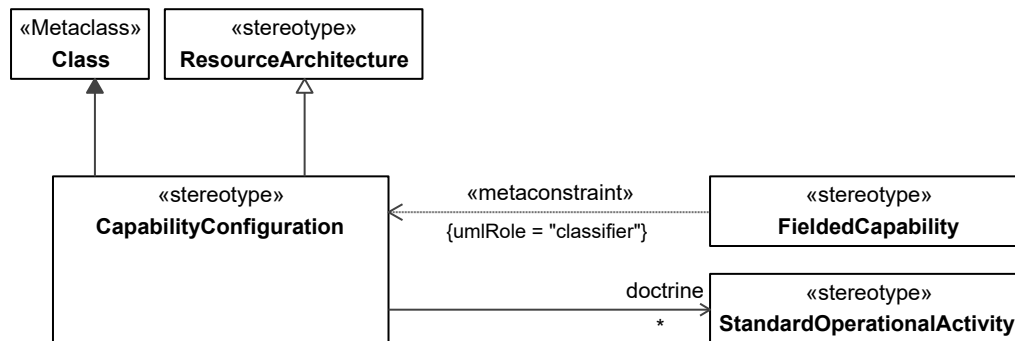


Figure 3:110 – CapabilityConfiguration

Associations

doctrine : StandardOperationalActivity[*] Represents the doctrinal line of development of the Capability.

NaturalResource

Package: Taxonomy

isAbstract: No

Generalization: [PhysicalResource](#)

Extension: Class

Description

Type of physical resource that occurs in nature such as oil, water, gas or coal.

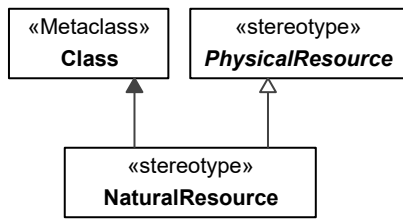


Figure 3:111 - NaturalResource

PhysicalResource

Package: Taxonomy

isAbstract: Yes

Generalization: [ResourcePerformer](#)

Extension: Class

Description

An abstract grouping that defines physical resources (i.e. OrganizationalResource, ResourceArtifact and NaturalResource).

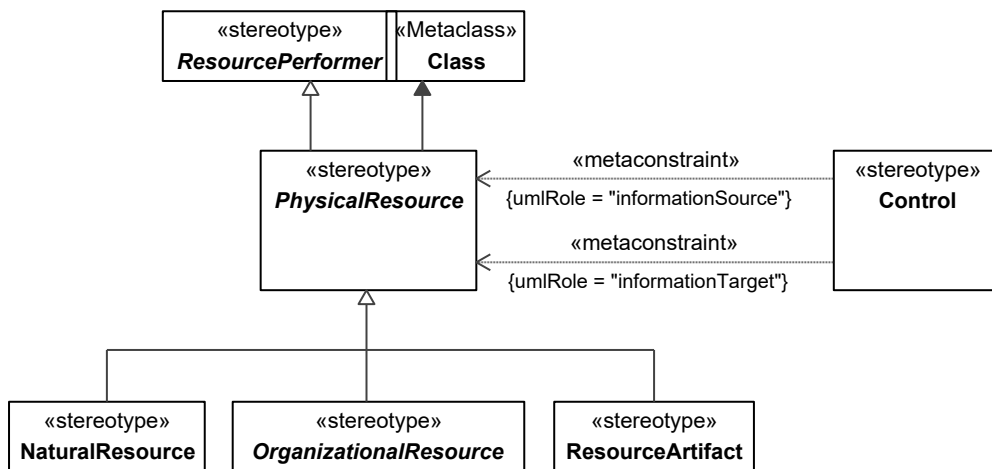


Figure 3:112 - PhysicalResource

ResourceArchitecture

Package: Taxonomy

isAbstract: No

Generalization: [ResourcePerformer](#), [Architecture](#)

Extension: Class

Description

An element used to denote a model of the Architecture, described from the ResourcePerformer perspective.

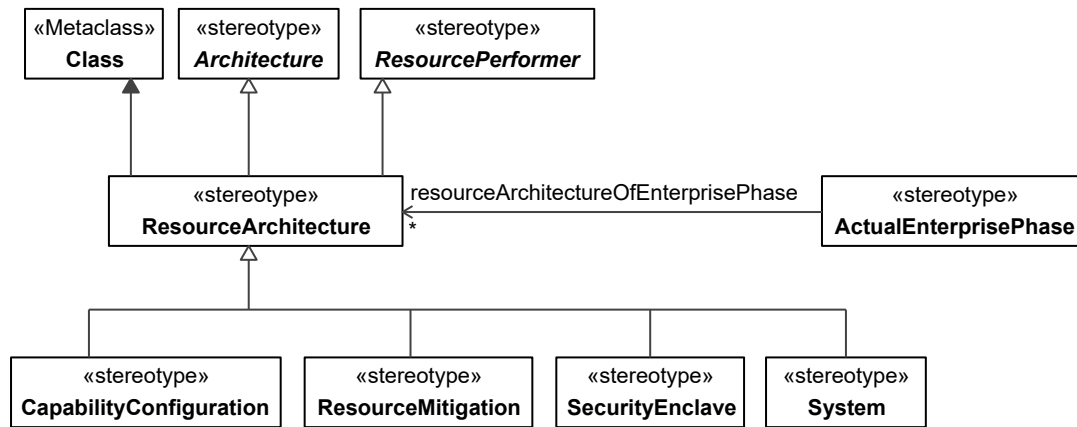


Figure 3:113 - ResourceArchitecture

ResourceArtifact

Package: Taxonomy

isAbstract: No

Generalization: [PhysicalResource](#)

Extension: Class

Description

A type of man-made object that contains no human beings (i.e., satellite, radio, petrol, gasoline, etc.).

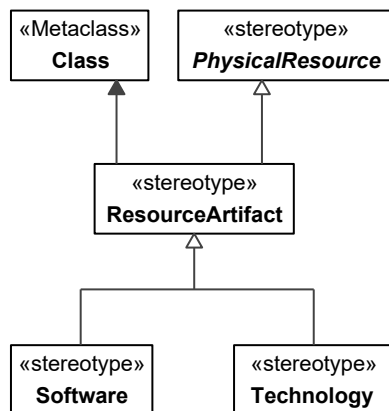


Figure 3:114 - ResourceArtifact

ResourcePerformer

Package: Taxonomy

isAbstract: Yes

Generalization: [ResourceAsset](#), [ResourceExchangeItem](#), [SubjectOfResourceConstraint](#), [VersionedElement](#), [CapableElement](#), [SubjectOfForecast](#), [OperationalExchangeItem](#), [Desirer](#)

Extension: Class

Description

An abstract grouping of elements that can perform Functions.

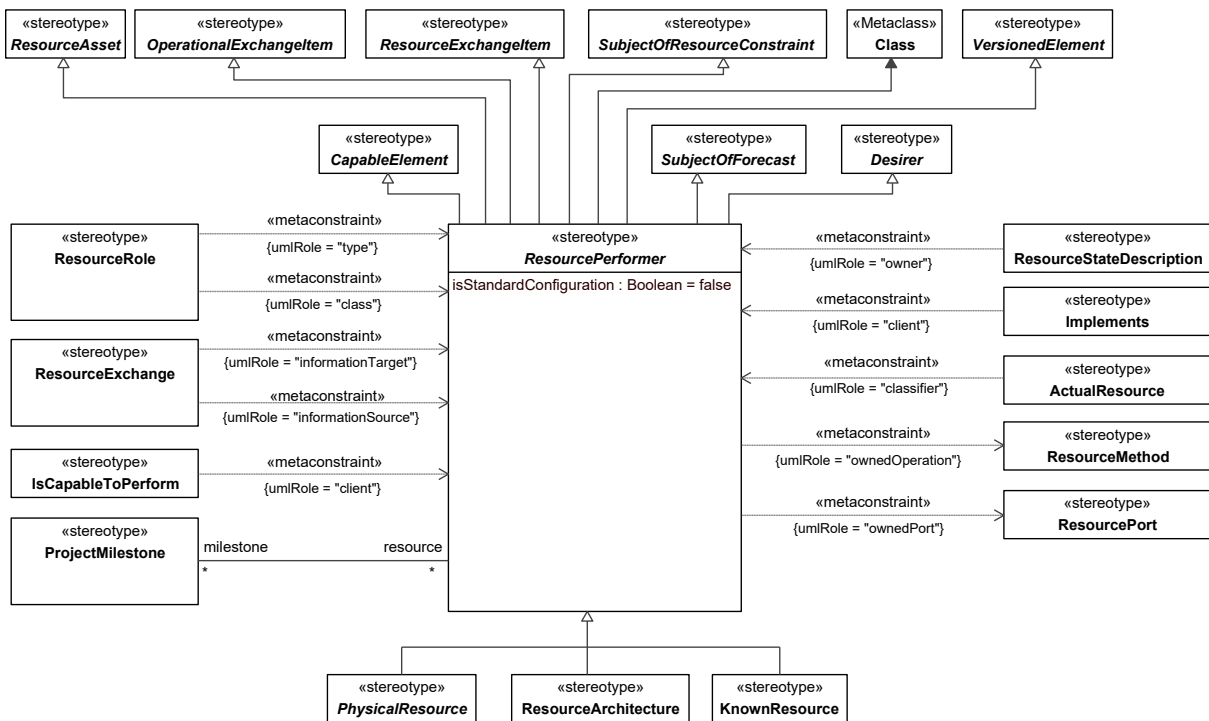


Figure 3:115 – ResourcePerformer

Attributes

isStandardConfiguration : Boolean[] Indicates if the ResourcePerformer is StandardConfiguration, default=false.

Associations

milestone : ProjectMilestone[*] Relates ResourcePerformer to ProjectMilestones that affect it.

Constraints

- | | |
|---|---|
| [1] ResourcePerformer.isCapableOfPerforming | Is capable of performing only «Function» elements or its specializations. |
| [2] ResourcePerformer.ownedOperation | Values for the ownedOperation metaproperty must be stereotyped «ResourceMethod» or its specializations. |
| [3] ResourcePerformer.ownedPort | Values for the ownedPort metaproperty must be stereotyped «ResourcePort» or its specializations. |

Software

Package: Taxonomy

isAbstract: No

Generalization: [ResourceArtifact](#)

Extension: Class

Description

A sub-type of ResourceArtifact that specifies an executable computer program.

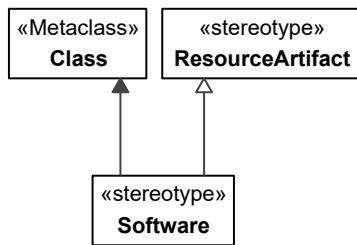


Figure 3:116 - Software

System

Package: Taxonomy

isAbstract: No

Generalization: [ResourceArchitecture](#)

Extension: Class

Description

An integrated set of elements, subsystems, or assemblies that accomplish a defined objective. These elements include products (hardware, software, firmware), processes, people, information, techniques, facilities, services, and other support elements (INCOSE SE Handbook V4, 2015).

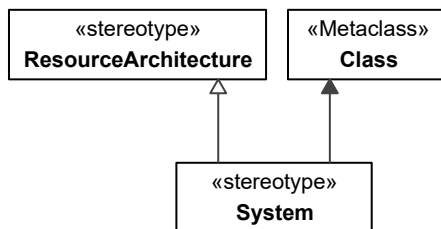


Figure 3:117 - System

UAF::Resources::Structure

Contains the elements that contribute to the Resources Structure Viewpoint.

ResourceMethod

Package: Structure

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Operation

Description

A behavioral feature of a ResourcePerformer whose behavior is specified in a Function.

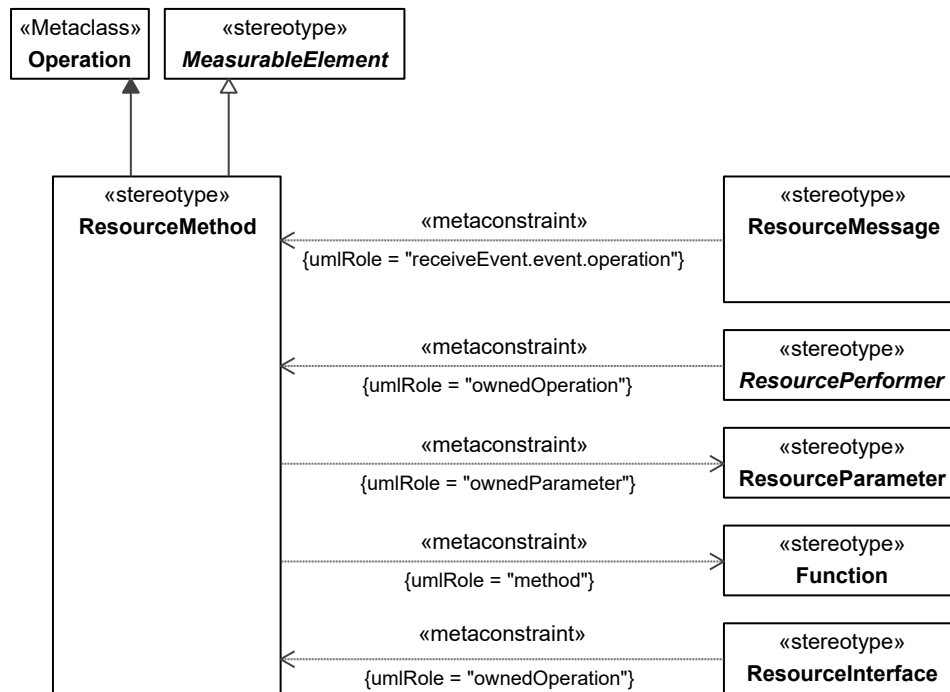


Figure 3:118 - ResourceMethod

Constraints

- | | |
|-----------------------------------|--|
| [1] ResourceMethod.method | Value for the method metaproperty must be stereotyped «Function» or its specializations. |
| [2] ResourceMethod.ownedParameter | The values for the ownedParameter metaproperty must be stereotyped «ResourceParameter». |

ResourceParameter

Package: Structure

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Parameter

Description

An element that represents inputs and outputs of a Function. It is typed by a ResourceInteractionItem.

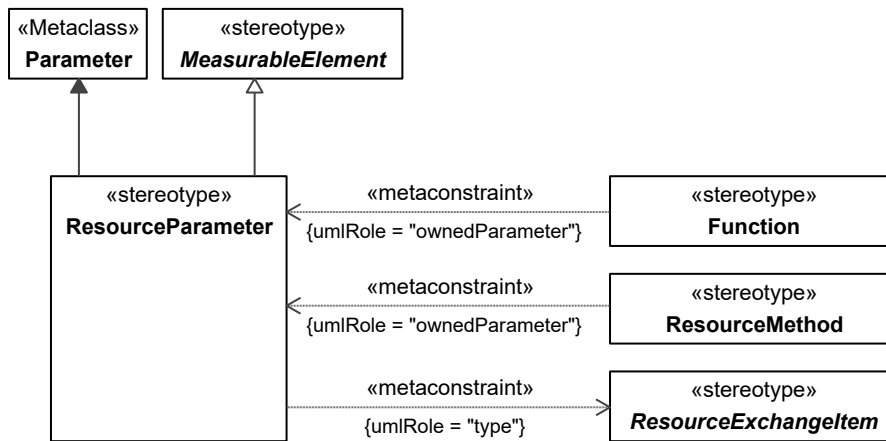


Figure 3:119 - ResourceParameter

Constraints

- [1] ResourceParameter.type Value for the type metaproperty must be stereotyped with a specialization of «ResourceInteractionItem».

ResourcePort

Package: Structure

isAbstract: No

Generalization: ProxyPort, [MeasurableElement](#), [ProtocolImplementation](#)

Extension: Port

Description

An interaction point for a ResourcePerformer through which it can interact with the outside environment and which is defined by a ResourceInterface.

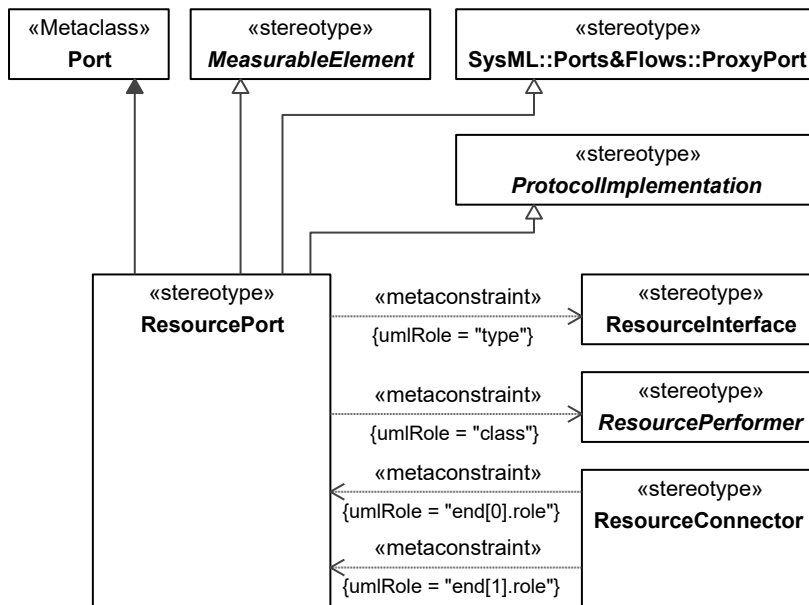


Figure 3:120 – ResourcePort

Constraints

- [1] ResourcePort.type Value for the type metaproperty must be stereotyped «ResourceInterface» or its specializations.
- [2] ResourcePort.class Value for the class metaproperty must be stereotyped by the specialization of «ResourcePerformer».

ResourceRole

Package: Structure

isAbstract: No

Generalization: [LocationHolder](#), [SubjectOfResourceConstraint](#), [MeasurableElement](#), [AssetRole](#)

Extension: Property

Description

Usage of a ResourcePerformer in the context of another ResourcePerformer. Creates a whole-part relationship.

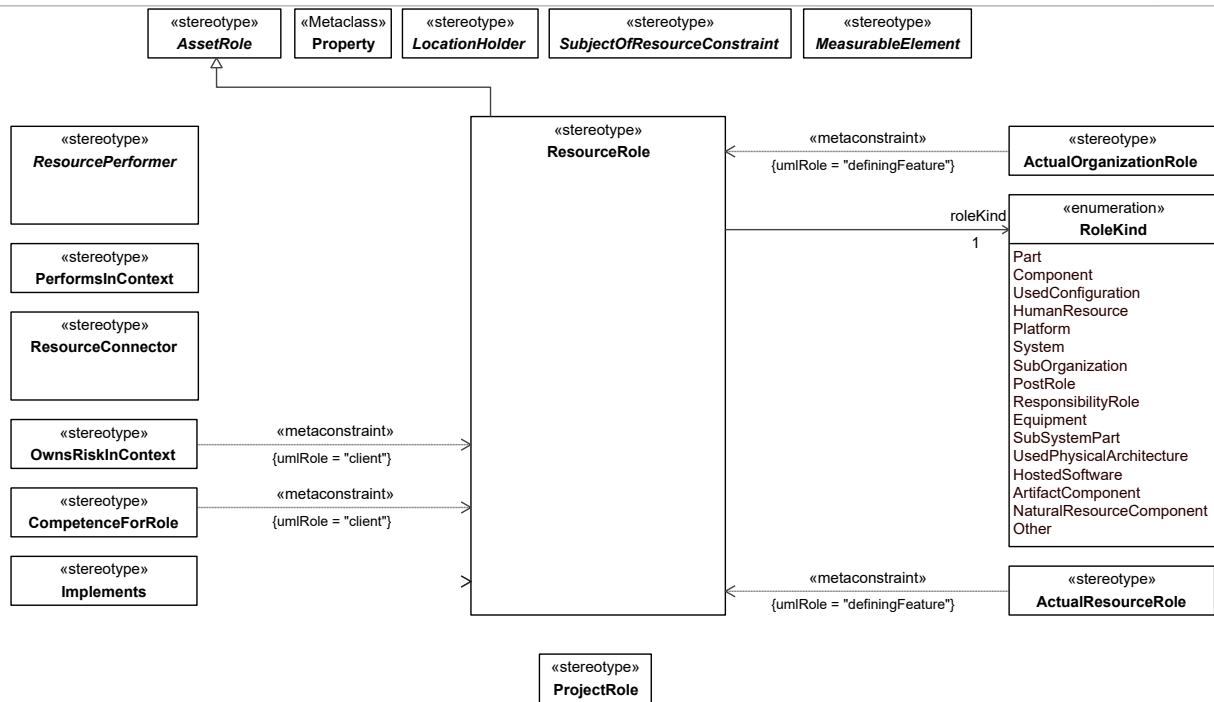


Figure 3:121 – ResourceRole

Associations

roleKind : RoleKind[1] Captures the kind of role a Resource can play.

Constraints

- [1] ResourceRole.type Value for the type metaproperty must be stereotyped by the specialization of «ResourcePerformer».
- [2] ResourceRole.class Value for the class metaproperty must be stereotyped by the specialization of «ResourcePerformer».

RoleKind

Package: Structure

isAbstract: No

Description

Enumeration of the possible kinds of roles that a ResourceRole may play in the context of a ResourcePerformer. Its enumeration literals are:

- Part - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of a ResourcePerformer that is used as a part of another ResourcePerformer.
- Component - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of Software that is used in the context of a ResourcePerformer.
- UsedConfiguration - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of existing CapabilityConfiguration that is used in the context of a ResourcePerformer.
- HumanResource - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of human resource that is used in the context of a ResourcePerformer.
- Platform - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of a ResourcePerformer that represents a platform (e.g. vessel, aircraft, etc.) that is used in the context of a SystemsResource.
- System - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of assembly of ResourcePerformers that is used in the context of another ResourcePerformer.
- SubOrganization - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of Organization that is typically the parent of another - e.g. a squadron may be part of a batallion, that is used in the context of a ResourcePerformer.
- PostRole - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of Post that is used in the context of a ResourcePerformer.
- ResponsibilityRole - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of Responsibility associated with a role that is used in the context of a ResourcePerformer.
- Equipment - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of man-made resource that is used to accomplish a task or function in the context of a ResourcePerformer.
- SubSystemPart - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of subsystem (represented as a ResourcePerformers) is part of another ResourcePerformer.
- UsedPhysicalArchitecture - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of existing PhysicalArchitecture that is used in the context of a ResourcePerformer.
- HostedSoftware - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of software that is used in the context of a ResourcePerformer.
- ArtifactComponent - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of non human resource that is used as a component in the context of a ResourcePerformer.
- NaturalResourceComponent - Indicates that the ResourceRole associated with the ResourceRoleKind is a kind of natural resource that is used as a component in the context of a ResourcePerformer.
- Other - Indicates that the ResourceRole associated with the ResourceRoleKind is another kind of RoleKind that is not on the enumerated list.

UAF::Resources::Connectivity

Contains the elements that contribute to the Resources Connectivity Viewpoint.

ResourceConnector

Package: Connectivity

isAbstract: No

Generalization: [MeasurableElement](#), [ProtocolImplementation](#)

Extension: Connector

Description

A channel for exchange between two ResourceRoles.

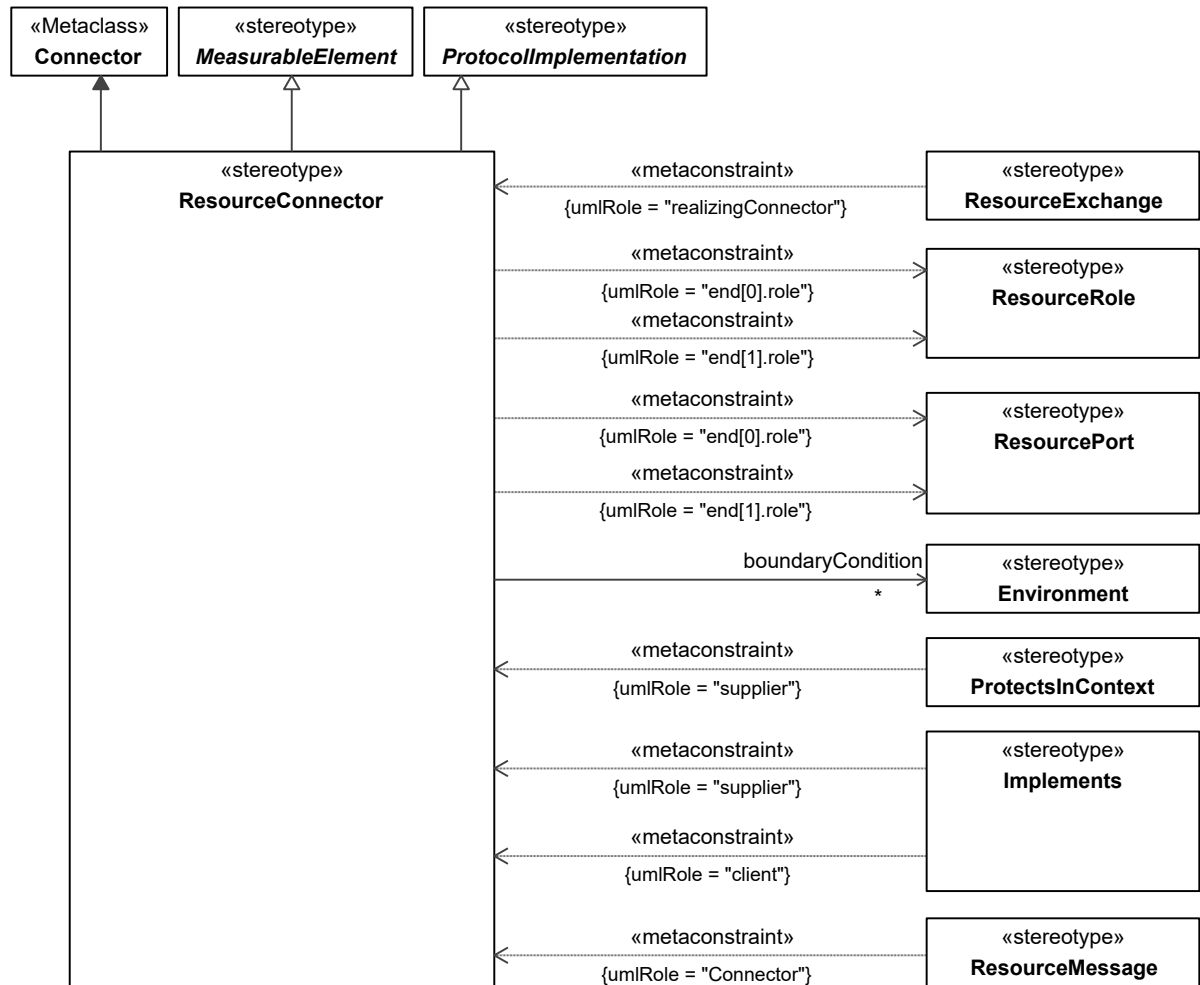


Figure 3:122 – ResourceConnector

Associations

boundaryCondition : **Environment**[*] Relates a ResourceConnector to the extremes of the Environment in which it is required to be made available.

Constraints

[1] **ResourceConnector.end** The value for the role metaproperty for the owned ConnectorEnd must be stereotype **ResourcePort**, **ResourceRole** or their specializations.

ResourceExchange

Package: Connectivity

isAbstract: No

Generalization: [Exchange](#)

Extension: InformationFlow

Description

Asserts that a flow can exist between ResourcePerformers (i.e., flows of data, people, materiel, or energy).

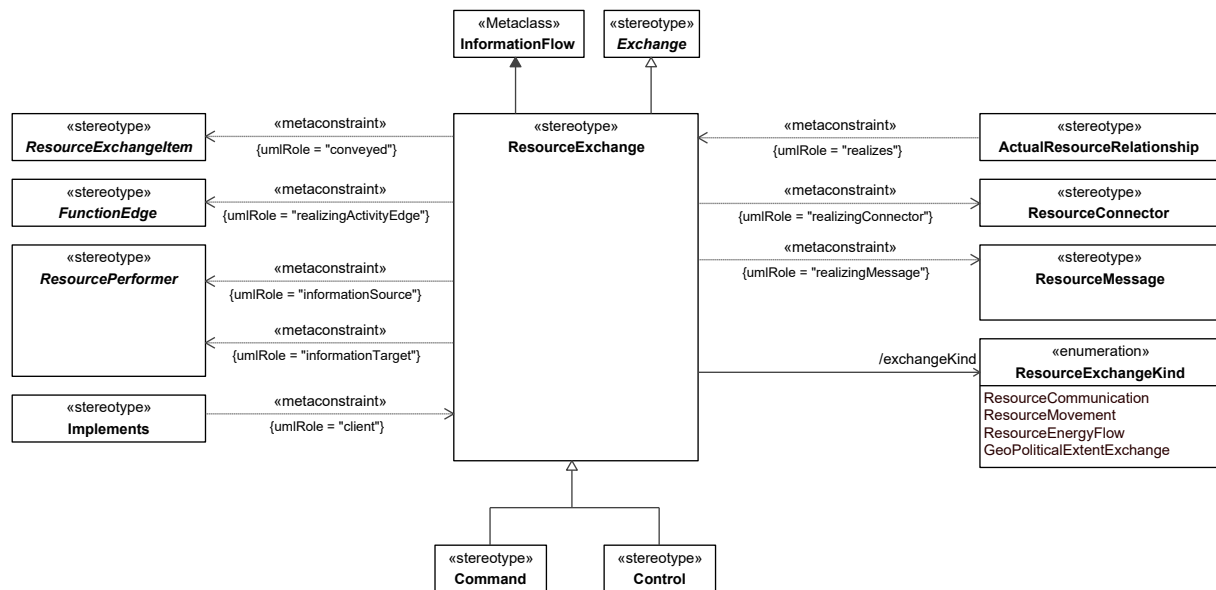


Figure 3:123 – ResourceExchange

Associations

exchangeKind : ResourceExchangeKind[] Captures the kind of ResourceExchange.

Constraints

[1] ResourceExchange.conveyed

In case of ResourceExchange.exchangeKind:
 = ResourceCommunication, the conveyed element must be stereotyped «DataElement» or its specializations,
 = ResourceMovement, the conveyed element must be stereotyped by the specialization of «PhysicalResource»,
 = ResourceEnergyFlow, the conveyed element must be stereotyped «NaturalResource» or its specializations,
 = GeoPoliticalExtentExchange, the conveyed element must be stereotyped «GeoPoliticalExtentType» or its specializations.

[2] ResourceInteraction.informationSource

Value for the informationSource metaproperty must be stereotyped by the specialization of «ResourcePerformer».

[3] ResourceInteraction.informationTarget

Value for the informationTarget metaproperty must be stereotyped by the specialization of «ResourcePerformer».

[4] ResourceInteraction.realizingActivityEdge

Value for the realizingActivityEdge metaproperty must be stereotyped by the specialization of «FunctionEdge».

[5] ResourceInteraction.realizingConnector

Value for the realizingConnector metaproperty must be stereotyped «ResourceConnector» or its specializations.

[6] ResourceInteraction.realizingMessage

Value for the realizingMessage metaproperty must be stereotyped «ResourceMessage» or its specializations.

ResourceExchangeItem

Package: Connectivity

isAbstract: Yes

Generalization: [Resource](#)

Description

An abstract grouping for elements that defines the types of elements that can be exchanged between ResourcePerformers and conveyed by a ResourceExchange.

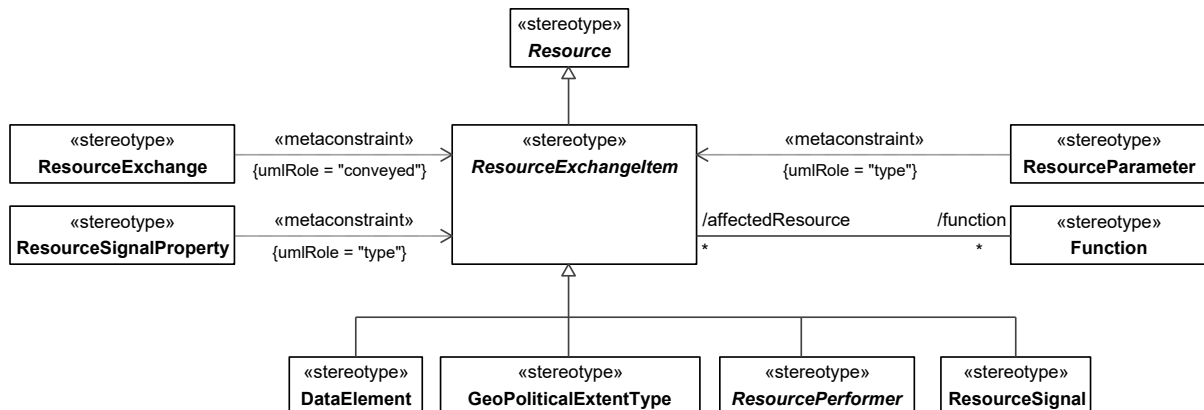


Figure 3:124 – ResourceExchangeItem

Associations

function : Function[*] Function using the ResourceExchangeItem internally.

ResourceExchangeKind

Package: Connectivity

isAbstract: No

Description

Enumeration of the possible kinds of resource exchange applicable to a ResourceExchange. Its enumeration literals are:

- ResourceCommunication - Indicates that the ResourceInteraction associated with the ResourceInteractionKind is an implementation of logical flow of data between Resources.
- ResourceMovement - Indicates that the ResourceInteraction associated with the ResourceInteractionKind is an implementation of logical flow of Resources between Resources.
- ResourceEnergyFlow - Indicates that the ResourceInteraction associated with the ResourceInteractionKind is an implementation of logical flow of natural resources between Resources.

GeoPoliticalExtentExchange - Indicates that the ResourceInteraction associated with the ResourceInteractionKind is an implementation of logical flow where GeoPoliticalExtents (i.e., Borders) flow from one place to another.

ResourceInterface

Package: Connectivity

isAbstract: No

Generalization: [PropertySet](#), InterfaceBlock

Extension: Class

Description

A declaration that specifies a contract between the ResourcePerformers it is related to and any other ResourcePerformers it can interact with. It is also intended to be an implementation of a specification of an Interface in the Business and/or Service layer.

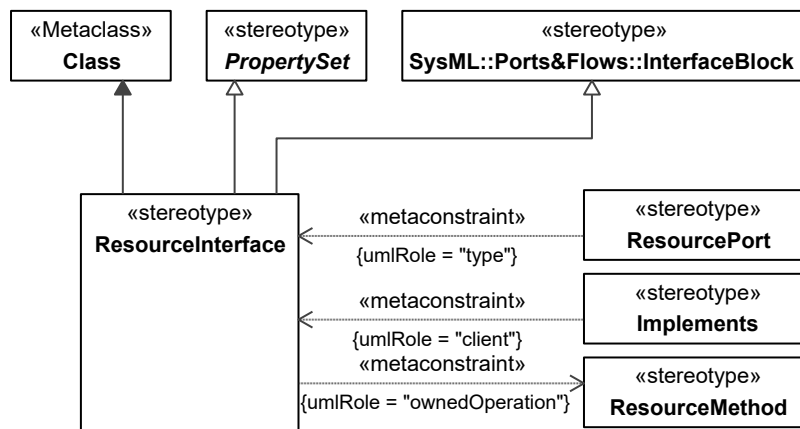


Figure 3:125 – ResourceInterface

Constraints

- [1] ResourceInterface.ownedOperation Values for ownedOperation metaproperty must be stereotyped «ResourceMethod» or its specializations.

ResourceSignal

Package: Connectivity

isAbstract: No

Generalization: [ResourceExchangeItem](#)

Extension: Signal

Description

A ResourceSignal is a specification of a kind of communication between resources (ResourcePerformers) in which a reaction is asynchronously triggered in the receiver without a reply.

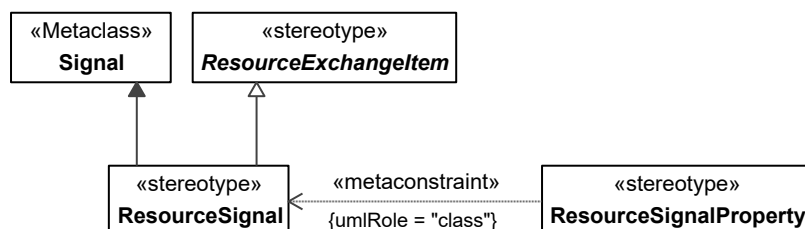


Figure 3:126 - ResourceSignal

ResourceSignalProperty

Package: Connectivity

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Property

Description

A property of a ResourceSignal typed by ResourceExchangeItem. It enables ResourceExchangeItem e.g., DataElement to be passed as arguments of the ResourceSignal.

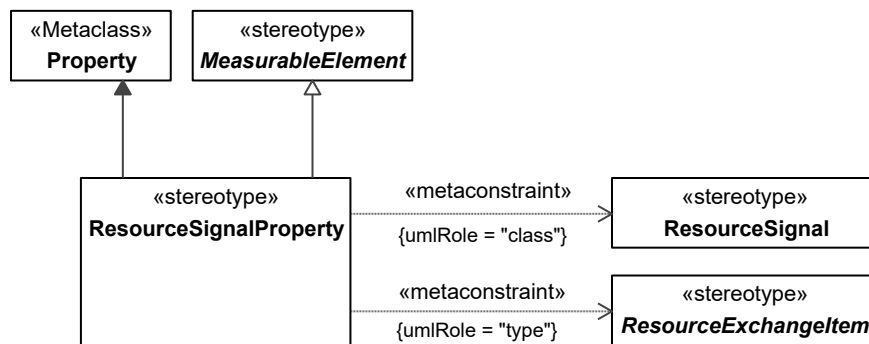


Figure 3:127 – ResourceSignalProperty

Constraints

- [1] ResourceSignalProperty.class Value for class metaproperty must be stereotyped «ResourceSignal» or its specializations.
- [2] ResourceSignalProperty.type Value for type metaproperty must be stereotyped by a specialization of «ResourceExchangeItem».

UAF::Resources::Processes

Contains the elements that contribute to the Resources Processes Viewpoint.

Function

Package: Processes

isAbstract: No

Generalization: [Activity](#), [SubjectOfResourceConstraint](#)

Extension: Activity

Description

An Activity which is specified in the context to the ResourcePerformer (human or machine) that IsCapableToPerform it.

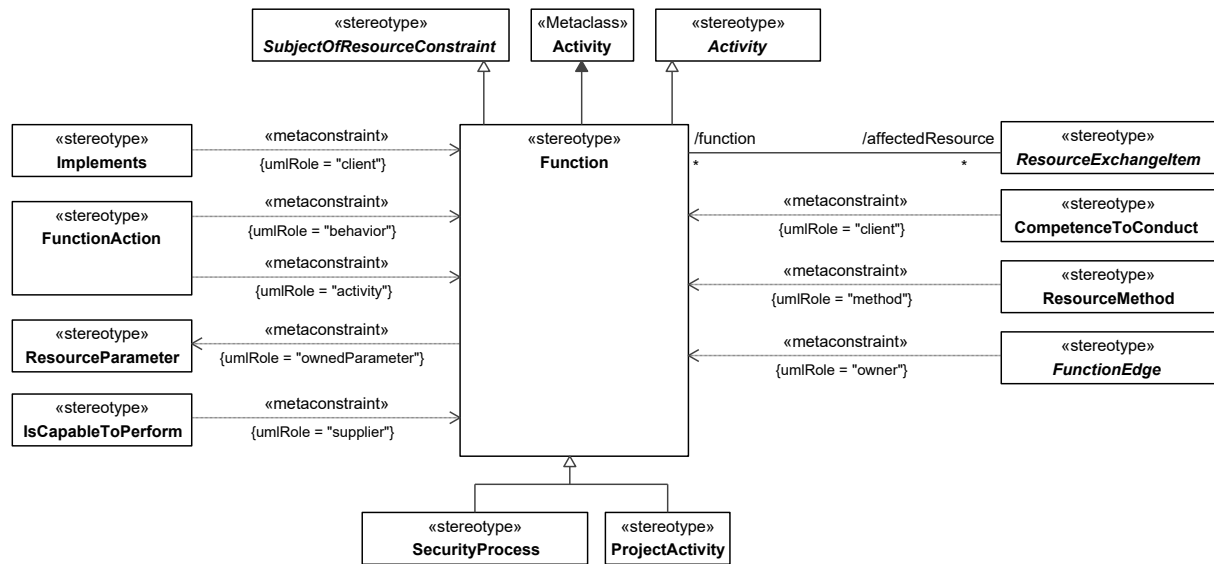


Figure 3:128 – Function

Associations

affectedResource : ResourceExchangeItem[*] ResourceExchangeItems consumed and produced internally within a Function.

Constraints

[1] Function.ownedParameter The values for the ownedParameter metaproperty must be stereotyped «ResourceParameter» or its specializations.

FunctionAction

Package: Processes

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: CallBehaviorAction

Description

A call of a Function indicating that the Function is performed by a ResourceRole in a specific context.

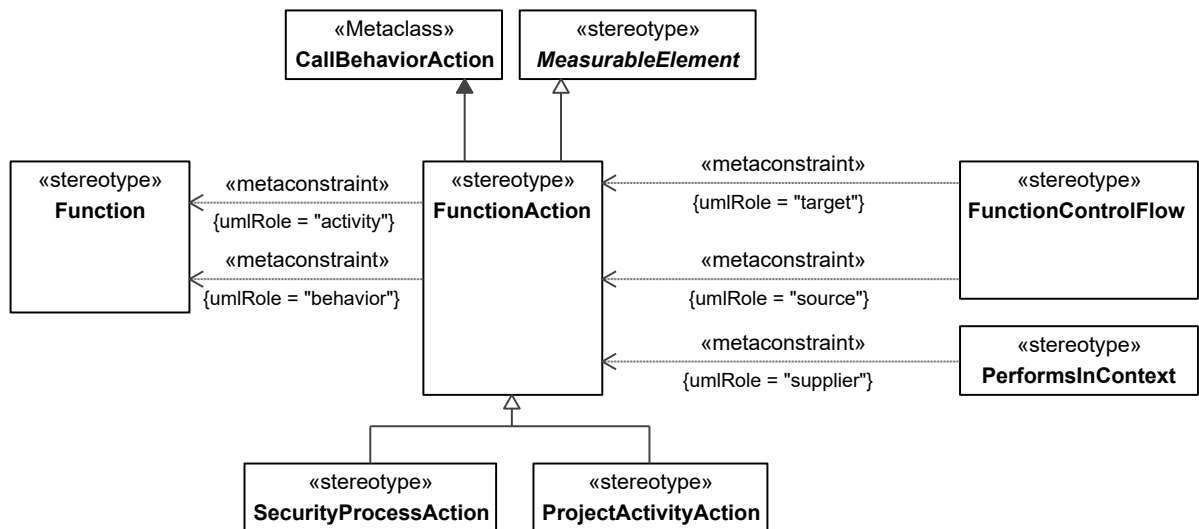


Figure 3:129 – FunctionAction

Constraints

- [1] FunctionAction.activity Value for the activity metaproperty must be stereotyped «Function» or its specializations.
- [2] FunctionAction.behavior Value for the behavior metaproperty must be stereotyped «Function» or its specializations.

FunctionControlFlow

Package: Processes

isAbstract: No

Generalization: [FunctionEdge](#)

Extension: ControlFlow

Description

An ActivityEdge that shows the flow of control between FunctionActions.

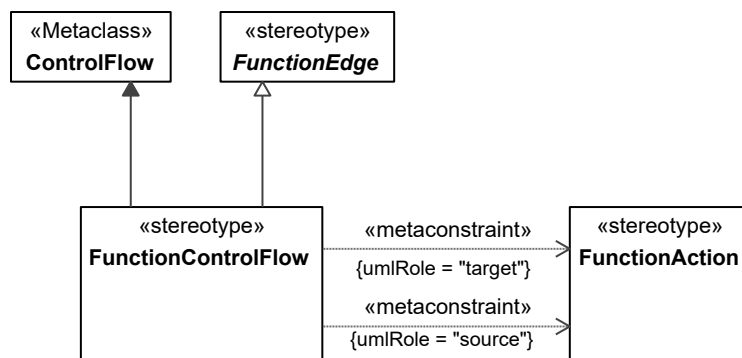


Figure 3:130 – FunctionControlFlow

Constraints

- [1] FunctionControlFlow.source Value for the source metaproperty must be stereotyped «FunctionAction» or its specializations.
- [2] FunctionControlFlow.target Value for the target metaproperty must be stereotyped «FunctionAction» or its specializations.

FunctionEdge

Package: Processes

isAbstract: Yes

Generalization: [MeasurableElement](#)

Extension: ActivityEdge

Description

Abstract grouping for FunctionControlFlow and FunctionObjectFlow.

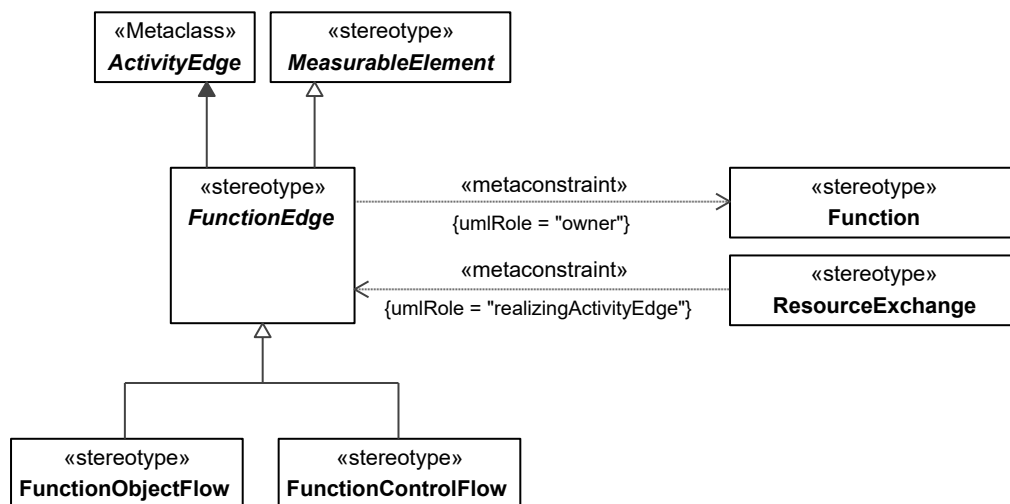


Figure 3:131 – FunctionEdge

Constraints

- [1] FunctionEdge.owner «FunctionEdge» must be owned directly or indirectly by «Function» or its specializations.

FunctionObjectFlow

Package: Processes

isAbstract: No

Generalization: [FunctionEdge](#)

Extension: ObjectFlow

Description

An ActivityEdge that shows the flow of Resources (objects/data) between FunctionActions.

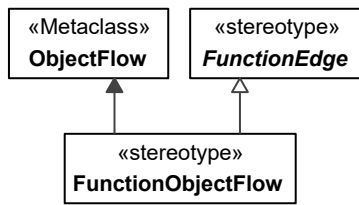


Figure 3:132 - FunctionObjectFlow

UAF::Resources::States

Contains the elements that contribute to the Resources States Viewpoint.

ResourceStateDescription

Package: States

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: StateMachine

Description

A state machine describing the behavior of a ResourcePerformer, depicting how the ResourcePerformer responds to various events and the actions.

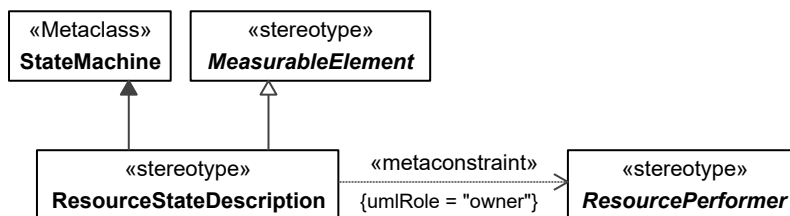


Figure 3:133 – ResourceStateDescription

Constraints

- [1] ResourceStateDescription.owner Values for the owner metaproperty must be stereotyped with the specialization of «ResourcePerformer».

UAF::Resources::Interaction Scenarios

Contains the elements that contribute to the Resources Interaction Scenarios Viewpoint.

ResourceMessage

Package: Interaction Scenarios

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Message

Description

Message for use in a Resource Event-Trace which carries any of the subtypes of ResourceExchange.

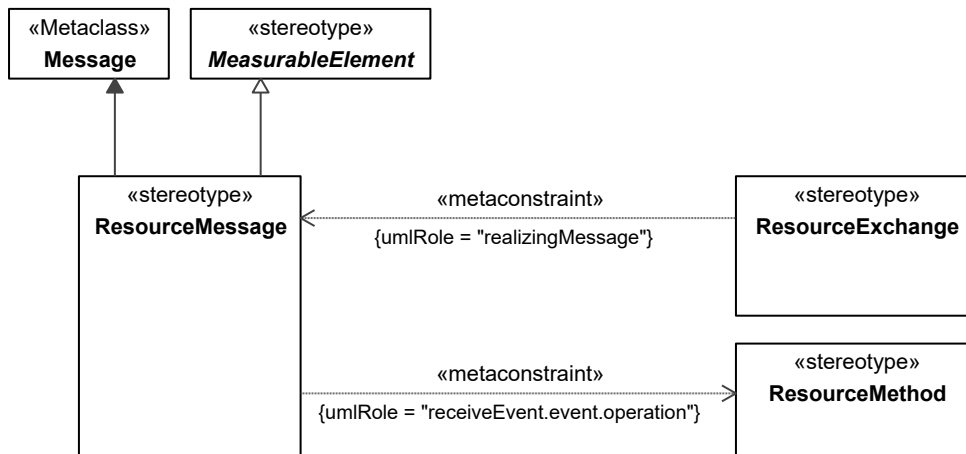


Figure 3:134 – ResourceMessage

Constraints

- [1] ResourceMessage.receiveEvent.event.operation Values for the receiveEvent.event.operation metaproperty must be stereotyped with «ResourceMethod» or its specializations.

UAF::Resources::Information

Contains the elements that contribute to the Resources Information Viewpoint.

DataElement

Package: Information

isAbstract: No

Generalization: [ResourceExchangeItem](#), [SubjectOfResourceConstraint](#), [ResourceAsset](#)

Extension: Class

Description

A formalized representation of data that is managed by or exchanged between systems.

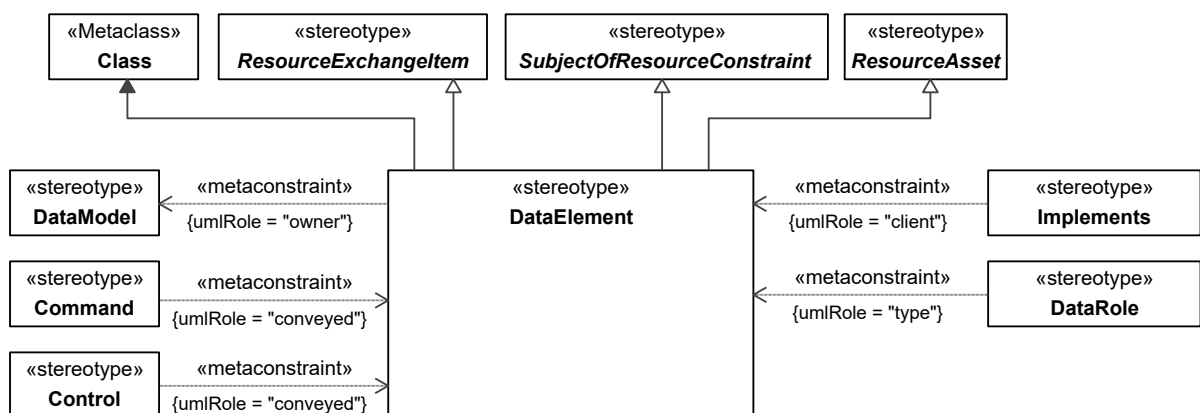


Figure 3:135 – DataElement

Constraints

[1] DataElement.owner Values for the owner metaproperty must be stereotyped «DataModel» or its specializations.

UAF::Resources::Constraints

Contains the elements that contribute to the Resources Constraints Viewpoint.

ResourceConstraint

Package: Constraints

isAbstract: No

Generalization: [Rule](#)

Extension: Constraint

Description

A rule governing the structural or functional aspects of an implementation.

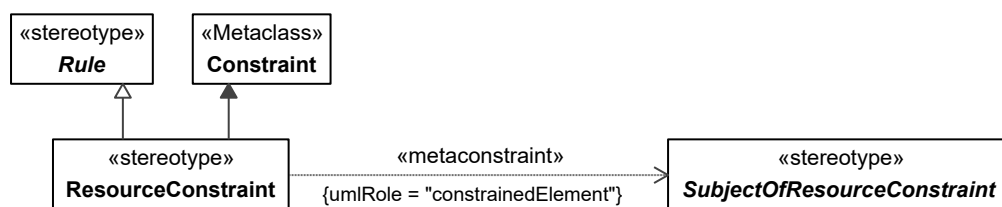


Figure 3:136 – ResourceConstraint

Constraints

[1] ResourceConstraint.constrainedElement Value for the constrainedElement metaproperty must be stereotyped by the specialization of «SubjectOfResourceConstraint».

SubjectOfResourceConstraint

Package: Constraints

isAbstract: Yes

Generalization: [UAFElement](#)

Extension: Element

Description

An abstract grouping of elements that can be the subject of a ResourceConstraint.

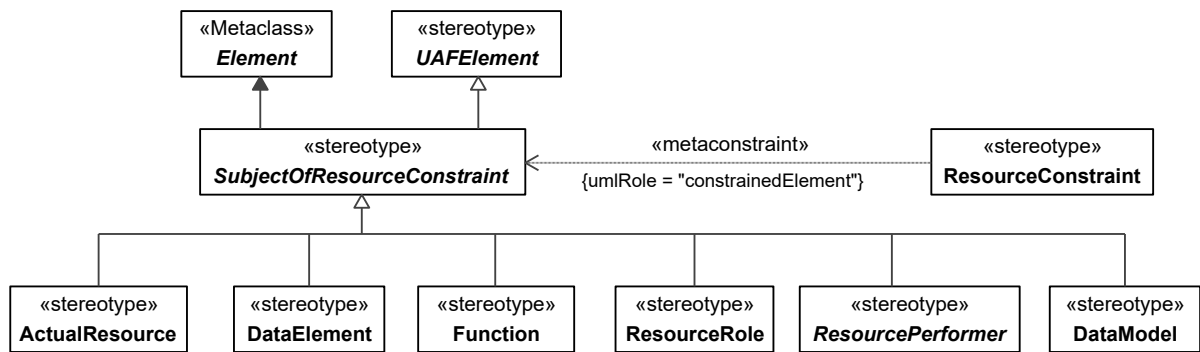


Figure 3:137 - SubjectOfResourceConstraint

UAF::Resources::Roadmap

Contains the elements that contribute to the Resources Roadmap Viewpoint.

Forecast

Package: Roadmap

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Dependency

Description

A dependency relationship that specifies a transition from one Asset, Standard, Competence to another future one. It is related to an ActualEnterprisePhase to give it a temporal context.

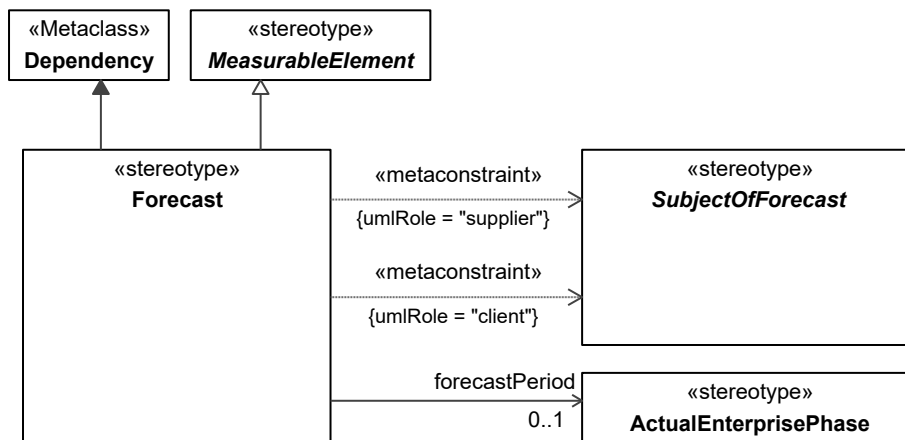


Figure 3:138 – Forecast

Associations

forecastPeriod : ActualEnterprisePhase[0..1] Relates the SubjectOfForecast to the ActualEnterprisePhase in which the SubjectOfForecast is expected to be provided.

Constraints

[1] Forecast.client Value for the client metaproperty must be stereotyped by the specialization of «SubjectOfForecast».

- [2] Forecast.pair Values for the client and supplier metaproperties must be stereotyped by the same specialization of «SubjectOfForecast» (e.g. «Software» to «Software», «Standard» to «Standard», etc).
- [3] Forecast.supplier Value for the supplier property must be stereotyped by the specialization of «SubjectOfForecast».

SubjectOfForecast

Package: Roadmap

isAbstract: Yes

Generalization: [UAFElement](#)

Extension: Class

Description

An abstract grouping of elements that can be the subject of a Forecast.

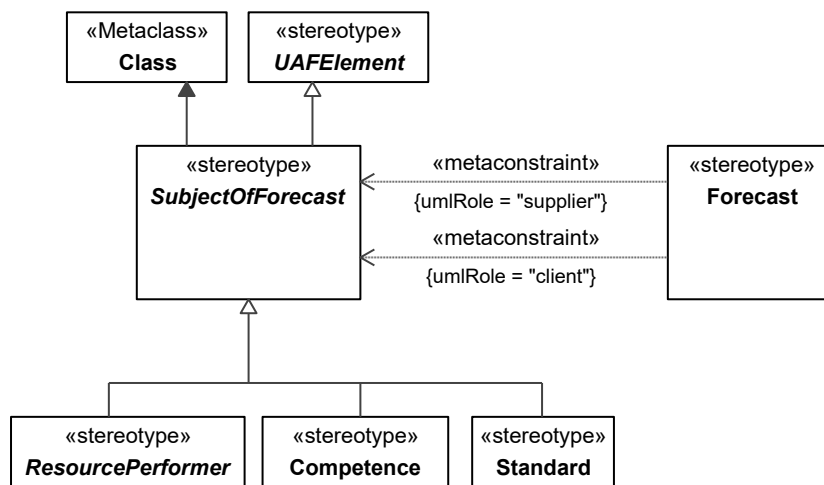


Figure 3:139 - SubjectOfForecast

Technology

Package: Roadmap

isAbstract: No

Generalization: [ResourceArtifact](#)

Extension: Class

Description

A sub type of ResourceArtifact that indicates a technology domain, i.e. nuclear, mechanical, electronic, mobile telephony etc.

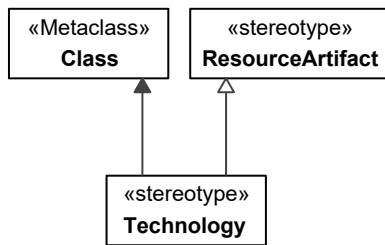


Figure 3:140 - Technology

VersionedElement

Package: Roadmap

isAbstract: Yes

Generalization: [UAFElement](#)

Extension: Class

Description

An abstract grouping of ResourcePerformer and ServiceSpecification that allows VersionOfConfiguration to be related to ActualProjectMilestones.

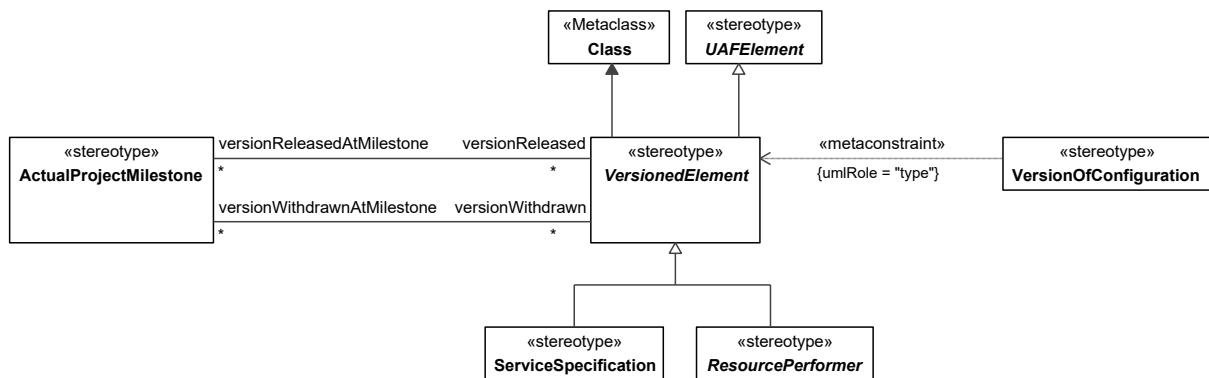


Figure 3:141 – VersionedElement

Associations

versionReleasedAtMilestone : ActualProjectMilestone[*]	Relates a VersionedElement to the ActualProjectMilestone. It indicates the ActualProjectMilestone at which the VersionedElement is released.
versionWithdrawnAtMilestone : ActualProjectMilestone[*]	Relates a VersionedElement to the ActualProjectMilestone. It indicates the ActualProjectMilestone at which the VersionedElement is withdrawn.

VersionOfConfiguration

Package: Roadmap

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Property

Description

A property of a WholeLifeConfiguration, used in version control of a VersionedElement. It asserts that a VersionedElement is a version of a WholeLifeConfiguration.

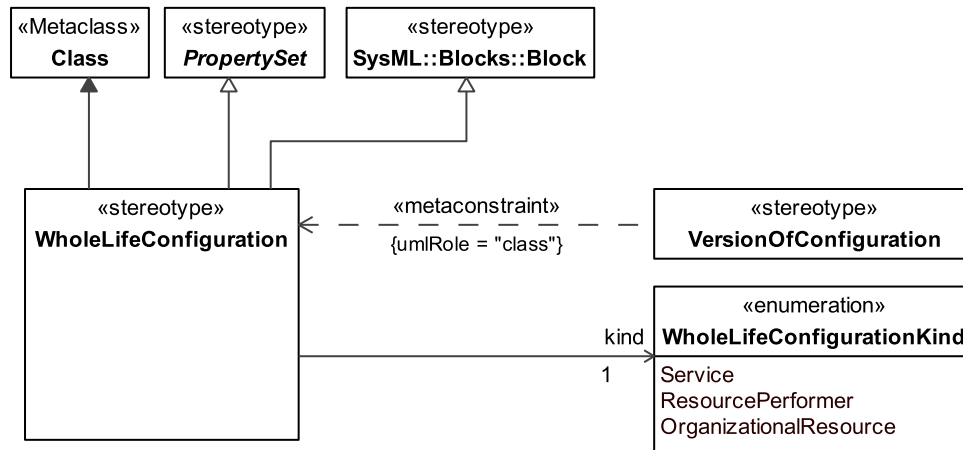


Figure 3:142 – VersionOfConfiguration

Constraints

- [1] VersionOfConfiguration.class Value for the class metaproperty must be stereotyped «WholeLifeConfiguration» or its specializations.
- [2] VersionOfConfiguration.type Value for the type metaproperty must be stereotyped by the specialization of «VersionedElement».

VersionSuccession

Package: Roadmap

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Dependency

Description

A dependency relationship between two VersionOfConfigurations that denotes that one VersionOfConfiguration follows from another.

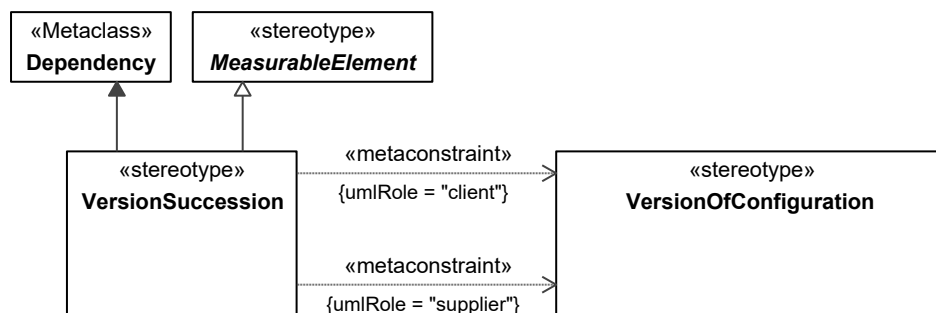


Figure 3:143 – VersionSuccession

Constraints

- [1] VersionSuccession.client Value for the client metaproperty must be stereotyped «VersionOfConfiguration» or its specializations.
- [2] VersionSuccession.supplier Value for the supplier metaproperty must be stereotyped «VersionOfConfiguration» or its specializations.

WholeLifeConfiguration

Package: Roadmap

isAbstract: No

Generalization: [PropertySet](#), Block

Extension: Class

Description

A set of VersionedElements.

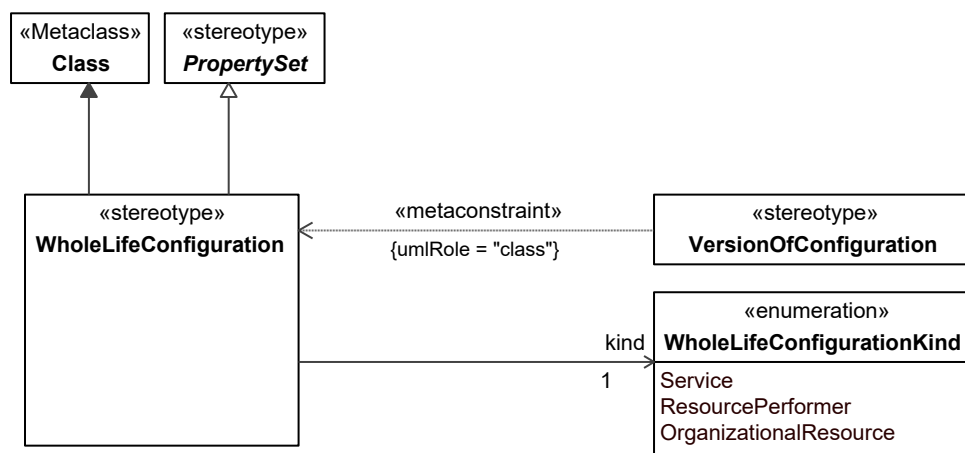


Figure 3:144 – WholeLifeConfiguration

Associations

kind : WholeLifeConfigurationKind[1] Captures the kind of WholeLifeConfiguration.

WholeLifeConfigurationKind

Package: Roadmap

isAbstract: No

Description

Enumeration of the possible kinds of WholeLifeConfiguration. Its enumeration literals are:

- Service - Indicates that the WholeLifeConfiguration associated with the WholeLifeConfigurationKind is the master specification from which Services are versioned.
- ResourcePerformer - Indicates that the WholeLifeConfiguration associated with the WholeLifeConfigurationKind is the master specification from which ResourcePerformers are versioned.
- OrganizationalResource - Indicates that the WholeLifeConfiguration associated with the WholeLifeConfigurationKind is the master specification from which OrganizationalResources are versioned.

UAF::Resources::Traceability

Contains the elements that contribute to the Resources Traceability Viewpoint.

ProtocolImplementation

Package: Traceability

isAbstract: Yes

Generalization: [UAFElement](#)

Extension: Element

Description

An abstract grouping of architectural elements that can implement Protocols.

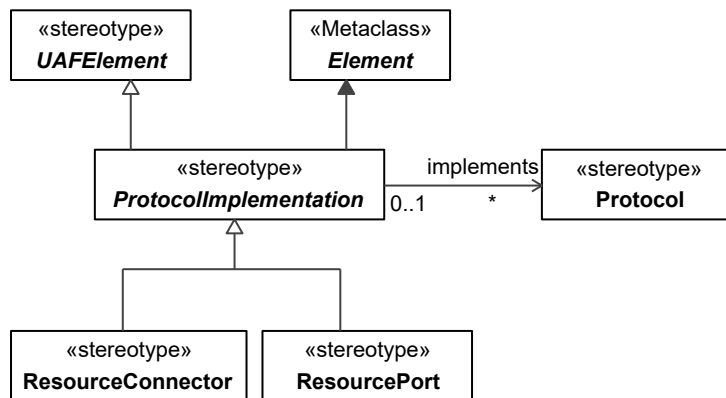


Figure 3:145 – ProtocolImplementation

Associations

implements : Protocol[*] Relates the ResourceConnector and ResourcePort to the Protocols that they can implement.

3.1.9 UAF::Security

Stakeholders: Security Architects, Security Engineers, Systems Engineers, Operational Architects.

Concerns: addresses the security constraints and information assurance attributes that exist on exchanges between resources and OperationalPerformers

Definition: illustrates the security assets, security constraints, security controls, families, and measures required to address specific security concerns.

UAF::Security::Taxonomy

Contains the elements that contribute to the Security Taxonomy Viewpoint.

Asset

Package: Taxonomy

isAbstract: Yes

Generalization: [ConceptItem](#), [PropertySet](#), [LocationHolder](#), [SubjectOfSecurityConstraint](#), Block

Extension: Class

Description

Asset as applied to Security views, an abstract element that indicates the types of elements that can be considered as a subject for security analysis.

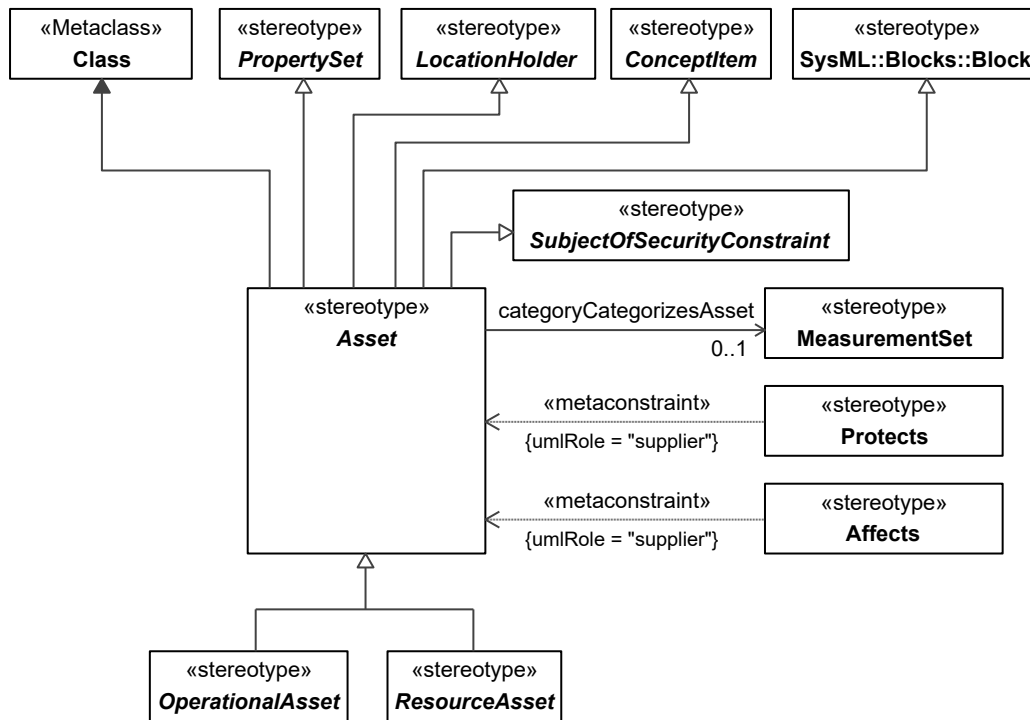


Figure 3:146 – Asset

Associations

categoryCategorizesAsset : MeasurementSet[0..1] Enables association of an Asset to the set of security related measurements (MeasurementSet).

OperationalAsset

Package: Taxonomy

isAbstract: Yes

Generalization: [Asset](#)

Extension: Class

Description

An abstract element used to group the elements of OperationalAgent and InformationElement allowing them to own InformationRoles.

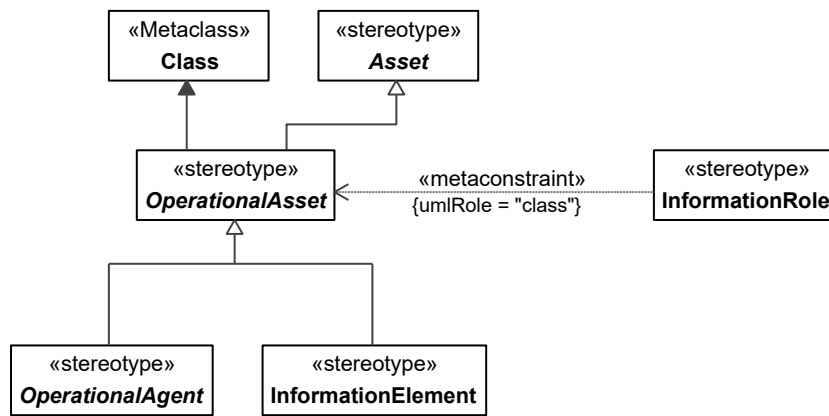


Figure 3:147 - OperationalAsset

OperationalMitigation

Package: Taxonomy

isAbstract: No

Generalization: [OperationalArchitecture](#)

Extension: Class

Description

A set of OperationalPerformers intended to address against specific operational risks.

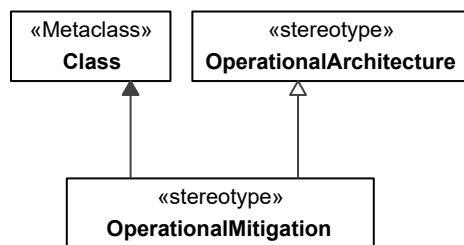


Figure 3:148 - OperationalMitigation

ResourceAsset

Package: Taxonomy

isAbstract: Yes

Generalization: [Asset](#)

Extension: Class

Description

An abstract element used to group the elements of ResourcePerformer and DataElement allowing them to own DataRoles.

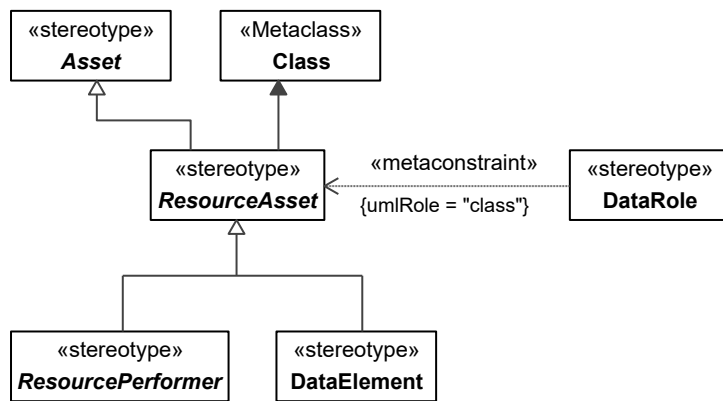


Figure 3:149 - ResourceAsset

ResourceMitigation

Package: Taxonomy

isAbstract: No

Generalization: [ResourceArchitecture](#)

Extension: Class

Description

A set of ResourcePerformers intended to address against specific risks.

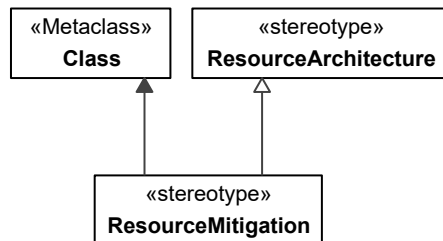


Figure 3:150 - ResourceMitigation

SecurityEnclave

Package: Taxonomy

isAbstract: No

Generalization: [ResourceArchitecture](#)

Extension: Class

Description

Collection of information systems connected by one or more internal networks under the control of a single authority and security policy. The systems may be structured by physical proximity or by function, independent of location.

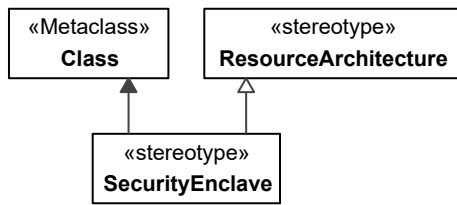


Figure 3:151 - SecurityEnclave

UAF::Security::Structure

Contains the elements that contribute to the Security Structure Viewpoint.

AssetRole

Package: Structure

isAbstract: Yes

Generalization: [MeasurableElement](#), [SubjectOfSecurityConstraint](#)

Extension: Property

Description

AssetRole as applied to Security views, an abstract element that indicates the type of elements that can be considered as a subject for security analysis in the particular context.

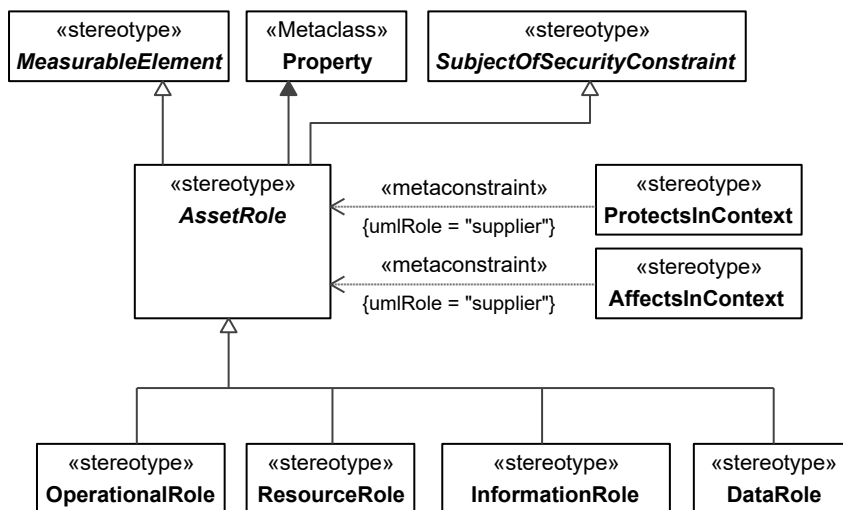


Figure 3:152 - AssetRole

DataRole

Package: Structure

isAbstract: No

Generalization: [AssetRole](#)

Extension: Property

Description

A usage of DataElement that exists in the context of a ResourceAsset. It also allows the representation of the whole-part aggregation of DataElements.

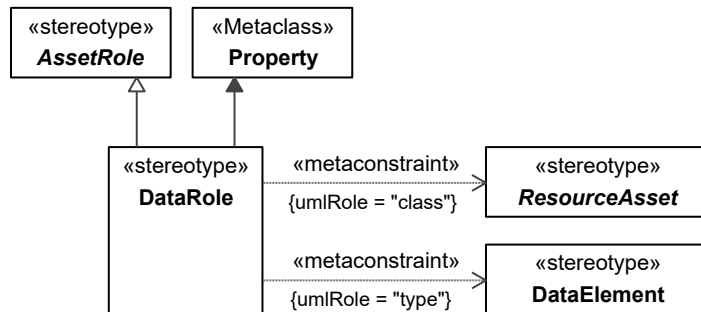


Figure 3:153 - DataRole

InformationRole

Package: Structure

isAbstract: No

Generalization: [AssetRole](#)

Extension: Property

Description

A usage of InformationElement that exists in the context of an OperationalAsset. It also allows the representation of the whole-part aggregation of InformationElements.

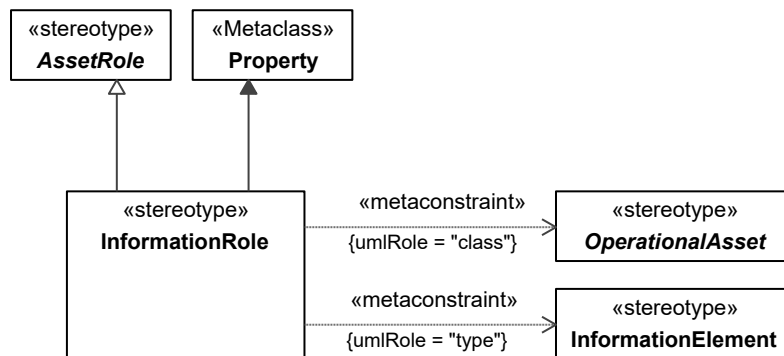


Figure 3:154 – InformationRole

Constraints

- | | |
|----------------------------|---|
| [1] SecurityProperty.class | Value for the class metaproperty must be stereotyped by the specialization of «Asset». |
| [2] SecurityProperty.type | In case of value for the class metaproperty is stereotyped:
a. by any of specializations of «OperationalAgent», values for the type metaproperty must be stereotyped «InformationElement» or its specializations,
b. by any of specializations of «ResourcePerformer», values for the type metaproperty must be stereotyped «DataElement» or its specializations,
c. «InformationElement», values for the type metaproperty must be stereotyped «InformationElement» or its specializations, |

d. «DataElement», values for the type metaproperty must be stereotyped «DataElement» or its specializations.

UAF::Security::Processes

Contains the elements that contribute to the Security Processes Viewpoint.

EnhancedSecurityControl

Package: Processes

isAbstract: No

Generalization: [SecurityControl](#)

Extension: Class

Description

Statement of security capability to: (i) build in additional but related, functionality to a basic control; and/or (ii) increase the strength of a basic control.

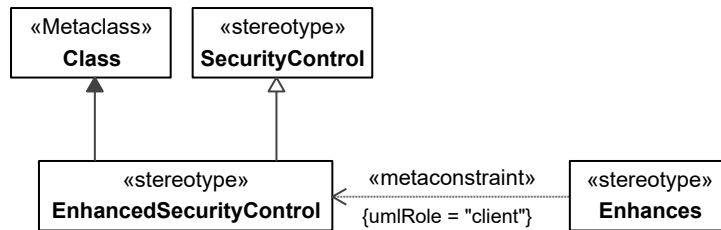


Figure 3:155 - EnhancedSecurityControl

Enhances

Package: Processes

isAbstract: No

Generalization: [DeriveReq](#), [MeasurableElement](#)

Extension: Abstraction, Connector

Description

A dependency relationship relating the EnhancedSecurityControl to a SecurityControl.

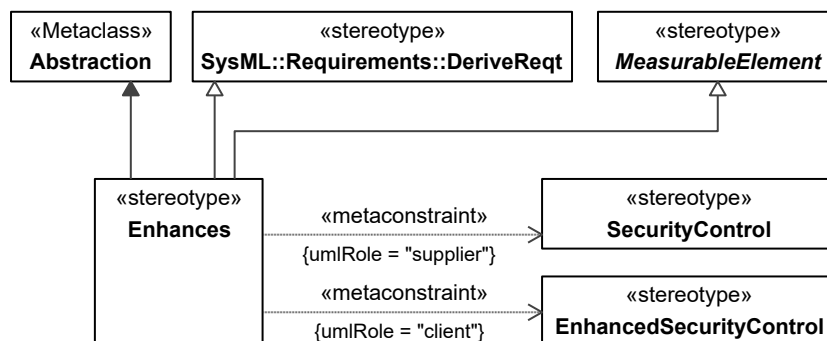


Figure 3:156 – Enhances

Constraints

- [1] Enhances.client Value for the client metaproperty must be stereotyped «EnhancedSecurityControl» or its specializations.
- [2] Enhances.supplier Value for the supplier metaproperty must be stereotyped «SecurityControl» or its specializations.

Protects

Package: Processes

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Dependency

Description

A dependency that asserts that a SecurityControl is required to protect an Asset.

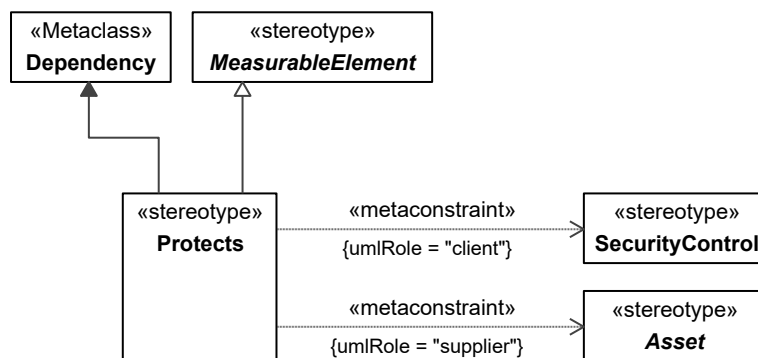


Figure 3:157 – Protects

Constraints

- [1] Protects.client Value for the client metaproperty must be stereotyped «SecurityControl» or its specializations.
- [2] Protects.supplier Value for the supplier metaproperty must be stereotyped by the specialization of «Asset».

ProtectsInContext

Package: Processes

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Dependency

Description

A dependency relationship that relates a SecurityControlAction to an OperationalRole, or a ResourceRole. It indicates that SecurityControl is required to protect an Asset in a specific context or configuration.

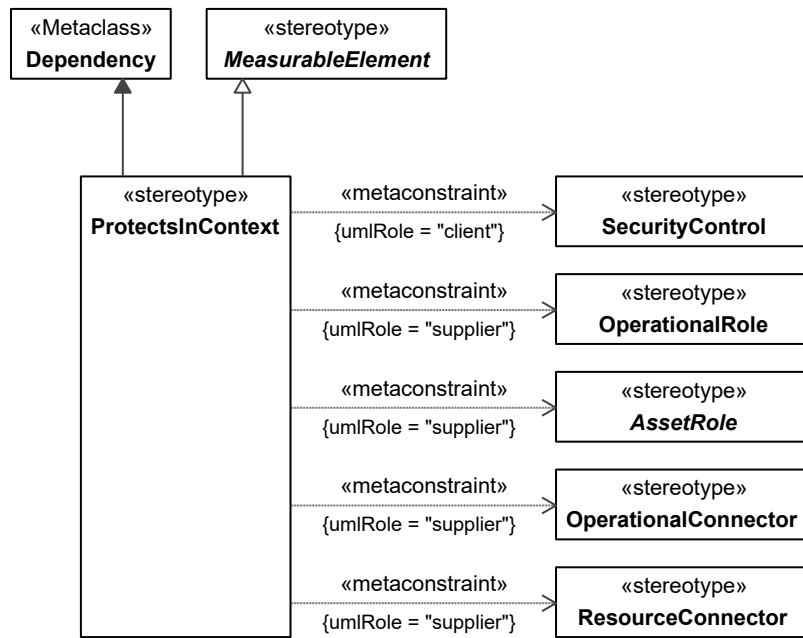


Figure 3:158 – ProtectsInContext

Constraints

- [1] ProtectsInContext.client Value for the client metaproperty must be stereotyped «SecurityControlAction» or its specializations.
- [2] ProtectsInContext.supplier Value for the supplier metaproperty must be stereotyped «OperationalRole», «ResourceRole», «OperationalConnector», «ResourceConnector», «SecurityProperty» or their specializations.

SecurityProcess

Package: Processes

isAbstract: No

Generalization: [OperationalActivity](#), [Function](#), [SubjectOfSecurityConstraint](#)

Extension: Activity

Description

The security-related procedure that satisfies the security control requirement.

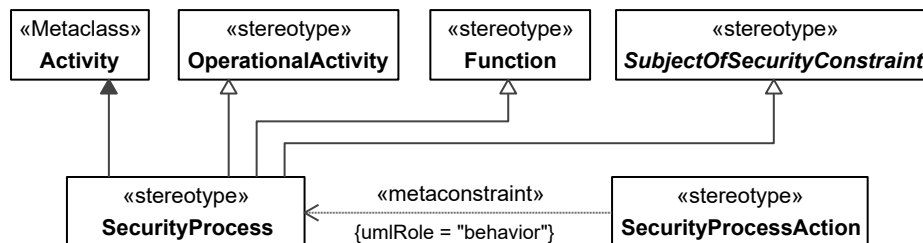


Figure 3:159 - SecurityProcess

SecurityProcessAction

Package: Processes

isAbstract: No

Generalization: [OperationalActivityAction](#), [FunctionAction](#)

Extension: CallBehaviorAction

Description

A call of a SecurityProcess in the context of another SecurityProcess.

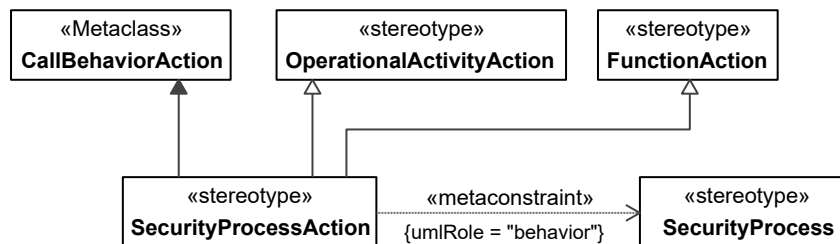


Figure 3:160 – SecurityProcessAction

Constraints

- [1] SecurityControlAction.behavior Value for behavior metaproperty must be stereotyped «SecurityControl» or its specializations.

UAF::Security::Constraints

Contains the elements that contribute to the Security Constraints Viewpoint.

ActualRisk

Package: Constraints

isAbstract: No

Generalization: [ActualPropertySet](#)

Extension: InstanceSpecification

Description

An instance of a Risk. A value holder for Risk Measurements.

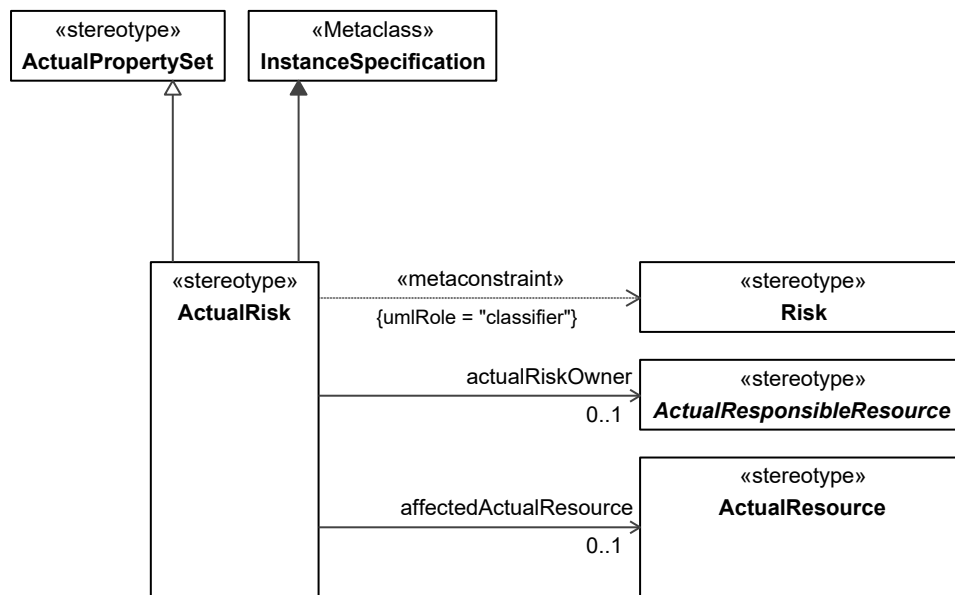


Figure 3:161 – ActualRisk

Associations

actualRiskOwner : ActualResponsibleResource[0..1]	Enables association of an ActualRisk to an actual organizational role that is responsible for executing the actual mitigation.
affectedActualResource : ActualResource[0..1]	Asserts that an ActualRisk is applicable to an ActualResource.

Risk

Package: Constraints

isAbstract: No

Generalization: [PropertySet](#), Block

Extension: Class

Description

A statement of the impact of an event on Assets. It represents a constraint on an Asset in terms of adverse effects, with an associated measure. The measure is used to capture the extent to which an entity is threatened by a potential circumstance or event. Risk is typically a function of: (i) the adverse impacts that would arise if the circumstance or event occurs; and (ii) the likelihood of occurrence.

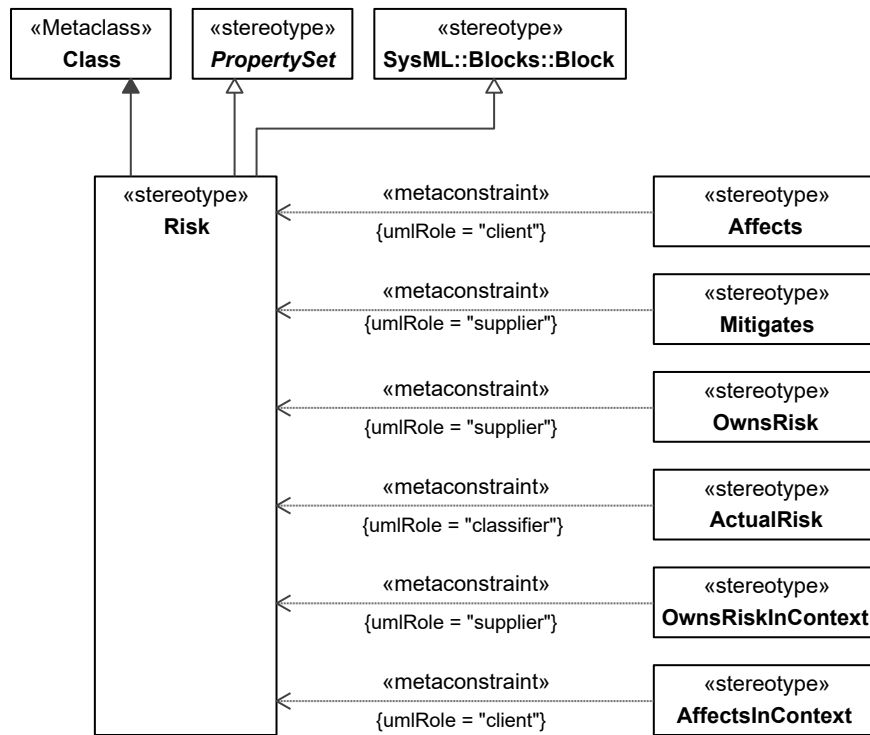


Figure 3:162 - Risk

SecurityConstraint

Package: Constraints

isAbstract: No

Generalization: [Rule](#)

Extension: Constraint

Description

A type of rule that captures a formal statement to define security laws, regulations, guidances, and policy.

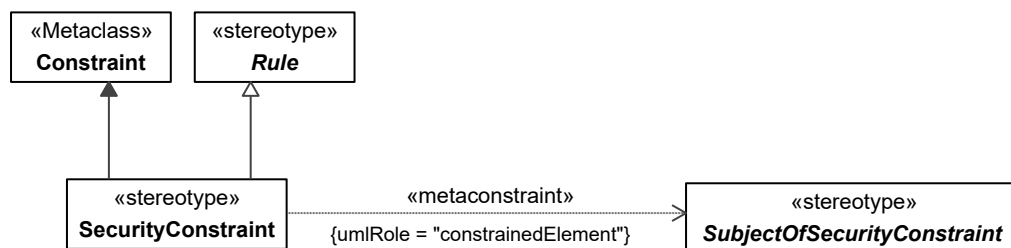


Figure 3:163 – SecurityConstraint

Constraints

- [1] Security.constrainedElement Value for the constrainedElement metaproperty must be stereotyped by the specialization of «SubjectOfSecurityConstraint».

SecurityControl

Package: Constraints

isAbstract: No

Generalization: Requirement, [MeasurableElement](#)

Extension: Class

Description

The management, operational, and technical control (i.e., safeguard or countermeasure) prescribed for an information system to protect the confidentiality, integrity, and availability of the system and its information [NIST SP 800-53].

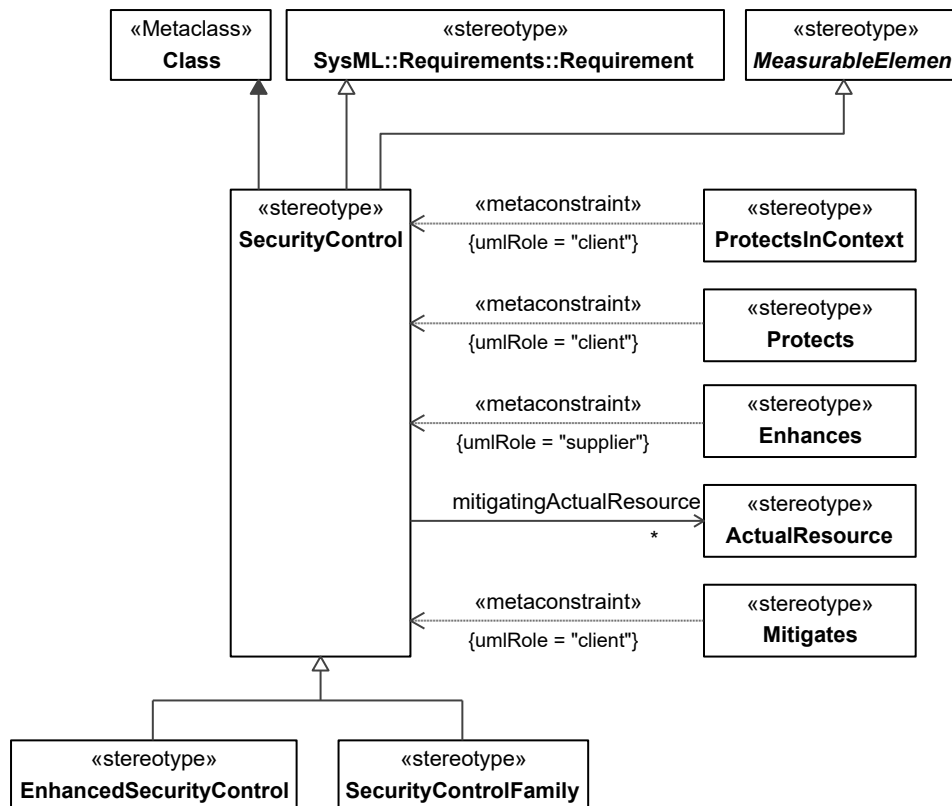


Figure 3:164 – SecurityControl

Associations

mitigatingActualResource : ActualResource[*] Relates an actual mitigation (an ActualResource for mitigating a Risk) to an ActualRisk.

SecurityControlFamily

Package: Constraints

isAbstract: No

Generalization: [SecurityControl](#)

Extension: Class

Description

An element that organizes security controls into a family. Each Security Control Family contains security controls related to the general security topic of the family.

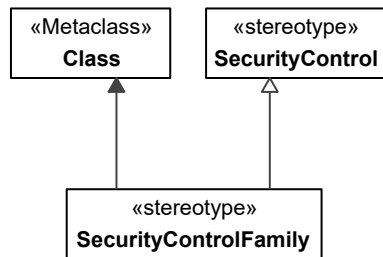


Figure 3:165 - SecurityControlFamily

SubjectOfSecurityConstraint

Package: Constraints

isAbstract: Yes

Generalization: [UAFElement](#)

Extension: Element

Description

An abstract grouping of elements that can be the subject of a SecurityConstraint.

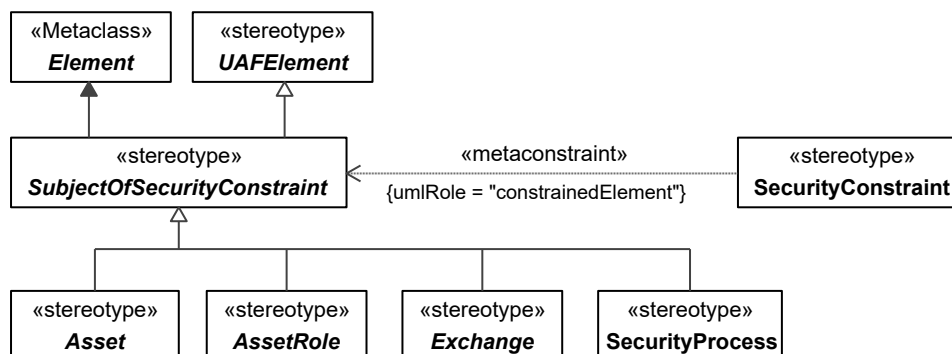


Figure 3:166 - SubjectOfSecurityConstraint

UAF::Security::Traceability

Contains the elements that contribute to the Security Traceability Viewpoint.

Affects

Package: Traceability

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Dependency

Description

A dependency that asserts that a Risk is applicable to an Asset.

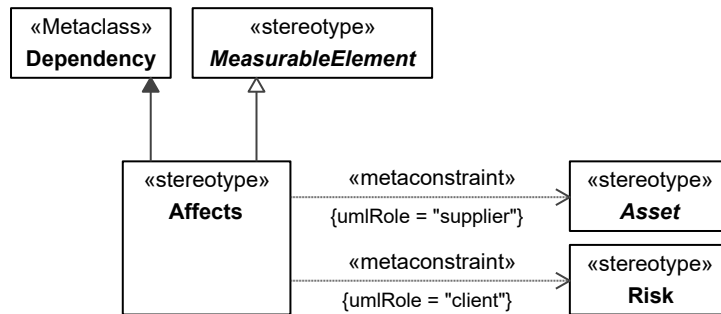


Figure 3:167 – Affects

Constraints

- [1] Affects.client Value for the client metaproperty must be stereotyped «Risk» or its specializations.
- [2] Affects.supplier Value for the supplier metaproperty must be stereotyped «Asset» or its specializations.

AffectsInContext

Package: Traceability

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Dependency

Description

A dependency that asserts that a Risk is applicable to an AssetRole in the specific context or configuration.

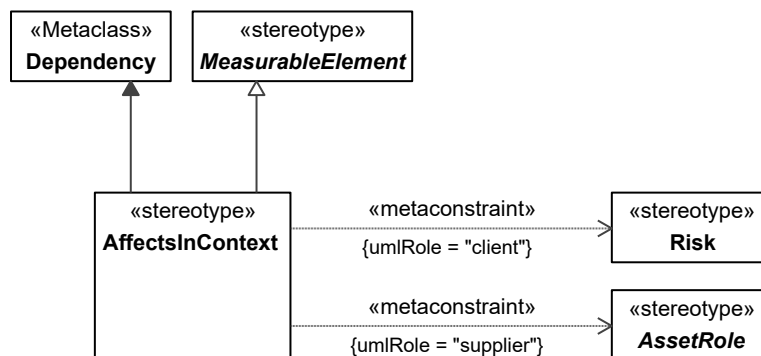


Figure 3:168 – AffectsInContext

Constraints

- [1] AffectsInContext.client Value for the client metaproperty must be stereotyped «Risk» or its specializations.
- [2] AffectsInContext.supplier Value for the supplier metaproperty must be stereotyped «AssetRole» or its specializations.

Mitigates

Package: Traceability

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Dependency

Description

A dependency relating a Security Control to a Risk. Mitigation is established to manage risk and could be represented as an overall strategy or through techniques (mitigation configurations) and procedures (SecurityProcesses).

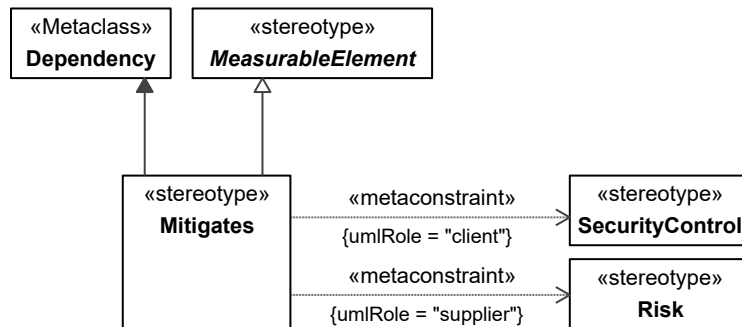


Figure 3:169 – Mitigates

Constraints

- [1] Mitigates.client Value for the client metaproperty must be stereotyped «SecurityControl» or its specializations.
- [2] Mitigates.supplier Value for the supplier metaproperty must be stereotyped «Risk» or its specializations.

OwnsRisk

Package: Traceability

isAbstract: No

Generalization: [MeasurableElement](#), Allocate

Extension: Abstraction

Description

An abstraction relating a Risk to an organizational resource that is responsible for executing the risk mitigation.

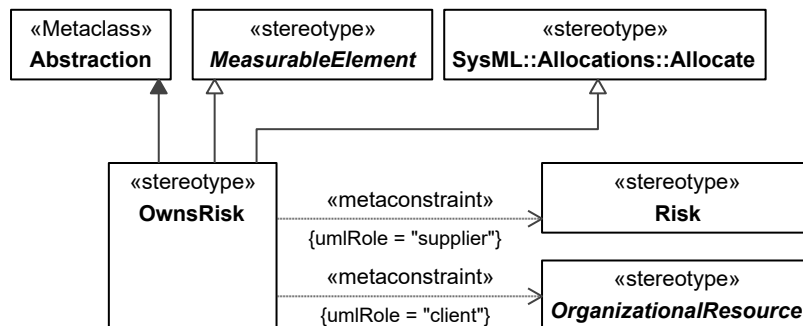


Figure 3:170 – OwnsRisk

Constraints

- [1] OwnsRisk.client Value for the client metaproperty must be stereotyped «OrganizationalResource» or its specializations.

[2] OwnsRisk.supplier Value for the supplier metaproperty must be stereotyped «Risk» or its specializations.

OwnsRiskInContext

Package: Traceability

isAbstract: No

Generalization: [MeasurableElement](#), Allocate

Extension: Abstraction

Description

An abstraction relating a Risk to an organizational role that is responsible for executing the risk mitigation in the specific context or configuration.

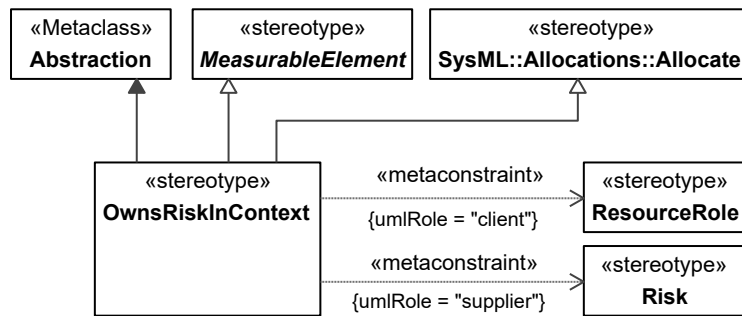


Figure 3:171 – OwnsRiskInContext

Constraints

- [1] OwnsRiskInContext.client Value for the client metaproperty must be stereotyped «ResourceRole» or its specializations.
- [2] OwnsRiskInContext.supplier Value for the supplier metaproperty must be stereotyped «Risk» or its specializations.

3.1.10 UAF::Projects

Stakeholders: PMs, Project Portfolio Managers, Enterprise Architects.

Concerns: project portfolio, projects and project milestones.

Definition: describes projects and project milestones, how those projects deliver capabilities, the organizations contributing to the projects and dependencies between projects.

UAF::Projects::Taxonomy

Contains the elements that contribute to the Project Taxonomy Viewpoint.

ActualMilestoneKind

Package: Taxonomy

isAbstract: No

Description

Enumeration of the possible kinds of ActualProjectMilestone. Its enumeration literals are:

- InService - Indicates that the ActualProjectMilestone associated with the ActualMilestoneKind is when the configuration goes into service.



Description

The role played by a ProjectMilestone in the context of a Project.

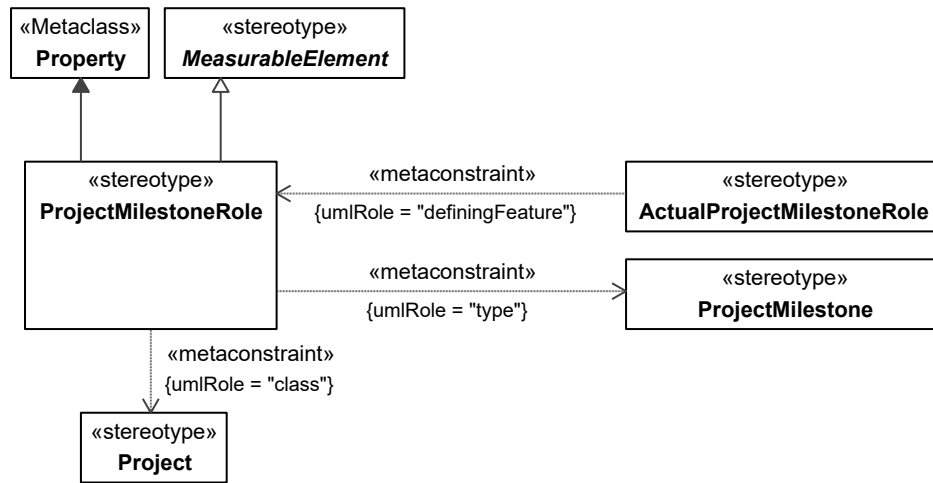


Figure 3:174 – ProjectMilestoneRole

Constraints

- [1] ProjectMilestoneRole.class Value for the class metaproperty must be stereotyped «Project» or its specializations.
- [2] ProjectMilestoneRole.type Value for the type metaproperty must be stereotyped «ProjectMilestone» or its specializations.

ProjectRole

Package: Structure

isAbstract: No

Generalization: [ResourceRole](#)

Extension: Property

Description

Usage of a Project in the context of another Project. Creates a whole-part relationship.

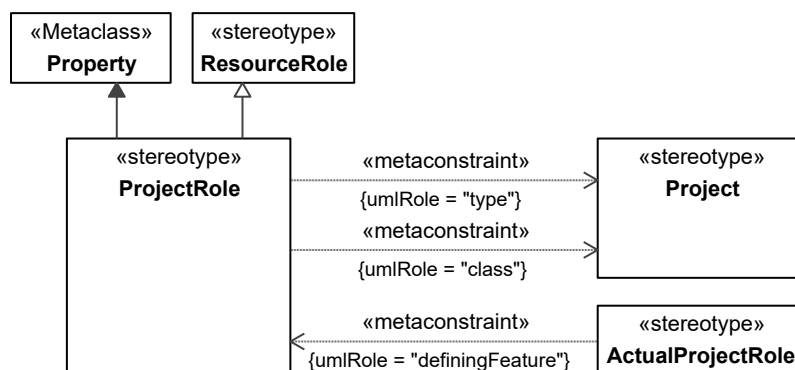


Figure 3:175 – ProjectRole

Constraints

- [1] ProjectRole.class Value for the class metaproperty must be stereotyped «Project» or its specializations.
- [2] ProjectRole.type Value for the type metaproperty must be stereotyped «Project» or its specializations.

ProjectStatus

Package: Structure

isAbstract: No

Generalization: [UAFElement](#)

Extension: Slot

Description

The status (i.e. level of progress) of a ProjectTheme for an ActualProject at the time of the ActualProjectMilestone.

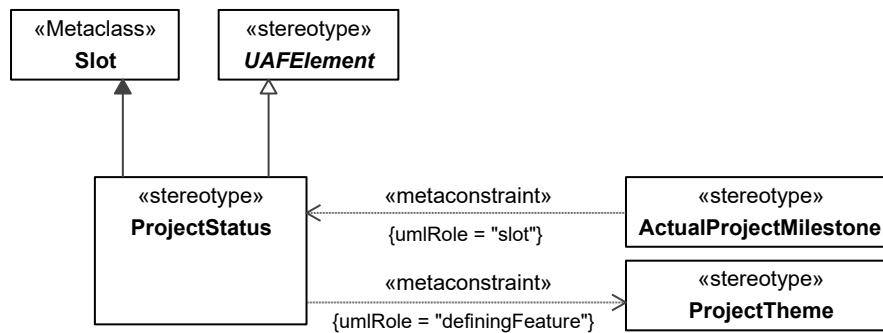


Figure 3:176 – ProjectStatus

Constraints

- [1] ProjectStatus.definingFeature Value for the DefiningFeature metaproperty must be stereotyped «ProjectTheme» or its specializations.

ProjectTheme

Package: Structure

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Property

Description

A property of a ProjectMilestone that captures an aspect by which the progress of ActualProjects may be measured.

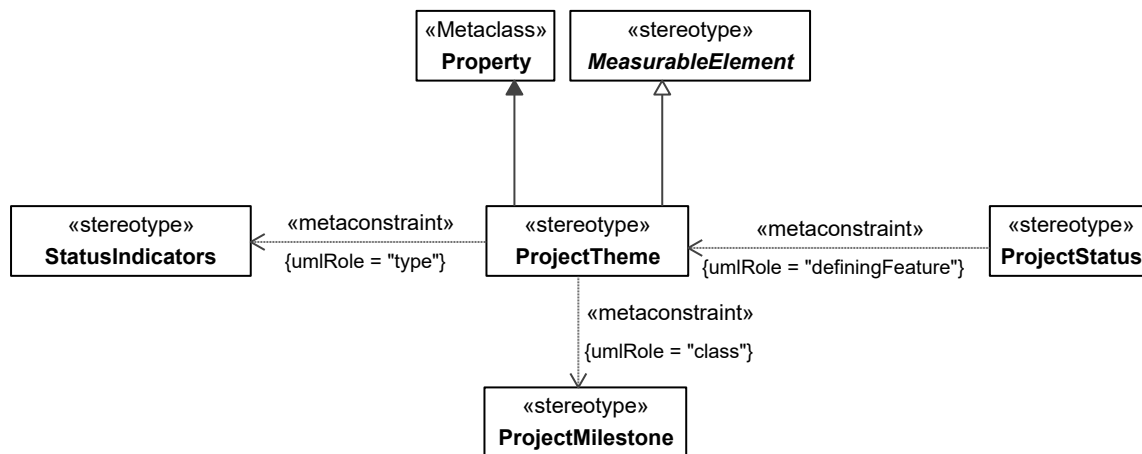


Figure 3:177 – ProjectTheme

Constraints

- [1] ProjectTheme.class Value for the class metaproperty must be stereotyped «ProjectMilestone» or its specializations.
- [2] ProjectTheme.type Value for the type metaproperty must be stereotyped «StatusIndicators» or its specializations.

StatusIndicators

Package: Structure

isAbstract: No

Generalization: [MeasurableElement](#), ValueType

Extension: Enumeration

Description

An enumerated type that specifies a status for a ProjectTheme.

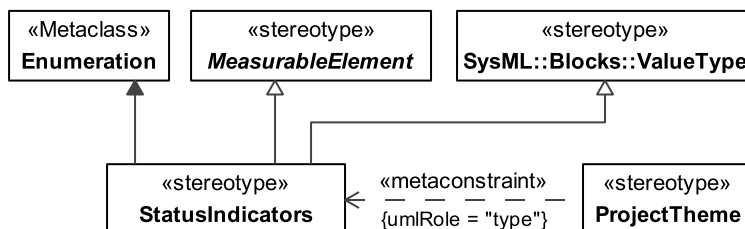


Figure 3:178 - StatusIndicators

UAF::Projects::Connectivity

Contains the elements that contribute to the Project Connectivity Viewpoint.

MilestoneDependency

Package: Connectivity

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Dependency

Description

A dependency relationship between two ActualProjectMilestones that denotes one ActualProjectMilestone follows from another.

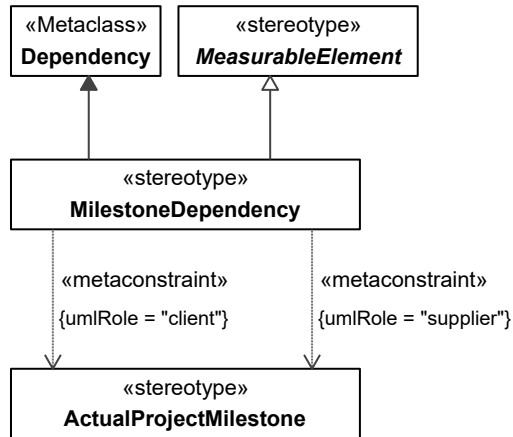


Figure 3:179 – MilestoneDependency

Constraints

- [1] MilestoneDependency.client Value for the client metaproperty must be stereotyped «ActualProjectMilestone» or its specializations.
- [2] MilestoneSequence.supplier Value for the supplier metaproperty must be stereotyped «ActualProjectMilestone» or its specializations.

ProjectSequence

Package: Connectivity

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Dependency

Description

A dependency relationship between two ActualProjects that denotes one ActualProject cannot start before the previous ActualProject is finished.

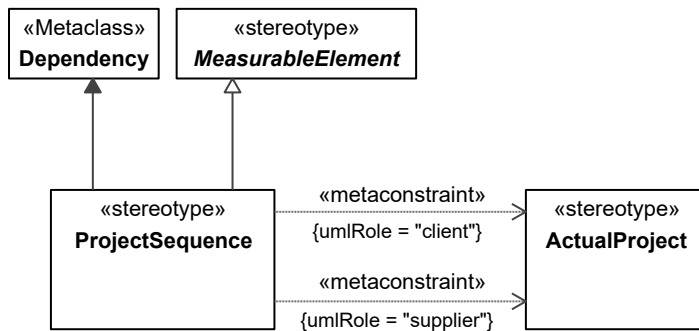


Figure 3:180 – ProjectSequence

Constraints

- [1] ProjectSequence.client Value for the client metaproperty must be stereotyped «ActualProject» or its specializations.
- [2] ProjectSequence.supplier Value for the supplier metaproperty must be stereotyped «ActualProject» or its specializations.

UAF::Projects::Processes

Contains the elements that contribute to the Project Processes Viewpoint.

ProjectActivity

Package: Processes

isAbstract: No

Generalization: [Function](#)

Extension: Activity

Description

An activity carried out during a project.

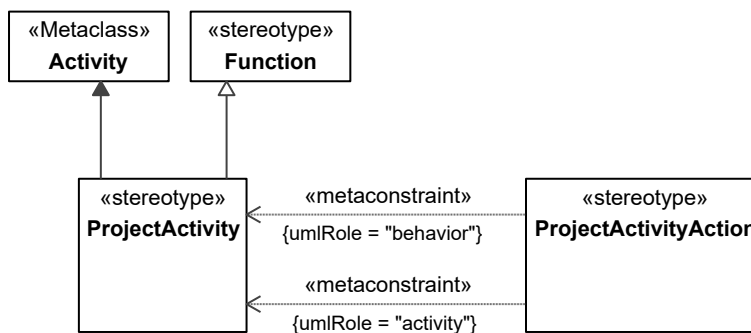


Figure 3:181 - ProjectActivity

ProjectActivityAction

Package: Processes

isAbstract: No

Generalization: [FunctionAction](#)

Extension: CallBehaviorAction

Description

The ProjectActivityAction is defined as a call behavior action that invokes the activity that needs to be preformed.

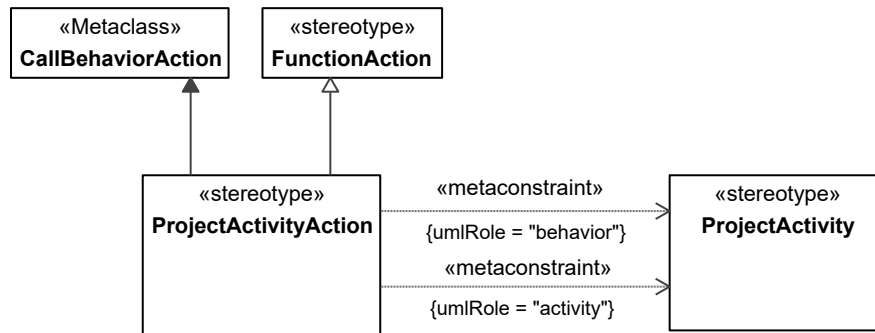


Figure 3:182 – ProjectActivityAction

Constraints

- | | |
|------------------------------------|---|
| [1] FunctionAction.behavior | Value for the behavior metaproperty must be stereotyped «ProjectActivity» or its specializations. |
| [2] ProjectActivityAction.activity | Value for the activity metaproperty must be stereotyped «ProjectActivity» or its specializations. |

UAF::Projects::Roadmap

Contains the elements that contribute to the Project Roadmap Viewpoint.

ActualProject

Package: Roadmap

isAbstract: No

Generalization: [ActualOrganizationalResource](#), [Achiever](#)

Extension: InstanceSpecification

Description

A time-limited endeavor to provide a specific set of ActualResources that meet specific Capability needs.

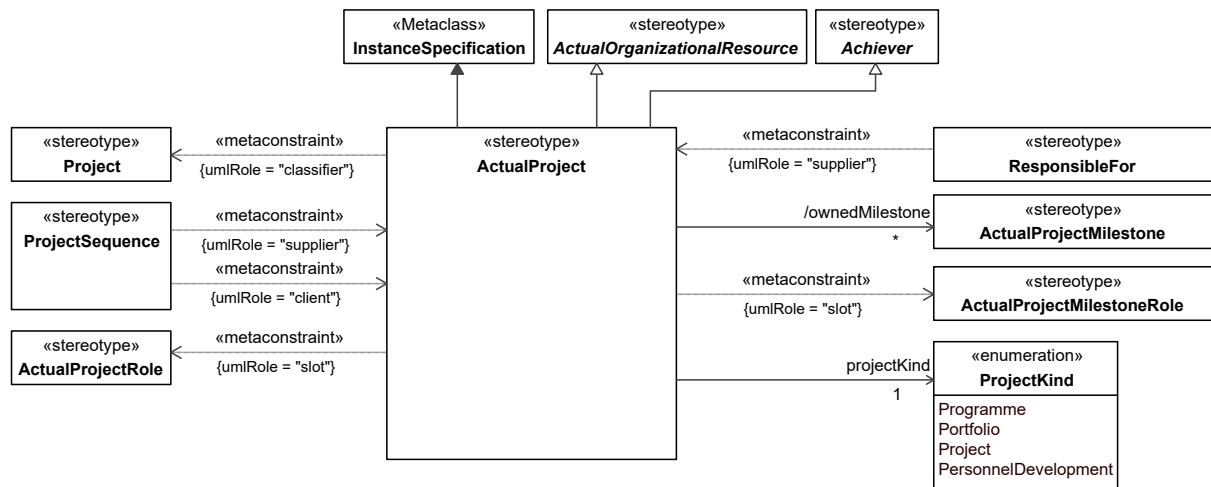


Figure 3:183 – ActualProject

Associations

ownedMilestone : ActualProjectMilestone[*] Relates the ActualProjectMilestones to the relevant ActualProject.
 projectKind : ProjectKind[1] Enumerated value describing the kind of ActualProject.

Constraints

- [1] ActualProject.classifier Value for the classifier metaproperty must be stereotyped «Project» or its specializations.
 [2] ActualProject.slot Value for the slot metaproperty must be stereotyped «ActualProjectRole», «ActualProjectMilestoneRole», or their specializations.

ActualProjectMilestone

Package: Roadmap

isAbstract: No

Generalization: [ActualPropertySet](#)

Extension: InstanceSpecification

Description

An event with a start date in an ActualProject from which progress is measured.

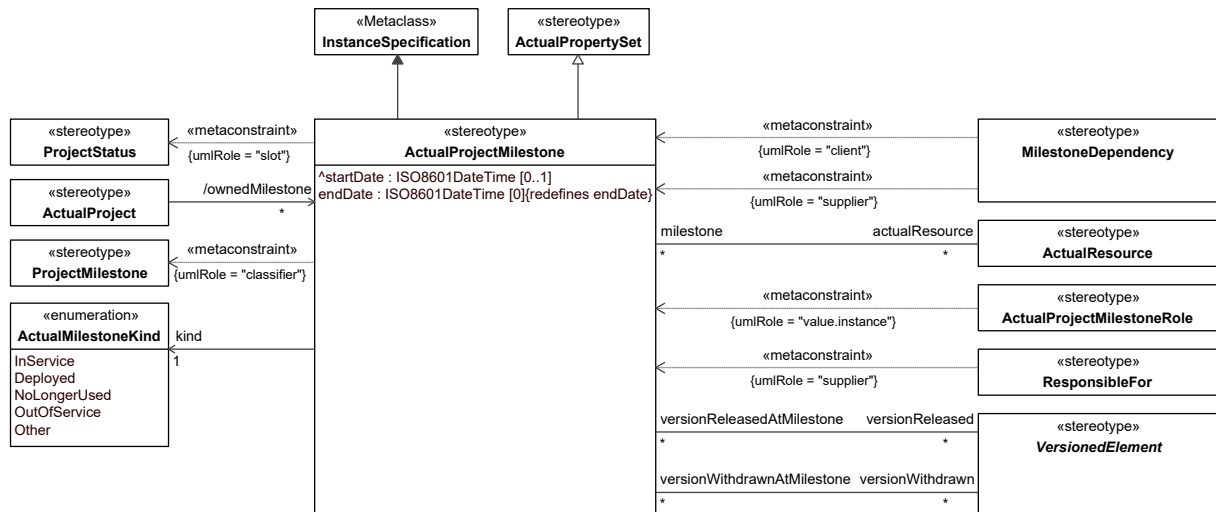


Figure 3:184 – ActualProjectMilestone

Attributes

endDate : ISO8601DateTime[0] End time for this ActualProjectMilestone.

Associations

actualResource : ActualResource[*] Relates an ActualProjectMilestone to the ActualResources that are affected by the milestone. It is used to describe aspects of the lifecycle of an ActualResource.

kind : ActualMilestoneKind[1] Enumerated value describing the kind of ActualProjectMilestone.

versionReleased : VersionedElement[*]

versionWithdrawn : VersionedElement[*]

Constraints

[1] ActualProjectMilestone.classifier Value for the classifier metaproperty must be stereotyped «ProjectMilestone» or its specializations.

ActualProjectMilestoneRole

Package: Roadmap

isAbstract: No

Generalization: [ActualState](#)

Extension: Slot

Description

An ActualProjectMilestone that is applied to a ProjectMilestoneRole.

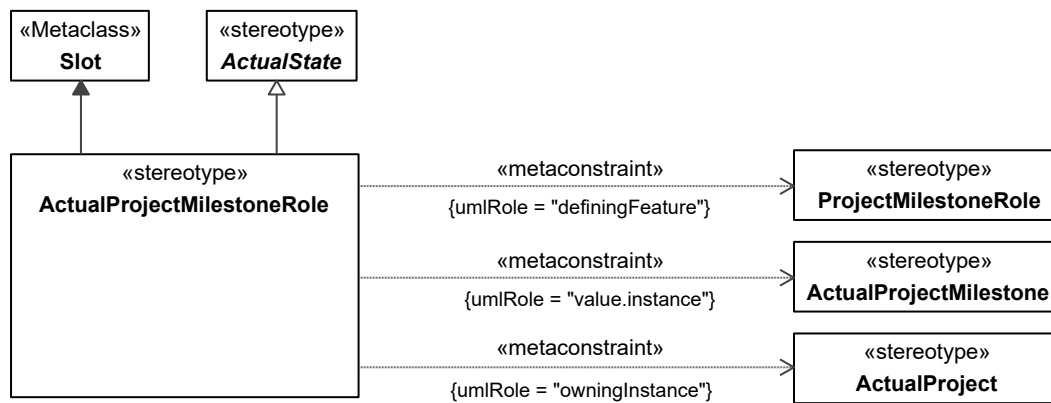


Figure 3:185 – ActualProjectMilestoneRole

Constraints

- | | |
|--|--|
| [1] ActualProjectMilestoneRole.definingFeature | Value for the definingFeature metaproperty has to be stereotyped «ProjectMilestoneRole» or its specializations. |
| [2] ActualProjectMilestoneRole.owningInstance | Value for the owningInstance metaproperty has to be stereotyped «ActualProject» or its specializations. |
| [3] ActualProjectMilestoneRole.value.instance | Value for the value.instance metaproperty has to be stereotyped «ActualProjectMilestone» or its specializations. |

ActualProjectRole

Package: Roadmap

isAbstract: No

Generalization: [ActualState](#)

Extension: Slot

Description

An ActualProject that is applied to a ProjectRole.

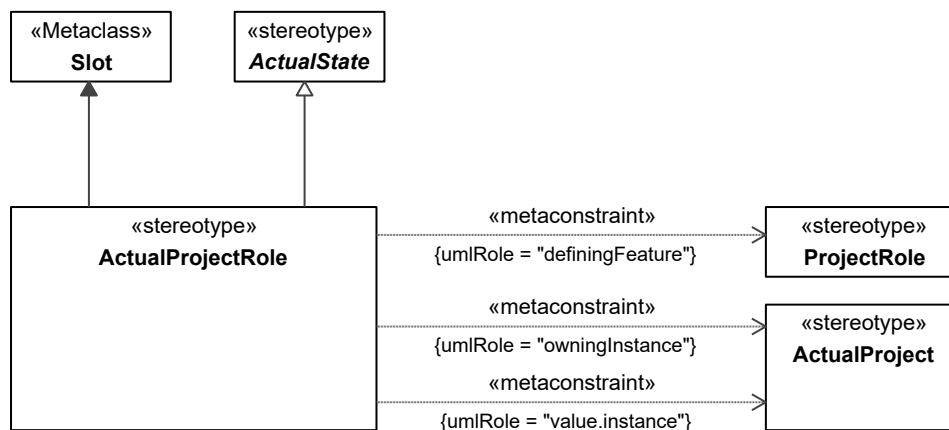


Figure 3:186 – ActualProjectRole

Constraints

- | | |
|---------------------------------------|---|
| [1] ActualProjectRole.definingFeature | Value for the definingFeature metaproperty has to be stereotyped «ProjectRole» or its specializations. |
| [2] ActualProjectRole.owningInstance | Value for the owningInstance metaproperty has to be stereotyped «ActualProject» or its specializations. |
| [3] ActualProjectRole.value.instance | Value for the value.instance metaproperty has to be stereotyped «ActualProject» or its specializations. |

3.1.11 UAF::Standards

Stakeholders: Solution Providers, Systems Engineers, Software Engineers, Systems Architects, Business Architects.

Concerns: technical and non-technical Standards applicable to the architecture.

Definition: shows the technical, operational, and business Standards applicable to the architecture. Defines the underlying current and expected Standards.

UAF::Standards::Taxonomy

Contains the elements that contribute to the Standards Taxonomy Viewpoint.

Protocol

Package: Taxonomy

isAbstract: No

Generalization: [Standard](#)

Extension: Class

Description

A Standard for communication over a network. Protocols may be composite, represented as a ProtocolStack made up of ProtocolLayers.

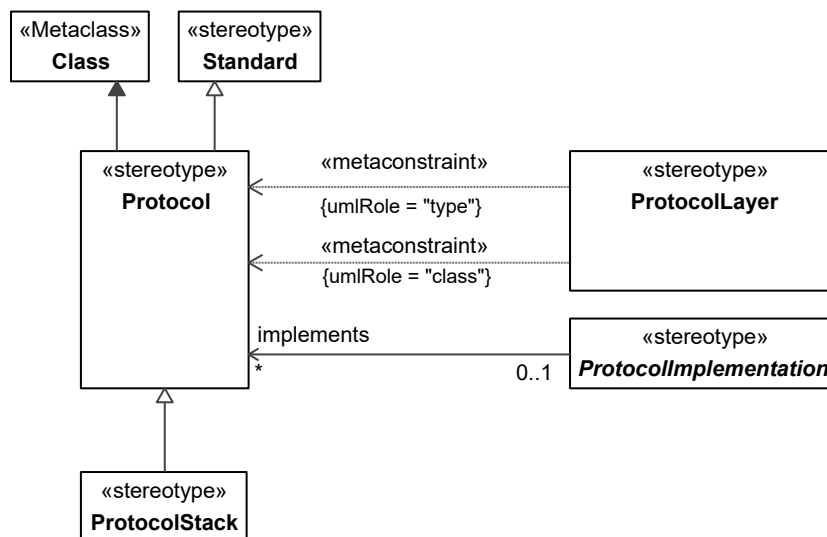


Figure 3:187 - Protocol

Attributes

mandatedDate : ISO8601DateTime[0..1] The date when this version of the Standard was published.

retiredDate : ISO8601DateTime[0..1] The date when this version of the Standard was retired.

Associations

ratifiedBy : ActualOrganization[*] Relates a Standard to the ActualOrganization that ratified the Standard.

UAF::Standards::Structure

Contains the elements that contribute to the Standards Structure Viewpoint.

ProtocolLayer

Package: Structure

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Property

Description

Usage of a Protocol in the context of another Protocol. Creates a whole-part relationship.

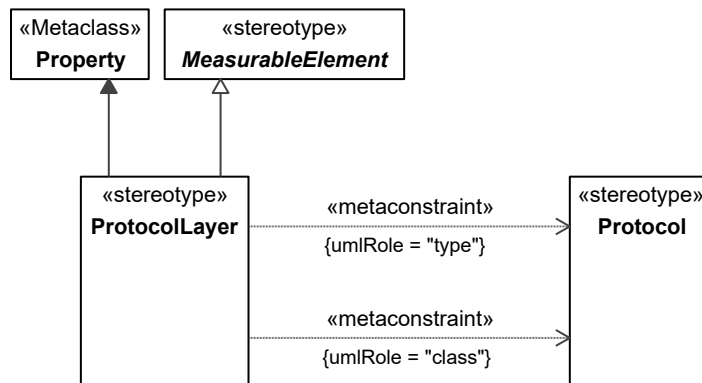


Figure 3:190 – ProtocolLayer

Constraints

[1] ProtocolLayer.class Value for the class metaproperty must be stereotyped «Protocol» or its specializations.

[2] ProtocolLayer.type Value for the type metaproperty must be stereotyped «Protocol» or its specializations.

3.1.12 UAF::Actual Resources

Stakeholders: Solution Providers, Systems Engineers, Business Architects, Human Resources.

Concerns: the analysis, e.g., evaluation of different alternatives, what-if, trade-offs, V&V on the actual resource configurations.

Definition: illustrates the expected or achieved actual resource configurations and actual relationships between them.

UAF::Actual Resources::Taxonomy

Contains the elements that contribute to the Actual Resources Taxonomy Viewpoint.

ActualOrganization

Package: Taxonomy

isAbstract: No

Generalization: [ActualResponsibleResource](#)

Extension: InstanceSpecification

Description

An actual formal or informal organizational unit, e.g., "Driving and Vehicle Licensing Agency", "UAF team Alpha".

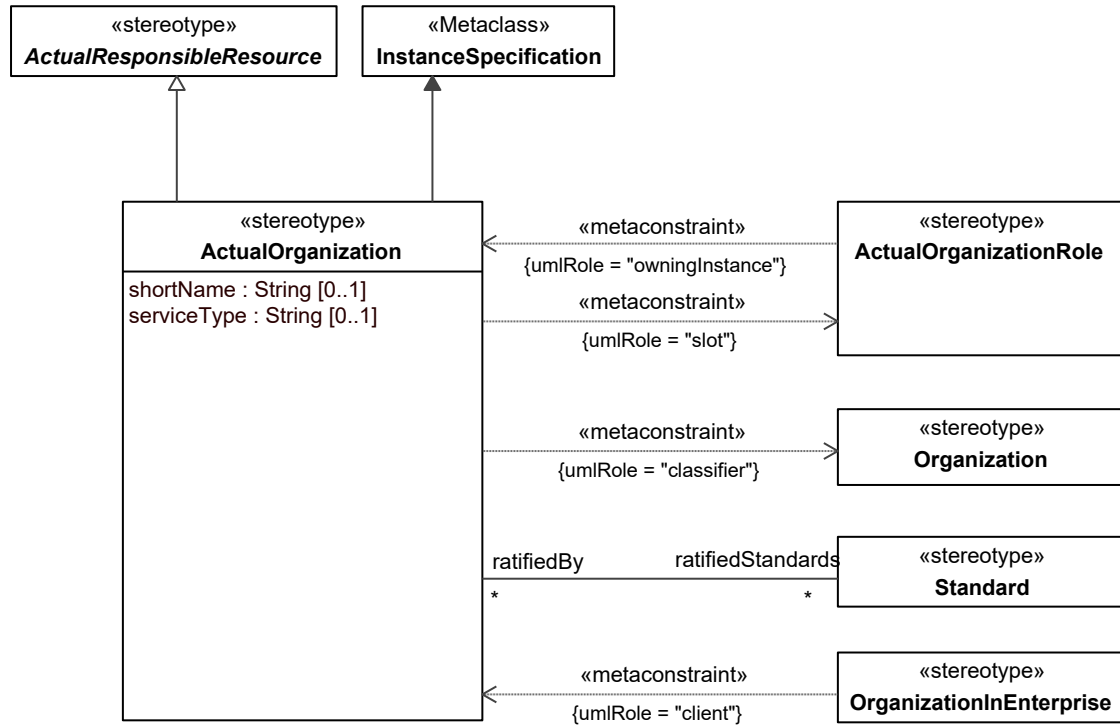


Figure 3:191 – ActualOrganization

Attributes

- serviceType : String[0..1] Service office code or symbol
- shortName : String[0..1] String providing a simplified means of identifying an ActualOrganization, i.e. SoftWareGroup could use SWG as the shortName.

Associations

- ratifiedStandards : Standard[*] Standards that were ratified by this ActualOrganization.

Constraints

- [1] ActualOrganization.classifier Classifier metaproperty value must be stereotyped «Organization» or its specializations.
- [2] ActualOrganization.slot Slot metaproperty value must be stereotyped «ActualOrganizationRole» or its specializations.

ActualOrganizationalResource

Package: Taxonomy

isAbstract: Yes

Generalization: [Stakeholder](#), [ActualResource](#)

Extension: InstanceSpecification

Description

Abstract element for an ActualOrganization, ActualPerson or ActualPost.

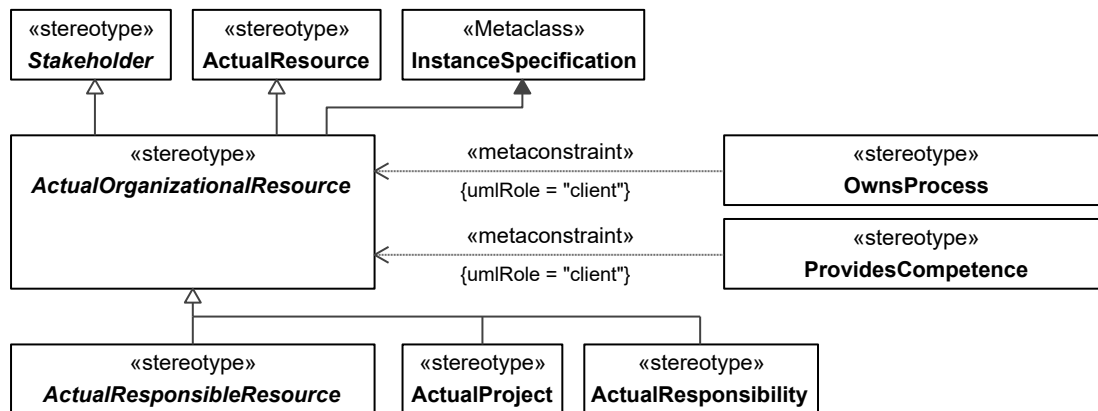


Figure 3:192 - ActualOrganizationalResource

ActualPerson

Package: Taxonomy

isAbstract: No

Generalization: [ActualResponsibleResource](#)

Extension: InstanceSpecification

Description

An individual human being.

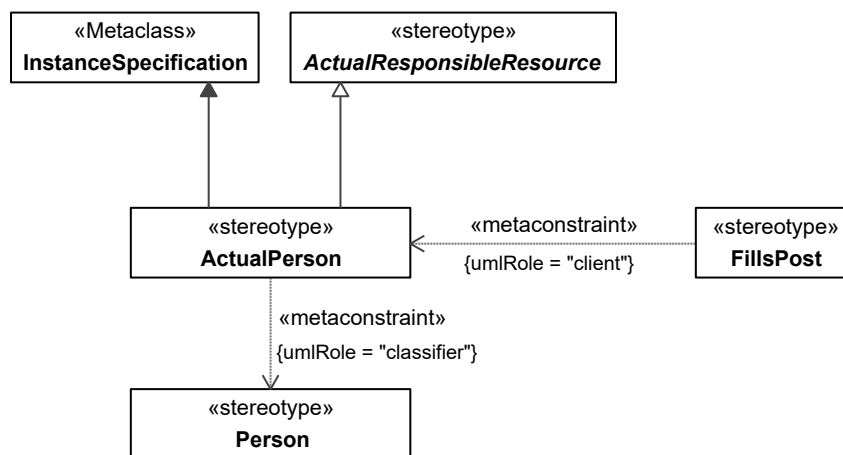


Figure 3:193 – ActualPerson

Constraints

- [1] ActualPerson.classifier Value for the classifier metaproperty has to be stereotyped «Person» or its specializations.

ActualPost

Package: Taxonomy

isAbstract: No

Generalization:

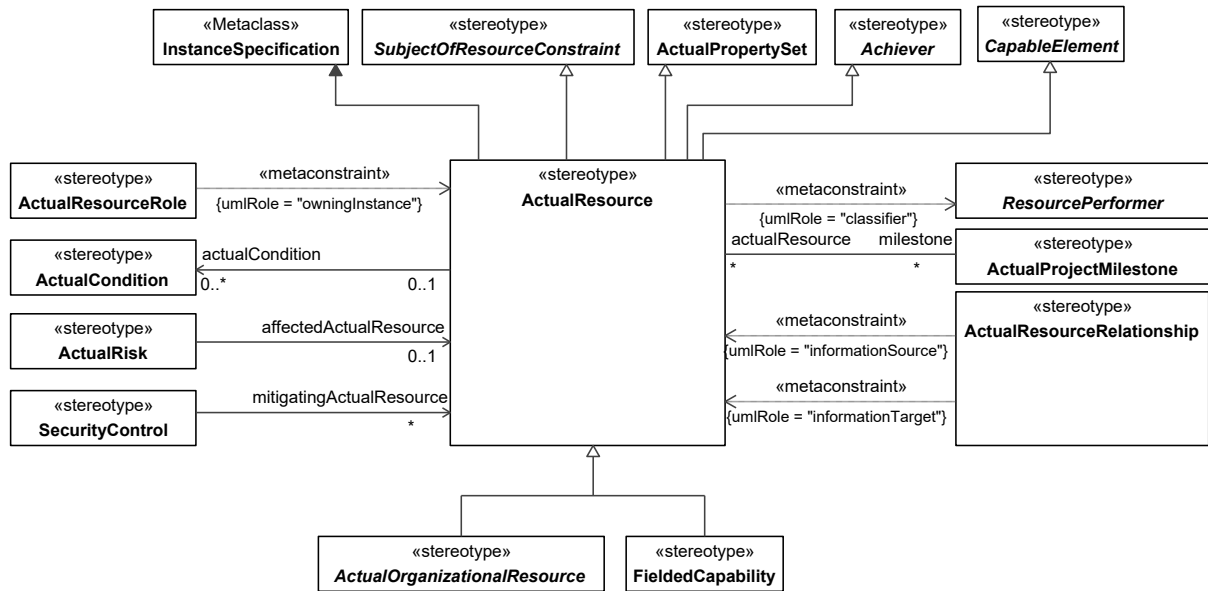


Figure 3:195 – ActualResource

Associations

- actualCondition : ActualCondition[0..*] Relates the ActualResource to the ActualStates of an environment or location describing its situation
- milestone : ActualProjectMilestone[*] Relates an ActualResource to the ActualProjectMilestones. It is used to describe aspects of the lifecycle of an ActualResource.

Constraints

- [1] ActualResource.classifier Classifier metaproperty value must be stereotyped by a specialization of «ResourcePerformer».

ActualResponsibility

Package: Taxonomy

isAbstract: No

Generalization: [ActualOrganizationalResource](#)

Extension: InstanceSpecification

Description

The duty required of a Person or Organization.

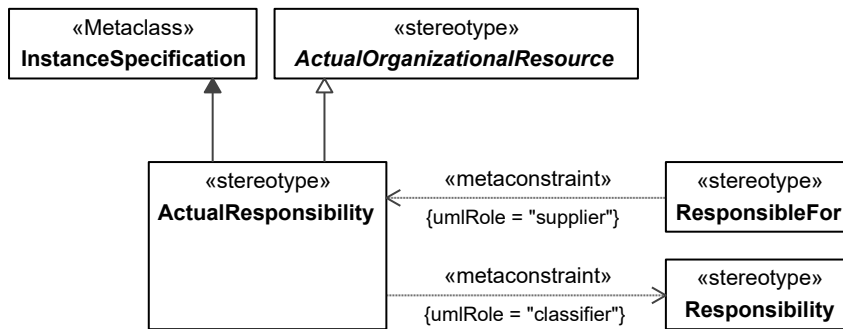


Figure 3:196 – ActualResponsibility

Constraints

- [1] ActualResponsibility.classifier Classifier metaproperty value must be stereotyped «Responsibility» or its specializations.

ActualResponsibleResource

Package: Taxonomy

isAbstract: Yes

Generalization: [ActualOrganizationalResource](#)

Extension: InstanceSpecification

Description

An abstract grouping of responsible OrganizationalResources.

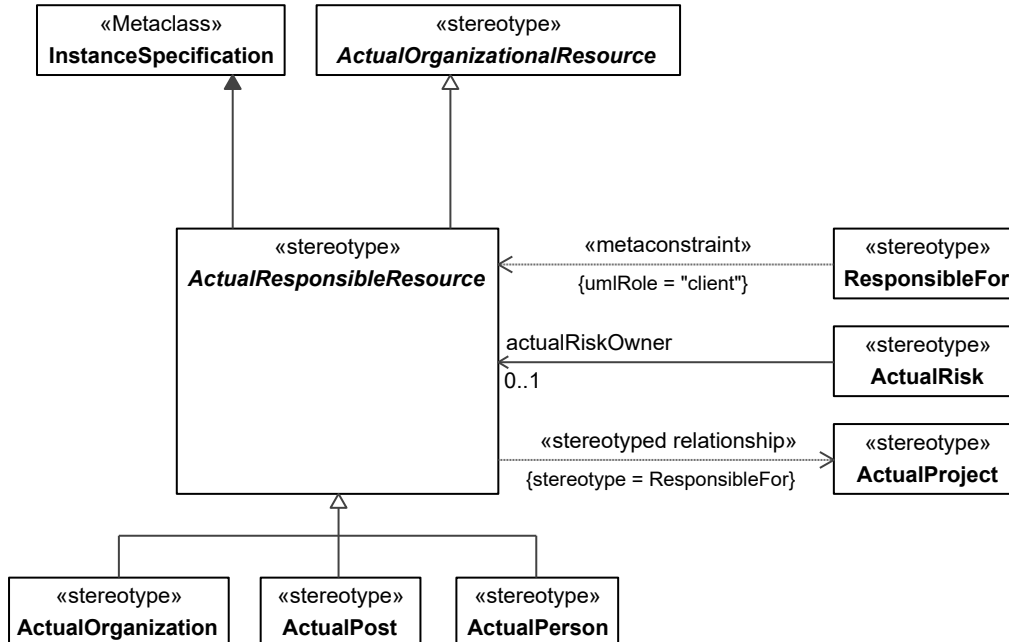


Figure 3:197 - ActualResponsibleResource

FieldedCapability

Package: Taxonomy

isAbstract: No

Generalization: [ActualResource](#)

Extension: InstanceSpecification

Description

An actual, fully-realized capability. A FieldedCapability is typed by a CapabilityConfiguration.

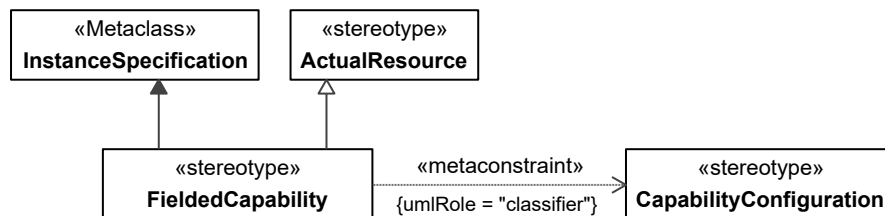


Figure 3:198 – FieldedCapability

Constraints

- [1] FieldedCapability.classifier Value for the classifier metaproperty must be stereotyped «CapabilityConfiguration» or its specializations.

UAF::Actual Resources::Structure

Contains the elements that contribute to the Actual Resources Structure Viewpoint.

ActualOrganizationRole

Package: Structure

isAbstract: No

Generalization: [ActualResourceRole](#)

Extension: Slot

Description

An ActualOrganizationalResource that is applied to a ResourceRole.

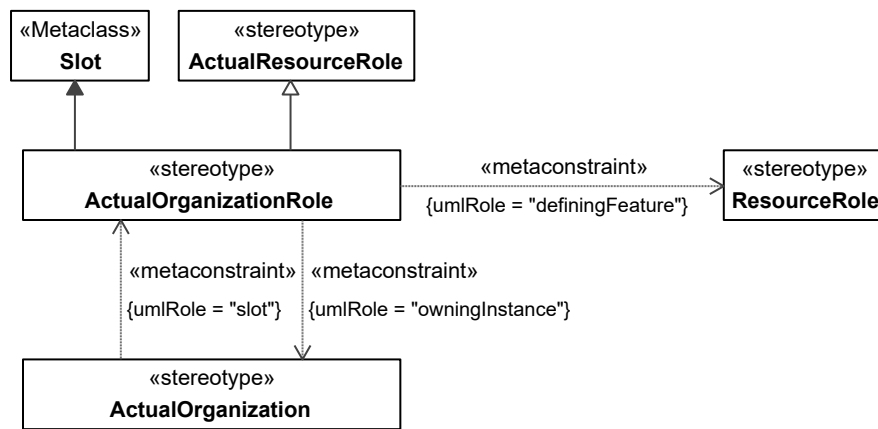


Figure 3:199 – ActualOrganizationRole

Constraints

- [1] ActualOrganizationRole.owningInstance Value for owningInstance metaproperty has to be stereotyped «ActualOrganization» or its specializations.

ActualResourceRole

Package: Structure

isAbstract: No

Generalization: [UAFEElement](#)

Extension: Slot

Description

An instance of a ResourcePerformer.

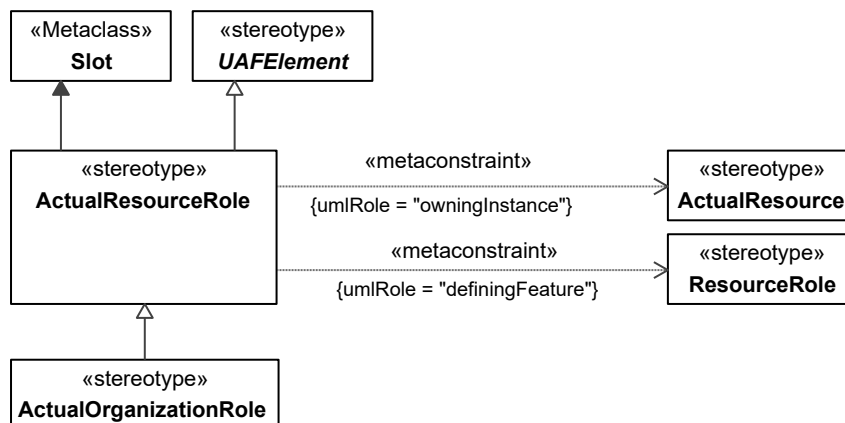


Figure 3:200 – ActualResourceRole

Constraints

- [1] ActualResourceRole.definingFeature Value for definingFeature metaproperty has to be stereotyped «ResourceRole» or its specializations.

«ster

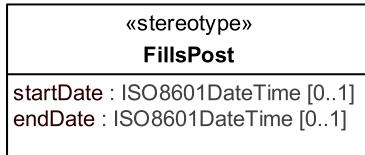


Figure 3:202 – FillsPost

Attributes

endDate : ISO8601DateTime[0..1] End

startDate : ISO8601DateTime[0..1]

Constraints

[1] FillsPost.client Value

[2] FillsPost.supplier Y

UAF::Actual P

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Constraints

- [1] ActualService.classifier Value for the classifier metaproperty must be stereotyped by «ServiceSpecification» or its specializations.

ProvidedServiceLevel

Package: Constraints

isAbstract: No

Generalization: [ActualService](#)

Extension: InstanceSpecification

Description

A sub type of ActualService that details a specific service level delivered by the provider.

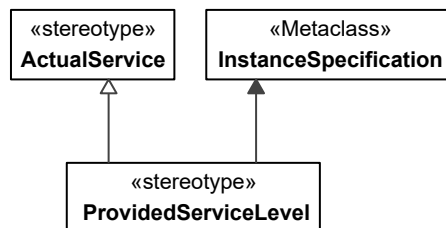


Figure 3:204 - ProvidedServiceLevel

ProvidesCompetence

Package: Constraints

isAbstract: No

Generalization: [MeasurableElement](#)

Extension: Dependency

Description

A dependency relationship that asserts that an ActualOrganizationalResource provides a specific set of Competencies.

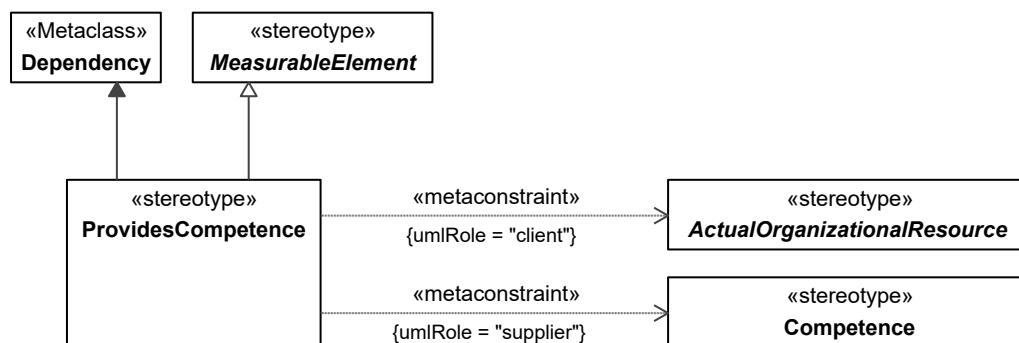


Figure 3:205 – ProvidesCompetence

Constraints

- [1] ProvidesCompetence.client Value for the client metaproperty must be stereotyped by a specialization of «ActualOrganizationalResource».

Constraints

- | | |
|--------------------------|--|
| [1] OwnsProcess.client | Value for the client metaproperty must be stereotyped «ActualOrganizationalResource» or its specializations. |
| [2] OwnsProcess.supplier | Value for the supplier metaproperty must be stereotyped «OperationalActivity» or its specializations. |

3.1.13 UAF::Summary and Overview

Stakeholders: Executives, PMs, Enterprise Architects.

Concerns: executive-level summary information in a consistent form.

Definition: provides executive-level summary information in a consistent form that allows quick reference and

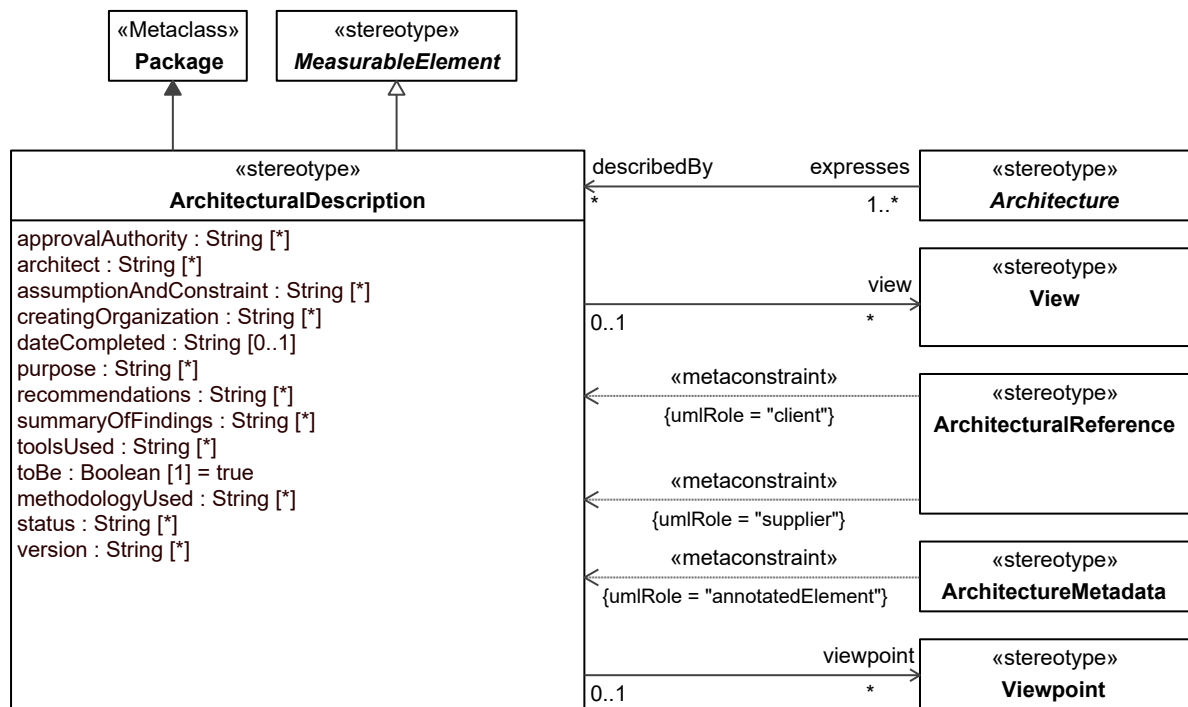


Figure 3:208 – ArchitecturalDescription

Attributes

approvalAuthority : String[*]	Someone or something that has the authority to approve the ArchitecturalDescription.
architect : String[*]	Someone responsible for the creation of ArchitecturalDescription.
assumptionAndConstraint : String[*]	Any assumptions, constraints, and limitations contained in the ArchitecturalDescription, including those affecting deployment, communications performance, information assurance environments, etc.
creatingOrganization : String[*]	The organization responsible for creating the ArchitecturalDescription.
dateCompleted : String[0..1]	Date that the ArchitecturalDescription was completed.
methodologyUsed : String[*]	The methodology used in developing the architecture.
purpose : String[*]	Explains the need for the Architecture, what it will demonstrate, the types of analyses that will be applied to it, who is expected to perform the analyses, what decisions are expected to be made on the basis of each form of analysis, who is expected to make those decisions, and what actions are expected to result.
recommendations : String[*]	States the recommendations that have been developed based on the architecture effort. Examples include recommended system implementations, and opportunities for technology insertion.
status : String[*]	Approval status of the architecture.
summaryOfFindings : String[*]	Summarizes the findings that have been developed so far. This may be updated several times during the development of the ArchitecturalDescription.
toBe : Boolean[1]	Indicates whether the ArchitecturalDescription represents an Architecture that exists or will exist in the future.
toolsUsed : String[*]	Identifies any tools used to develop the ArchitecturalDescription as well as file names and formats if appropriate.

version : String[*] Version number of the architecture.

Associations

architectureFramework : String[1] Indicates the type of framework used.

view : View[*] Indicates which views are used in the ArchitecturalDescription.

viewpoint : Viewpoint[*]

Architecture

Package: Summary and Overview

isAbstract: Yes

Generalization: [UAFElement](#)

Extension: Class

Description

An abstract type that represents a generic architecture. Subtypes are OperationalArchitecture and PhysicalArchitecture.

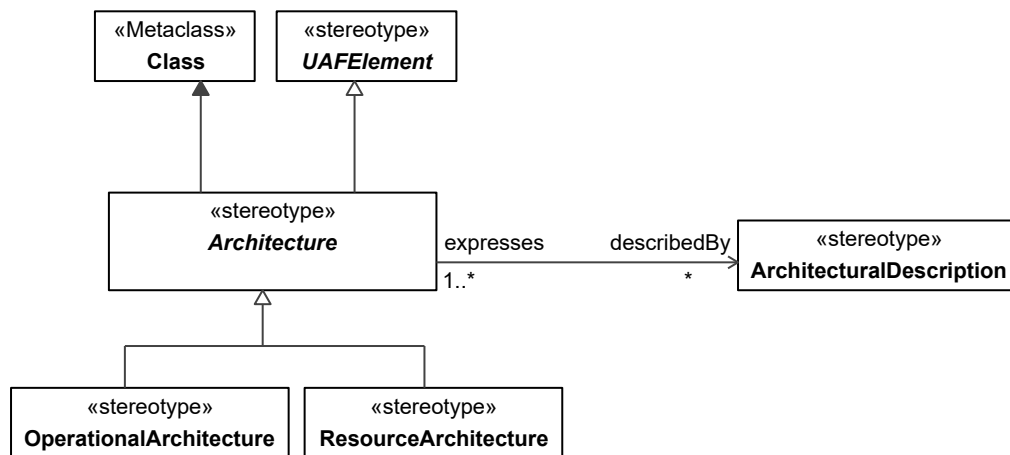


Figure 3:209 – Architecture

Associations

describedBy : ArchitecturalDescription[*] The description of an Architecture.

Concern

Package: Summary and Overview

isAbstract: No

Generalization: [PropertySet](#), Block

Extension: Class

Description

Interest in an EnterprisePhase (EnterprisePhase is synonym for System in ISO 42010) relevant to one or more of its stakeholders.

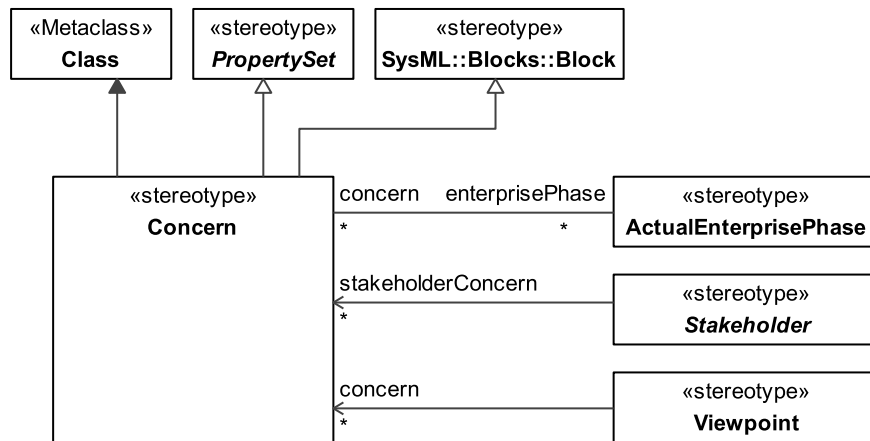


Figure 3:210 – Concern

Associations

enterprisePhase : ActualEnterprisePhase[*] Relates a Concern to the ActualEnterprisePhase that addresses that concern (ActualEnterprisePhase is synonym for System in ISO 42010).

Stakeholder

Package: Summary and Overview

isAbstract: Yes

Generalization: [UAFElement](#)

Extension: Element

Description

individual, team, organization, or classes thereof, having an interest in an EnterprisePhase [ISO/IEC/IEEE 42010:2011].

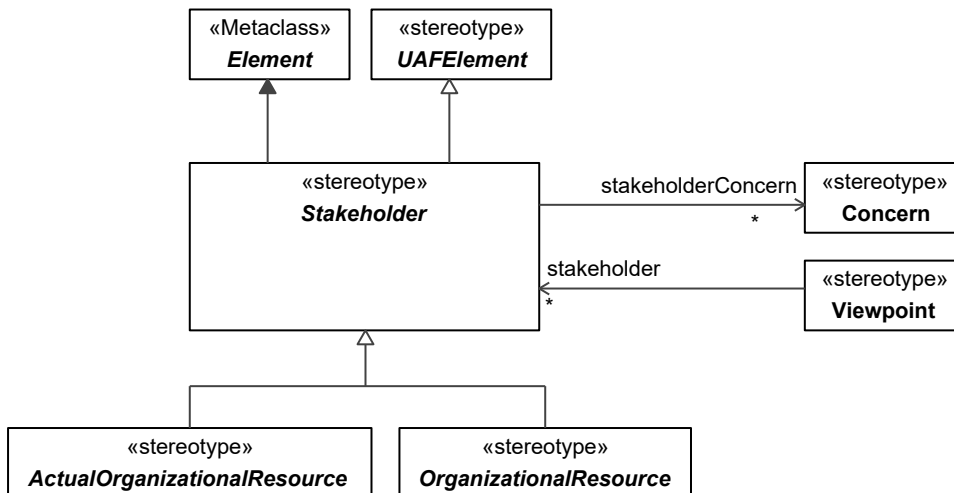


Figure 3:211 – Stakeholder

Associations

stakeholderConcern : Concern[*] Relates a Stakeholder to a Concern.

UAFElement

Package: Summary and Overview

isAbstract: Yes

Extension: Element

Description

Abstract super type for all of the UAF elements. It provides a way for all of the UAF elements to have a common set of properties.

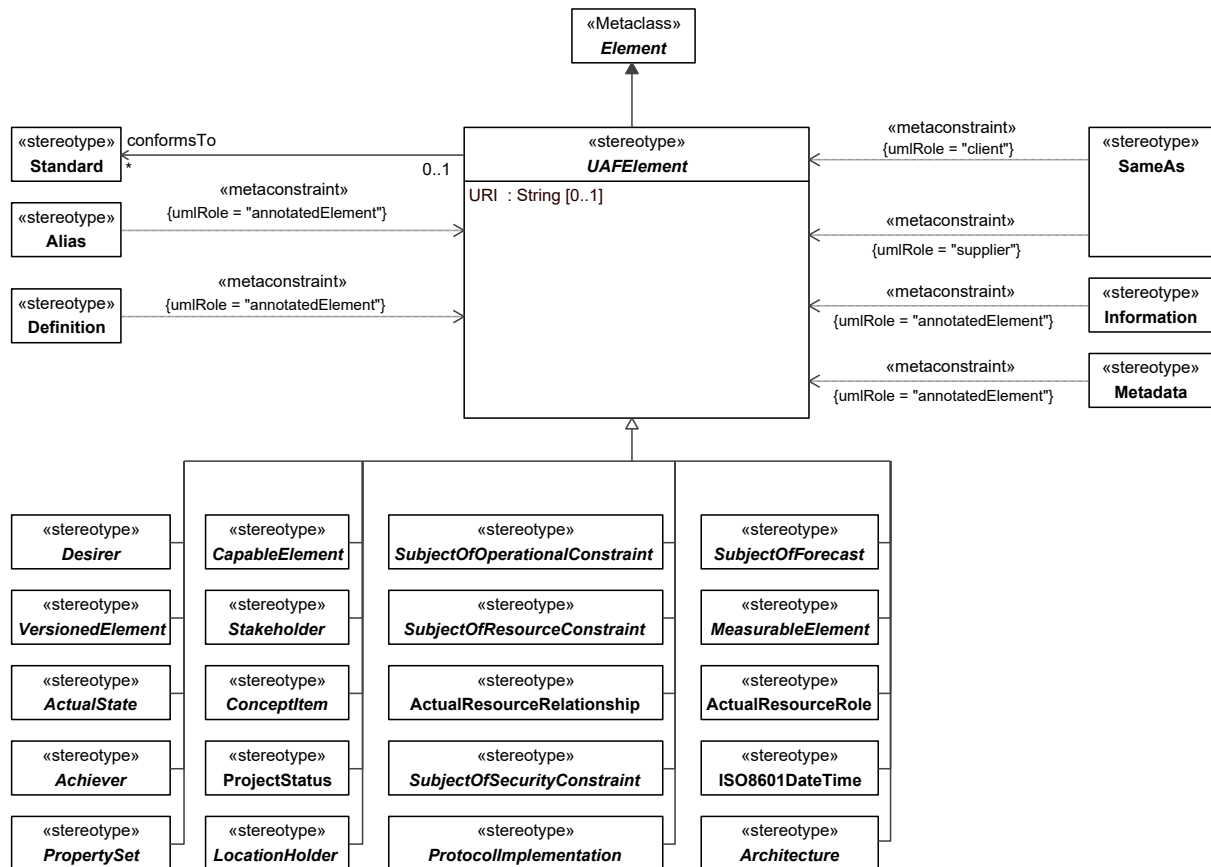


Figure 3:212 – UAFElement

Attributes

URI : String[0..1] Captures Unique identifier for the element.

Associations

conformsTo : Standard[*] Relates a UAFElement to the Standard that the UAFElement is conforming to.

View

Package: Summary and Overview

isAbstract: No

Generalization: [PropertySet](#), View

Extension: Class

Description

An architecture view expresses the architecture of the system-of-interest in accordance with an architecture viewpoint (or simply, viewpoint). [ISO/IEC/IEEE 42010:2011(E)].

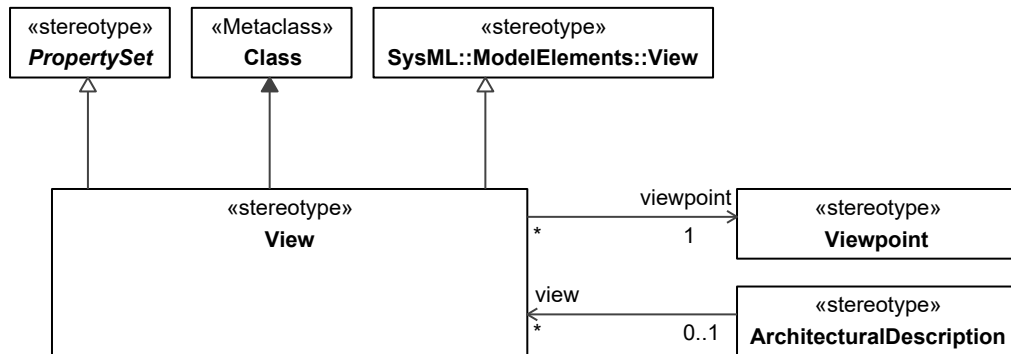


Figure 3:213 - View

Associations

viewpoint : Viewpoint[1] Relates the View to the Viewpoint that the View conforms to.

Viewpoint

Package: Summary and Overview

isAbstract: No

Generalization: [PropertySet](#), Viewpoint

Extension: Class

Description

An architecture viewpoint frames (to formulate or construct in a particular style or language) one or more concerns. A concern can be framed by more than one viewpoint. [ISO/IEC/IEEE 42010:2011(E)].

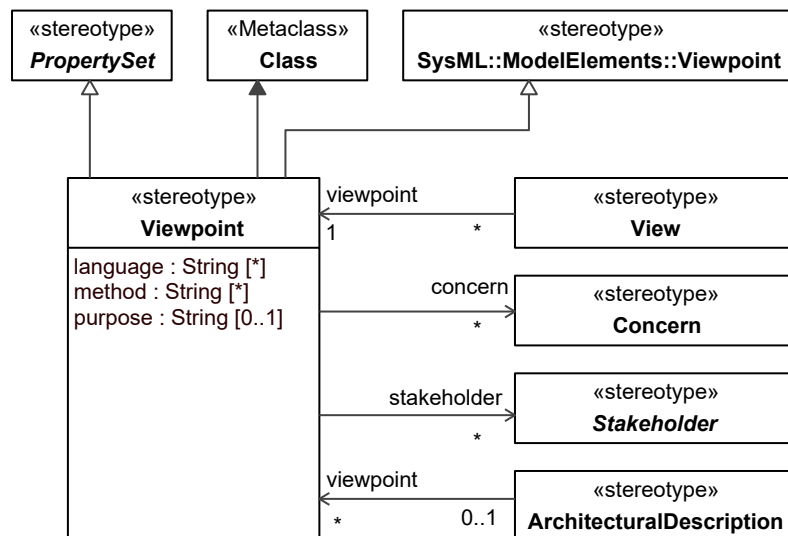


Figure 3:214 – Viewpoint

Attributes

language : String[*] The languages used to express the Viewpoint.
method : String[*] The methods employed in the development of the Viewpoint.
purpose : String[0..1] The purpose of the Viewpoint.

Associations

concern : Concern[*] Relates the Viewpoint to the Concerns that the Viewpoint addresses.
stakeholder : Stakeholder[*] Relates the Viewpoint to the Stakeholders whose Concerns are being addressed by the Viewpoint.

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4 UAF View Specifications

This paragraph is intended as normative guidance for developers and users as to what UAF stereotypes and metaconstraints are applicable for each of the UAF view specifications.

4.1 View Specifications

This section describes the normative stereotypes and metaconstraints needed to define UAF view specifications.

4.1.1 View Specifications::Strategic

Stakeholders: Capability Portfolio Managers.

Concerns: capability management process.

Definition: describe capability taxonomy, composition, dependencies and evolution.

View Specifications::Strategic::Taxonomy

Stakeholders: PMs, Enterprise Architects, Executives.

Concerns: capability needs.

Definition: shows the taxonomy of capabilities.

Recommended Implementation: SysML Block Definition Diagram.

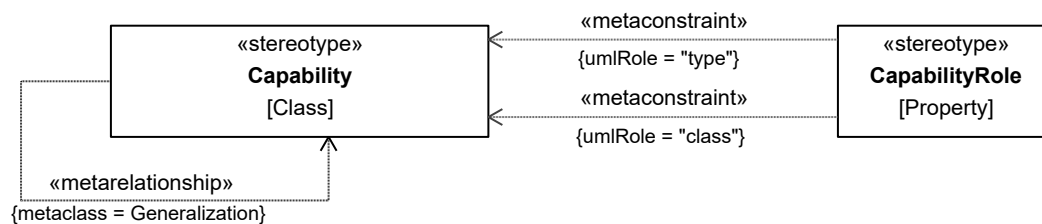


Figure 4:1 - Strategic Taxonomy

Elements

- [Capability](#)
- [CapabilityRole](#)

View Specifications::Strategic::Structure

Stakeholders: PMs, Enterprise Architects, Executives.

Concerns: capability needs.

Definition: shows the relationship between EnterprisePhases and the Capabilities that are intended to be developed during the enterprise phases, and the organizations involved in the enterprise.

Recommended Implementation: SysML Block Definition Diagram.

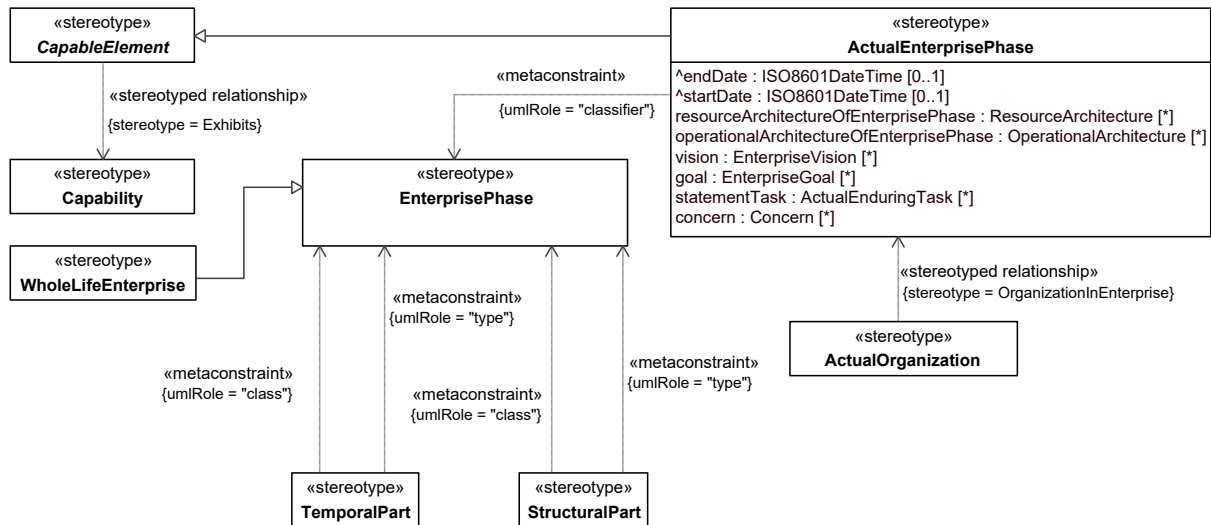


Figure 4:2 - Strategic Structure

Elements

- [ActualEnterprisePhase](#)
- [ActualOrganization](#)
- [Capability](#)
- [CapableElement](#)
- [EnterprisePhase](#)
- [StructuralPart](#)
- [TemporalPart](#)
- [WholeLifeEnterprise](#)

View Specifications::Strategic::Connectivity

Stakeholders: PMs, Executives, Enterprise Architects.

Concerns: capability dependencies.

Definition: describes the dependencies between planned capabilities.

Recommended Implementation: SysML Block Definition Diagram. SysML Internal Block Diagram.

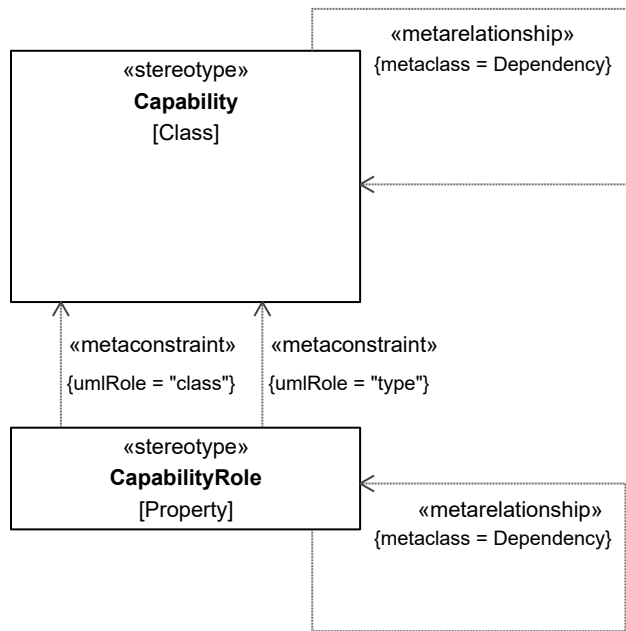


Figure 4:3 - Strategic Connectivity

Elements

- [Capability](#)
- [CapabilityRole](#)

View Specifications::Strategic::States

Stakeholders: PMs, Enterprise Architects.

Concerns: effects that the implementation(s) of capabilities are expected to deliver.

Definition: captures the relationships between capability(ies) and desired effect(s) that implementation(s) of capability(ies) should achieve.

Recommended Implementation: SysML Block Definition Diagram.

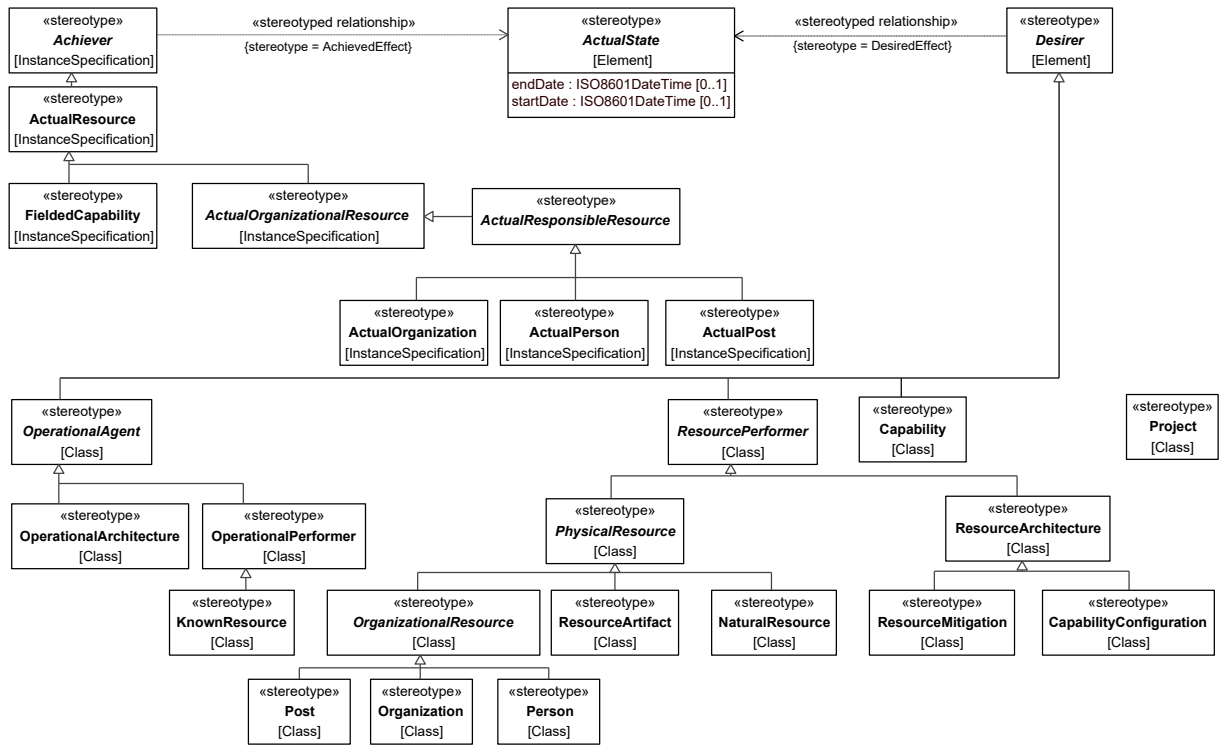


Figure 4:4 - Strategic States

Elements

- [Achiever](#)
- [ActualOrganization](#)
- [ActualOrganizationalResource](#)
- [ActualPerson](#)
- [ActualPost](#)
- [ActualResource](#)
- [ActualResponsibleResource](#)
- [ActualState](#)
- [Capability](#)
- [CapabilityConfiguration](#)
- [Desirer](#)
- [FieldedCapability](#)
- [KnownResource](#)
- [NaturalResource](#)
- [OperationalAgent](#)
- [OperationalArchitecture](#)
- [OperationalPerformer](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [Project](#)
- [ResourceArchitecture](#)
- [ResourceArtifact](#)

- [ResourceMitigation](#)
- [ResourcePerformer](#)

View Specifications::Strategic::Constraints

Stakeholders: PMs, Enterprise Architects.

Concerns: capability constraints.

Definition: details the measurements that set performance requirements constraining capabilities.

Recommended Implementation: tabular format, SysML Block Definition Diagram.

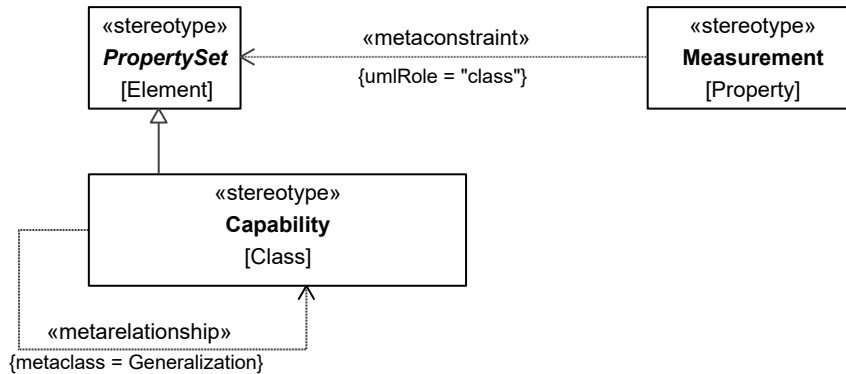


Figure 4:5 - Strategic Constraints

Elements

- [Capability](#)
- [Measurement](#)
- [PropertySet](#)

View Specifications::Strategic::Roadmap

Stakeholders: PMs, Executives, Enterprise Architects.

Concerns: capability deployment to organizations over time.

Definition: addresses the deployment of capability(ies) to actual organizations over time.

Recommended Implementation: timeline, tabular format, SysML Block Definition Diagram.

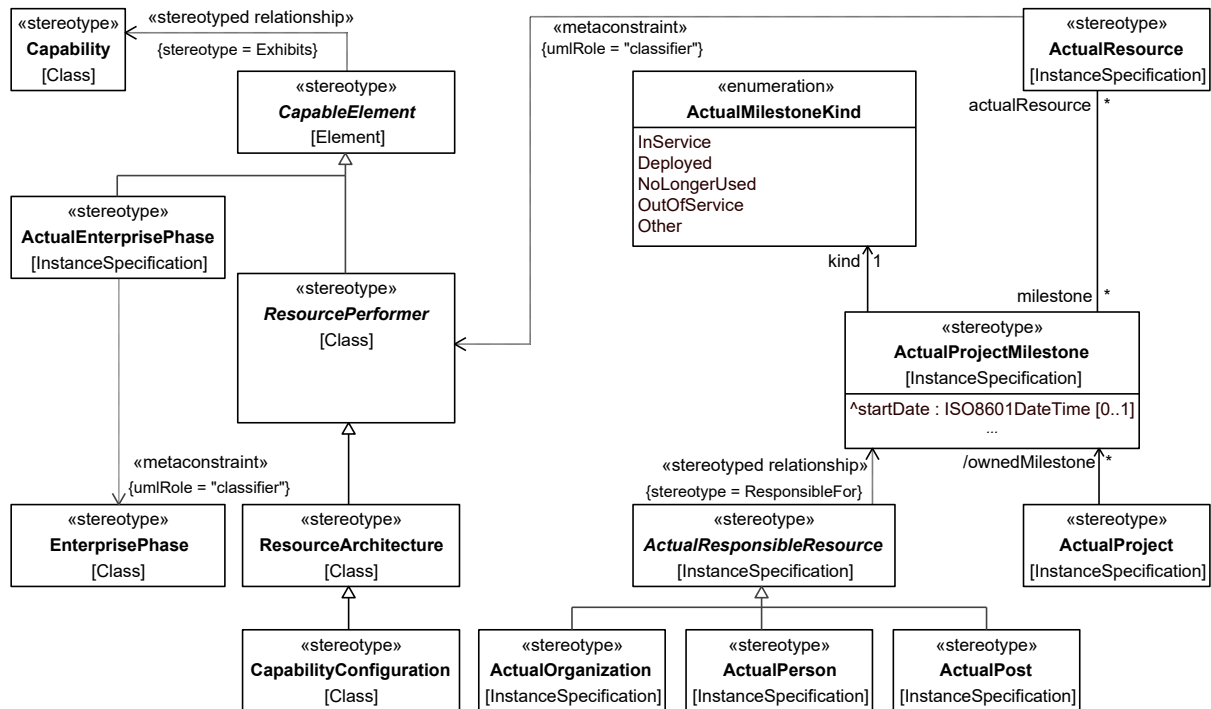


Figure 4:6 - Strategic Roadmap: Deployment

Elements

- [ActualEnterprisePhase](#)
- [ActualMilestoneKind](#)
- [ActualOrganization](#)
- [ActualPerson](#)
- [ActualPost](#)
- [ActualProject](#)
- [ActualProjectMilestone](#)
- [ActualResource](#)
- [ActualResponsibleResource](#)
- [Capability](#)
- [CapabilityConfiguration](#)
- [CapableElement](#)
- [EnterprisePhase](#)
- [ResourceArchitecture](#)
- [ResourcePerformer](#)

Stakeholders: PMs, Executives, Enterprise Architects.

Concerns: capability(ies) achievement over time.

Definition: the planned achievement of capability(ies) at different points in time or during specific periods of time.

Recommended Implementation: timeline, tabular format, SysML Block Definition Diagram.



Figure 4:7 - Strategic Roadmap: Phasing

Elements

- [ActualMilestoneKind](#)
- [ActualProject](#)
- [ActualProjectMilestone](#)
- [ActualResource](#)

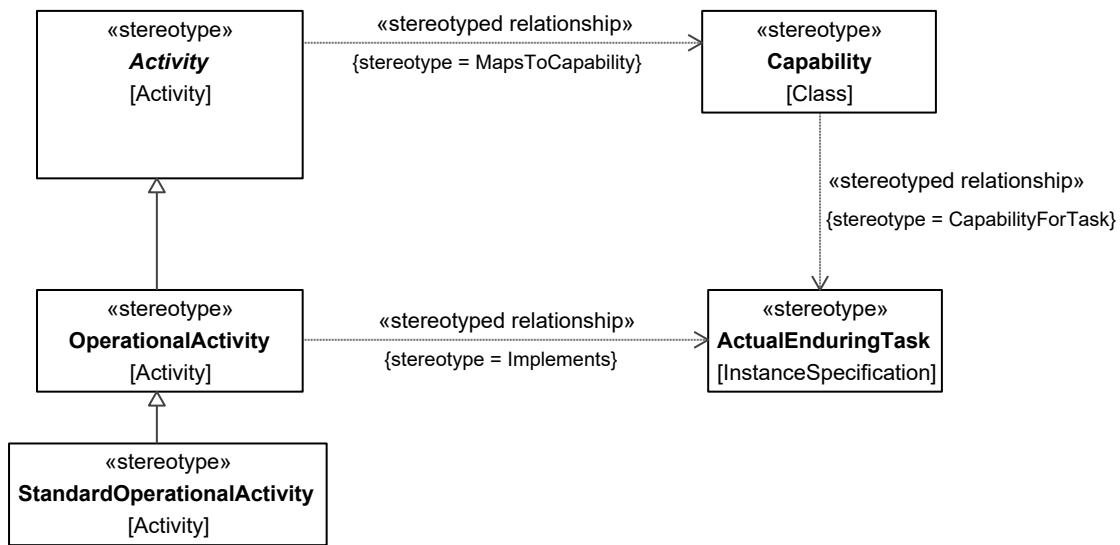


Figure 4:8 - Strategic Traceability

Elements

- [Activity](#)
- [ActualEnduringTask](#)
- [Capability](#)
- [OperationalActivity](#)
- [StandardOperationalActivity](#)

4.1.2 View Specifications::Operational

Stakeholders: Business Architects, Executives

Concerns: illustrate the Logical Architecture of the enterprise.

Definition: describe the requirements, operational behavior, structure, and exchanges required to support (exhibit) capabilities. Defines all operational elements in an implementation/solution independent manner.

View Specifications::Operational::Taxonomy

Stakeholders: Business Architects, Systems Engineers, Enterprise Architects, Owners responsible for Operational Agents.

Concerns: OperationalAgent types.

Definition: shows the taxonomy of types of OperationalAgents.

Recommended Implementation: SysML Block Definition Diagram, SysML Internal Block Diagram.

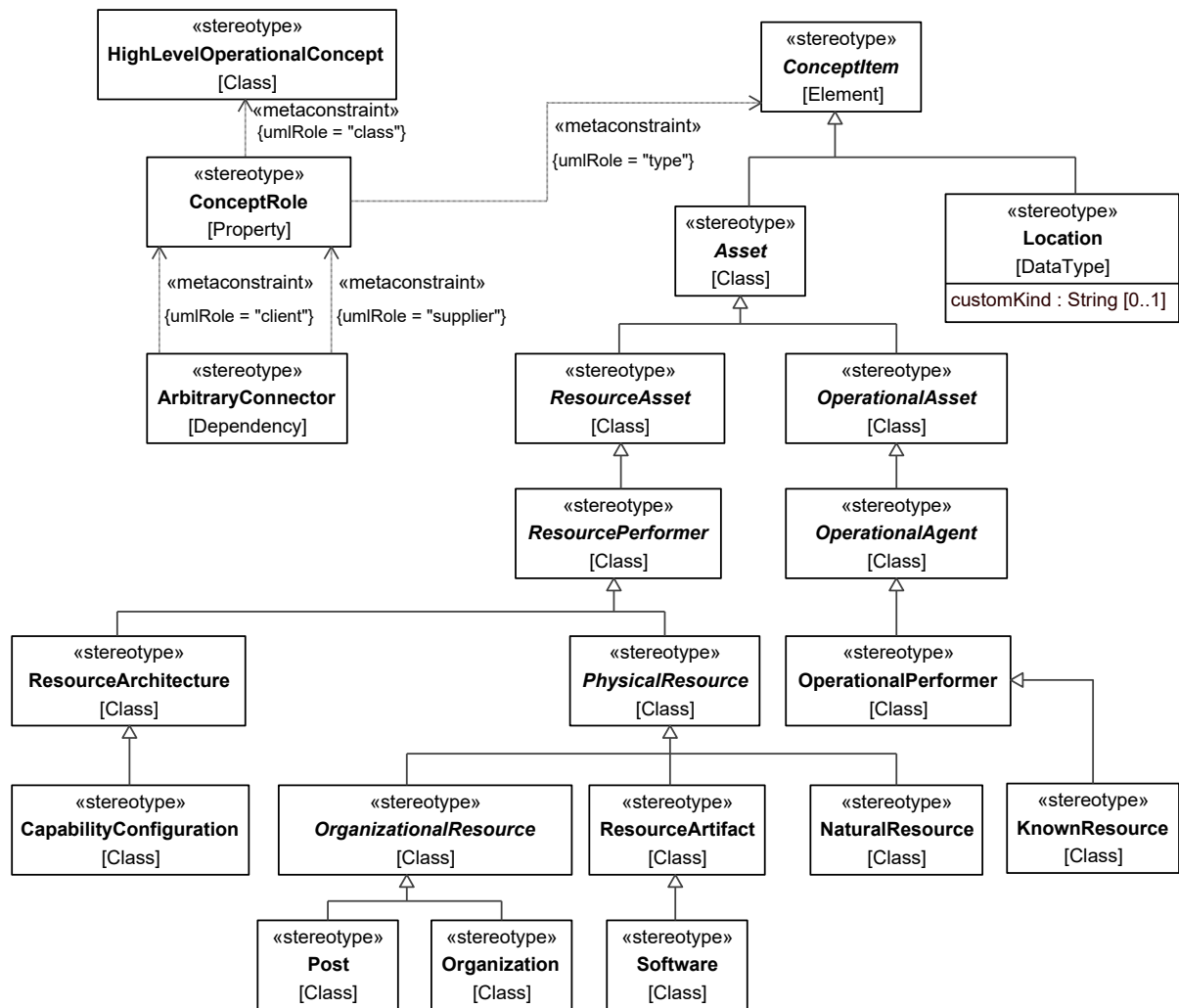


Figure 4:9 - Operational Taxonomy

Elements

- [ArbitraryConnector](#)
- [Asset](#)
- [CapabilityConfiguration](#)
- [ConceptItem](#)
- [ConceptRole](#)
- [HighLevelOperationalConcept](#)
- [KnownResource](#)
- [Location](#)
- [NaturalResource](#)
- [OperationalAgent](#)
- [OperationalAsset](#)
- [OperationalPerformer](#)
- [Organization](#)
- [OrganizationalResource](#)
- [PhysicalResource](#)
- [Post](#)

- [ResourceArchitecture](#)
- [ResourceArtifact](#)
- [ResourceAsset](#)
- [ResourcePerformer](#)
- [Software](#)

View Specifications::Operational::Structure

Stakeholders: Business Architects, Systems Engineers, Enterprise Architects, Owners responsible for Operational Agents.

Concerns: identifies the operational exchange requirements between OperationalPerformers.

Definition: defines operational architecture and exchange requirements necessary to support a specific set of Capability(ies).

Recommended Implementation: SysML Block Definition Diagram, SysML Internal Block Diagram.

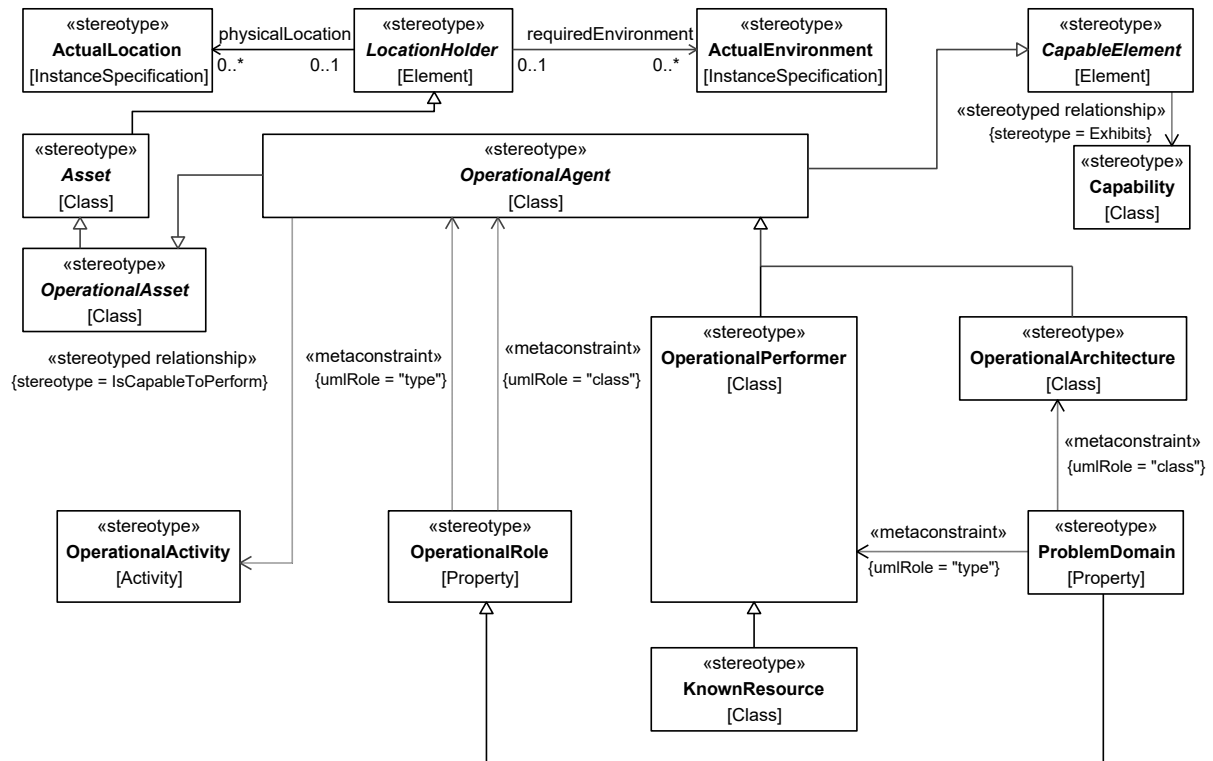


Figure 4:10 - Operational Structure

Elements

- [ActualEnvironment](#)
- [ActualLocation](#)
- [Asset](#)
- [Capability](#)
- [CapableElement](#)
- [KnownResource](#)
- [LocationHolder](#)
- [OperationalActivity](#)
- [OperationalAgent](#)
- [OperationalArchitecture](#)
- [OperationalAsset](#)
- [OperationalPerformer](#)

- [OperationalRole](#)
- [ProblemDomain](#)

View Specifications::Operational::Connectivity

Stakeholders: Systems Engineers, Architects, Solution Providers.

Concerns: capture the interfaces between OperationalPerformers.

Definition: summarizes logical exchanges between OperationalPerformers of information, systems, personnel, energy etc. and the logical activities that produce and consume them. Measurements can optionally be included.

Recommended Implementation: SysML Internal Block Diagram, tabular format.

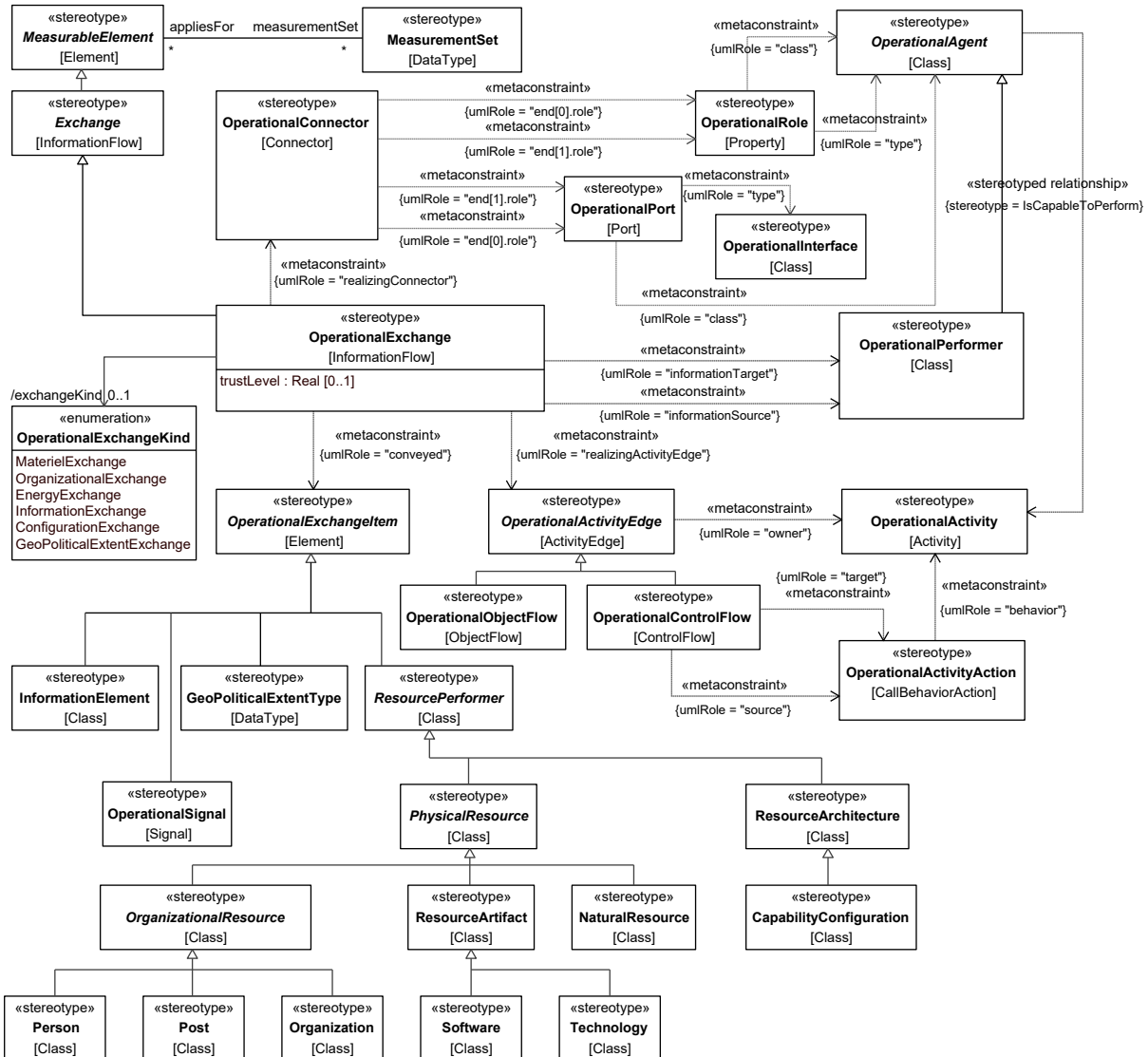


Figure 4:11 - Operational Connectivity

Elements

- [CapabilityConfiguration](#)
- [Exchange](#)
- [GeoPoliticalExtentType](#)
- [InformationElement](#)
- [MeasurableElement](#)

- [MeasurementSet](#)
- [NaturalResource](#)
- [OperationalActivity](#)
- [OperationalActivityAction](#)
- [OperationalActivityEdge](#)
- [OperationalAgent](#)
- [OperationalConnector](#)
- [OperationalControlFlow](#)
- [OperationalExchange](#)
- [OperationalExchangeItem](#)
- [OperationalExchangeKind](#)
- [OperationalInterface](#)
- [OperationalObjectFlow](#)
- [OperationalPerformer](#)
- [OperationalPort](#)
- [OperationalRole](#)
- [OperationalSignal](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourceArchitecture](#)
- [ResourceArtifact](#)
- [ResourcePerformer](#)
- [Software](#)
- [Technology](#)

View Specifications::Operational::Processes

Stakeholders: Business Architect, Systems Engineers, Enterprise Architects

Concerns: captures activity based behavior and flows.

Definition: describes the activities that are normally conducted in the course of achieving business goals that support a capability. It describes operational activities, their Inputs/Outputs, operational activity actions and flows between them.

Recommended Implementation: SysML Activity Diagram, SysML Block Definition Diagram, BPMN Process Diagram.

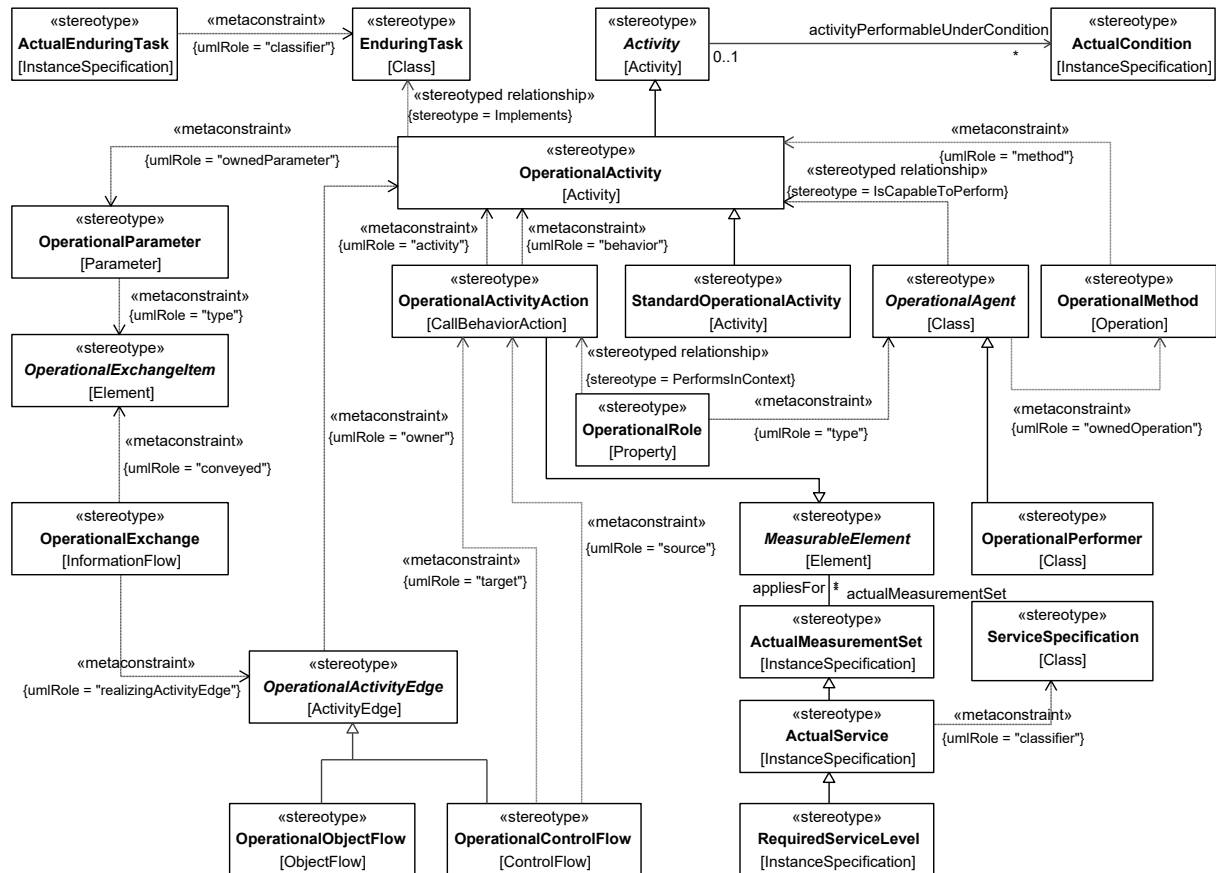


Figure 4:12 - Operational Processes

Elements

- [Activity](#)
- [ActualCondition](#)
- [ActualEnduringTask](#)
- [ActualMeasurementSet](#)
- [ActualService](#)
- [EnduringTask](#)
- [MeasurableElement](#)
- [OperationalActivity](#)
- [OperationalActivityAction](#)
- [OperationalActivityEdge](#)
- [OperationalAgent](#)
- [OperationalControlFlow](#)
- [OperationalExchange](#)
- [OperationalExchangeItem](#)
- [OperationalMethod](#)
- [OperationalObjectFlow](#)
- [OperationalParameter](#)
- [OperationalPerformer](#)
- [OperationalRole](#)

- [RequiredServiceLevel](#)
- [ServiceSpecification](#)
- [StandardOperationalActivity](#)

View Specifications::Operational::States

Stakeholders: Systems Engineers, Software Engineers.

Concerns: capture state-based behavior of an operational OperationalPerformer.

Definition: it is a graphical representation of states of an operational OperationalPerformer and how that operational OperationalPerformer responds to various events and actions.

Recommended Implementation: SysML State Machine Diagram.

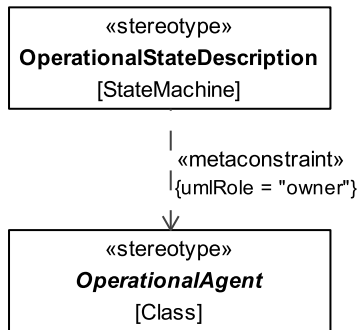


Figure 4:13 - Operational States

Elements

- [OperationalAgent](#)
- [OperationalStateDescription](#)

View Specifications::Operational::Interaction Scenarios

Stakeholders: Systems Engineers, Business Architects.

Concerns: express a time ordered examination of the operational exchanges as a result of a particular operational scenario.

Definition: provides a time-ordered examination of the operational exchanges between participating nodes (OperationalPerformer roles) as a result of a particular operational scenario.

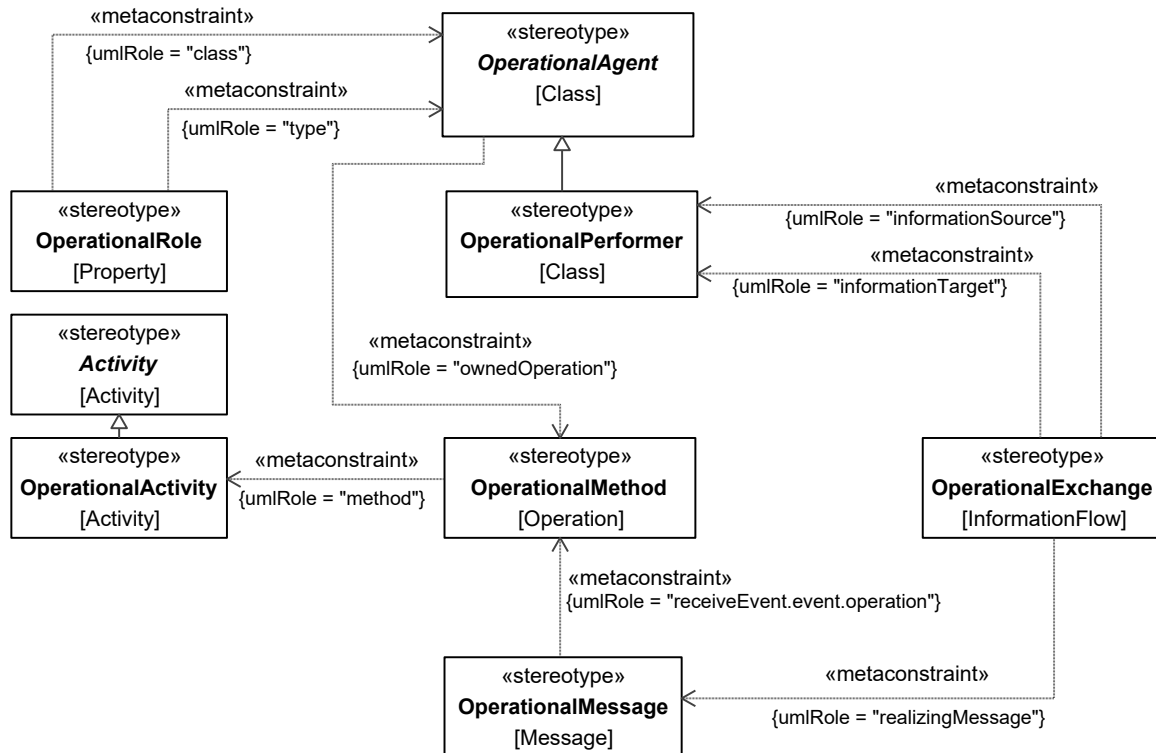


Figure 4:14 - Operational Interaction Scenarios

Elements

- [Activity](#)
- [OperationalActivity](#)
- [OperationalAgent](#)
- [OperationalExchange](#)
- [OperationalMessage](#)
- [OperationalMethod](#)
- [OperationalPerformer](#)
- [OperationalRole](#)

View Specifications::Operational::Constraints

Stakeholders: Systems Engineers, Architects, Program Sponsors

Concerns: define operational limitations, constraints and performance parameters for the enterprise.

Definition: specifies traditional textual operational or business rules that are constraints on the way that business is done in the enterprise. The addition of SysML parametrics provides a computational means of defining operational constraints across the enterprise or within a specific operational context.

Recommended Implementation: tabular format, SysML Block Definition Diagram, SysML Parametric Diagram.

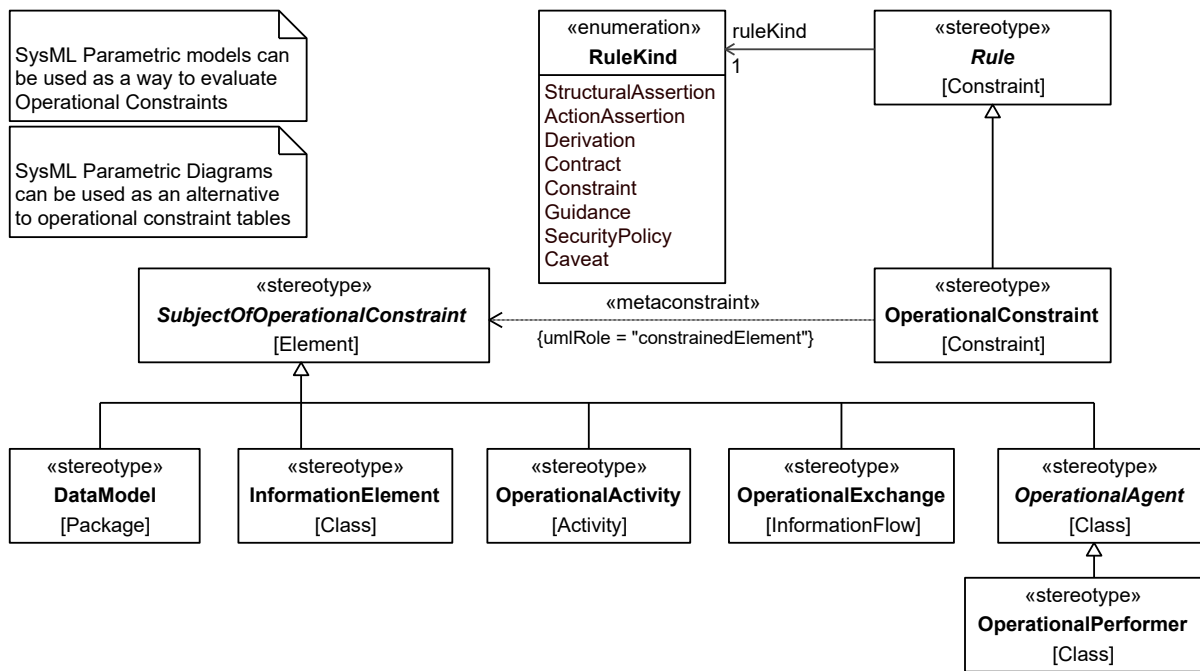


Figure 4:15 - Operational Constraints

Elements

- [DataModel](#)
- [InformationElement](#)
- [OperationalActivity](#)
- [OperationalAgent](#)
- [OperationalConstraint](#)
- [OperationalExchange](#)
- [OperationalPerformer](#)
- [Rule](#)
- [RuleKind](#)
- [SubjectOfOperationalConstraint](#)

View Specifications::Operational::Traceability

Stakeholders: PMs, Enterprise Architects, Business Architects.

Concerns: traceability between capabilities and operational activities and capabilities and operational agents.

Definition: describes the mapping between the capabilities required by an Enterprise and the supporting operational activities and operational agents.

Recommended Implementation: matrix format, SysML Block Definition Diagram.

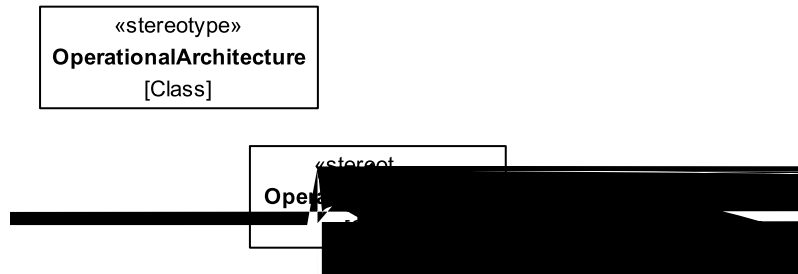


Figure 4:16 - Operational Traceability

Elements

- [Activity](#)
- [Capability](#)
- [CapableElement](#)
- [OperationalActivity](#)
- [OperationalAgent](#)
- [OperationalArchitecture](#)
- [OperationalPerformer](#)

4.1.3 View Specifications::Services

Stakeholders: Enterprise Architects, Solution Providers, Systems Engineers, Software Architects, Business Architects..

Concerns: specifications of services required to exhibit a Capability.

Definition: shows Service Specifications and required and provided service levels of these specifications required to exhibit a Capability or to support an Operational Activity.

View Specifications::Services::Taxonomy

Stakeholders: Enterprise Architects, Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: service specification types and required and provided service levels of these types.

Definition: shows the taxonomy of types of services and the level of service that they are expected to provide or are required to meet through the display of ActualMeasurements associated with the Provided and Required Service Level.

Recommended Implementation: SysML Block Definition Diagram.

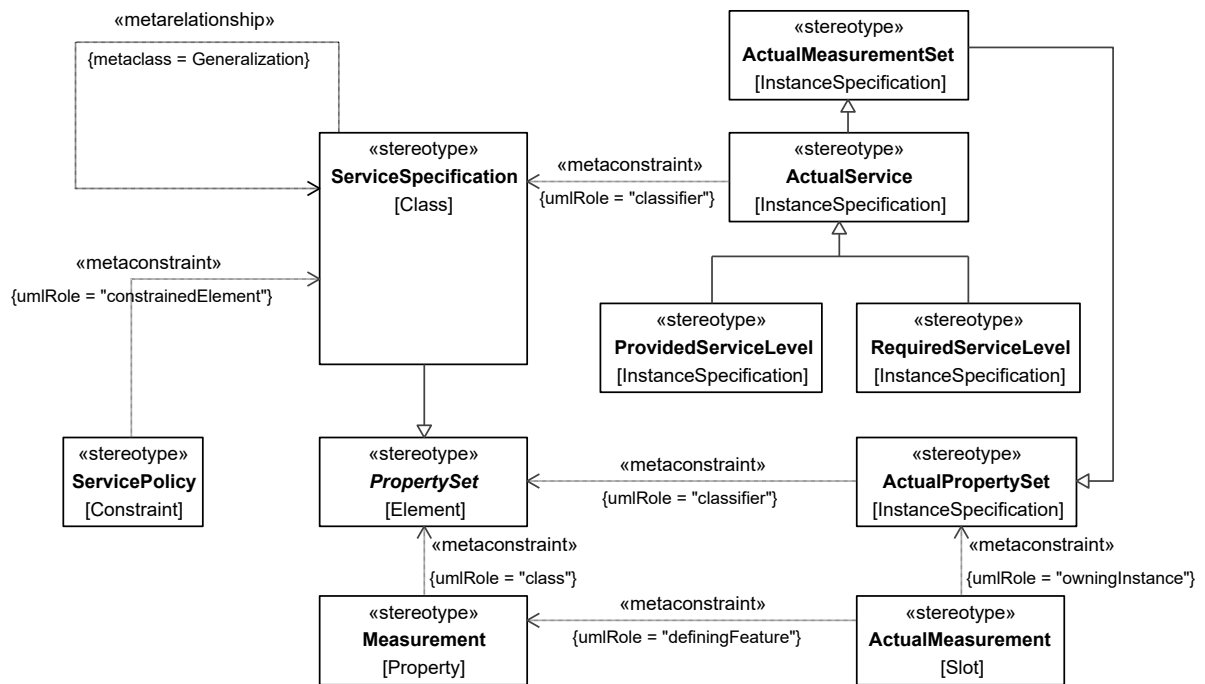


Figure 4:17 - Services Taxonomy

Elements

- [ActualMeasurement](#)
- [ActualMeasurementSet](#)
- [ActualPropertySet](#)
- [ActualService](#)
- [Measurement](#)
- [PropertySet](#)
- [ProvidedServiceLevel](#)
- [RequiredServiceLevel](#)
- [ServicePolicy](#)
- [ServiceSpecification](#)

View Specifications::Services::Structure

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: combination of services required to exhibit a capability.

Definition: shows the composition of services and how services are combined into a higher level service required to exhibit a capability or support an operational activity.

Recommended Implementation: SysML Block Definition Diagram, SysML Internal Block Diagram.

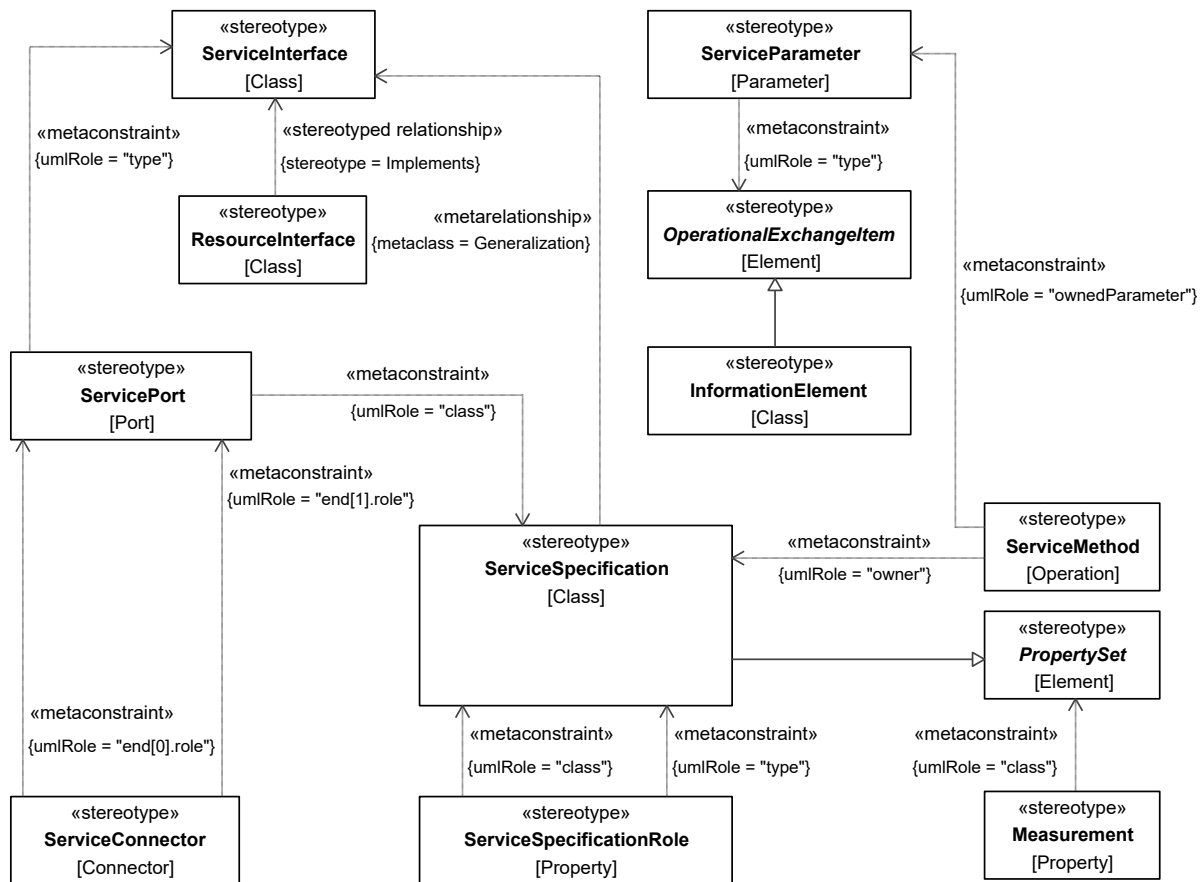


Figure 4:18 - Services Structure

Elements

- [InformationElement](#)
- [Measurement](#)
- [OperationalExchangeItem](#)
- [PropertySet](#)
- [ResourceInterface](#)
- [ServiceConnector](#)
- [ServiceInterface](#)
- [ServiceMethod](#)
- [ServiceParameter](#)
- [ServicePort](#)
- [ServiceSpecification](#)
- [ServiceSpecificationRole](#)

View Specifications::Services::Connectivity

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: interoperability among services

Definition: specifies service interfaces, e.g. provided and required service methods, signal receptions, and/or flow properties, to ensure compatibility and reusability of services.

Recommended Implementation: SysML Block Definition Diagram, SysML Internal Block Diagram, tabular format.

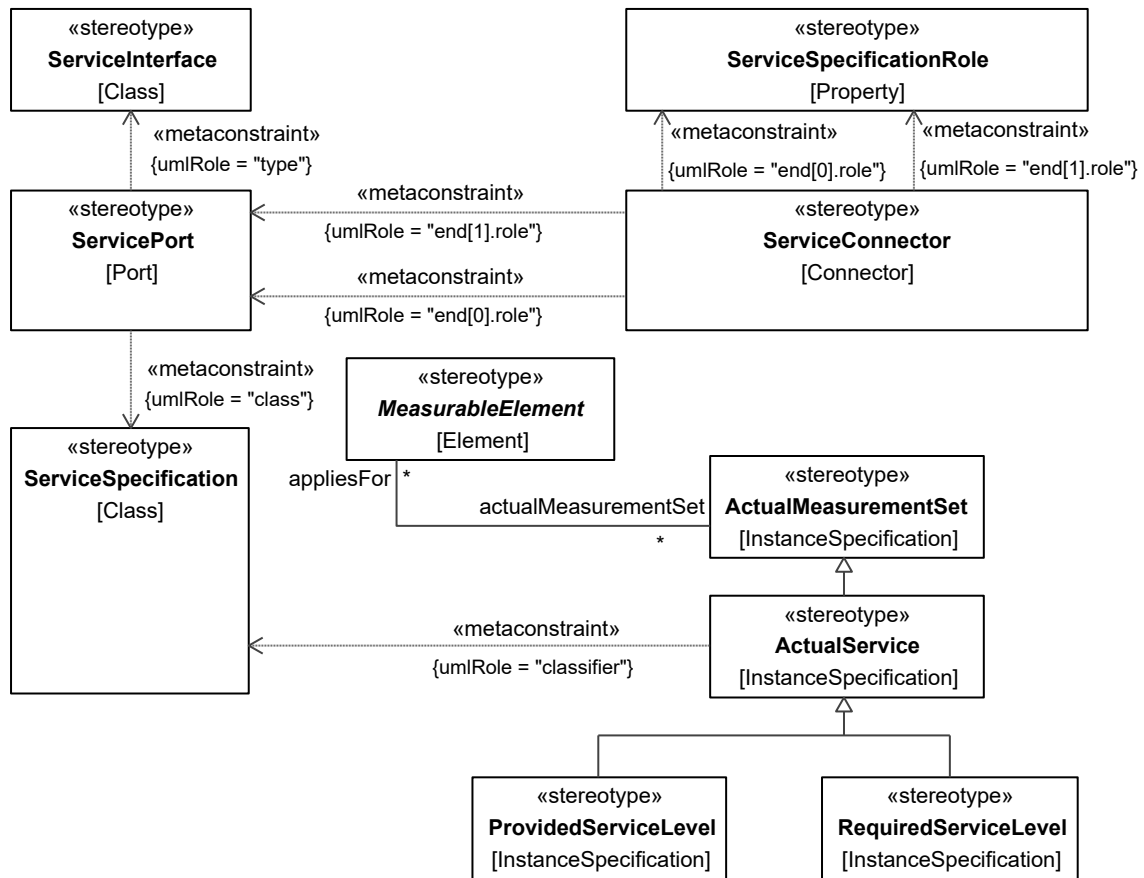


Figure 4:19 - Services Connectivity

Elements

- [ActualMeasurementSet](#)
- [ActualService](#)
- [MeasurableElement](#)
- [ProvidedServiceLevel](#)
- [RequiredServiceLevel](#)
- [ServiceConnector](#)
- [ServiceInterface](#)
- [ServicePort](#)
- [ServiceSpecification](#)
- [ServiceSpecificationRole](#)

View Specifications::Services::Processes

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: the behavior of a service in terms of the operational activities it is expected to support.

Definition: provides detailed information regarding the allocation of service functions to service specifications, and data flows between service functions.

Recommended Implementation: SysML Activity Diagram, BPMN Process Diagram, SysML Block Definition Diagram.

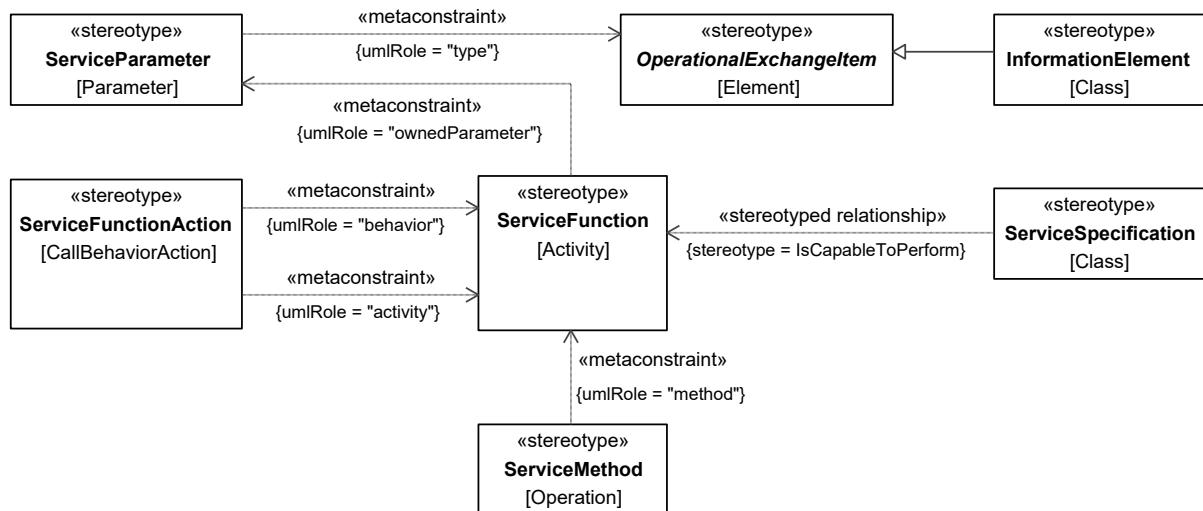


Figure 4:20 - Services Processes

Elements

- [InformationElement](#)
- [OperationalExchangeItem](#)
- [ServiceFunction](#)
- [ServiceFunctionAction](#)
- [ServiceMethod](#)
- [ServiceParameter](#)
- [ServiceSpecification](#)

View Specifications::Services::States

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: the behavior of a service specification in terms of states and events causing transitions between states.

Definition: specifies the possible states a service specification may have, and the possible transitions between those states.

Recommended Implementation: SysML State Machine Diagram.

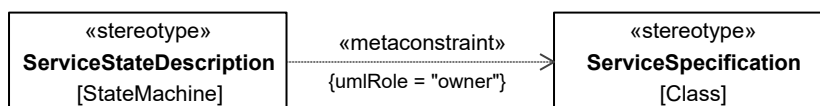


Figure 4:21 - Services States

Elements

- [ServiceSpecification](#)
- [ServiceStateDescription](#)

View Specifications::Services::Interaction Scenarios

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: the behavior of a service specification in terms of expected time-ordered examination of the interactions between service roles.

Definition: specifies how a service roles interact with each other, service providers and consumers, and the sequence and dependencies of those interactions.

Recommended Implementation: SysML Sequence Diagram.

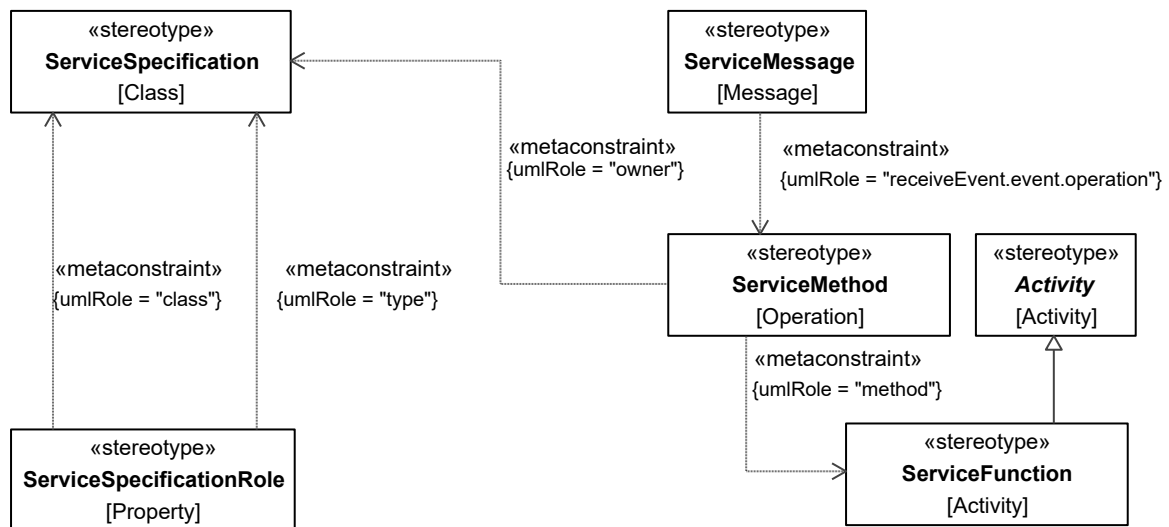


Figure 4:22 - Services Interaction Scenarios

Elements

- [Activity](#)
- [ServiceFunction](#)
- [ServiceMessage](#)
- [ServiceMethod](#)
- [ServiceSpecification](#)
- [ServiceSpecificationRole](#)

View Specifications::Services::Constraints

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: service policies that apply to implementations of service specifications.

Definition: specifies traditional textual service policies that are constraints on the way that service specifications are implemented within resources. The addition of SysML parametrics provide a computational means of defining service policies across the enterprise or within a specific service configuration.

Recommended Implementation: tabular format, SysML Parametric Diagram.

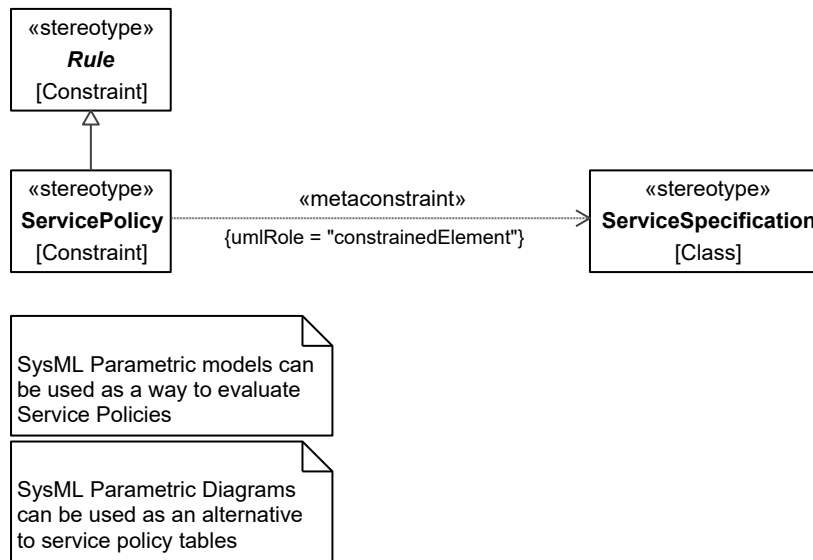


Figure 4:23 - Services Constraints

Elements

- [Rule](#)
- [ServicePolicy](#)
- [ServiceSpecification](#)

View Specifications::Services::Roadmap

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: service specification changes over time.

Definition: provides an overview of how a service specification changes over time. It shows the combination of several service specifications mapped against a timeline.

Recommended Implementation: timeline, SysML Block Definition Diagram, SysML Internal Block Diagram.

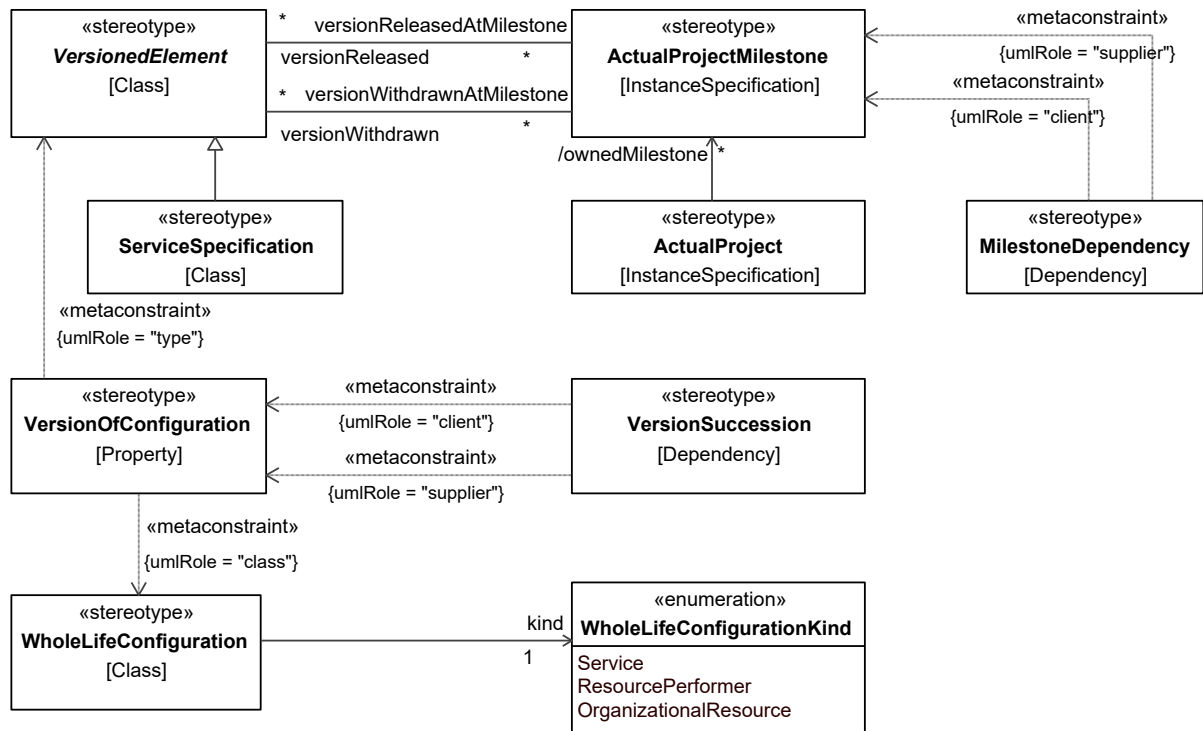


Figure 4:24 - Services Roadmap

Elements

- [ActualProject](#)
- [ActualProjectMilestone](#)
- [MilestoneDependency](#)
- [ServiceSpecification](#)
- [VersionedElement](#)
- [VersionOfConfiguration](#)
- [VersionSuccession](#)
- [WholeLifeConfiguration](#)
- [WholeLifeConfigurationKind](#)

View Specifications::Services::Traceability

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: traceability between operational activities and service specifications that support them.

Definition: depicts the mapping of service specifications to operational activities and how service specifications contribute to the achievement of a capability.

Recommended Implementation: tabular or matrix format.

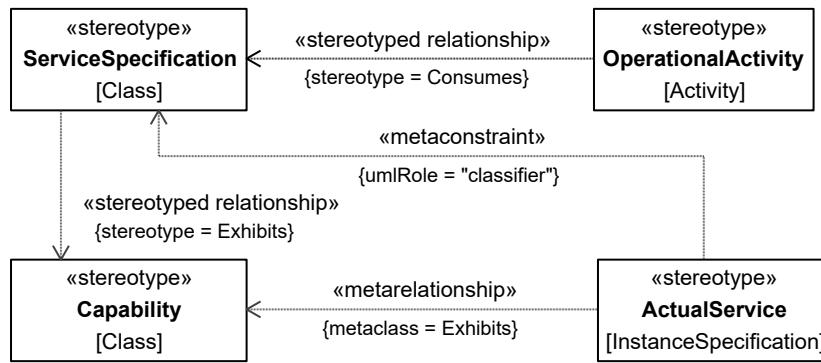


Figure 4:25 - Services Traceability

Elements

- [ActualService](#)
- [Capability](#)
- [OperationalActivity](#)
- [ServiceSpecification](#)

4.1.4 View Specifications::Personnel

Stakeholders: Human resources, Solution Providers, PMs.

Concerns: human factors.

Definition: aims to clarify the role of Human Factors (HF) when creating architectures in order to facilitate both Human Factors Integration (HFI) and systems engineering (SE).

View Specifications::Personnel::Taxonomy

Stakeholders: Human resources, Solution Providers, PMs.

Concerns: organizational resource types.

Definition: shows the taxonomy of types of organizational resources.

Recommended Implementation: SysML Block Definition Diagram.

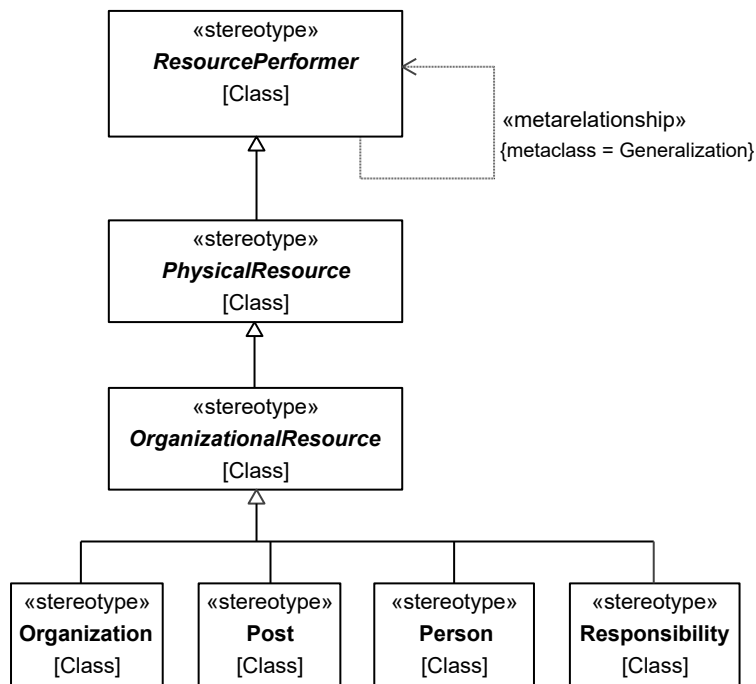


Figure 4:26 - Personnel Taxonomy

Elements

- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourcePerformer](#)
- [Responsibility](#)

View Specifications::Personnel::Structure

Stakeholders: Human resources, Solution Providers, PMs.

Concerns: typical organizational structure used to support a capability(ies).

Definition: shows organizational structures and possible interactions between organizational resources.

Recommended Implementation: SysML Block Definition Diagram, SysML Internal Block Diagram.

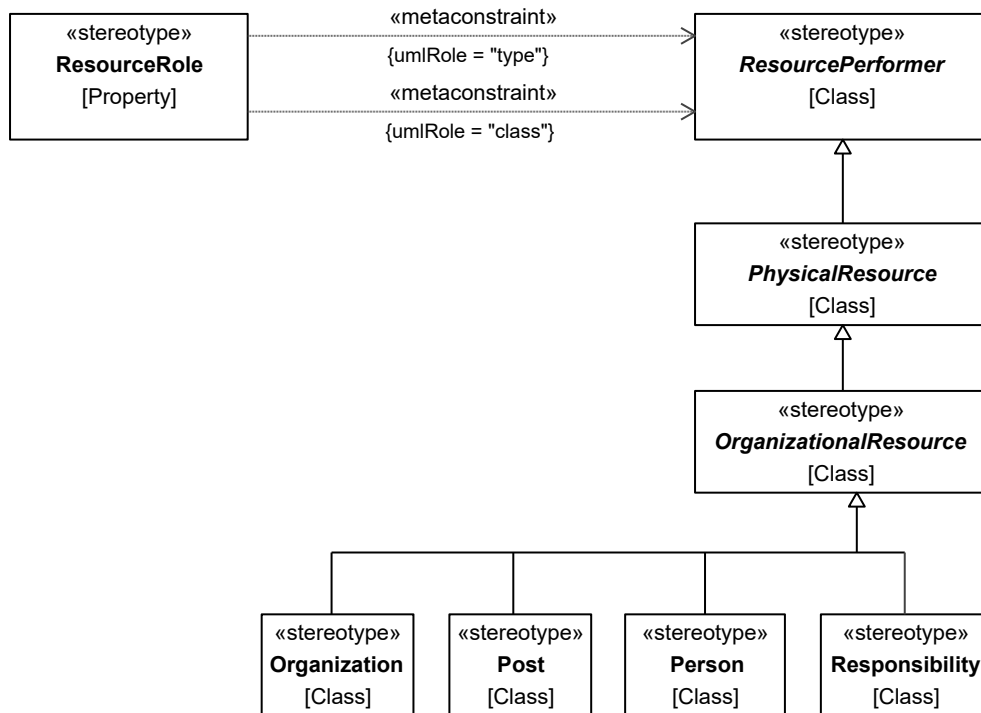


Figure 4:27 - Personnel Structure

Elements

- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- [Responsibility](#)

View Specifications::Personnel::Connectivity

Stakeholders: Solution providers.

Concerns: interaction of organizational resources.

Definition: captures the possible interactions between organizational resources, including command and control relationships. Interactions typically illustrate the fundamental roles and management responsibilities.

Recommended Implementation: tabular format.

- [ResourceInterface](#)
- [ResourceMessage](#)
- [ResourceMethod](#)
- [ResourcePerformer](#)
- [ResourcePort](#)
- [ResourceRole](#)
- [Responsibility](#)

View Specifications::Personnel::Processes

Stakeholders: Systems engineers, Solution providers.

Concerns: functions that have to be carried out by organizational resources.

Definition: specifies organizational resource functions in relation to resource definitions.

Recommended Implementation: SysML Activity Diagram, SysML Block Definition Diagram, BPMN Process Diagram.

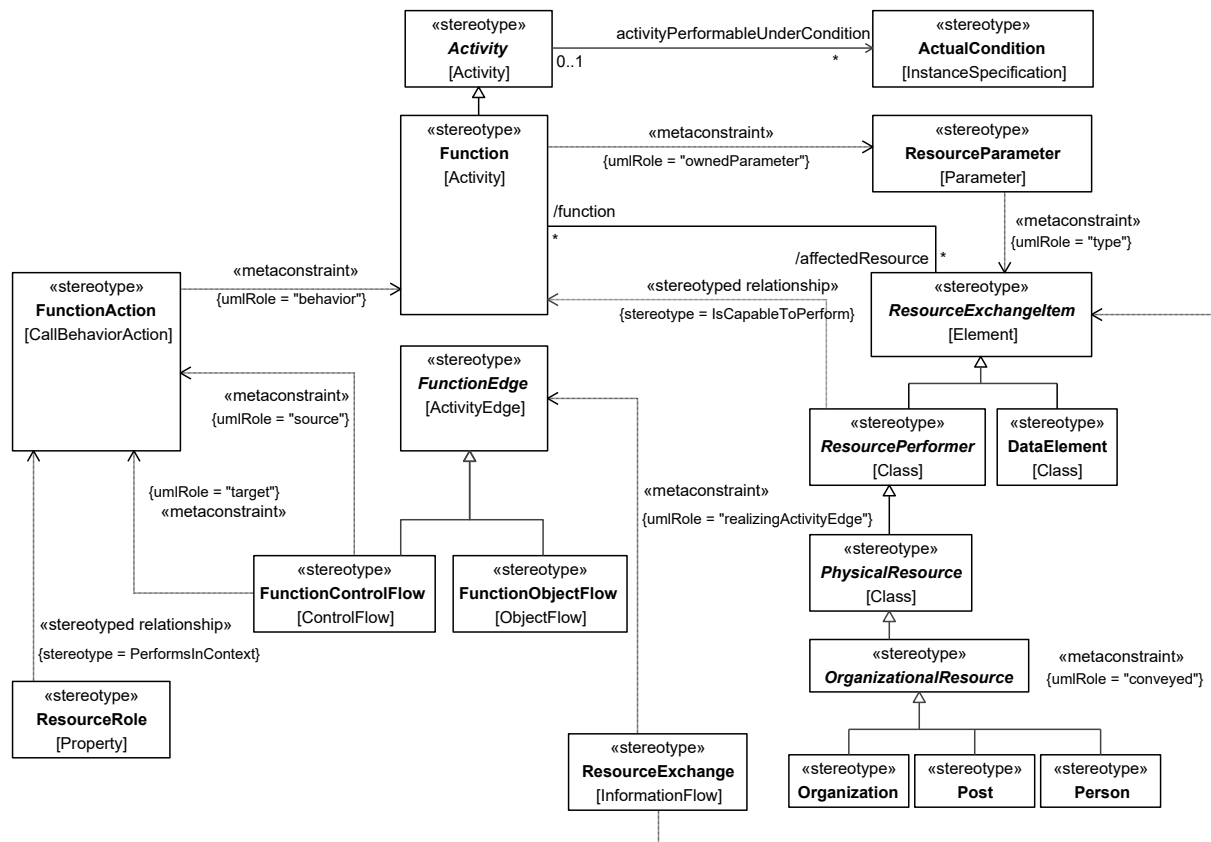


Figure 4:29 - Personnel Processes

Elements

- [Activity](#)
- [ActualCondition](#)
- [DataElement](#)
- [Function](#)
- [FunctionAction](#)
- [FunctionControlFlow](#)
- [FunctionEdge](#)
- [FunctionObjectFlow](#)
- [Organization](#)

- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourceExchange](#)
- [ResourceExchangeItem](#)
- [ResourceParameter](#)
- [ResourcePerformer](#)
- [ResourceRole](#)

View Specifications::Personnel::States

Stakeholders: Systems Engineers, Software Engineers.

Concerns: capture state-based behavior of an organizational resource.

Definition: it is a graphical representation of states of an organizational resource and how that organizational resource responds to various events and actions.

Recommended Implementation: SysML State Machine Diagram.

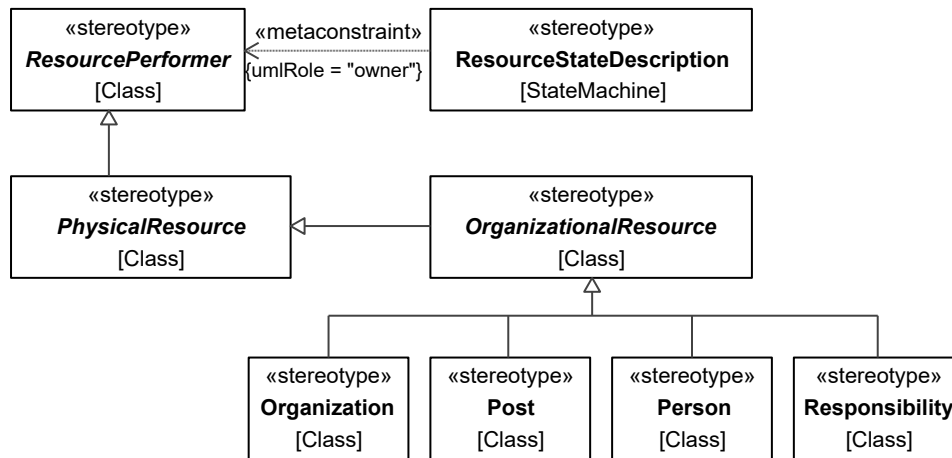


Figure 4:30 - Personnel States

Elements

- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourcePerformer](#)
- [ResourceStateDescription](#)
- [Responsibility](#)

View Specifications::Personnel::Interaction Scenarios

Stakeholders: Software Engineers, Systems Engineers.

Concerns: interactions between organizational resources (roles).

Definition: provides a time-ordered examination of the interactions between organizational resources.

Recommended Implementation: SysML Sequence Diagram, BPMN Collaboration Diagram.

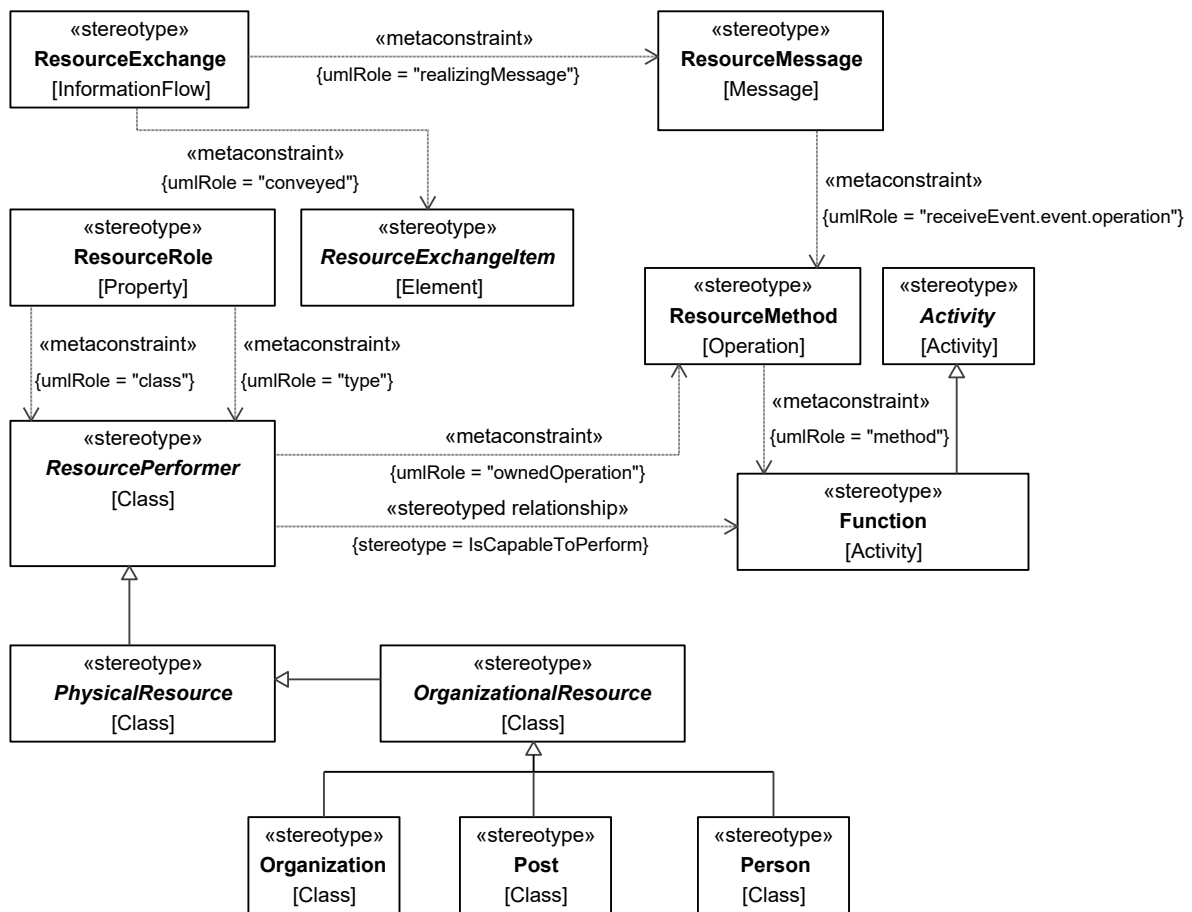


Figure 4:31 - Personnel Interaction Scenarios

Elements

- [Activity](#)
- [Function](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourceExchange](#)
- [ResourceExchangeItem](#)
- [ResourceMessage](#)
- [ResourceMethod](#)
- [ResourcePerformer](#)
- [ResourceRole](#)

View Specifications::Personnel::Constraints

Stakeholders: Systems engineers, Solution providers.

Concerns: allocation of competencies to actual posts.

Definition: specifies requirements for actual organizational resources – by linking competencies and actual posts.

Recommended Implementation: SysML Block Definition Diagram.

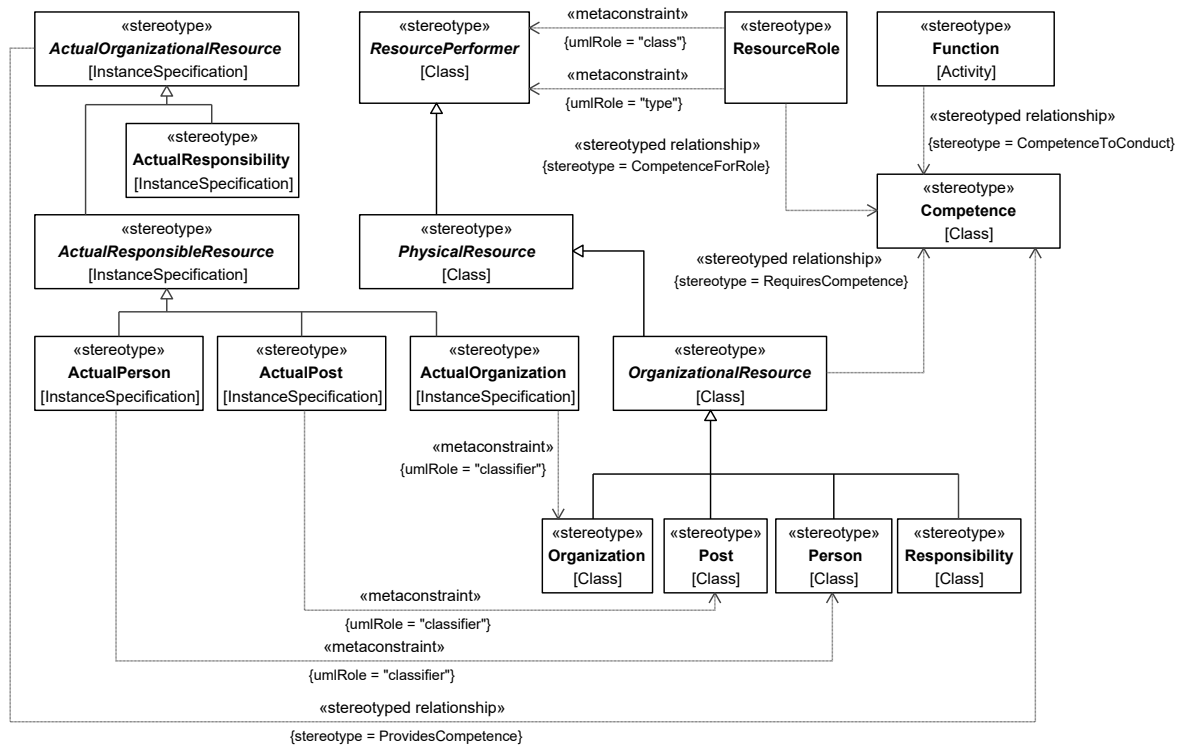


Figure 4:32 - Personnel Constraints: Competence

Elements

- [ActualOrganization](#)
- [ActualOrganizationalResource](#)
- [ActualPerson](#)
- [ActualPost](#)
- [ActualResponsibility](#)
- [ActualResponsibleResource](#)
- [Competence](#)
- [Function](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- [Responsibility](#)

Stakeholders: Systems engineers, Solution providers, Human resources.

Concerns: optimization of organizational resource behavior.

Definition: captures the factors that affect, constrain and characterize organizational resource behavior as the basis for performance predictions at the level of actual persons and actual organizations. It creates a bridge between static architectural definitions and behavior predictions through executable models.

Recommended Implementation: tabular format, SysML Parametric Diagram, SysML Block Definition Diagram.

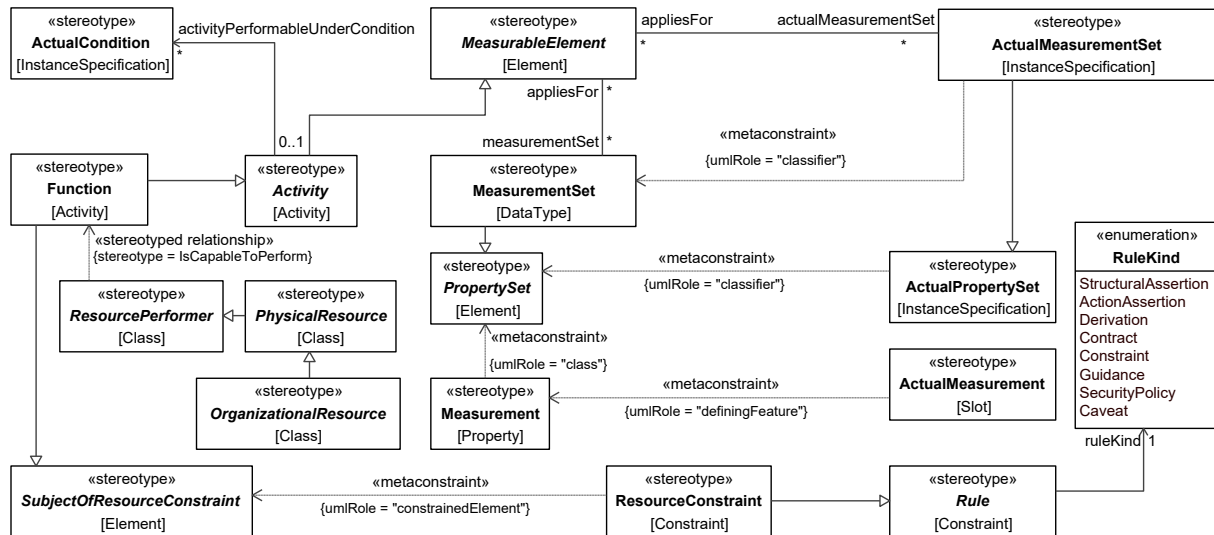


Figure 4:33 - Personnel Constraints: Drivers

Elements

- [Activity](#)
- [ActualCondition](#)
- [ActualMeasurement](#)
- [ActualMeasurementSet](#)
- [ActualPropertySet](#)
- [Function](#)
- [MeasurableElement](#)
- [Measurement](#)
- [MeasurementSet](#)
- [OrganizationalResource](#)
- [PhysicalResource](#)
- [PropertySet](#)
- [ResourceConstraint](#)
- [ResourcePerformer](#)
- [Rule](#)
- [RuleKind](#)
- [SubjectOfResourceConstraint](#)

Stakeholders: Human resources, solution providers.

Concerns: how well an actual organizational resource matches the needs of the actual organization.

Definition: provides a repository for human-related measures (i.e., quality objectives and performance criteria (HFI values)), targets and competences.

Recommended Implementation: SysML Block Definition Diagram.

- [Post](#)
- [PropertySet](#)
- [ResourcePerformer](#)

View Specifications::Personnel::Roadmap

Stakeholders: Human Resources, Training, Logisticians, Solution Providers.

Concerns: the staffing and training of resources.

Definition: defines the requirements and functions to ensure that actual persons with the right competencies, and in the right numbers, are available to fulfill actual posts.

Recommended Implementation: Timeline, SysML Block Definition Diagram.

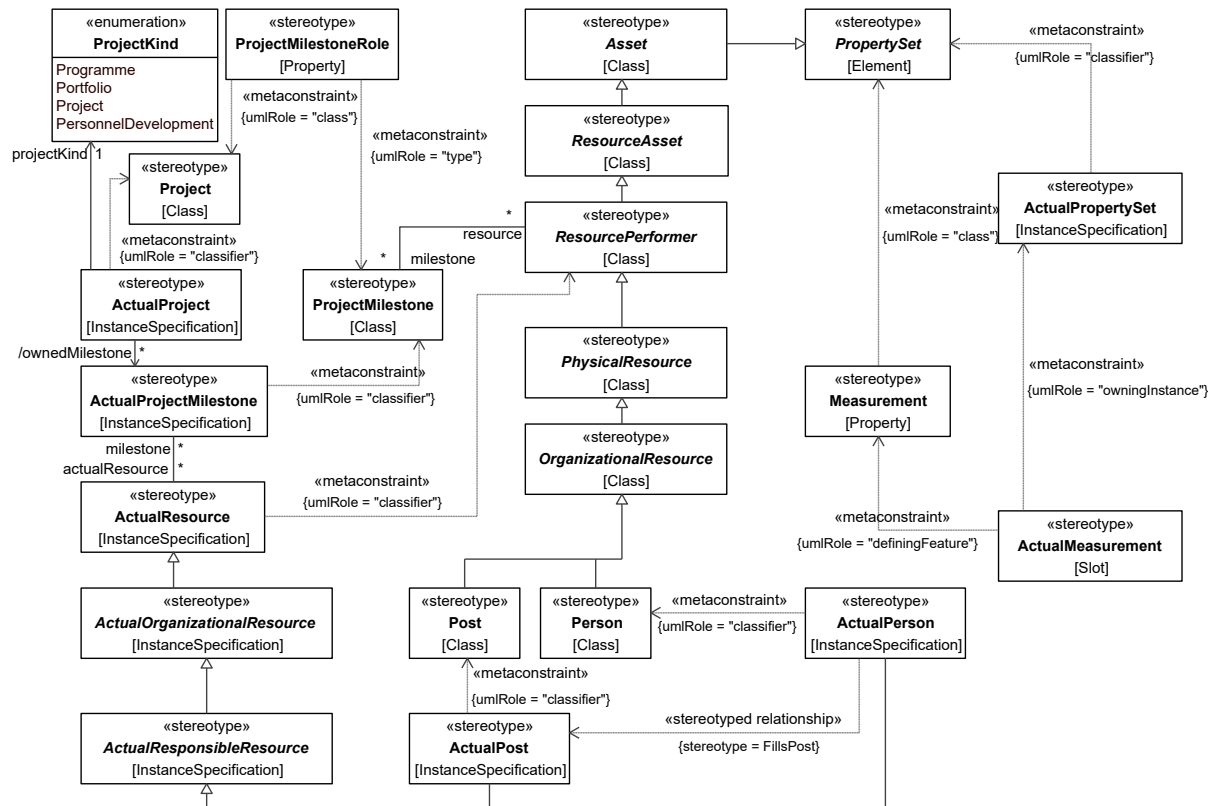


Figure 4:35 - Personnel Roadmap: Availability

Elements

- [ActualMeasurement](#)
- [ActualOrganizationalResource](#)
- [ActualPerson](#)
- [ActualPost](#)
- [ActualProject](#)
- [ActualProjectMilestone](#)
- [ActualPropertySet](#)
- [ActualResource](#)
- [ActualResponsibleResource](#)
- [Asset](#)
- [Measurement](#)
- [OrganizationalResource](#)
- [Person](#)

- [PhysicalResource](#)
- [Post](#)
- [Project](#)
- [ProjectKind](#)
- [ProjectMilestone](#)
- [ProjectMilestoneRole](#)
- [PropertySet](#)
- [ResourceAsset](#)
- [ResourcePerformer](#)

Stakeholders: Human resources, Solution Providers.

Concerns: organizational structure changes over time.

Definition: provides an overview of how an organizational structure changes over time. It shows the structure of several organizational structures mapped against a timeline.

Recommended Implementation: timeline, SysML Block Definition Diagram, SysML Internal Block Diagram.

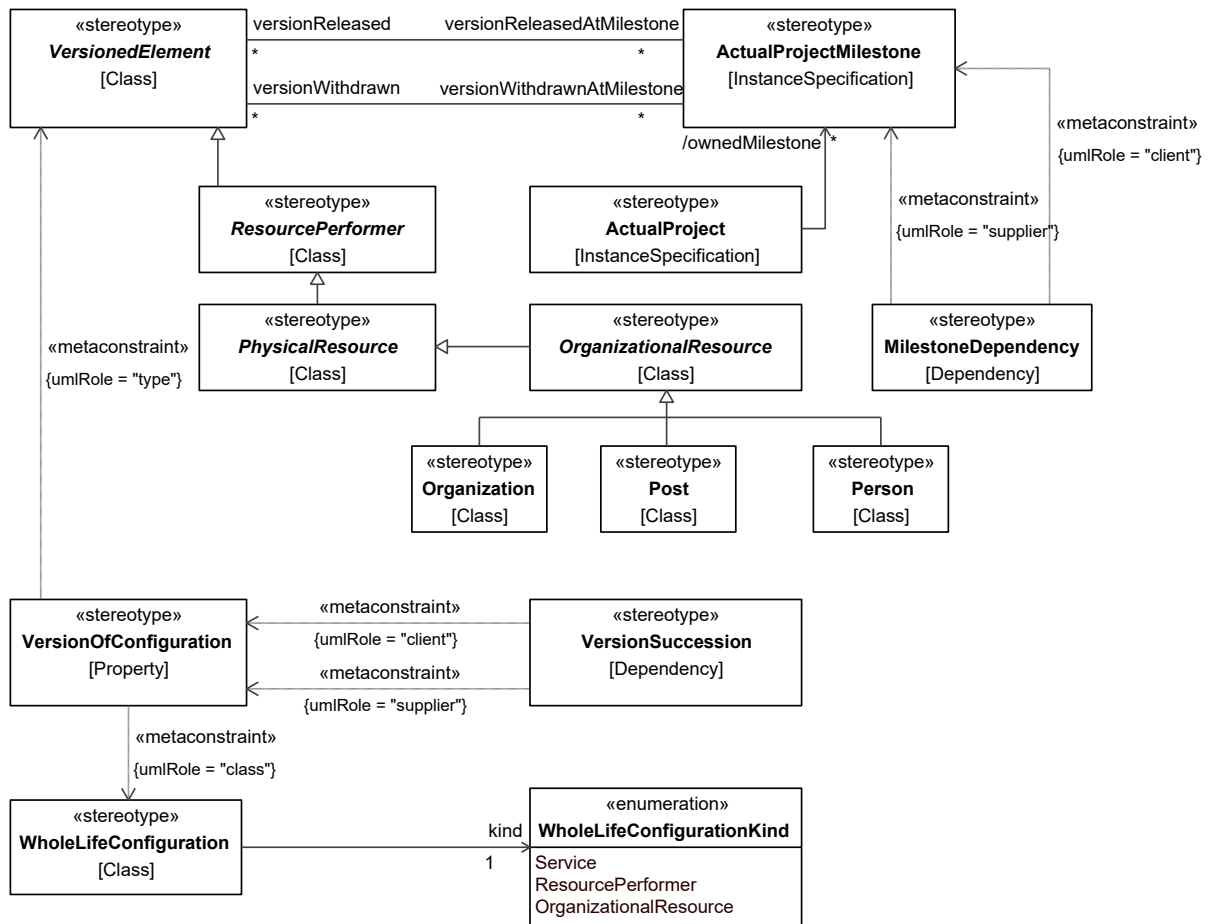


Figure 4:36 - Personnel Roadmap: Evolution

Elements

- [ActualProject](#)
- [ActualProjectMilestone](#)
- [MilestoneDependency](#)
- [Organization](#)

- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourcePerformer](#)
- [VersionedElement](#)
- [VersionOfConfiguration](#)
- [VersionSuccession](#)
- [WholeLifeConfiguration](#)
- [WholeLifeConfigurationKind](#)

Stakeholders: Human resources, Logisticians, Solution Providers.
Concerns: competencies and skills.

(v)-.7

Definition: depicts the mapping of functions (performed by organizational resources) to operational activities and thus identifies the transformation of an operational need into a purposeful function performed by an organizational resource or solution.

Recommended Implementation: Matrix format, SysML Block Definition Diagram.

Figure 4:38 - Personnel Traceability

Elements

- [Function](#)
- [OperationalActivity](#)
- [ServiceFunction](#)

4.1.5 View Specifications::Resources

Stakeholders: Systems Engineers, Resource Owners, Implementers, Solution Providers, IT Architects.

Concerns: definition of solution architectures to implement operational requirements.

Definition: captures a solution architecture consisting of resources, e.g., organizational, software, artifacts, capability configurations, natural resources that implement the operational requirements. Further design of a resource is typically detailed in SysML or UML.

View Specifications::Resources::Taxonomy

Stakeholders: Solution Providers, Systems Engineers, IT Architects, Implementers.

Concerns: resource ty9.9 (-)1.2 (n)-eg (r)-4.3 (e)2333.234 0 Td(C)Tj±0.003 Tc 0.00 -028-34.311 -1.15 Td55 (l)0.2 (e)-1.5 (f)-455 ()

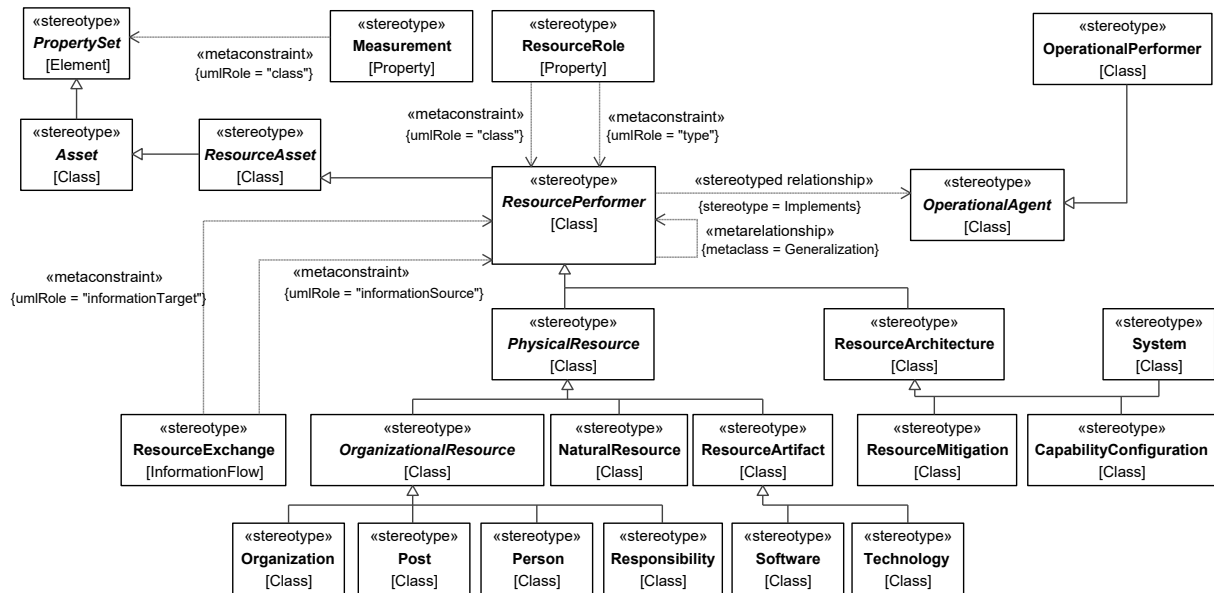


Figure 4:39 - Resources Taxonomy

Elements

- [Asset](#)
- [CapabilityConfiguration](#)
- [Measurement](#)
- [NaturalResource](#)
- [OperationalAgent](#)
- [OperationalPerformer](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [PropertySet](#)
- [ResourceArchitecture](#)
- [ResourceArtifact](#)
- [ResourceAsset](#)
- [ResourceExchange](#)
- [ResourceMitigation](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- [Responsibility](#)
- [Software](#)
- [System](#)
- [Technology](#)

View Specifications::Resources::Structure

Stakeholders: Systems Engineers, Resource Owners, Implementers, Solution Providers.

Concerns: reference the resource structure, connectors and interfaces in a specific context.

Definition: defines the physical resources, e.g., capability configuration(s)/system(s) and interactions necessary to implement a specific set of OperationalPerformer(s). Can be used to represent communications networks and pathways

that link communications resources and provides details regarding their configuration.
Recommended Implementation: SysML Internal Block Diagram, SysML Block Definition Diagram.

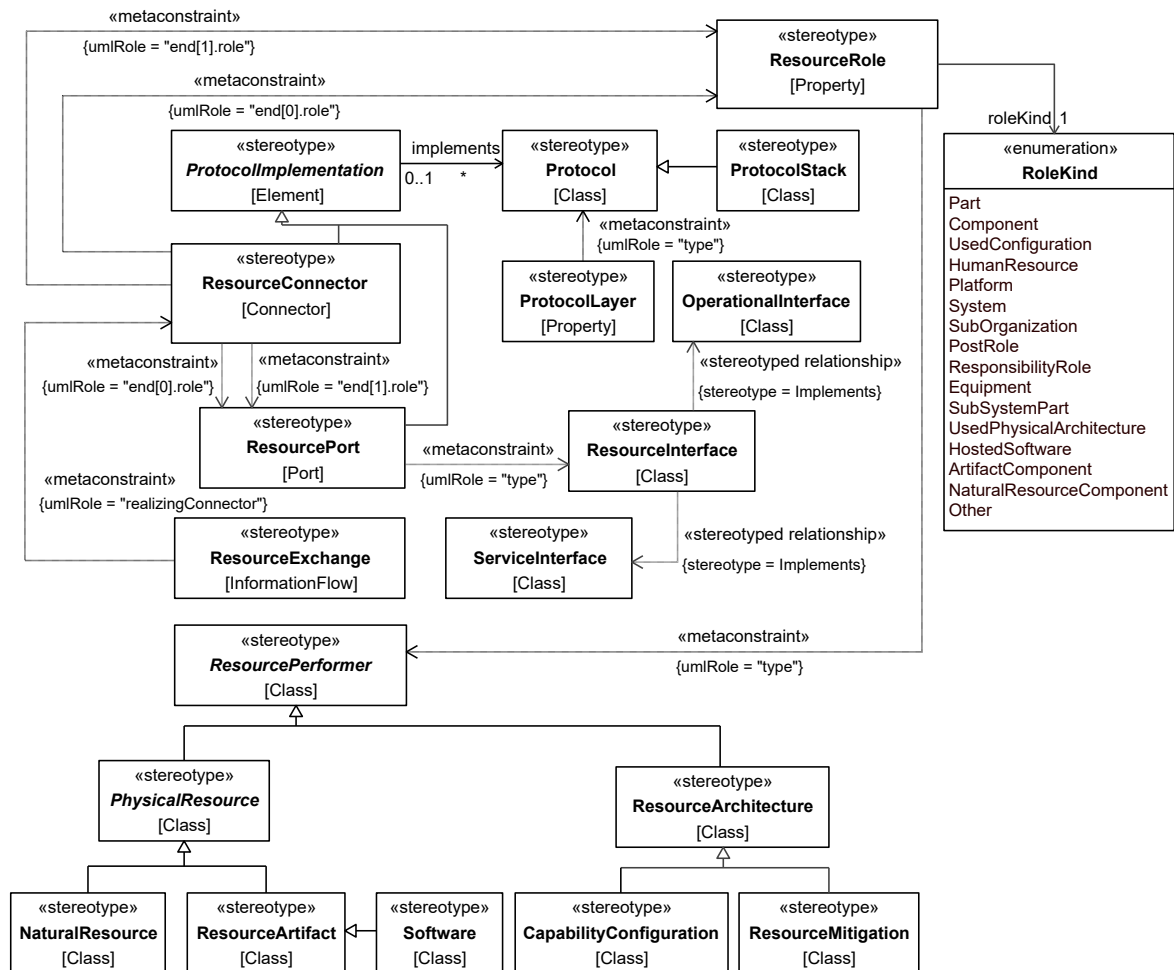


Figure 4:40 - Resources Structure

Elements

- [CapabilityConfiguration](#)
- [NaturalResource](#)
- [OperationalInterface](#)
- [PhysicalResource](#)
- [Protocol](#)
- [ProtocolImplementation](#)
- [ProtocolLayer](#)
- [ProtocolStack](#)
- [ResourceArchitecture](#)
- [ResourceArtifact](#)
- [ResourceConnector](#)
- [ResourceExchange](#)
- [ResourceInterface](#)
- [ResourceMitigation](#)
- [ResourcePerformer](#)
- [ResourcePort](#)

- [ResourceRole](#)
- [RoleKind](#)
- [ServiceInterface](#)
- [Software](#)

View Specifications::Resources::Connectivity

Stakeholders: Systems Engineers, IT Architects, Solution Providers, Implementers.

Concerns: capture the interactions between resources.

Definition: summarizes interactions between resources of information, systems, personnel, natural resources etc. and the functions that produce and consume them. Measurements can optionally be included.

Recommended Implementation: SysML Internal Block Diagram, tabular format.

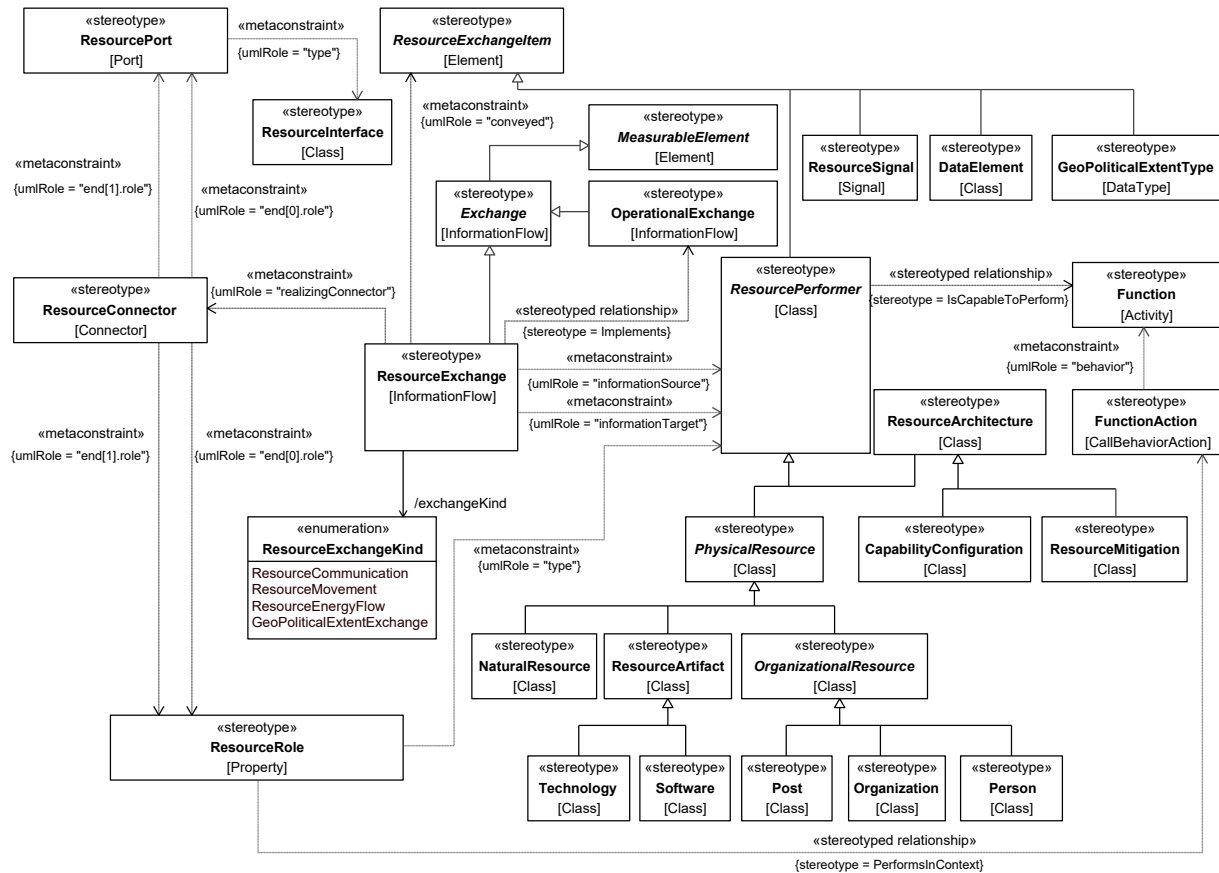


Figure 4:41 - Resources Connectivity

Elements

- [CapabilityConfiguration](#)
- [DataElement](#)
- [Exchange](#)
- [Function](#)
- [FunctionAction](#)
- [GeoPoliticalExtentType](#)
- [MeasurableElement](#)
- [NaturalResource](#)
- [OperationalExchange](#)
- [Organization](#)
- [OrganizationalResource](#)

- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourceArchitecture](#)
- [ResourceArtifact](#)
- [ResourceConnector](#)
- [ResourceExchange](#)
- [ResourceExchangeItem](#)
- [ResourceExchangeKind](#)
- [ResourceInterface](#)
- [ResourceMitigation](#)
- [ResourcePerformer](#)
- [ResourcePort](#)
- [ResourceRole](#)
- [ResourceSignal](#)
- [Software](#)
- [Technology](#)

View Specifications::Resources::Processes

S 10.8 (t)064 (a)-1.1 ke ls:S 10.8 s36.7,mS3lsB685t :gs ts

a2693685f125365037D488694d413654369405 [(s8175 .s)]TJ0 T3 09 T 1256943654369405 TjE003 T36900d Tw-236934e3694 ft

74(t)1064 h(a3094 m)8155 .s

794Mhe36.3 A:)05 [cs3694 t

Elements

- [Activity](#)
- [ActualCondition](#)
- [DataElement](#)
- [Function](#)
- [FunctionAction](#)
- [FunctionControlFlow](#)
- [FunctionEdge](#)
- [FunctionObjectFlow](#)
- [OperationalActivity](#)
- [PhysicalResource](#)
- [ResourceArchitecture](#)
- [ResourceExchange](#)
- [ResourceExchangeItem](#)
- [ResourceParameter](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- [ServiceFunction](#)

View Specifications::Resources::States

Stakeholders: Systems Engineers, Software Engineers.

Concerns: capture state-based behavior of a resource.

Definition: it is a graphical representation of states of a resource and how that resource responds to various events and actions.

Recommended Implementation: SysML State Machine Diagram.

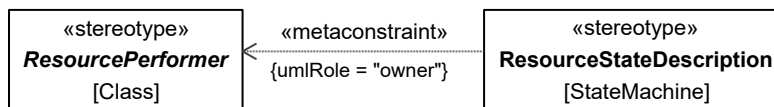


Figure 4:43 - Resources States

Elements

- [ResourcePerformer](#)
- [ResourceStateDescription](#)

View Specifications::Resources::Interaction Scenarios

Stakeholders: Software Engineers, Systems Engineers.

Concerns: interactions between resources (roles).

Definition: provides a time-ordered examination of the interactions between resources.

Recommended Implementation: SysML Sequence Diagram.

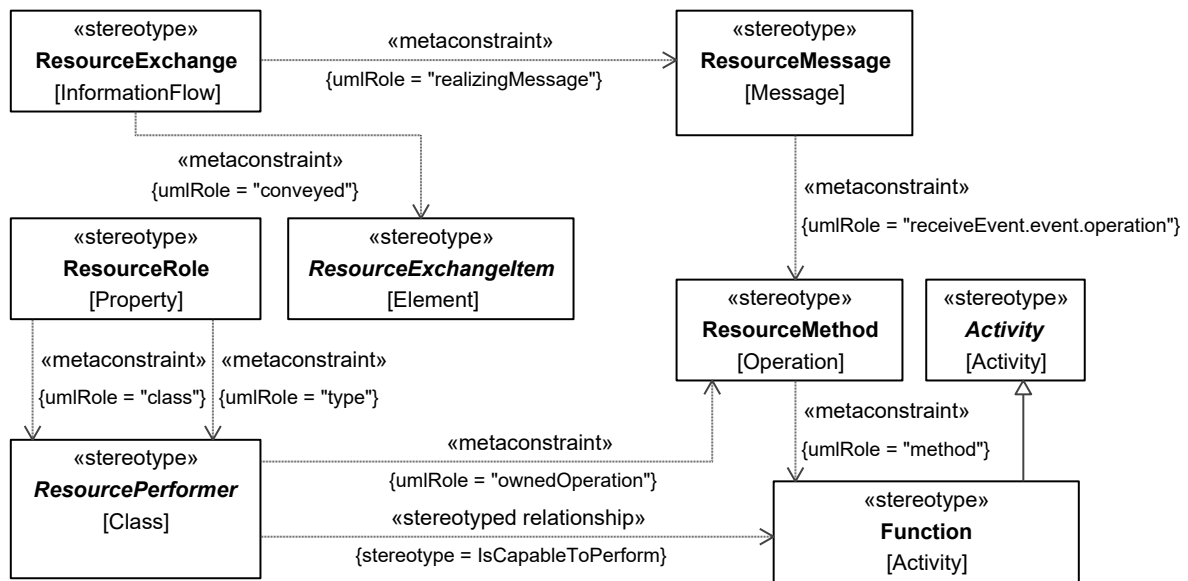


Figure 4:44 - Resources Interaction Scenarios

Elements

- [Activity](#)
- [Function](#)
- [ResourceExchange](#)
- [ResourceExchangeItem](#)
- [ResourceMessage](#)
- [ResourceMethod](#)
- [ResourcePerformer](#)
- [ResourceRole](#)

View Specifications::Resources::Constraints

Stakeholders: Systems Engineers, IT Architects, Solution Providers, Implementers.

Concerns: define limitations, constraints and performance parameters for resources, their interactions, performed functions, and data.

Definition: specifies traditional textual rules/non-functional requirements that are constraints on resources, their interactions, performed functions, and data. The addition of SysML parametrics provide a computational means of defining resource constraints within a specific context.

Recommended Implementation: tabular format, SysML Block Definition Diagram, SysML Parametric Diagram, OCL.

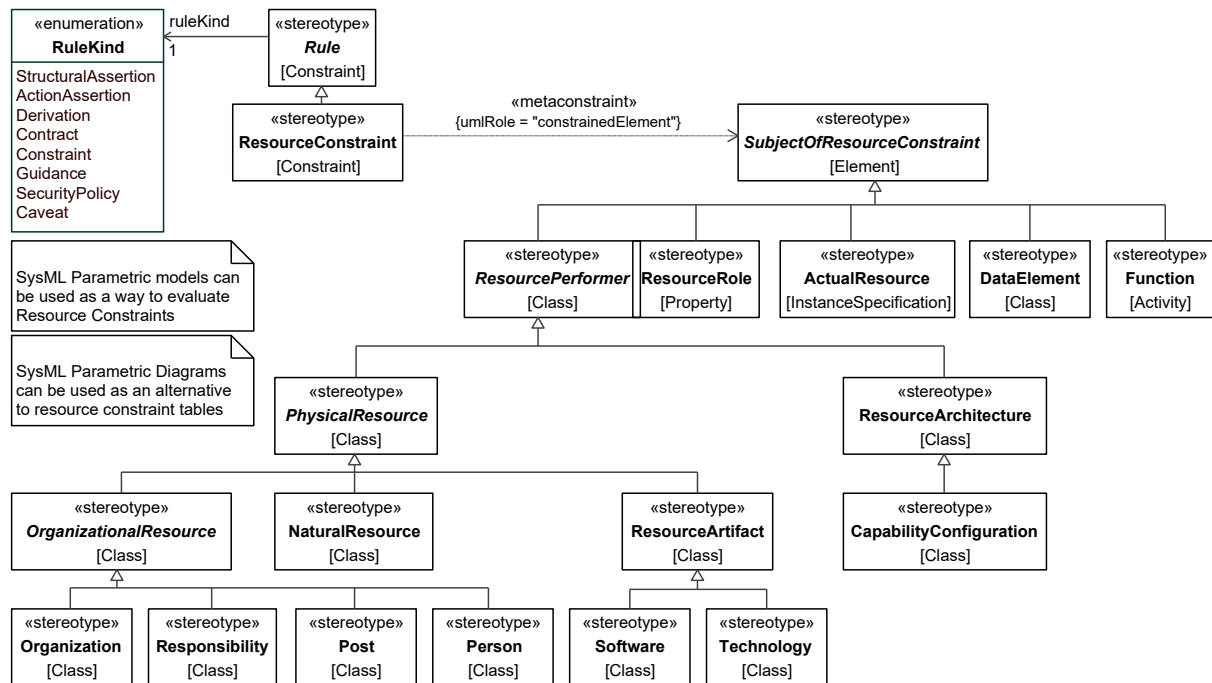


Figure 4:45 - Resources Constraints

Elements

- [ActualResource](#)
- [CapabilityConfiguration](#)
- [DataElement](#)
- [Function](#)
- [NaturalResource](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourceArchitecture](#)
- [ResourceArtifact](#)
- [ResourceConstraint](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- [Responsibility](#)
- [Rule](#)
- [RuleKind](#)
- [Software](#)
- [SubjectOfResourceConstraint](#)
- [Technology](#)

View Specifications::Resources::Roadmap

Stakeholders: Systems Engineers, IT Architects, Solution Providers, Implements.

Concerns: resource structure changes over time.

Definition: provides an overview of how a resource structure changes over time. It shows the structure of several

resources mapped against a timeline.

Recommended Implementation: timeline, SysML Block Definition Diagram, SysML Internal Block Diagram.

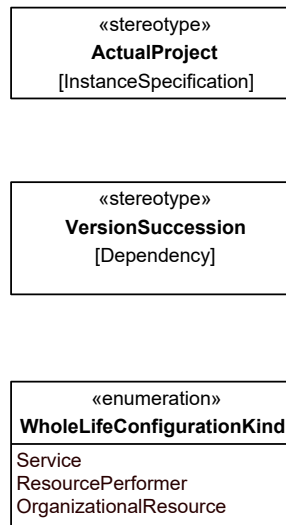


Figure 4:46 - Resources Roadmap: Evolution

Elements

- [ActualProject](#)
- [ActualProjectMilestone](#)
- [MilestoneDependency](#)
- [ResourcePerformer](#)
- [VersionedElement](#)
- [VersionOfConfiguration](#)
- [VersionSuccession](#)
- [WholeLifeConfiguration](#)
- [WholeLifeConfigurationKind](#)

Stakeholders: Solution Providers, Systems Engineers, IT Architects.

Concerns: technology forecast.

Definition: defines the underlying current and expected supporting technologies. Expected supporting technologies are those that can be reasonably forecast given the current state of technology, and expected improvements / trends.

Recommended Implementation: timeline, tabular format, SysML Block Definition Diagram.

e

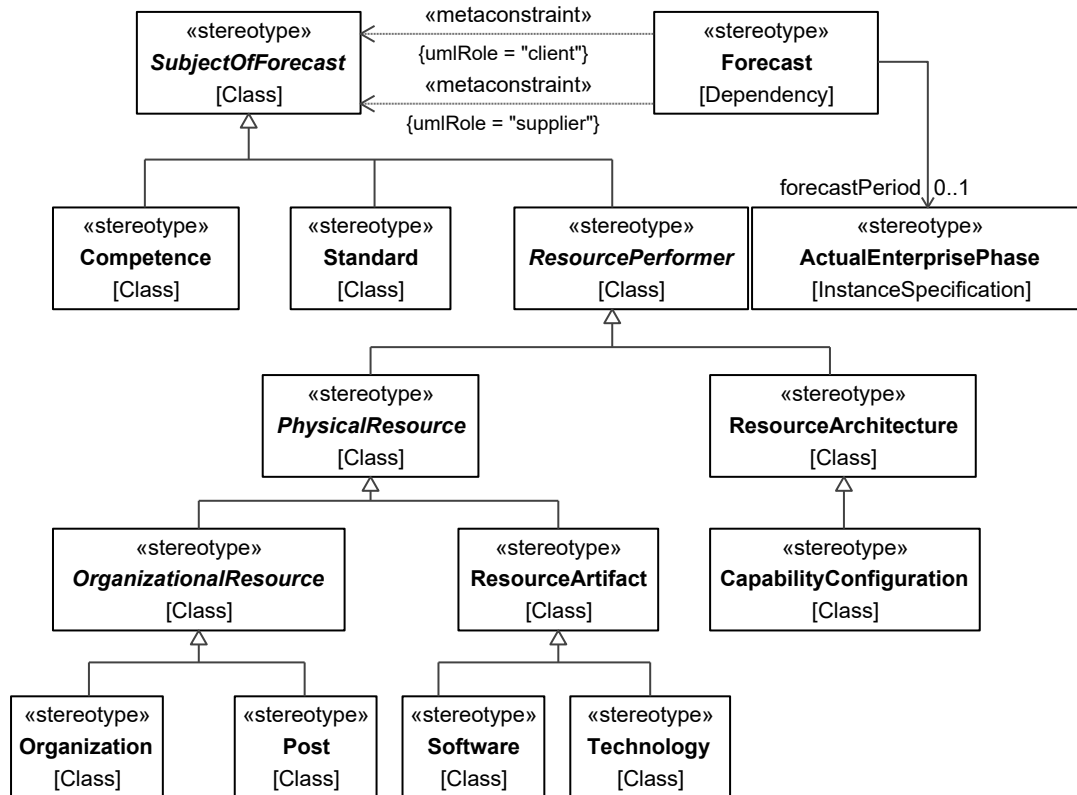


Figure 4:47 - Resources Roadmap: Forecast

Elements

- [ActualEnterprisePhase](#)
- [CapabilityConfiguration](#)
- [Competence](#)
- [Forecast](#)
- [Organization](#)
- [OrganizationalResource](#)
- [PhysicalResource](#)
- [Post](#)
- [ResourceArchitecture](#)
- [ResourceArtifact](#)
- [ResourcePerformer](#)
- [Software](#)
- [Standard](#)
- [SubjectOfForecast](#)
- [Technology](#)

View Specifications::Resources::Traceability

Stakeholders: Systems Engineers, Enterprise Architects, Solution Providers, Business Architects.

Concerns: traceability between operational activities and functions that implements them.

Definition: depicts the mapping of functions to operational activities and thus identifies the transformation of an operational need into a purposeful function performed by a resource or solution.

Recommended Implementation: Matrix format, SysML Block Definition Diagram.

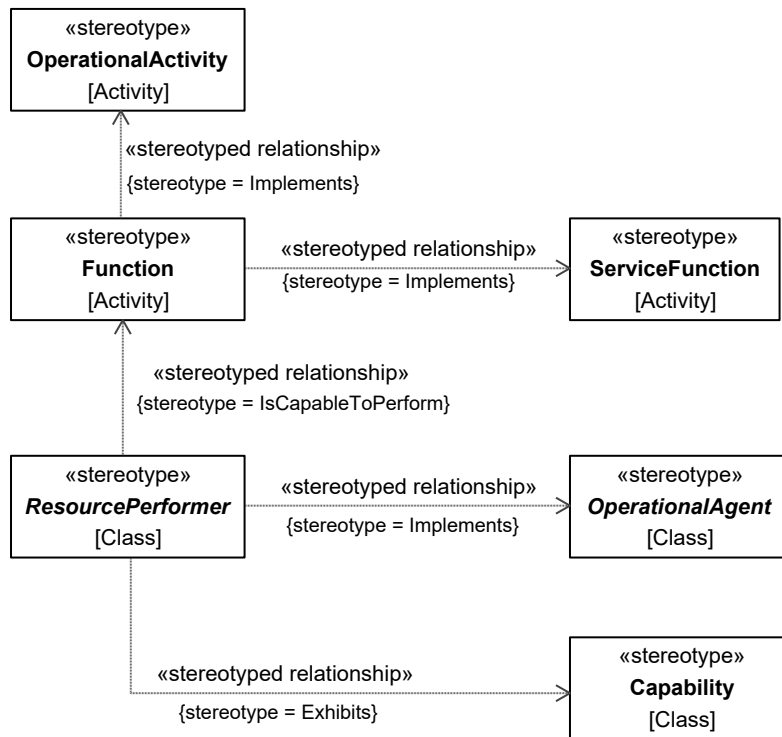


Figure 4:48 - Resources Traceability

Elements

- [Capability](#)
- [Function](#)
- [OperationalActivity](#)
- [OperationalAgent](#)
- [ResourcePerformer](#)
- [ServiceFunction](#)

4.1.6 View Specifications::Security

View Specifications::Security::Taxonomy

Stakeholders: Security Architects, Security Engineers.

Concerns: Security assets and security enclaves.

Definition: Defines the hierarchy of security assets and asset owners that are available to implement security, security constraints (policy, guidance, laws and regulations) and details where they are located (security enclaves).

Recommended Implementation: tabular format, SysML Block Definition Diagram.

- [ResourceArchitecture](#)
- [ResourceArtifact](#)
- [ResourceAsset](#)
- [ResourcePerformer](#)
- [Responsibility](#)
- [SecurityEnclave](#)
- SecurityImpactProperties
- [Software](#)
- [System](#)
- [Technology](#)

View Specifications::Security::Structure

Stakeholders: Security Architects, Security Engineers.

Concerns: The structure of security information and where it is used at the operational and resource level.

Definition: Captures the allocation of assets (operational and resource, information and data) across the security enclaves, shows applicable security controls necessary to protect organizations, systems and information during processing, while in storage (bdd), and during transmission (flows on an ibd). This view also captures Asset Aggregation and allocates the usage of the aggregated information at a location through the use of the SecurityProperty.

Recommended Implementation: SysML Internal Block Diagram, SysML Block Definition Diagram.

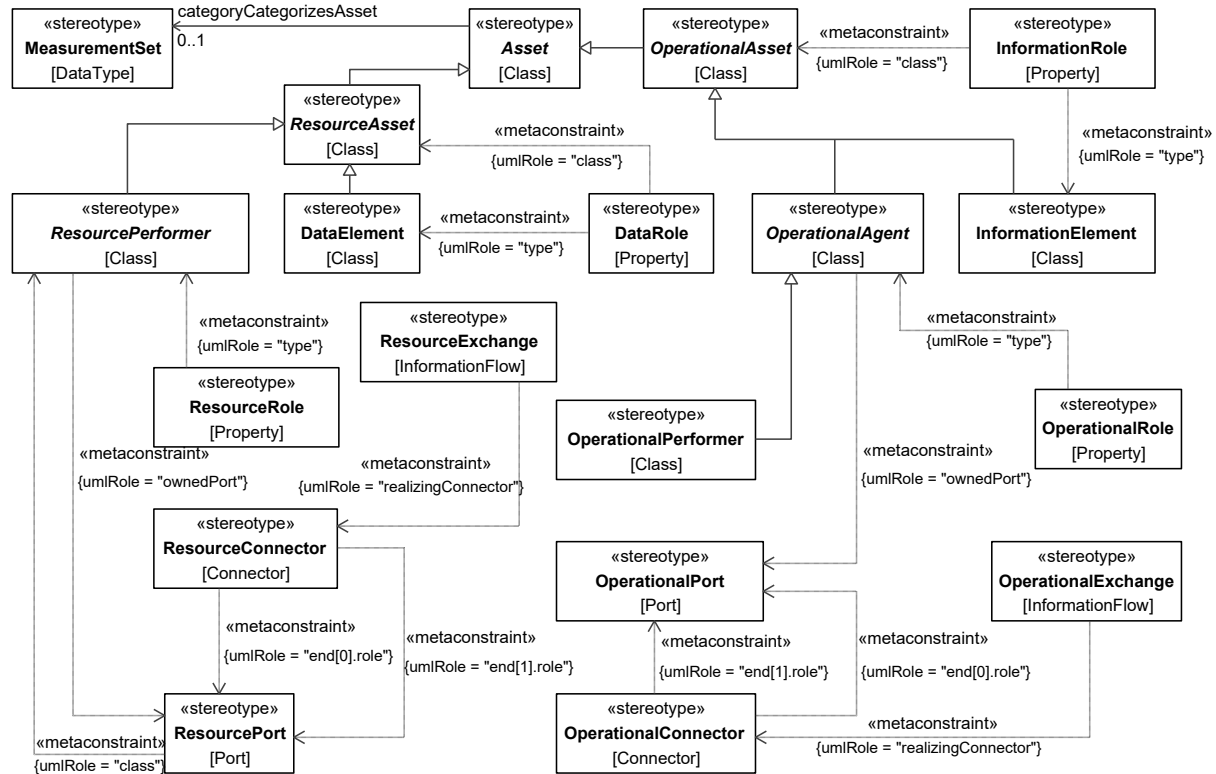


Figure 4:50 - Security Structure

Elements

- [Asset](#)
- [DataElement](#)
- [DataRole](#)
- [InformationElement](#)
- [InformationRole](#)

Elements

- [Asset](#)
- [DataElement](#)
- [Exchange](#)
- [InformationElement](#)
- [MeasurableElement](#)
- [Measurement](#)
- [MeasurementSet](#)
- [OperationalAgent](#)
- [OperationalAsset](#)
- [OperationalConnector](#)
- [OperationalExchange](#)
- [OperationalExchangeItem](#)
- [OperationalPerformer](#)
- [OperationalPort](#)
- [OperationalRole](#)
- [PropertySet](#)
- [ResourceAsset](#)
- [ResourceConnector](#)
- [ResourceExchange](#)
- [ResourceExchangeItem](#)
- [ResourceInterface](#)
- [ResourcePerformer](#)
- [ResourcePort](#)
- [ResourceRole](#)
- [SubjectOfSecurityConstraint](#)

View Specifications::Security::Processes

Stakeholders: Security Architects, Security Engineers.

Concerns: The specification of the Security Control families, security controls, and measures required to address a specific security baseline.

Definition: Provides a set of Security Controls and any possible enhancements as applicable to assets. The activity diagram describes operational or resource level processes that apply (operational level) or implement (resource level) security controls/enhancements to assets located in enclaves and across enclaves. This Security Process view can be instantiated either as a variant of an activity/flow diagram or as a hierarchical work breakdown structure.

Recommended Implementation: SysML Activity Diagram, SysML Block Definition Diagram.

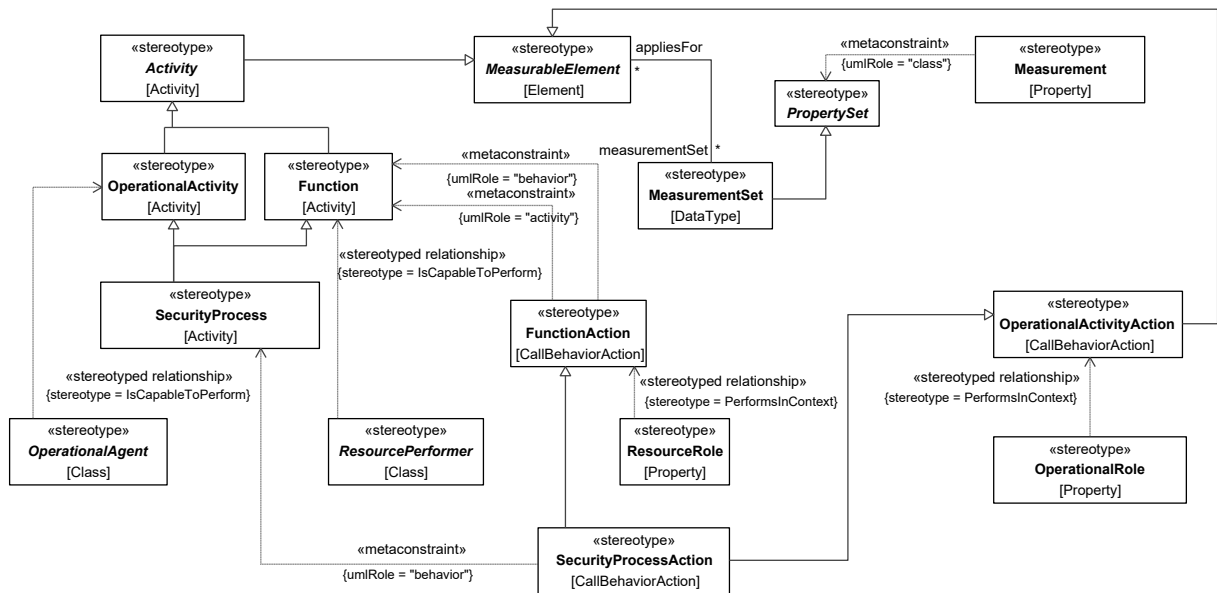


Figure 4:52 - Security Processes

Elements

- [Activity](#)
- [Function](#)
- [FunctionAction](#)
- [MeasurableElement](#)
- [Measurement](#)
- [MeasurementSet](#)
- [OperationalActivity](#)
- [OperationalActivityAction](#)
- [OperationalAgent](#)
- [OperationalRole](#)
- [PropertySet](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- [SecurityProcess](#)
- [SecurityProcessAction](#)

View Specifications::Security::Constraints

Stakeholders: Security Architects, Security Engineers, Risk Analysts.

Concerns: (i) Security-related policy, guidance, laws and regulations as applicable to assets, (ii) threats, vulnerabilities, and risk assessments as applicable to assets.

Definition: (i) Specifies textual rules/non-functional requirements that are security constraints on resources, information and data (e.g. security-related in the form of rules (e.g. access control policy). A common way of representing access control policy is through the use of XACML (eXtensible Access Control Markup Language), it is expected that implementations of UAF allow users to link security constraints to external files represented in XACML. (ii) Identifies risks, specifies risk likelihood, impact, asset criticality, other measurements and enables risk assessment.

Recommended Implementation: tabular or Matrix format, SysML Block Definition Diagram, SysML Parametric Diagram, or OCL.

- [SecurityControl](#)
- [SecurityControlFamily](#)
- [SecurityProcess](#)
- [SubjectOfSecurityConstraint](#)

View Specifications::Security::Traceability

Stakeholders: Security Architects, Security Engineers, Risk Analysts.

Concerns: traceability between risk and risk owner, risk mitigations, and affected asset roles.

Definition: depicts the mapping of a risk to each of the following: risk owner, risk mitigations, and affected asset roles.

Recommended Implementation: Matrix format, SysML Block Definition Diagram.

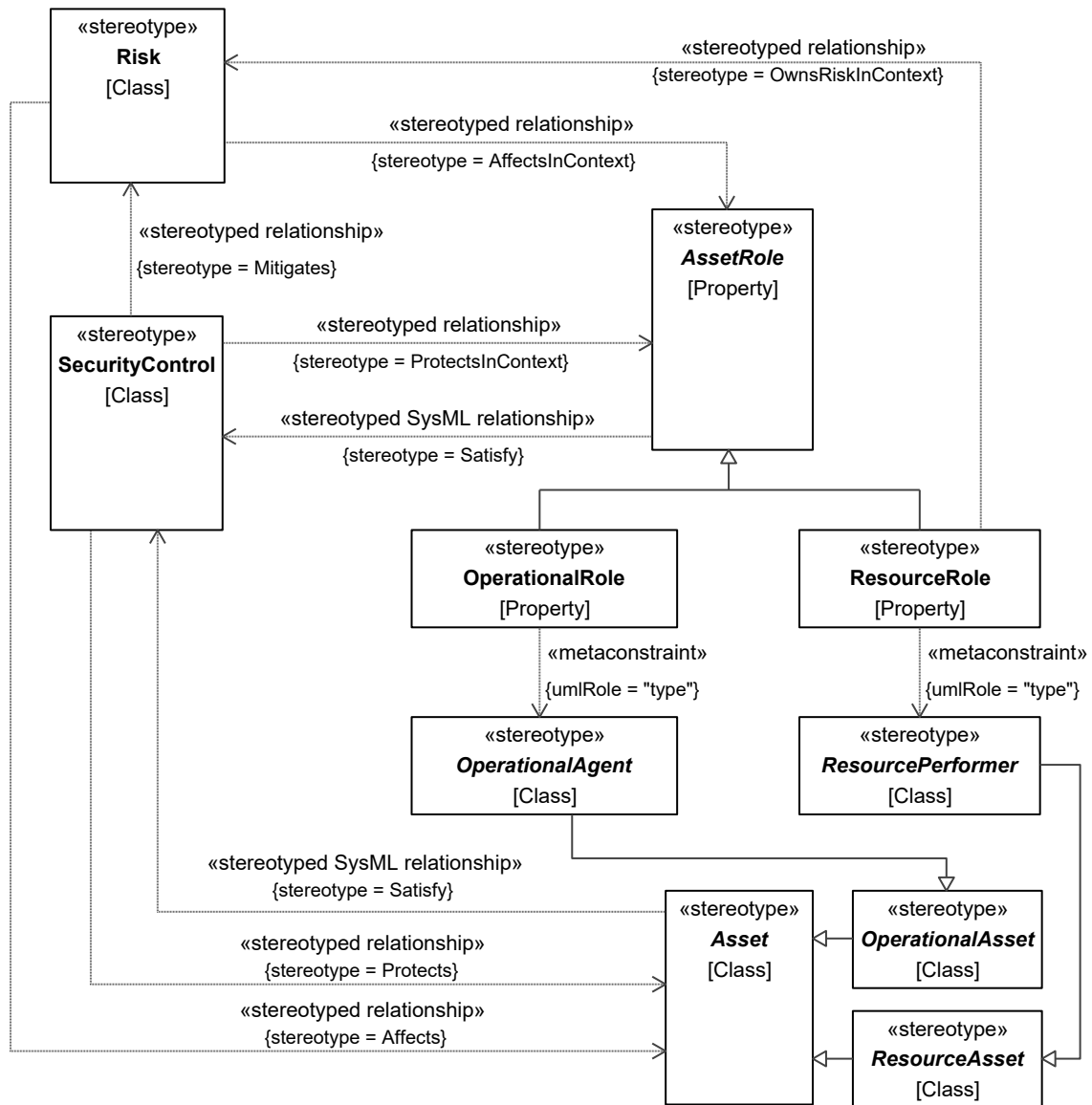


Figure 4:54 - Security Traceability

Elements

- [Asset](#)
- [AssetRole](#)
- [OperationalAgent](#)
- [OperationalAsset](#)
- [OperationalRole](#)
- [ResourceAsset](#)
- [ResourcePerformer](#)
- [ResourceRole](#)
- [Risk](#)
- [SecurityControl](#)

4.1.7 View Specifications::Projects

Stakeholders: PMs, Project Portfolio Managers, Enterprise Architects.

Concerns: project portfolio, projects and project milestones.

Definition: describes projects and project milestones, how those projects deliver capabilities, the organizations contributing to the projects and dependencies between projects.

View Specifications::Projects::Taxonomy

Stakeholders: PMs, Project Portfolio Managers, Enterprise Architects.

Concerns: types of projects and project milestones.

Definition: shows the taxonomy of types of projects and project milestones.

Recommended Implementation: SysML Block Definition Diagram.

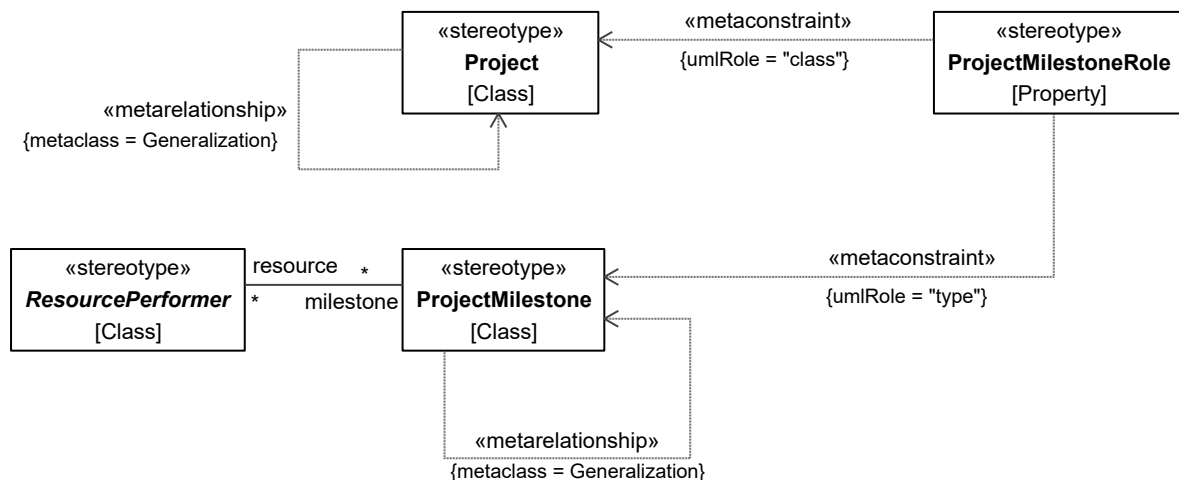


Figure 4:55 - Project Taxonomy

Elements

- [Project](#)
- [ProjectMilestone](#)
- [ProjectMilestoneRole](#)
- [ResourcePerformer](#)

View Specifications::Projects::Structure

Stakeholders: PMs.

Concerns: relationships between types of projects and project milestones.

Definition: provides a template for an actual project(s) road map(s) to be implemented.

Recommended Implementation: SysML Block Definition Diagram.

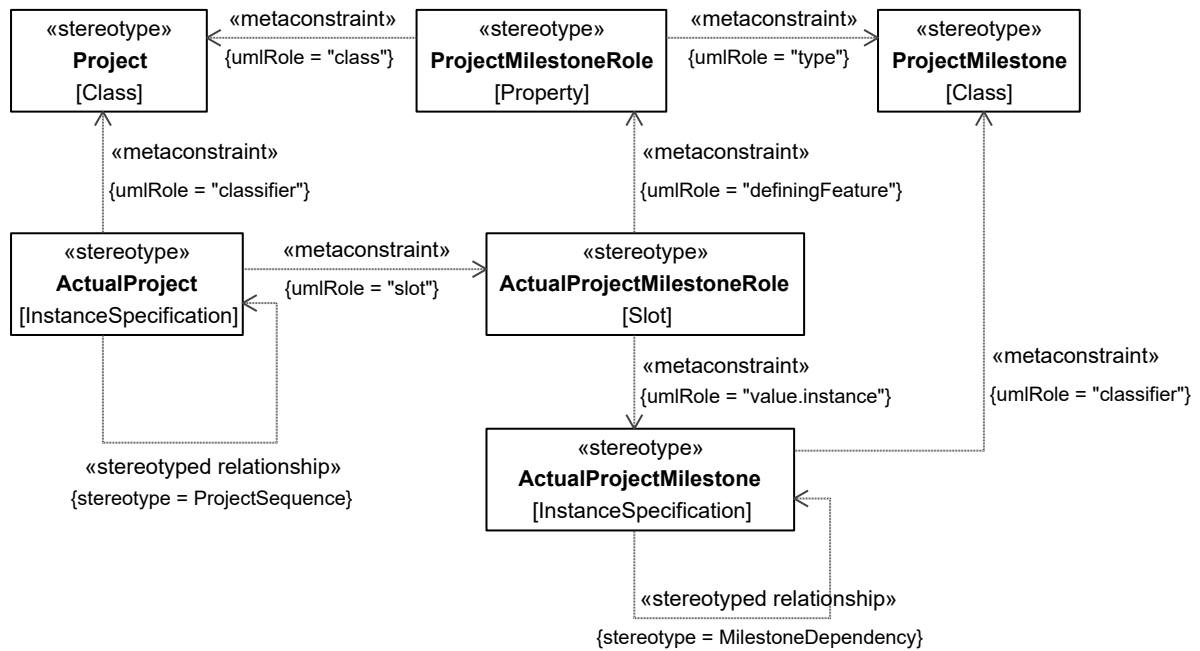


Figure 4:57 - Project Connectivity

Elements

- [ActualProject](#)
- [ActualProjectMilestone](#)
- [ActualProjectMilestoneRole](#)
- [Project](#)
- [ProjectMilestone](#)
- [ProjectMilestoneRole](#)

View Specifications::Projects::Processes

Stakeholders: PMs.

Concerns: captures project tasks (ProjectActivities) and flows between them.

Definition: describes the ProjectActivities that are normally conducted in the course of projects to support capability(ies) and implement resources. It describes the ProjectActivities, their Inputs/Outputs, ProjectActivityActions and flows between them.

Recommended Implementation: SysML Activity Diagram, SysML Block Definition Diagram.

- [ActualProjectRole](#)
- [ActualResource](#)
- [ActualResponsibleResource](#)
- [CapabilityConfiguration](#)
- [FieldedCapability](#)
- [MilestoneDependency](#)
- [Project](#)
- [ProjectMilestone](#)
- [ProjectMilestoneRole](#)
- [ProjectRole](#)
- [ProjectSequence](#)
- [ProjectStatus](#)
- [ProjectTheme](#)
- [ResourceArchitecture](#)
- [ResourcePerformer](#)
- [StatusIndicators](#)

View Specifications::Projects::Traceability

Stakeholders: PMs, Project Portfolio Managers, Enterprise Architects.

Concerns: traceability between capabilities and projects that deliver them.

Definition: depicts the mapping of projects to capabilities and thus identifies the transformation of a capability(ies) into a purposeful implementation via projects.

Recommended Implementation: Matrix format, SysML Block Definition Diagram.

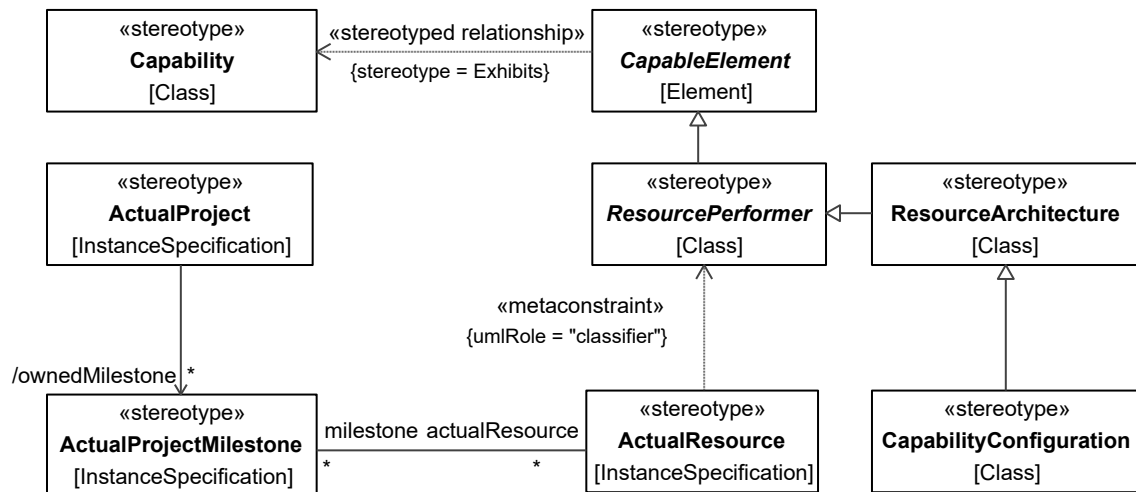


Figure 4:60 - Project Traceability

Elements

- [ActualProject](#)
- [ActualProjectMilestone](#)
- [ActualResource](#)
- [Capability](#)
- [CapabilityConfiguration](#)
- [CapableElement](#)
- [ResourceArchitecture](#)
- [ResourcePerformer](#)

4.1.8 View Specifications::Standards

Stakeholders: Solution Providers, Systems Engineers, Software Engineers, Systems Architects, Business Architects.

Concerns: technical and non-technical Standards applicable to the architecture.

Definition: shows the technical, operational, and business Standards applicable to the architecture. Defines the underlying current and expected Standards.

View Specifications::Standards::Taxonomy

Stakeholders: Solution Providers, Systems Engineers, Software Engineers, Systems Architects, Business Architects.

Concerns: technical and non-technical standards, guidance and policy applicable to the architecture.

Definition: shows the taxonomy of types of technical, operational, and business standards, guidance and policy applicable to the architecture.

Recommended Implementation: SysML Block Definition Diagram.

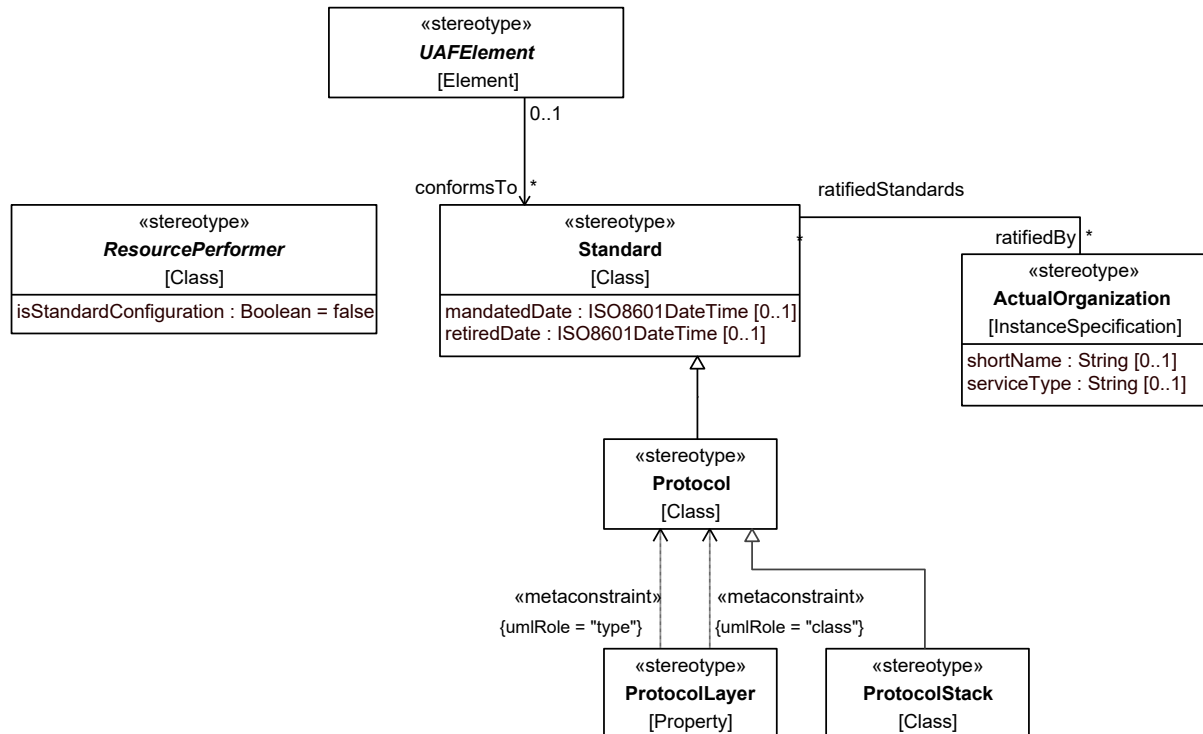


Figure 4:61 - Standards Taxonomy

Elements

- [ActualOrganization](#)
- [Protocol](#)
- [ProtocolLayer](#)
- [ProtocolStack](#)
- [ResourcePerformer](#)
- [Standard](#)
- [UAFElement](#)

View Specifications::Standards::Structure

Stakeholders: Solution Providers, Systems Engineers, Software Engineers, Systems Architects.

Concerns: the specification of the protocol stack used in the architecture.

Definition: shows the composition of standards required to achieve the architecture's objectives.

Recommended Implementation: SysML Internal Block Diagram.

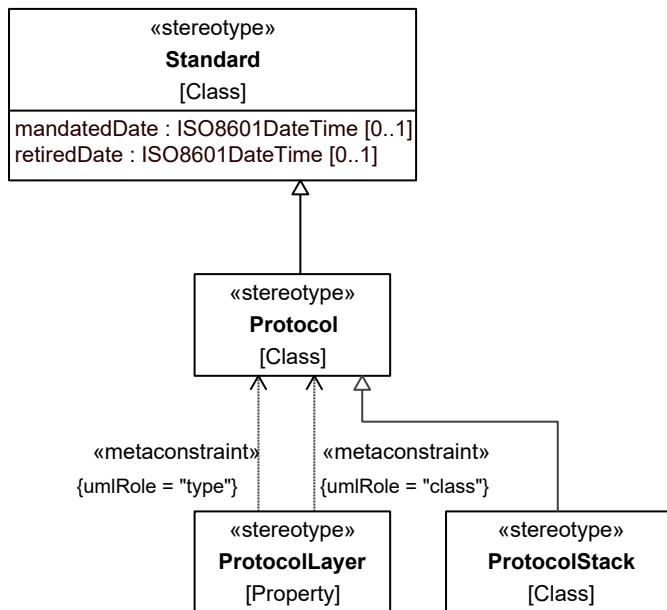


Figure 4:62 - Standards Structure

Elements

- [Protocol](#)
- [ProtocolLayer](#)
- [ProtocolStack](#)
- [Standard](#)

View Specifications::Standards::Roadmap

Stakeholders: Solution Providers, Systems Engineers, Systems Architects, Software Engineers, Business Architects.

Concerns: expected changes in technology-related standards and conventions, operational standards, or business standards and conventions.

Definition: defines the underlying current and expected standards. Expected standards are those that can be reasonably forecast given the current state of technology, and expected improvements / trends.

Recommended Implementation: timeline, tabular format, SysML Block Definition Diagram.

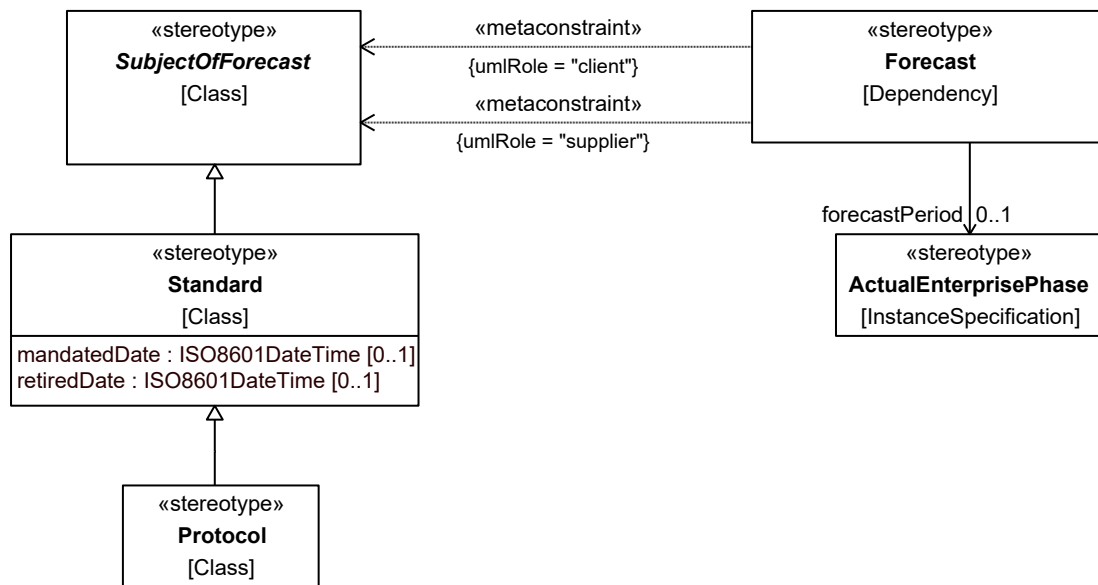


Figure 4:63 - Standards Roadmap

Elements

- [ActualEnterprisePhase](#)
- [Forecast](#)
- [Protocol](#)
- [Standard](#)
- [SubjectOfForecast](#)

View Specifications::Standards::Traceability

Stakeholders: Solution Providers, Systems Engineers, Software Engineers, Systems Architects, Business Architects.

Concerns: standards that need to be taken in account to ensure the interoperability of the implementation of architectural elements.

Definition: shows the applicability of standards to specific elements in the architecture.

Recommended Implementation: tabular format, matrix format, SysML Block Definition Diagram.

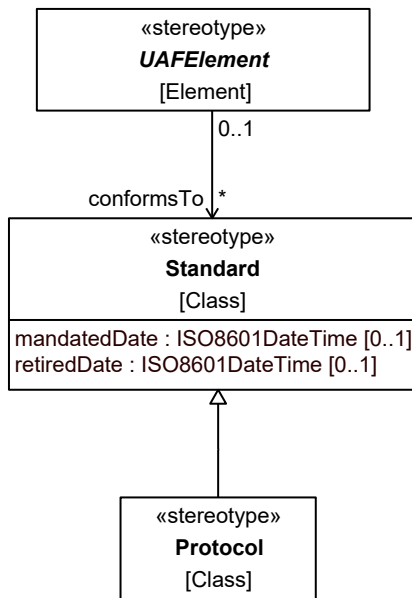


Figure 4:64 - Standards Traceability

Elements

- [Protocol](#)
- [Standard](#)
- [UAFElement](#)

4.1.9 View Specifications::Actual Resources

View Specifications::Actual Resources::Structure

Stakeholders: Solution Providers, Systems Engineers, Business Architects.

Concerns: the analysis, e.g., evaluation of different alternatives, what-if, trade-offs, V&V on the actual resource configurations as it provides a means to capture different solution architectures. The detailed analysis (trade-off, what-if etc.) is carried out using the Resource Constraints view.

Definition: illustrates the expected or achieved actual resource configurations required to meet an operational need.

Recommended Implementation: SysML Block Definition Diagram.

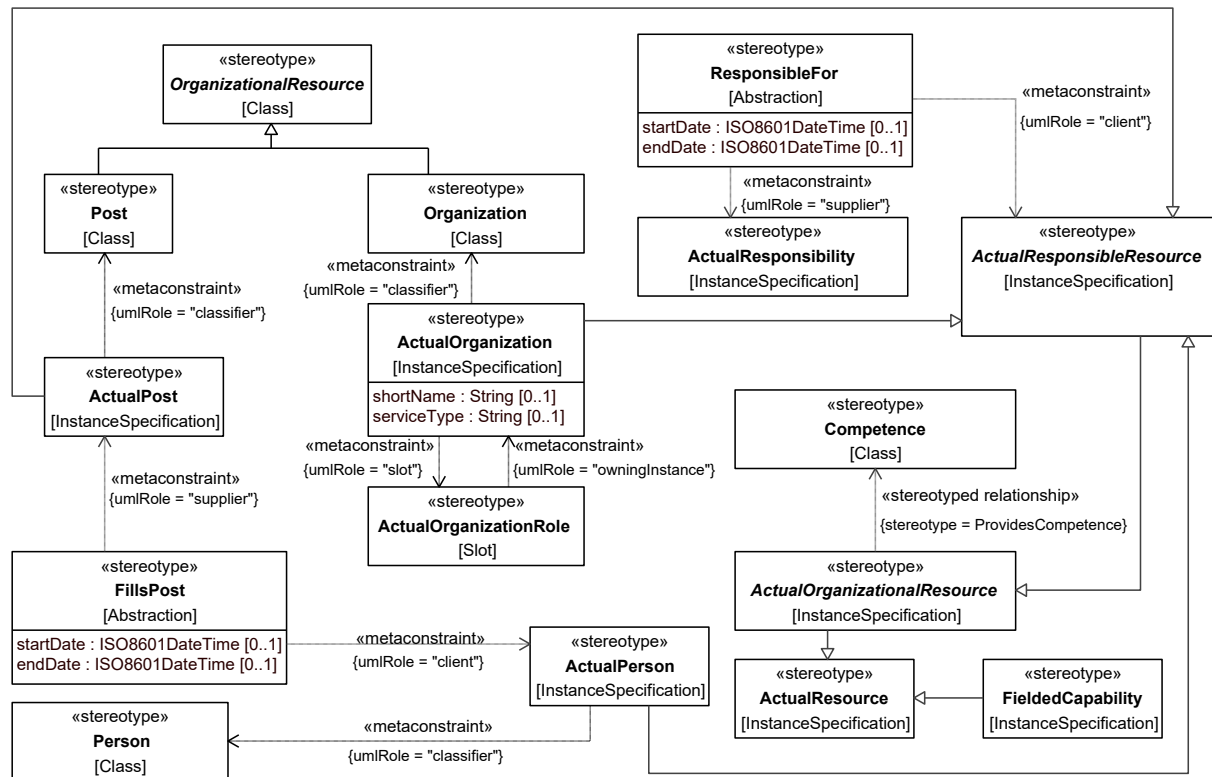


Figure 4:65 - Actual Resources Structure

Elements

- [ActualOrganization](#)
- [ActualOrganizationalResource](#)
- [ActualOrganizationRole](#)
- [ActualPerson](#)
- [ActualPost](#)
- [ActualResource](#)
- [ActualResponsibility](#)
- [ActualResponsibleResource](#)
- [Competence](#)
- [FieldedCapability](#)
- [FillsPost](#)
- [Organization](#)
- [OrganizationalResource](#)
- [Person](#)
- [Post](#)
- [ResponsibleFor](#)

View Specifications::Actual Resources::Connectivity

Stakeholders: Solution Providers, Systems Engineers, Business Architects.

Concerns: the communication of actual resource.

Definition: illustrates the actual resource configurations and actual relationships between them.

Recommended Implementation: tabular format, SysML Block Definition Diagram, SysML Internal Block Diagram, SysML Sequence Diagram.

Figure 4:66 - Actual Resources Connectivity

Elements

- [ActualOrganization](#)
- [ActualOrganizationalResource](#)
- [ActualPerson](#)
- [ActualPost](#)
- [ActualResource](#)
- [ActualResourceRelationship](#)
- [ActualResponsibility](#)
- [ActualResponsibleResource](#)
- [FieldedCapability](#)

View Specifications::Actual Resources::Traceability

Stakeholders: Systems Engineers, Enterprise Architects, Solution Providers, Business Architects.

Concerns: traceability between operational activities and functions that implements them.

4.1.10 View Specifications::Dictionary

Stakeholders: Architects, users of the architecture, Capability Owners, Systems Engineers, Solution Providers.

Concerns: Definitions for all the elements in the architecture, libraries of environments and measurements.

Definition: Presents all the elements used in an architecture. Can be used specifically to capture:

a. elements and relationships that are involved in defining the environments applicable to capability, operational concept or set of systems.

b. measurable properties that can be used to support analysis such as KPIs, MoEs, TPIs etc.

Recommended Implementation: Tabular format, SysML Block Definition Diagram.

View Specifications::Dictionary::Dictionary

Stakeholders: Solution Providers, Systems Engineers, Software Architects, Business Architects.

Concerns: provides a central reference for a given architecture's data and metadata. It enables the set of architecture description to stand alone, with minimal reference to outside resources.

Definition: contains definitions of terms used in the given architecture. It consists of textual definitions in the form of a glossary, their taxonomies, and their metadata (i.e., data about architecture data), including metadata for any custom-tailored views. Architects should use standard terms where possible (i.e., terms from existing, approved dictionaries, glossaries, and lexicons).

Recommended Implementation: text, table format.

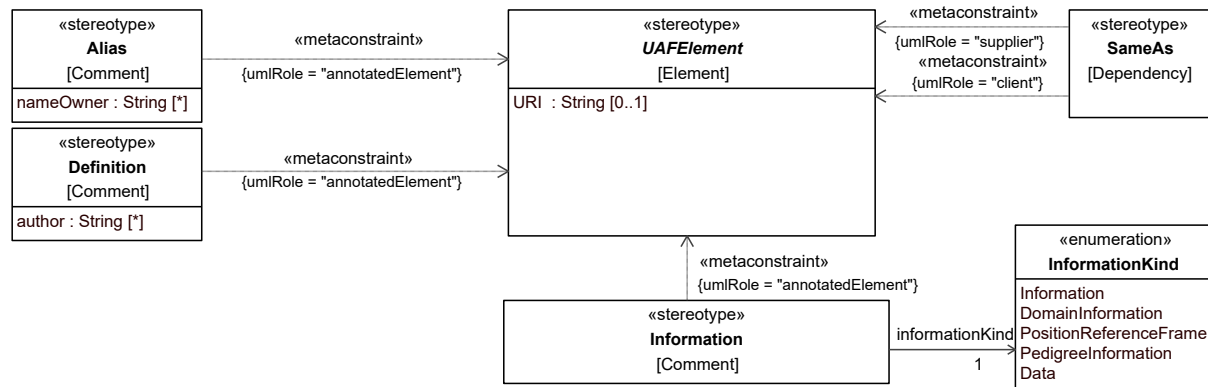


Figure 4:68 – Dictionary

Elements

- [Alias](#)
- [Definition](#)
- [Information](#)
- [InformationKind](#)
- [SameAs](#)
- [UAFEElement](#)

4.1.11 View Specifications::Requirements

View Specifications::Requirements::Requirements

Stakeholders: Requirement Engineers, Solution Providers, Systems Engineers, Software Engineers, Systems Architects, Business Architects.

Concerns: provides a central reference for a set of stakeholder needs expressed as requirements, their relationship (via traceability) to more detailed requirements and the solution described by the architecture that will meet those requirements.

Definition: used to represent requirements, their properties, and relationships (trace, verify, satisfy, refine) between each other and to UAF architectural elements.

Recommended Implementation: SysML Requirement Diagram, tabular format, matrix format.

Elements

- [ActualEnterprisePhase](#)
- [ActualOrganizationalResource](#)
- [ArchitecturalDescription](#)
- [ArchitecturalReference](#)
- [Architecture](#)
- [ArchitectureMetadata](#)
- [Concern](#)
- [EnterprisePhase](#)
- [Metadata](#)
- [OperationalArchitecture](#)
- [OrganizationalResource](#)
- [ResourceArchitecture](#)
- [Stakeholder](#)
- [View](#)
- [Viewpoint](#)
- [WholeLifeEnterprise](#)

4.1.13 View Specifications::Information

View Specifications::Information::Information Model

Stakeholders: Data Modelers, Software Engineers, Systems Engineers

Concerns: address the information perspective on operational, service, and resource architectures.

Definition: allows analysis of an architecture's information and data definition aspect, without consideration of implementation specific issues.

Recommended Implementation: SysML Block Definition Diagram.

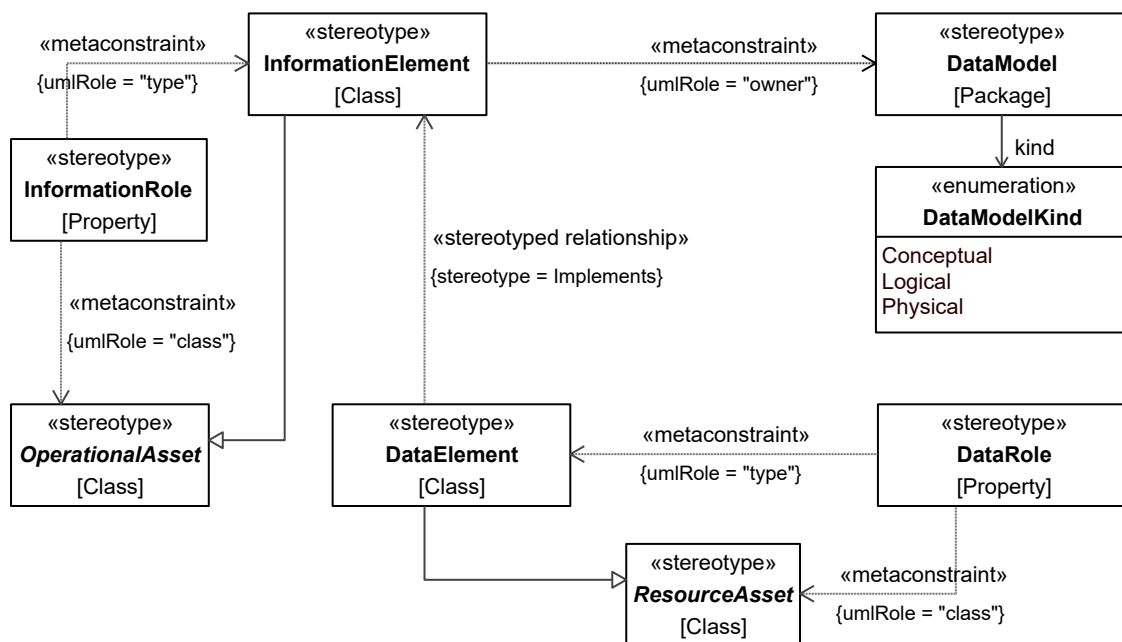


Figure 4:71 - Information Model

Elements

- [DataElement](#)
- [DataModel](#)
- [DataModelKind](#)
- [DataRole](#)
- [InformationElement](#)
- [InformationRole](#)
- [OperationalAsset](#)
- [ResourceAsset](#)

4.1.14 View Specifications::Parameters

Stakeholders: Capability owners, Systems Engineers, Solution Providers.

Concerns: identifies measurable properties that can be used to support engineering analysis and environment for the Capabilities

Definition: Shows the measurable properties of something in the physical world and elements and relationships that are involved in defining the environments applicable to capability, operational concept or set of systems.

View Specifications::Parameters::Parameters: Environment

Stakeholders: Capability owners, Systems Engineers, Solution Providers.

Concerns: defines the environment for the capabilities.

Definition: shows the elements and relationships that are involved in defining the environments applicable to capability, operational concept or set of systems.

Recommended Implementation: SysML Block Definition Diagram.

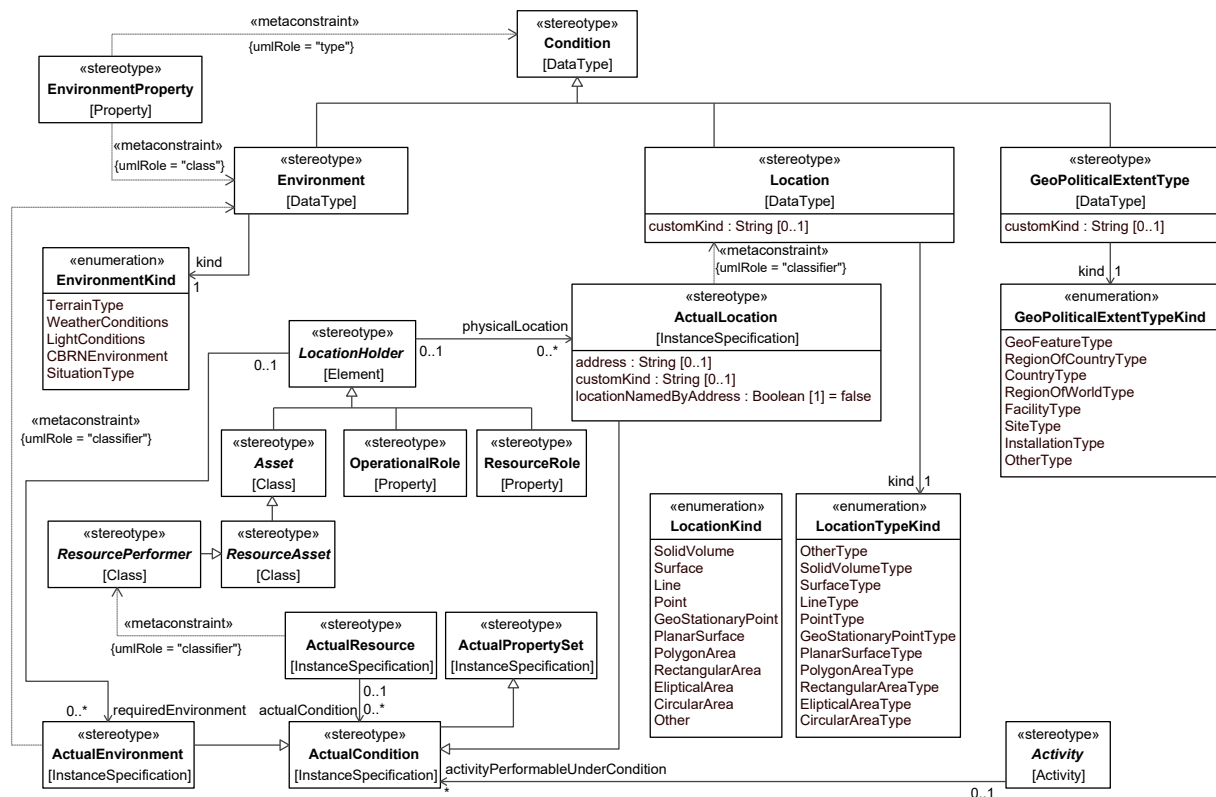


Figure 4:72 - Parameters: Environment

Elements

- [Activity](#)
- [ActualCondition](#)
- [ActualEnvironment](#)
- [ActualLocation](#)
- [ActualPropertySet](#)
- [ActualResource](#)
- [Asset](#)
- [Condition](#)
- [Environment](#)
- [EnvironmentKind](#)
- [EnvironmentProperty](#)
- [GeoPoliticalExtentType](#)
- [GeoPoliticalExtentTypeKind](#)
- [Location](#)
- [LocationHolder](#)
- [LocationKind](#)
- [LocationTypeKind](#)
- [OperationalRole](#)
- [ResourceAsset](#)
- [ResourcePerformer](#)
- [ResourceRole](#)

View Specifications::Parameters::Parameters: Measurements

Stakeholders: Capability owners, Systems Engineers, Solution Providers.

Concerns: identifies measurable properties that can be used to support analysis such as KPIs, MOs, TPIs etc.

Definition: Shows the measurable properties of something in the physical world, expressed in amounts of a unit of measure that can be associated with any element in the architecture.

Recommended Implementation: SysML Block Definition Diagram.

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5 Measurement Library

A library of Measurement Sets.

BillingItem

Package: Class Library

isAbstract: No

Description

Properties indicating the assurance of a piece of information.

Attributes

cost : Cost[1]	Details the cost of the BillingItem.
id : String[0..1]	Details the unique identifier of the BillingItem.
numberOfUses : Integer[0..1]	Details the numberOfUses of the BillingItem.
paymentLocation : String[0..1]	Details the location where payment should be made of the BillingItem.
paymentModality : PricingType[1]	Details if a payment is based upon Quantity, Time or Use.
paymentPeriod : Periodicity[1]	Details the frequency of a payment period.
paymentTimeDuration : Duration[*]	Details the length of time the payments should be made i.e., 1 year.
periodDuration : Duration[0..1]	Details the time period between payments.
quantity : String[0..1]	Details the number of units to be delivered.
unit : String[0..1]	Details the units used for the BillingItem e.g., 1 gross.

ClassificationAttributes

Package: Class Library

isAbstract: No

Description

W3C XML Schema for the Intelligence Community Metadata Standard for Information Security Marking (IC-ISM), which is part of the IC standards for Information Assurance.

Attributes

classificationReason : String[]	One or more reason indicators or explanatory text describing the basis for an original classification decision.
classifiedBy : String[]	Details The identity, by name or personal identifier, and position title of the original classification authority for a resource.
dateOfExemptedSource : String[]	Details the specific year, month, and day of publication or release of a source document, or the most recent source document, that was itself marked with a declassification constraint. This element is always used in conjunction with typeOfExemptedSource element.
declassDate : String[]	Details a specific year, month, and day upon which the information shall be automatically declassified if not properly exempted from automatic declassification.
declassException : String[]	Details a single indicator describing an exemption to the nominal 25-year point for automatic declassification. This element is used in conjunction with the Declassification Date or Declassification Event.
DeclassManualReview : String[]	Details a true/false indicator that a manual review is required for declassification. Use this attribute to force the appearance of "//MR" in the header and footer marking titles. Use this attribute ONLY when it is necessary to override the business logic applied to classification and control markings in the document to determine whether manual review is required.

derivedFrom : String[]	Details a citation of the authoritative source or reference to multiple sources of the classification markings used in a classified resource.
DisseminationControls : String[]	Details one or more indicators identifying the expansion or limitation on the distribution of information.
FGISourceOpen : String[]	Details one or more indicators identifying information which qualifies as foreign government information for which the source(s) of the information is not concealed.
FGISourceProtected : String[]	Details a single indicator that information qualifies as foreign government information for which the source(s) of the information must be concealed. Within protected internal organizational spaces this element may be used to maintain a record of the one or more indicators identifying information which qualifies as foreign government information for which the source(s) of the information must be concealed. Measures must be taken prior to dissemination of the information to conceal the source(s) of the foreign government information.
nonICmarkings : String[]	Details one or more indicators of the expansion or limitation on the distribution of an information resource or portion within the domain of information originating from non-intelligence components.
ownerProducer : String[]	Details one or more indicators identifying the national government or international organization that have purview over the classification marking of an information resource or portion therein. This element is always used in conjunction with the Classification element. Taken together, the two elements specify the classification category and the type of classification (US, non-US, or Joint). Within protected internal organizational spaces this element may include one or more indicators identifying information which qualifies as foreign government information for which the source(s) of the information must be concealed. Measures must be taken prior to dissemination of the information to conceal the source(s) of the foreign government information.
releasableTo : String[]	Details one or more indicators identifying the country or countries and/or international organization(s) to which classified information may be released based on the determination of an originator in accordance with established foreign disclosure procedures. This element is used in conjunction with the Dissemination Controls element.
SARIdentifier : String[]	Details the Authorized Special Access Required (SAR) program digraph(s) or trigraph(s) preceded by "SAR-". Either (a) a single digraph or trigraph or (b) a space-delimited list of digraphs or trigraphs. Example: "SAR-ABC SAR-DEF ..."
SCIControls : String[]	Details one or more indicators identifying sensitive compartmented information control system(s).
typeOfExemptedSource : String[]	Details a declassification marking of a source document that causes the current, derivative document to be exempted from automatic declassification. This element is always used in conjunction with the Date Of Exempted Source element.

Associations

taxonomy : String[]	Details a single indicator of the highest level of classification applicable to an information resource or portion within the domain of classified national security information. The Classification element is always used in conjunction with the Owner Producer element. Taken together, the two elements specify the classification category and the type of classification (US, non-US, or Joint).
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CommunicationsLinkProperties

Package: Class Library

isAbstract: No

Description

Properties detailing aspects of Resource Interfaces.

Attributes

capacity : String[]	Details how much information can be passed on the Communications Link.
infrastructureTechnology : String[]	Details the technology to be used to provide the communications infrastructure.

DataElementProperties

Package: Class Library

isAbstract: No

Description

Properties detailing the aspects of a DataElement.

Attributes

accuracy : String[]	Details the accuracy of the data.
content : String[]	Specifies content of the data element (i.e., actual data to be exchanged).
formatType : String[]	Details the format of the data.
mediaType : String[]	Details the media used to transmit the data.
scope : String[]	Details in text a description of the extent or range of the data element content.
unitOfMeasurement : String[]	Details the units of measurement of the data.

Duration

Package: Class Library

isAbstract: No

Description

Properties detailing aspects OperationalActivities.

Attributes

timeUnit : String[0..1]	Details the units of time e.g. second, hour, day.
value : Integer[0..1]	Details the value of the duration.

ExchangeProperties

Package: Class Library

isAbstract: No

Description

Properties detailing aspects of exchange for Operational Exchange and/or Resource Interaction.

Attributes

accountability : String[*]	Details who or what is responsible for the exchange.
periodicity : String[*]	Details the frequency of the exchange.
size : String[*]	Details the size (in KB) of data that be exchanged.
throughput : String[*]	Details how much information can be exchanged.
timeliness : String[*]	Details the allowable time of delay this system data can tolerate and still be relevant to the receiving system.
transactionType : String[*]	Details the type of transactions used by the exchange.

InformationElementProperties

Package: Class Library

isAbstract: No

Description

Predefined additional DoDAF properties for InformationElement.

Attributes

- | | |
|----------------------|---|
| accuracy : String[*] | Details the degree to which the information conforms to actual fact as required by the information producer and consumer. |
| content : String[*] | Specifies content of the information element (i.e., actual information to be exchanged). |
| language : String[*] | Details the language used to capture the information. |
| scope : String[*] | Details in text a description of the extent or range of the information element content. |

OperationalActivityProperties

Package: Class Library

isAbstract: No

Description

Properties detailing aspects OperationalActivities.

Attributes

- | | |
|-----------------|----------------------------------|
| cost : String[] | Details the cost of an activity. |
|-----------------|----------------------------------|

Periodicity

Package: Class Library

isAbstract: No

Description

Enumeration of how often the information exchange occurs; may be an average or a worst-case estimate and may include conditions. Its enumeration literals are:

- OnceAMonth - Indicates that an event of some sort may occur monthly.
- OnceAWeek - Indicates that an event of some sort may occur weekly.
- Anytime - Indicates that an event of some sort may occur at anytime.
- OnRequest - Indicates that an event of some sort may occur on request.

PricingType

Package: Class Library

isAbstract: No

Description

Enumeration of a unit of measure of a resource. Its enumeration literals are:

- perTime - Indicates that the unit of measure of a resource is based on a unit of time.
- perUse - Indicates that the unit of measure of a resource is based upon how often the resource is used.
- perQuantity - Indicates that the unit of measure of a resource is based on a quantity.

SecurityControlAssessmentProperties

Package: Class Library

isAbstract: No

Description

Properties detailing aspects of the Assessment and Authorization process.

Attributes

- | | |
|---|--|
| coverageOfSecurityControlAssessment : String[*] | Security controls assessment method that addresses the scope or breadth of the assessment objects included in the assessment (e.g., types of objects to be assessed and the number of objects to be assessed by type). |
|---|--|

depthOfSecurityControlAssessment : String[*]	Security controls assessment method that addresses the rigor and level of detail associated with the application of the method.
effectivenessOfSecurityControl : String[*]	Details if security control is satisfactory or not as assessed.

SecurityControlProperties

Package: Class Library

isAbstract: No

Description

Properties detailing aspects of Security Controls.

Attributes

securityControlApplicability : String[1]	Details how applicable a security control is to a given security objective.
securityControlImportance : String[1]	Details how important a security control is to a given security objective.

SecurityImpactProperties

Package: Class Library

isAbstract: No

Description

Properties detailing aspects of Security Categories.

Attributes

securityAvailabilityImpact : String[*]	Details the potential impact on organization or individuals if the information is not available to those who need to access it.
securityClassification : String[*]	Details a classification for the exchange.
securityConfidentialityImpact : String[*]	Details the potential impact on organization or individuals if the information is not available to those who need to access it.